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(54) **FLUID END ASSEMBLY**

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Related U.S. Application Data

(63) Continuation of application No. 18/933,692, filed on Oct. 31, 2024, now Pat. No. 12,345,254, which is a continuation-in-part of application No. 18/490,864, filed on Oct. 20, 2023, now Pat. No. 12,135,024, which is a continuation-in-part of application No. 17/884,712, filed on Aug. 10, 2022, now Pat. No. 12,188,458, which is a continuation-in-part of application No. 17/550,552, filed on Dec. 14, 2021, now Pat. No. 11,644,018, which is a continuation-in-part of application No. 17/515,707, filed on Nov. 1, 2021, now Pat. No. 11,359,615, which is a continuation of application No. 16/951,741, filed on Nov. 18, 2020, now Pat. No. 11,162,479, which is a continuation-in-part of application No. 18/488,435, filed on Oct. 17, 2023, now Pat. No. 12,000,257.

(60) Provisional application No. 62/936,789, filed on Nov. 18, 2019, provisional application No. 62/940,513, filed on Nov. 26, 2019, provisional application No. 62/953,763, filed on Dec. 26, 2019, provisional application No. 62/957,489, filed on Jan. 6, 2020, provisional application No. 62/959,570, filed on Jan. 10, 2020, provisional application No. 62/960,194, filed on Jan. 13, 2020, provisional application No.

(Continued)

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F04B 1/0452 (2020.01)

F04B 1/0538 (2020.01)

F04B 7/00 (2006.01)

F04B 7/02 (2006.01)

F04B 15/02 (2006.01)

F04B 17/00 (2006.01)

F04B 19/04 (2006.01)

F04B 19/22 (2006.01)

F04B 53/00 (2006.01)

F04B 53/10 (2006.01)

F04B 53/14 (2006.01)

F04B 53/16 (2006.01)

F04B 53/18 (2006.01)

F16J 15/26 (2006.01)

F16K 1/42 (2006.01)

F16K 11/10 (2006.01)

F16L 55/11 (2006.01)

(52) **U.S. Cl.**

CPC **F04B 53/22** (2013.01); **F04B 1/0421**

(2013.01); **F04B 15/02** (2013.01); **F04B**

53/007 (2013.01); **F04B 53/1022** (2013.01);

F04B 53/1032 (2013.01); **F04B 53/14**

(2013.01); **F04B 53/164** (2013.01); **F16J**

15/26 (2013.01); **F16K 1/42** (2013.01); **F16K**

11/105 (2013.01); **E21B 43/2607** (2020.05);

F04B 1/0452 (2013.01); **F04B 1/0538**

(2013.01); **F04B 7/0003** (2013.01); **F04B**

7/0084 (2013.01); **F04B 7/0088** (2013.01);

F04B 7/0208 (2013.01); **F04B 7/0266**

(2013.01); **F04B 17/00** (2013.01); **F04B 19/04**

(2013.01); **F04B 19/22** (2013.01); **F04B**

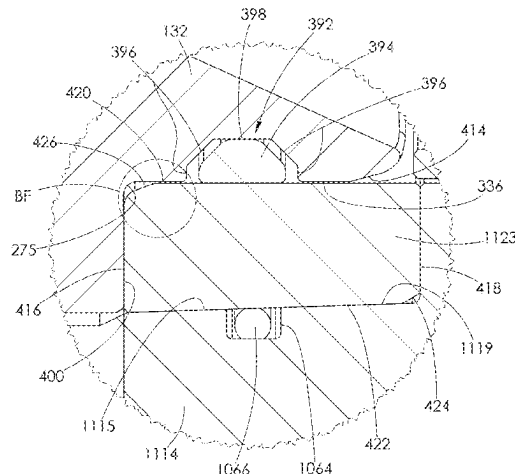
53/006 (2013.01); **F04B 53/18** (2013.01);

F16L 55/11 (2013.01)

(57)

ABSTRACT

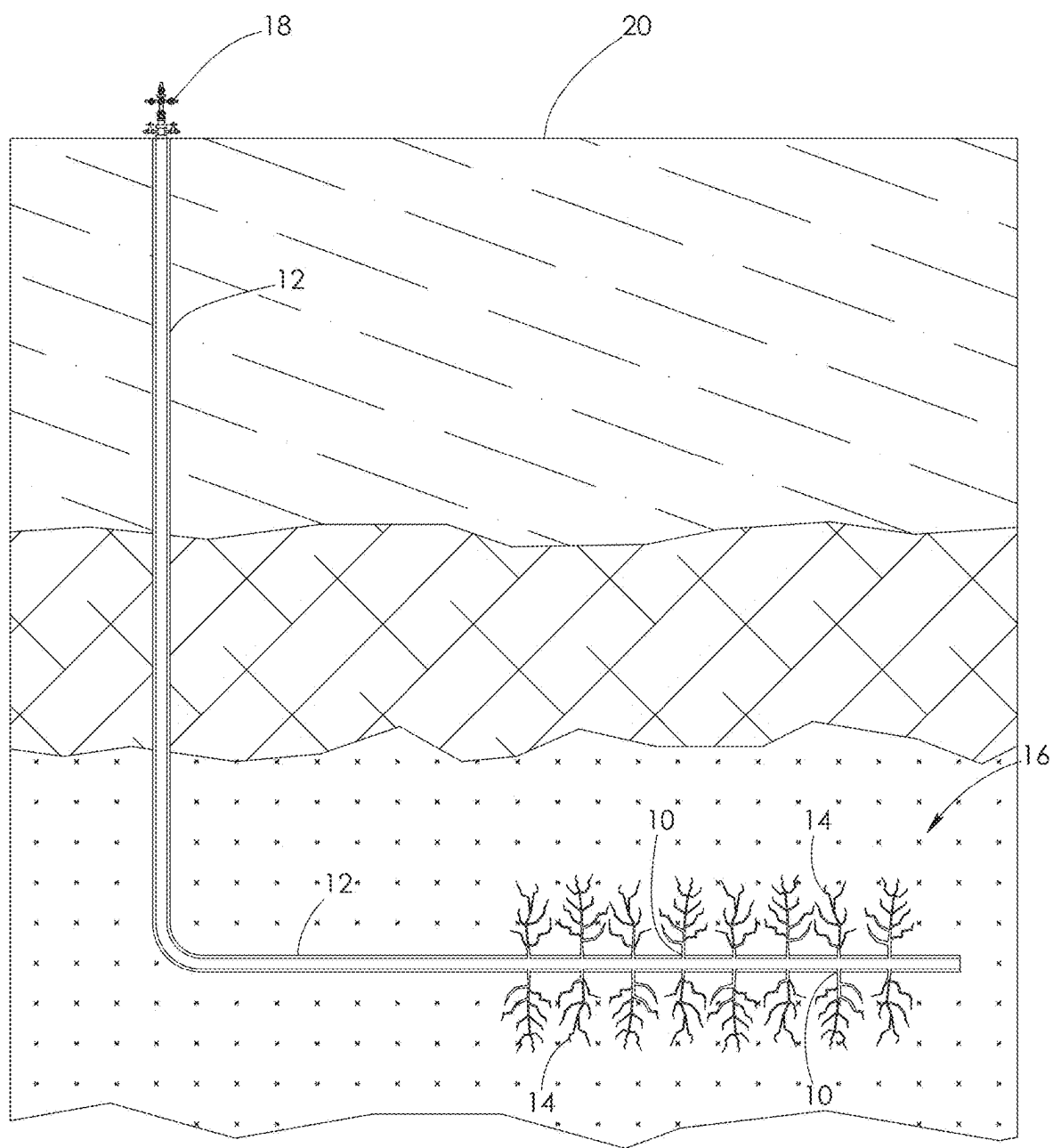
A fluid end assembly having a plurality of fluid end sections positioned in a side-by-side relationship. Each fluid end section has a housing made of multiple-piece construction. One or more pieces of the housing are configured to have a plurality of stay rods attached thereto. The stay rods interconnect the fluid end assembly and a power end assembly.



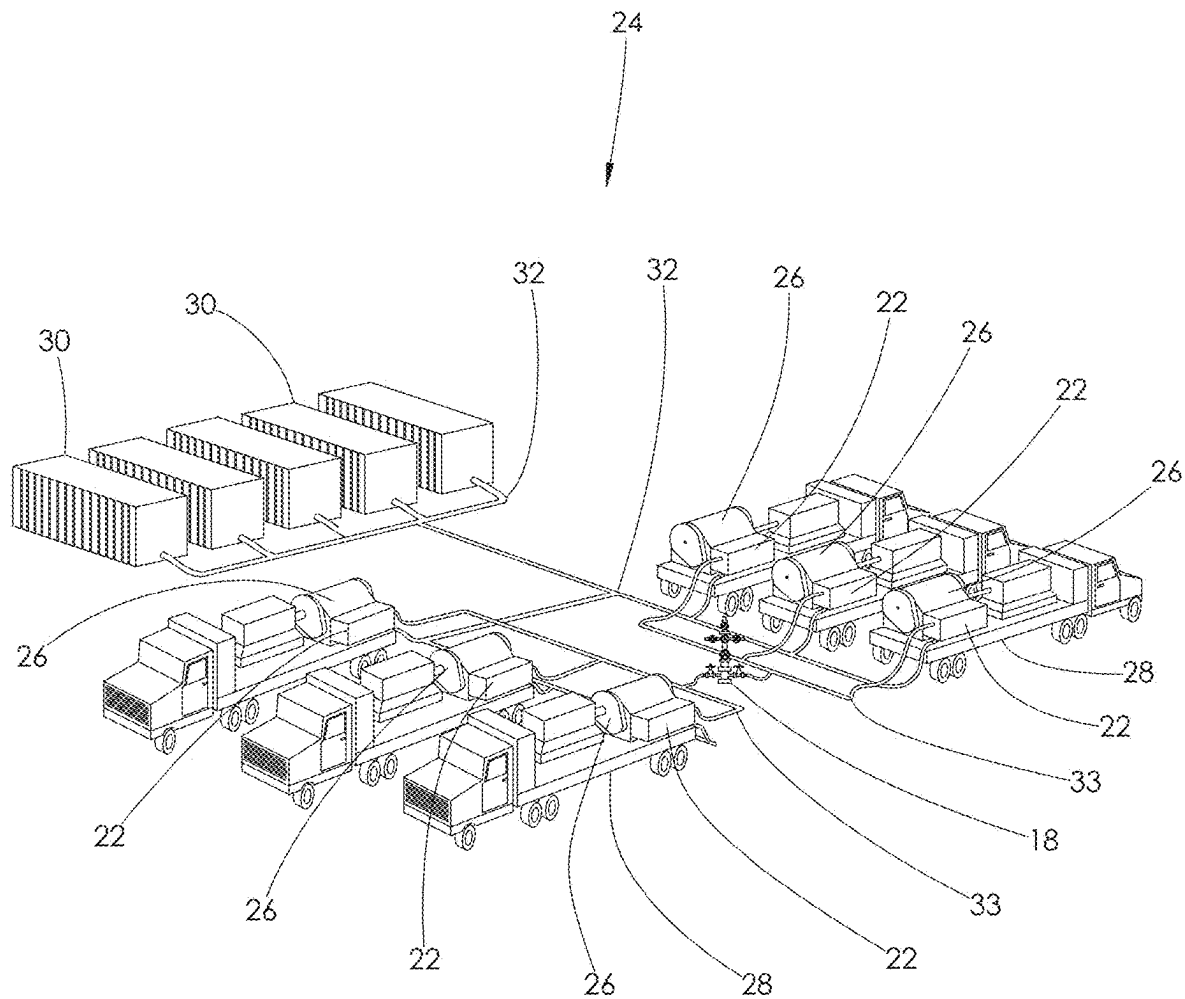
Related U.S. Application Data

62/960,366, filed on Jan. 13, 2020, provisional application No. 62/968,634, filed on Jan. 31, 2020, provisional application No. 62/990,817, filed on Mar. 17, 2020, provisional application No. 63/008,036, filed on Apr. 10, 2020, provisional application No. 63/018,021, filed on Apr. 30, 2020, provisional application No. 63/019,789, filed on May 4, 2020, provisional application No. 63/027,584, filed on May 20, 2020, provisional application No. 63/033,244, filed on Jun. 2, 2020, provisional application No. 63/040,086, filed on Jun. 17, 2020, provisional application No. 63/046,826, filed on Jul. 1, 2020, provisional application No. 63/053,797, filed on Jul. 20, 2020, provisional application No. 63/076,587, filed on Sep. 10, 2020, provisional application No. 63/089,882, filed on Oct. 9, 2020, provisional application No. 63/125,459, filed on Dec. 15, 2020, provisional application No. 63/148,065, filed on Feb. 10, 2021, provisional application No. 63/150,340, filed on Feb. 17, 2021, provisional application No. 63/155,835, filed on Mar. 3, 2021,

provisional application No. 63/168,364, filed on Mar. 31, 2021, provisional application No. 63/233,241, filed on Aug. 14, 2021, provisional application No. 63/235,251, filed on Aug. 20, 2021, provisional application No. 63/240,889, filed on Sep. 4, 2021, provisional application No. 63/246,099, filed on Sep. 20, 2021, provisional application No. 63/283,487, filed on Nov. 28, 2021, provisional application No. 63/301,524, filed on Jan. 21, 2022, provisional application No. 63/304,070, filed on Jan. 28, 2022, provisional application No. 63/309,168, filed on Feb. 11, 2022, provisional application No. 63/310,269, filed on Feb. 15, 2022, provisional application No. 63/312,541, filed on Feb. 22, 2022, provisional application No. 63/416,644, filed on Oct. 17, 2022, provisional application No. 63/422,637, filed on Nov. 4, 2022, provisional application No. 63/422,637, filed on Nov. 4, 2022, provisional application No. 63/506,222, filed on Jun. 5, 2023, provisional application No. 63/506,222, filed on Jun. 5, 2023, provisional application No. 63/508,577, filed on Jun. 16, 2023, provisional application No. 63/508,577, filed on Jun. 16, 2023.



PRIOR ART
FIG. 1



PRIOR ART
FIG. 2

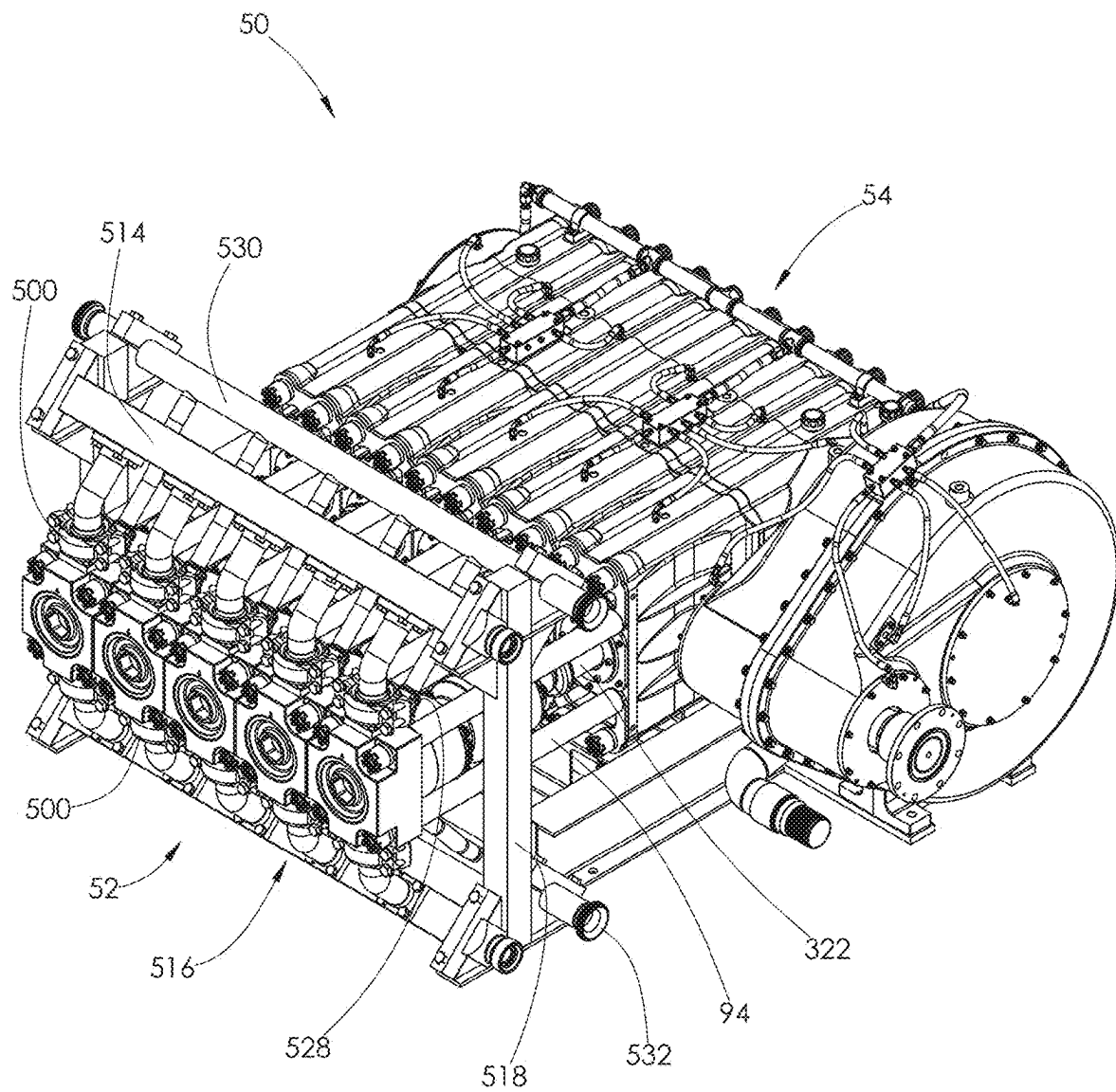


FIG. 3

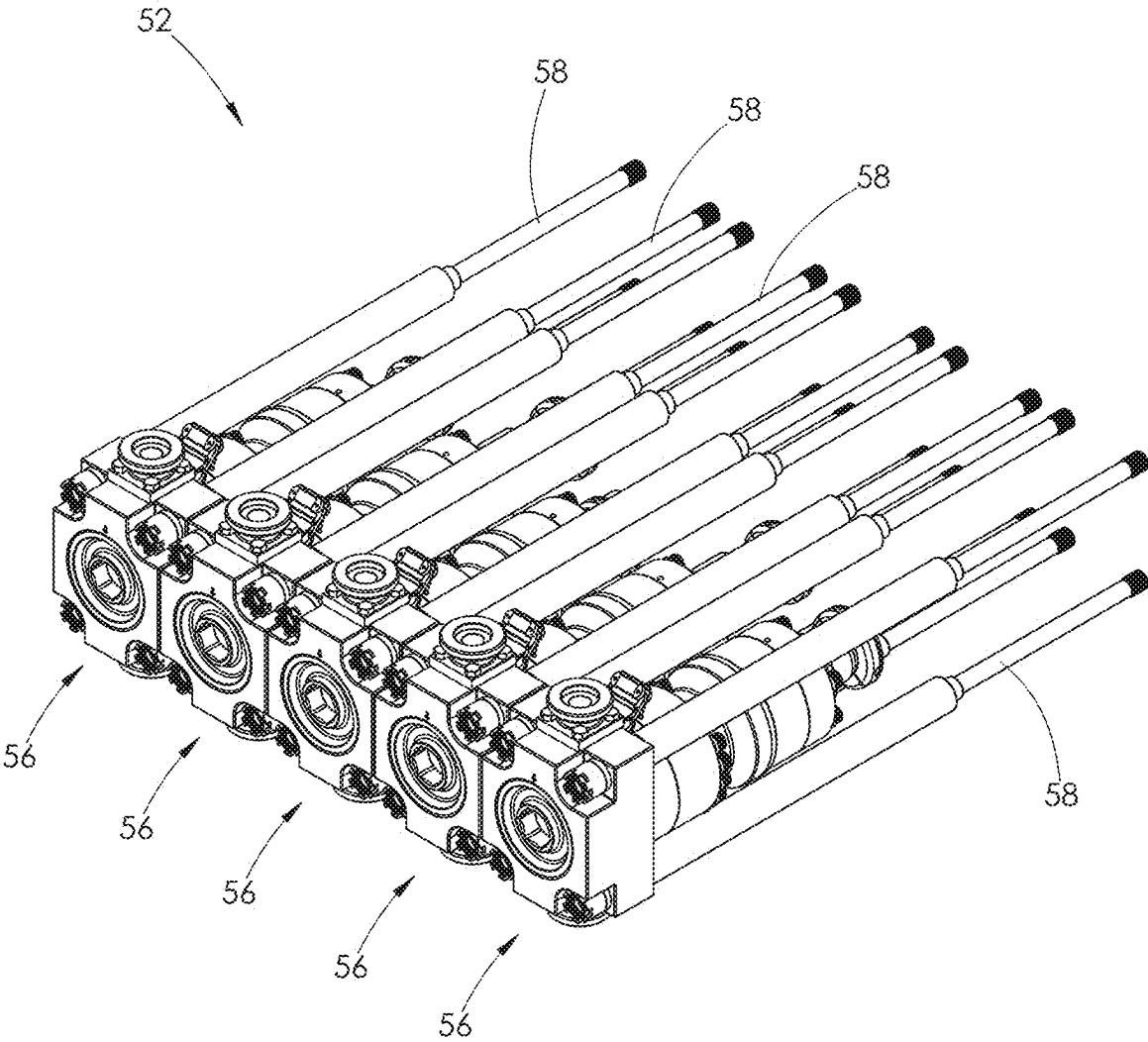


FIG. 4

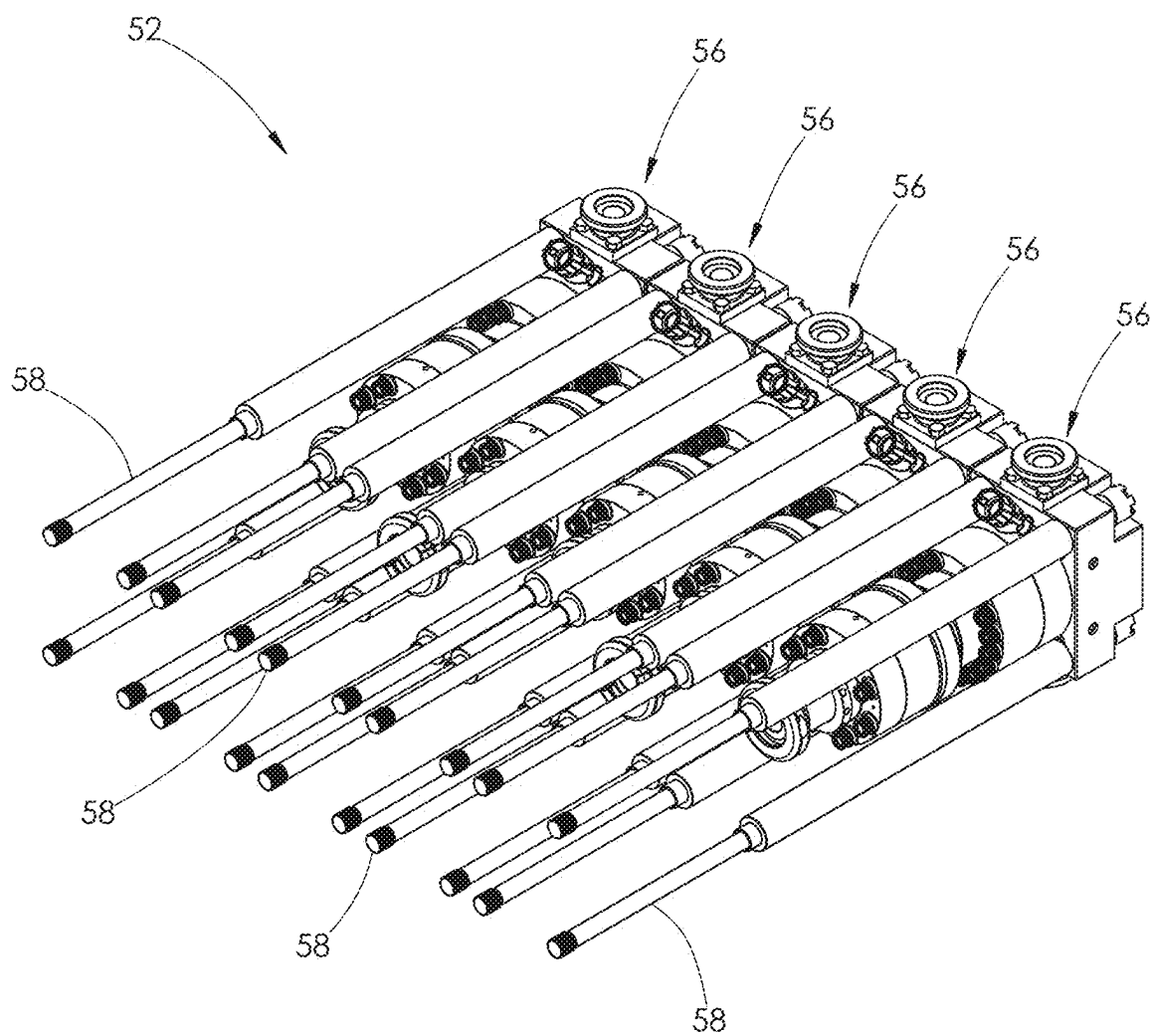


FIG. 5

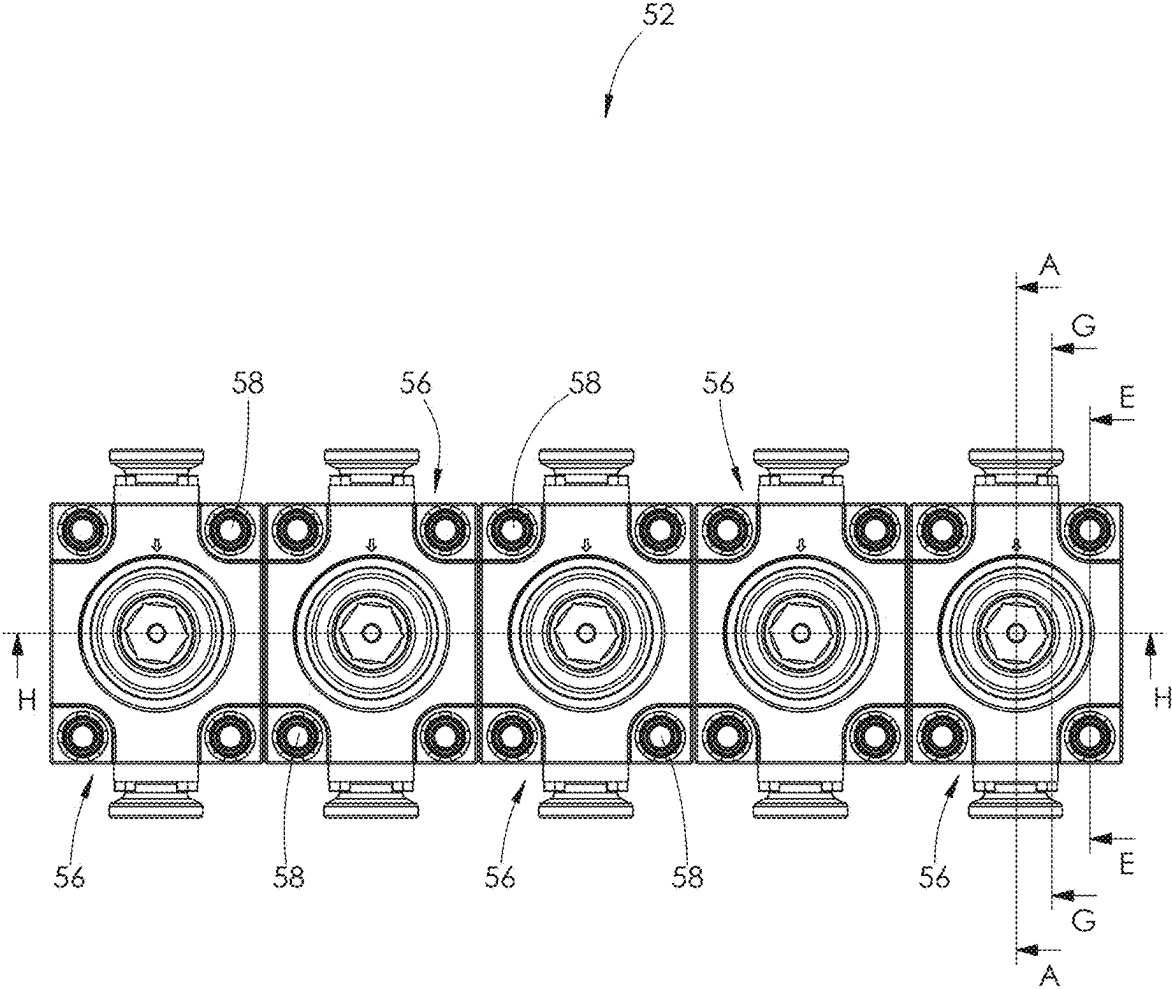


FIG. 6

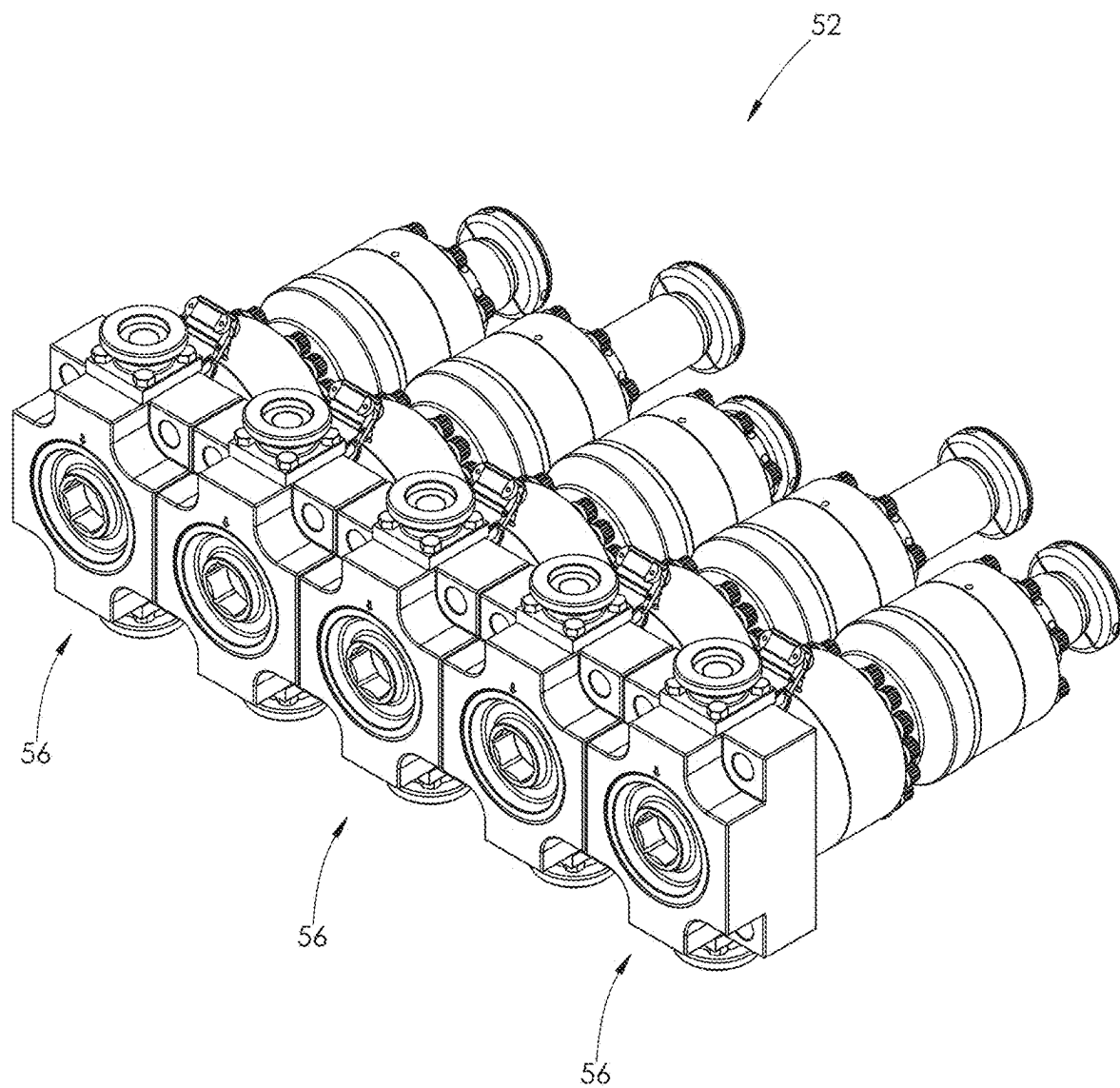


FIG. 7

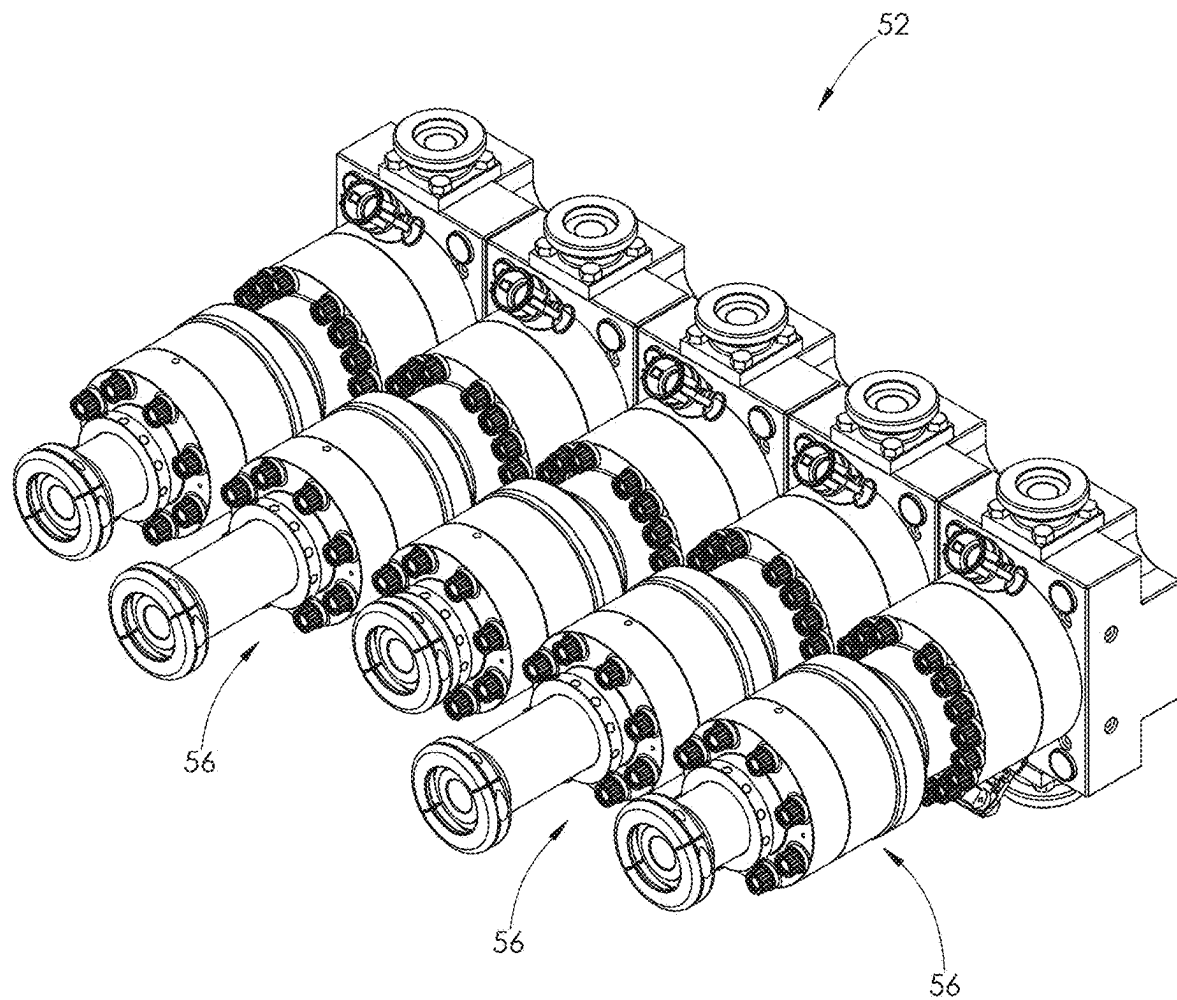


FIG. 8

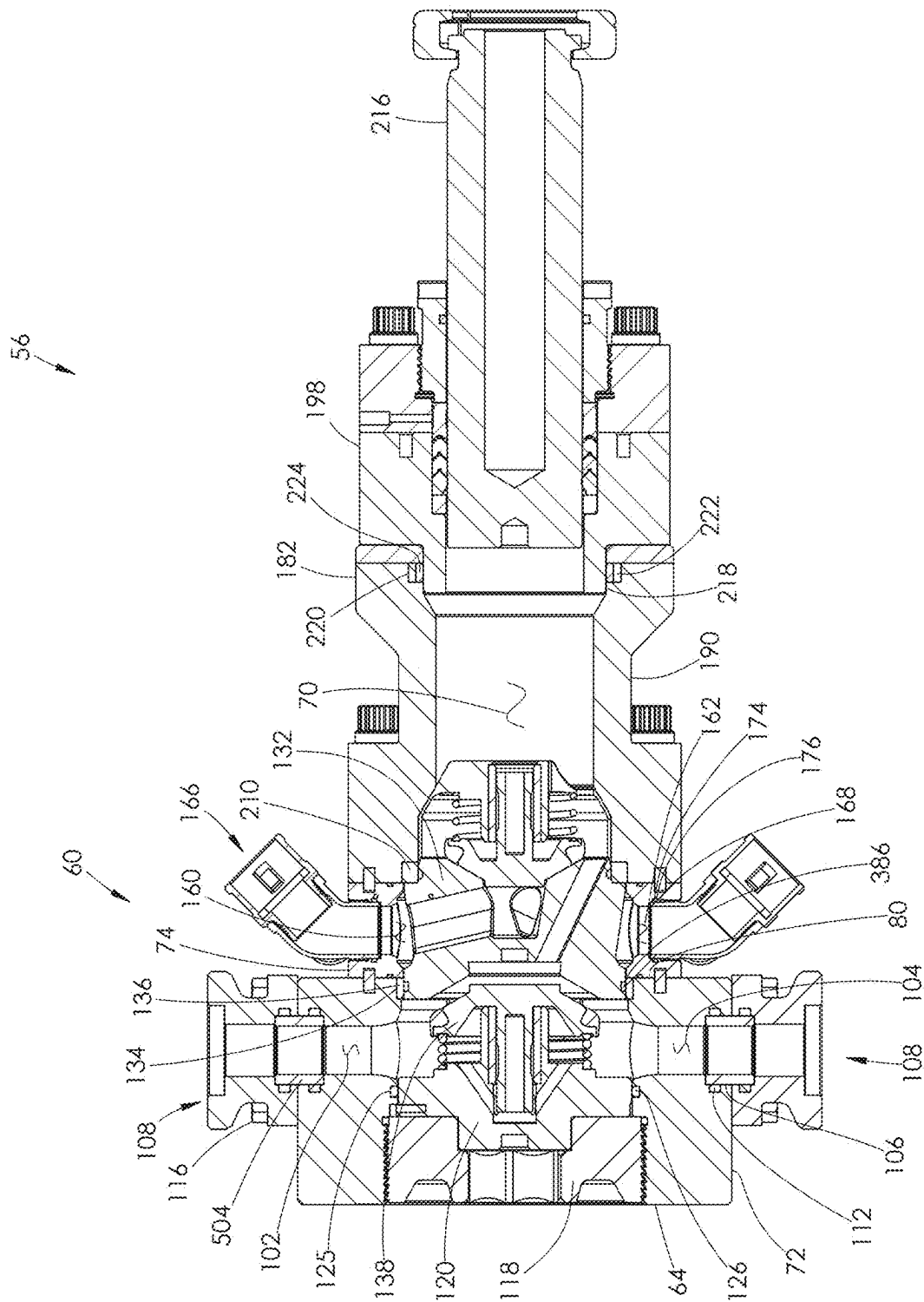


FIG. 9

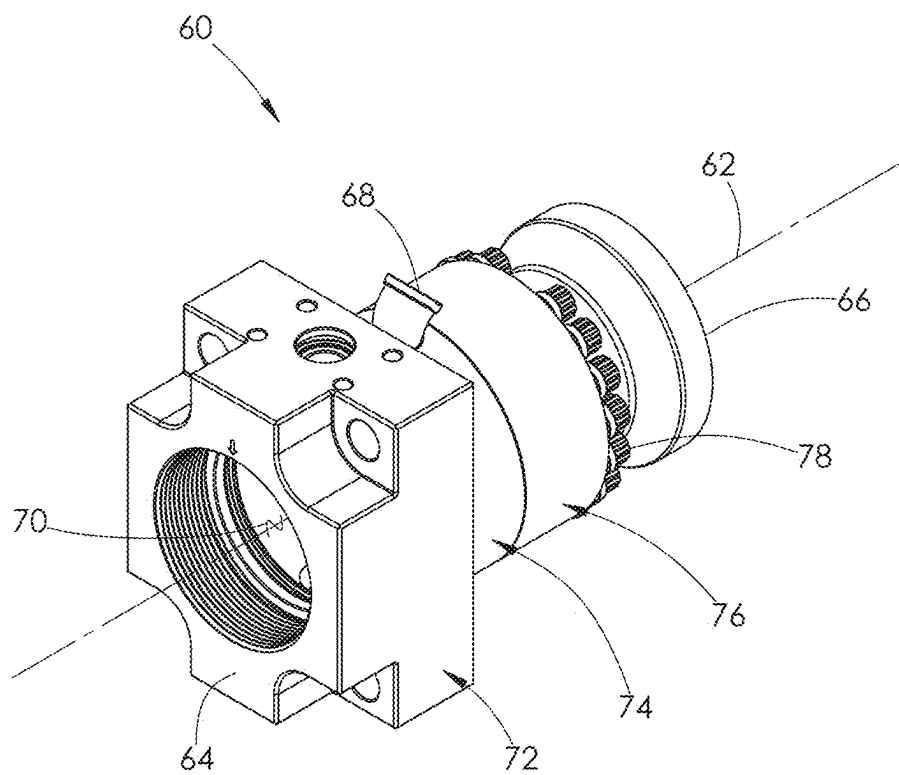


FIG. 10

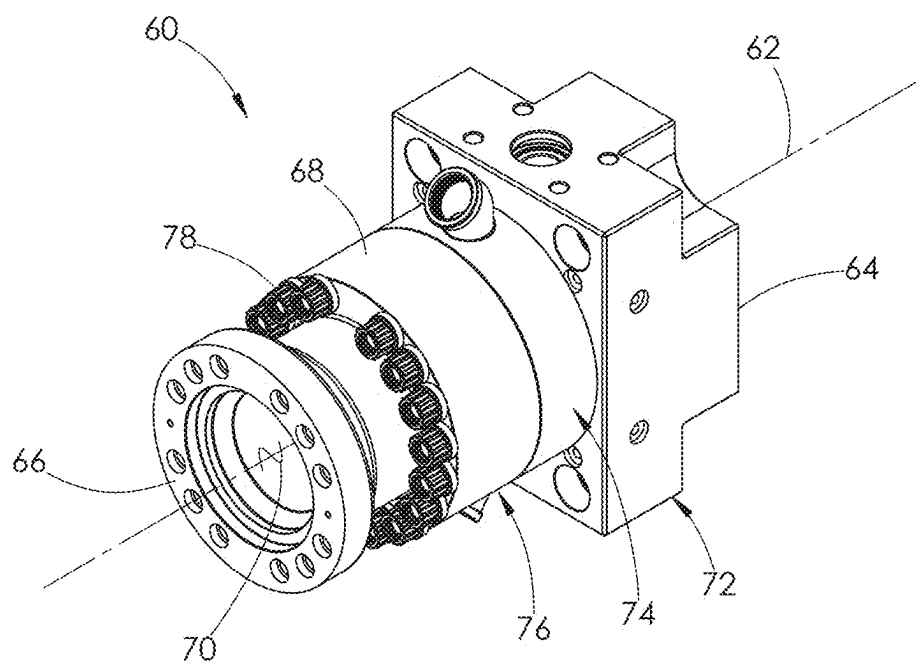


FIG. 11

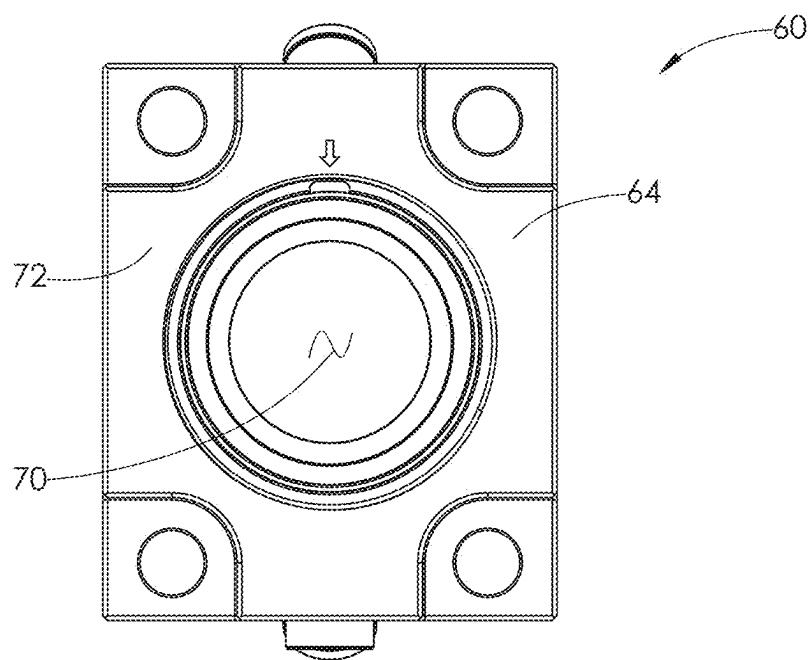


FIG. 12

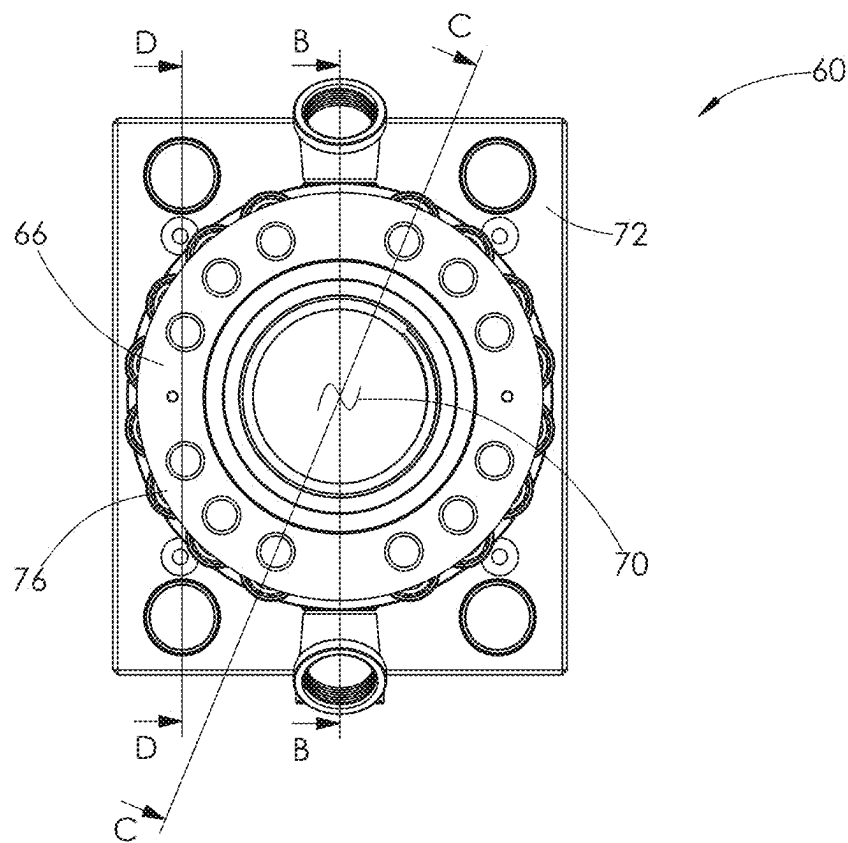


FIG. 13

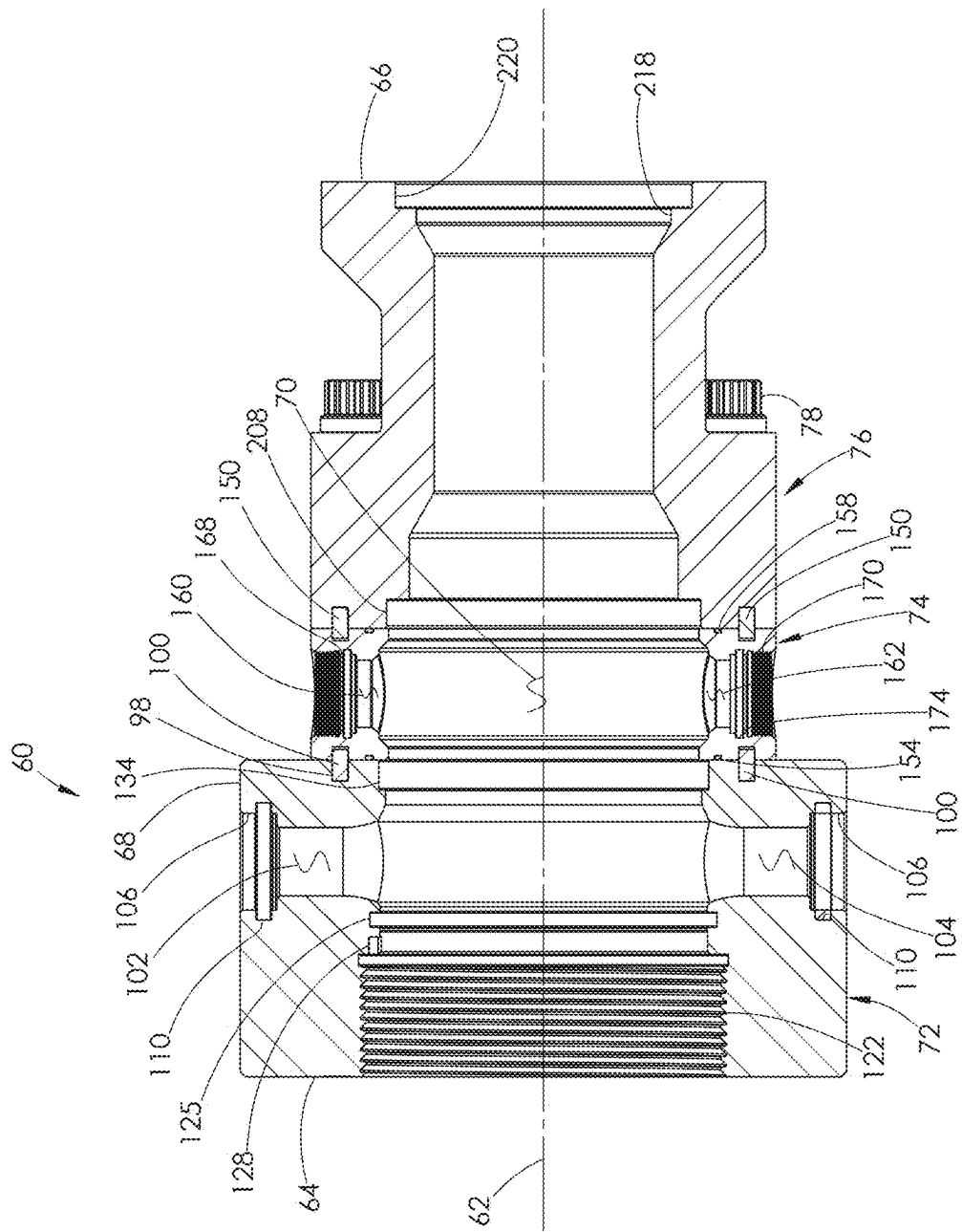


FIG. 14

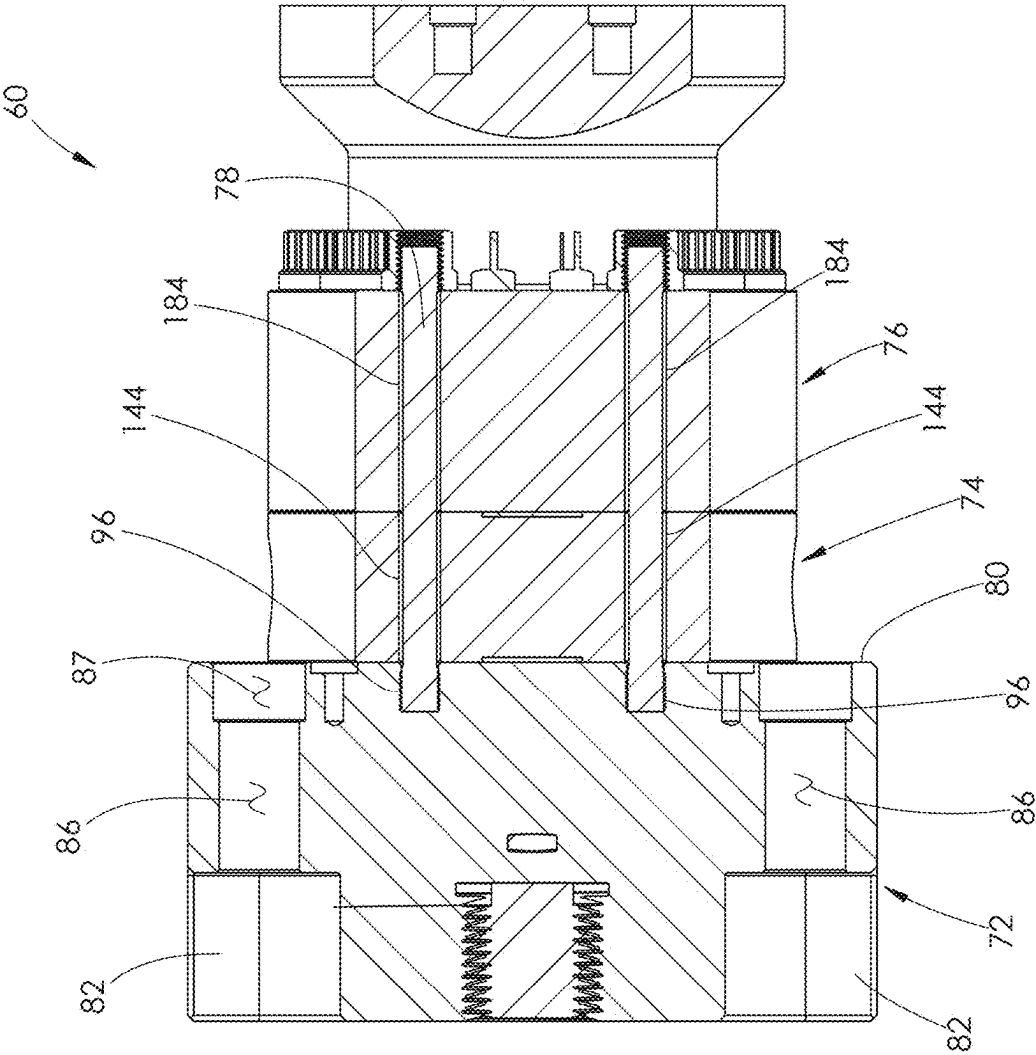


FIG. 16

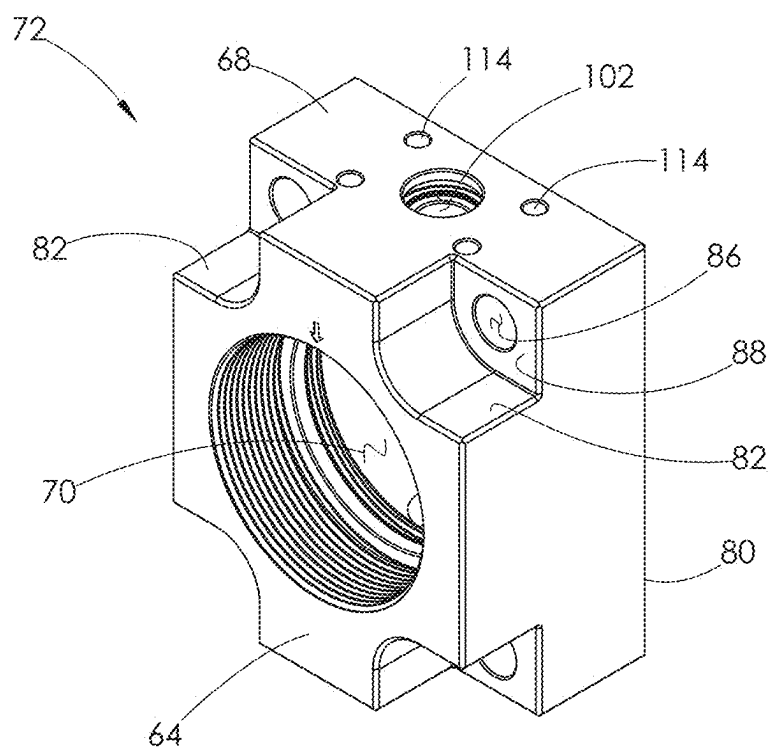


FIG. 17

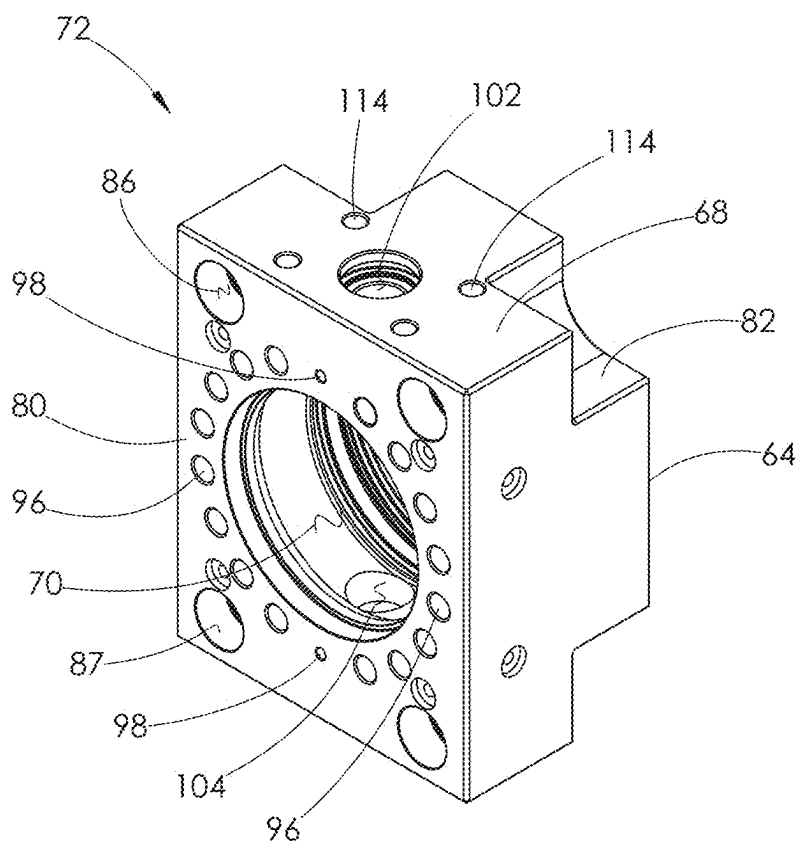
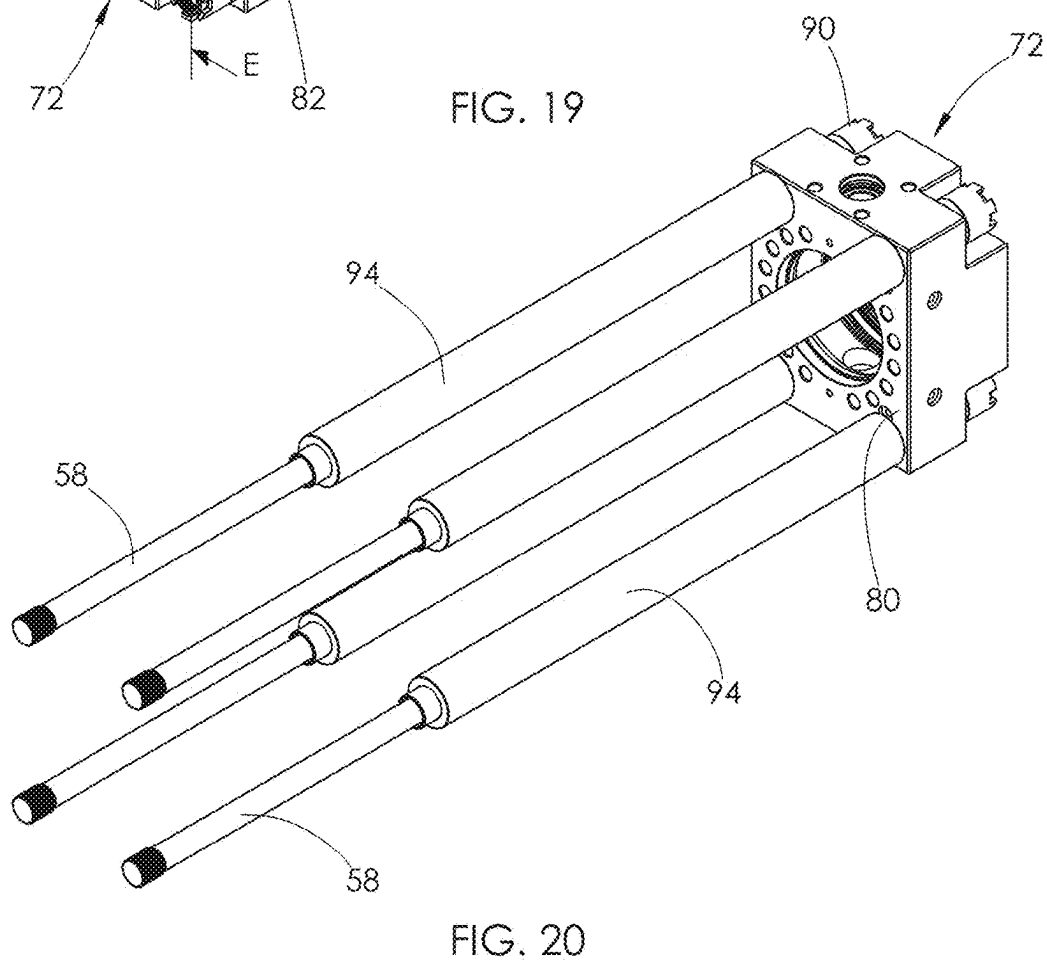
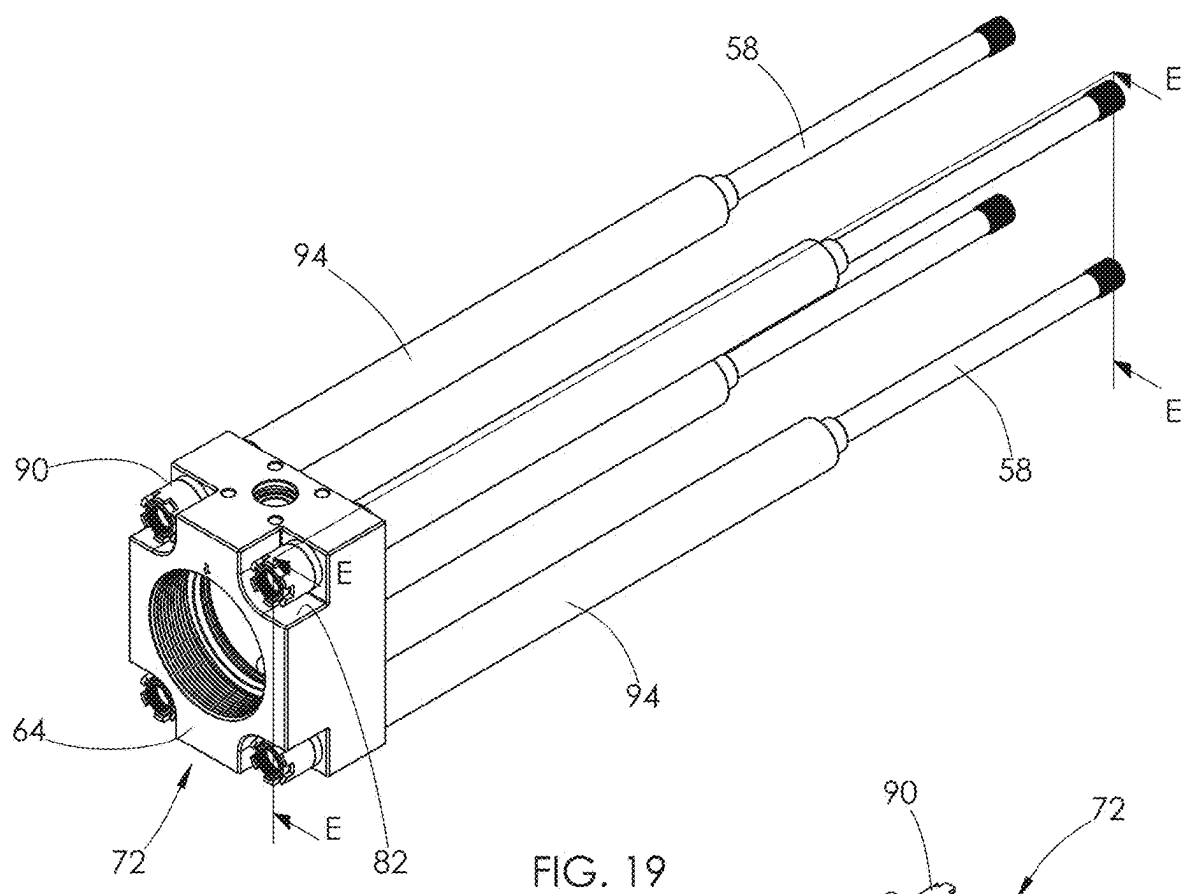


FIG. 18



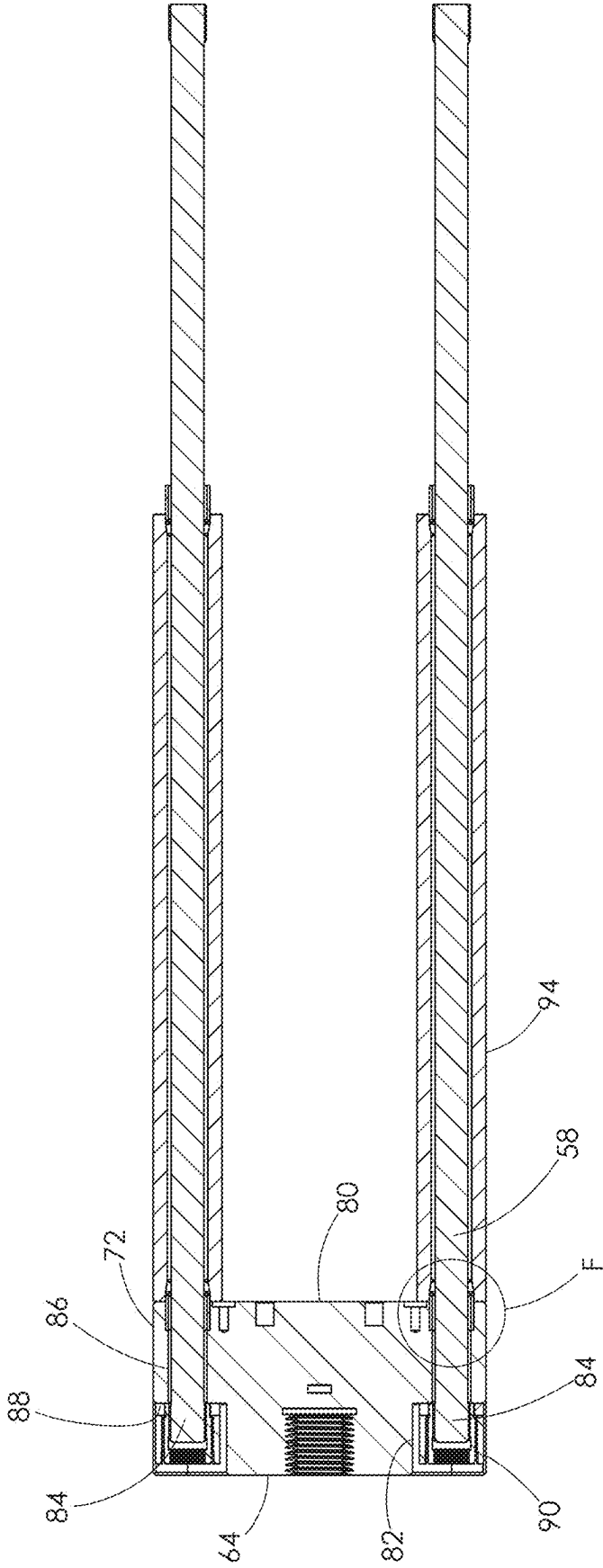


FIG. 21

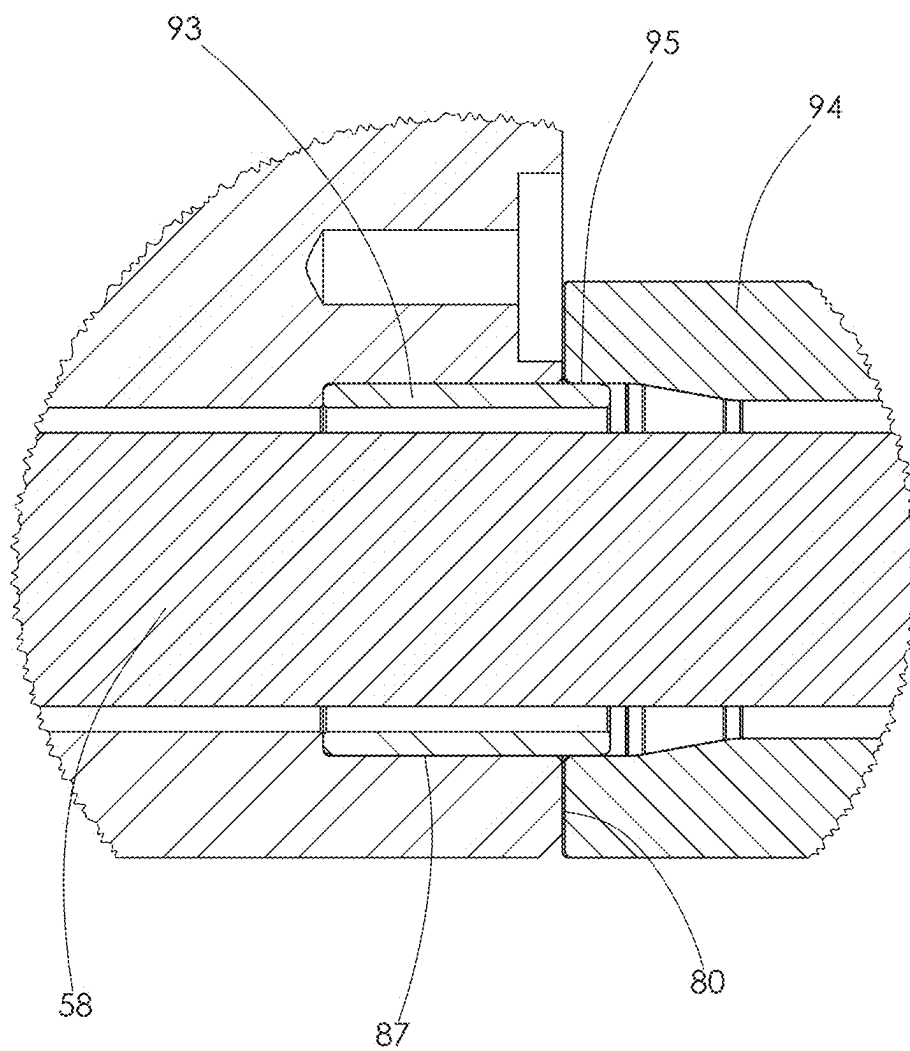
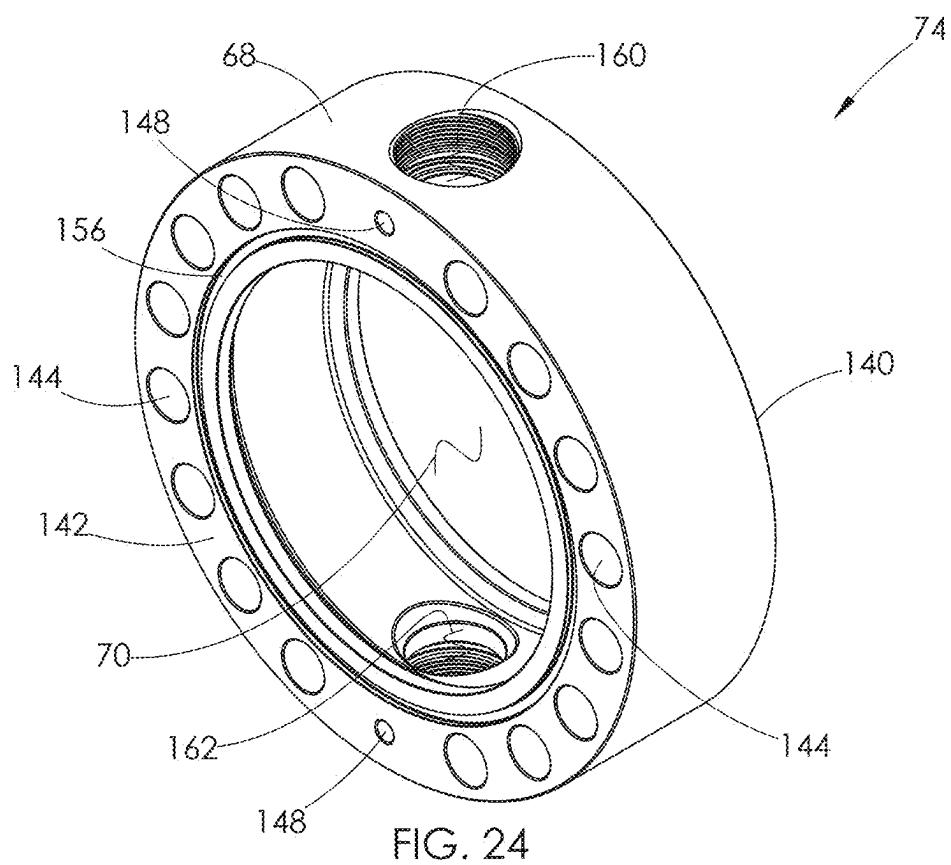
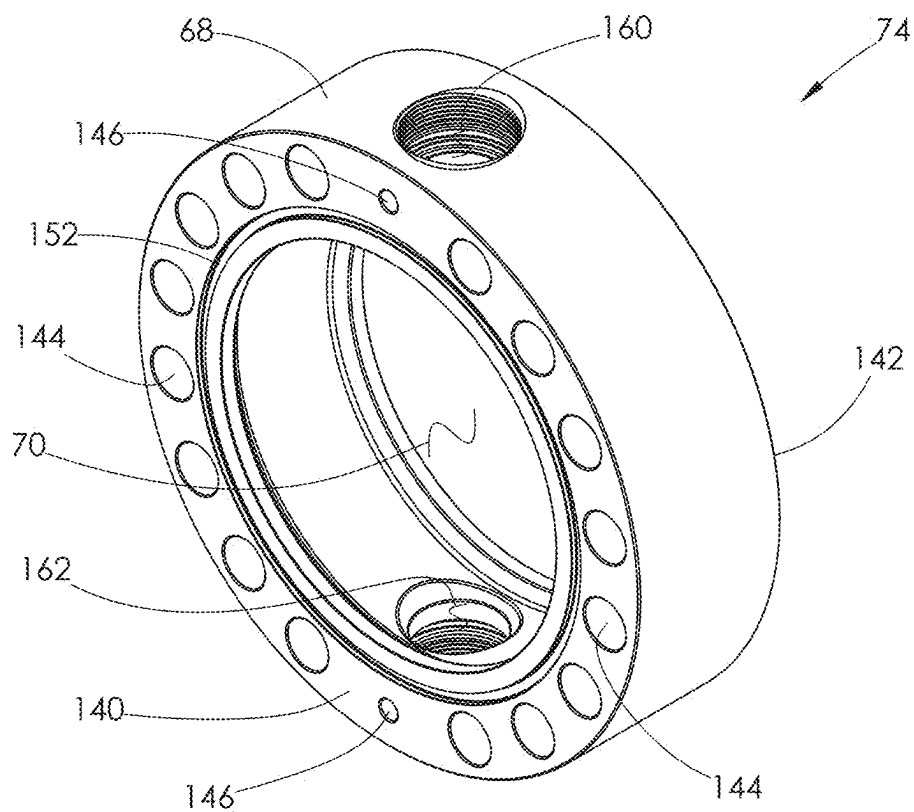


FIG. 22



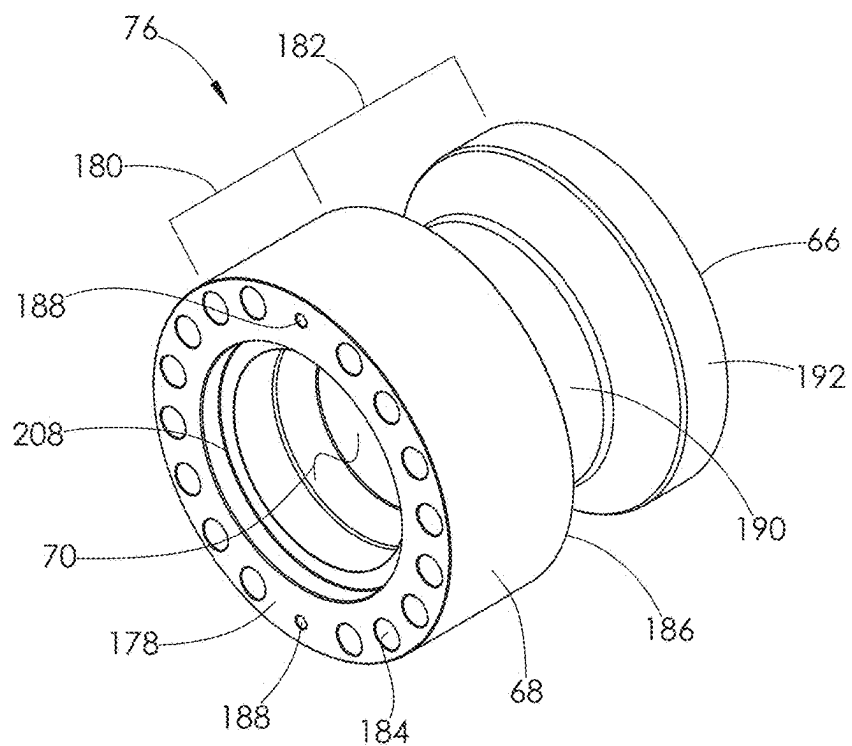


FIG. 25

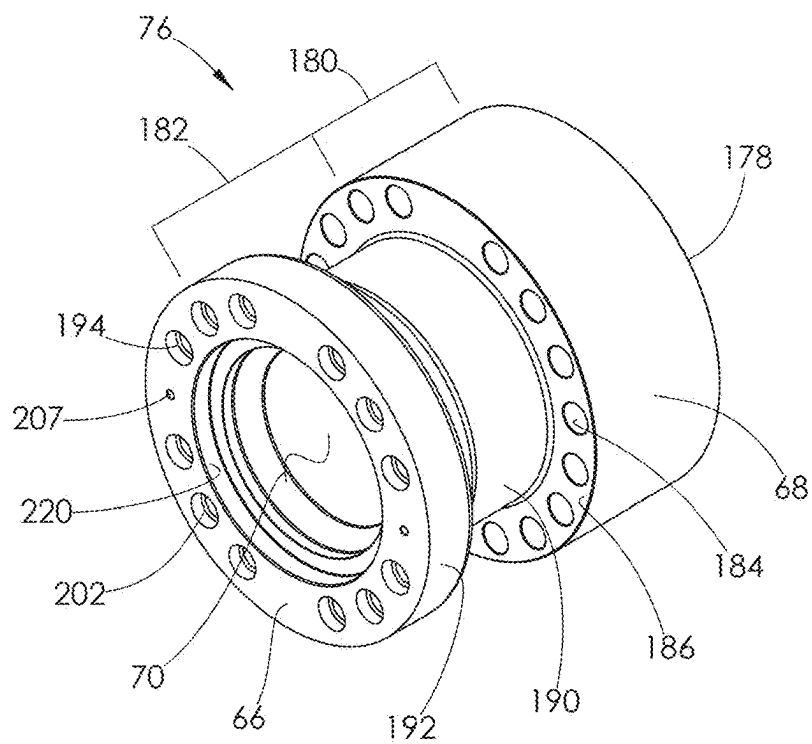


FIG. 26

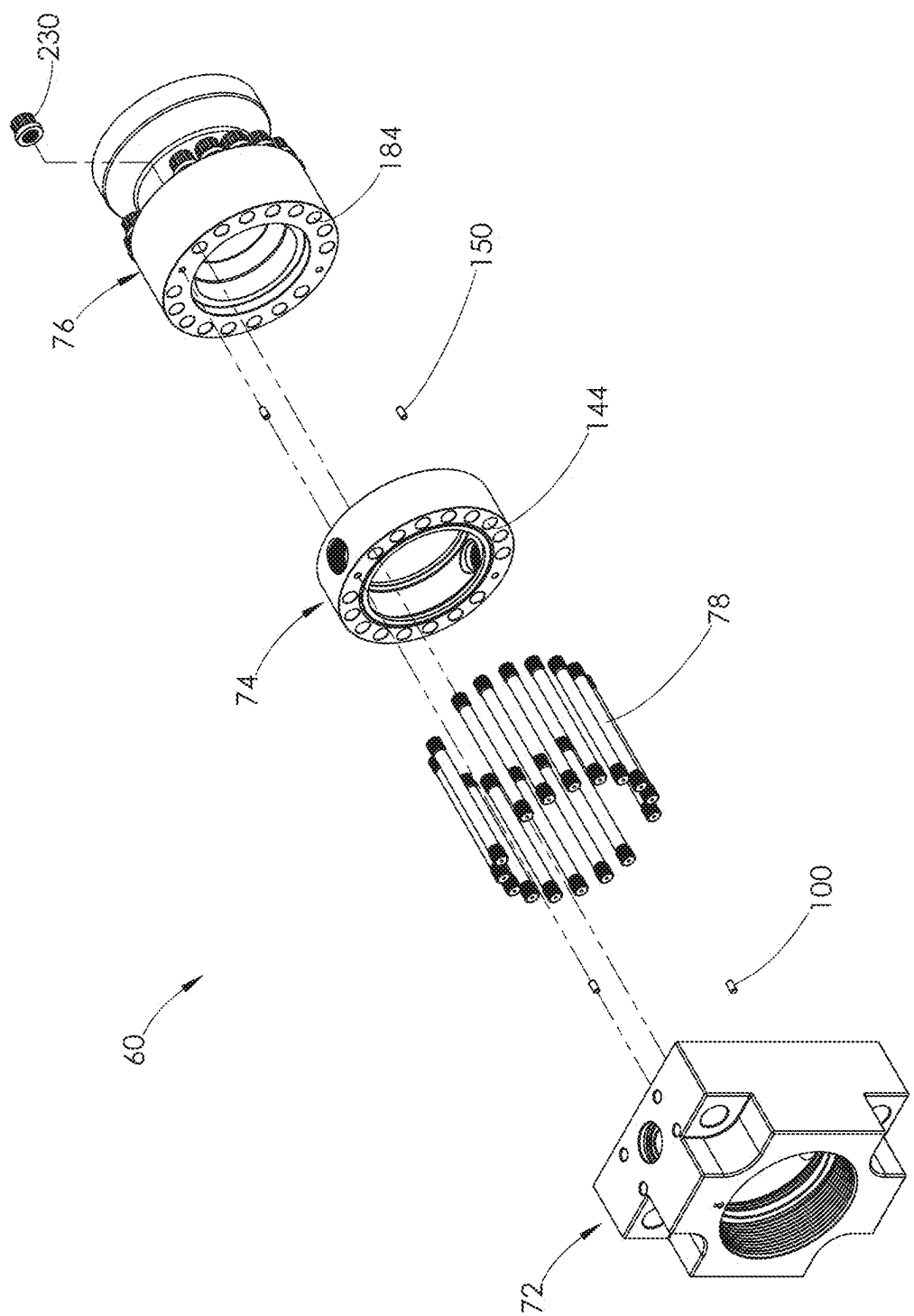


FIG. 27

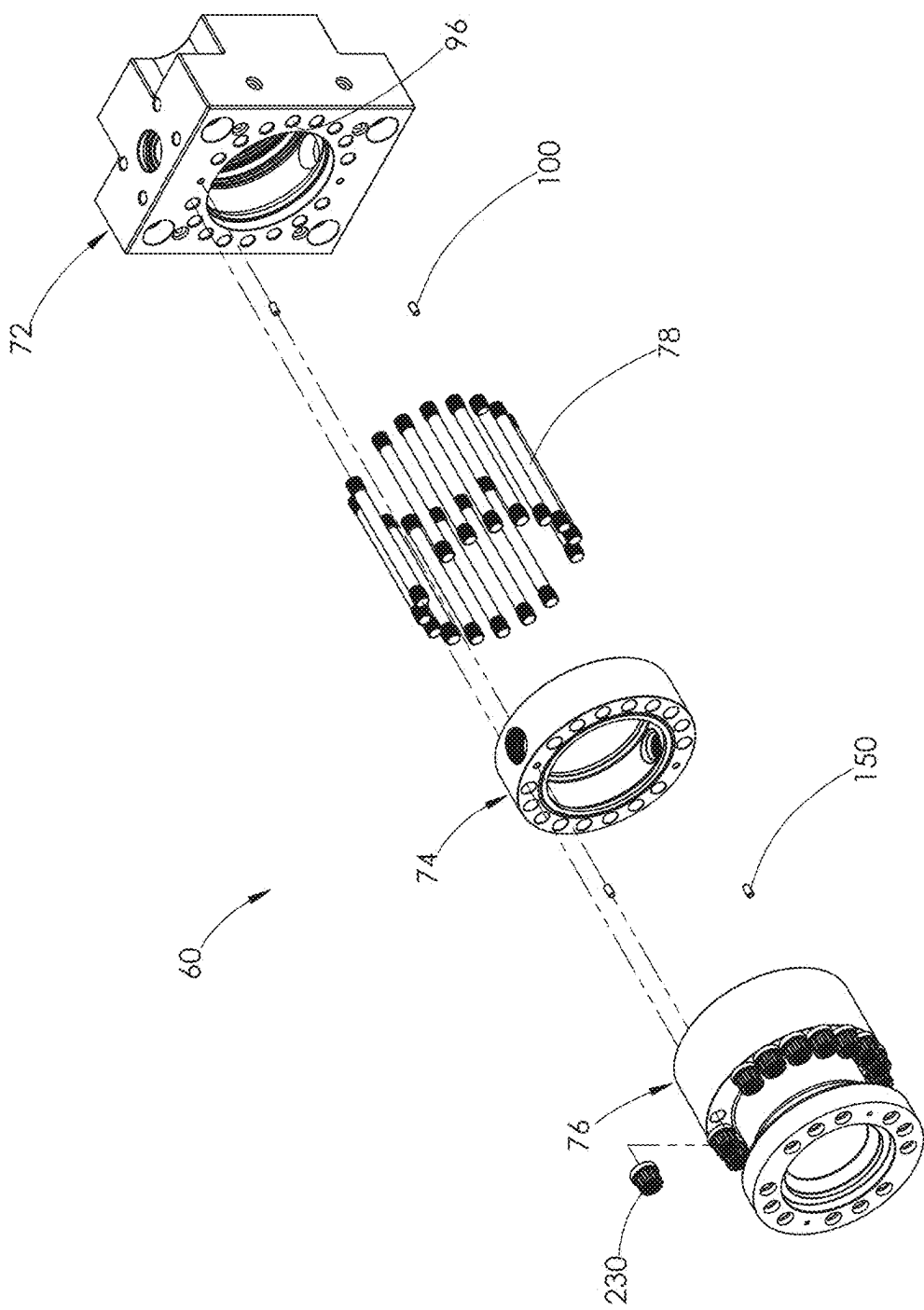


FIG. 28

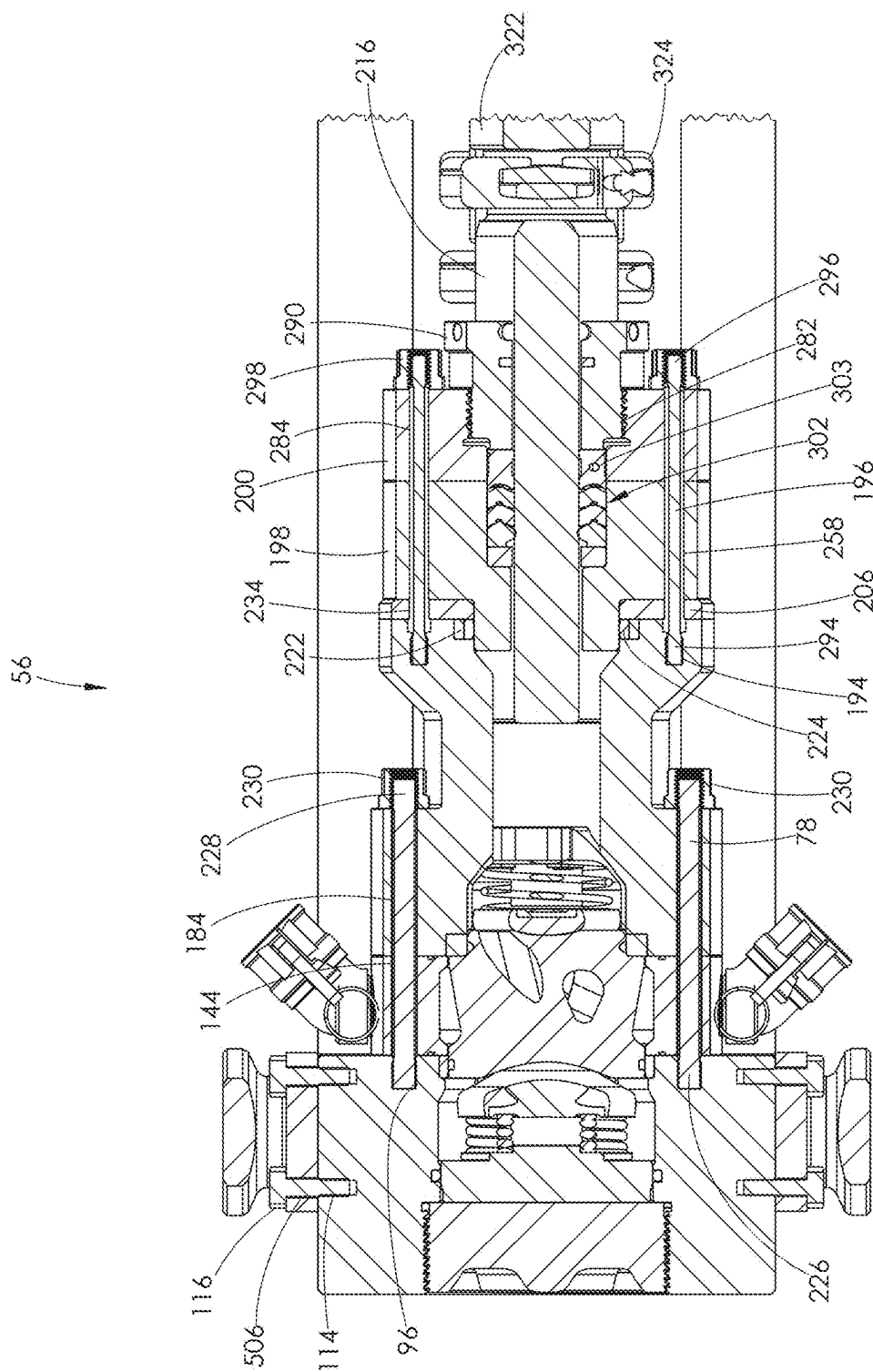


FIG. 29

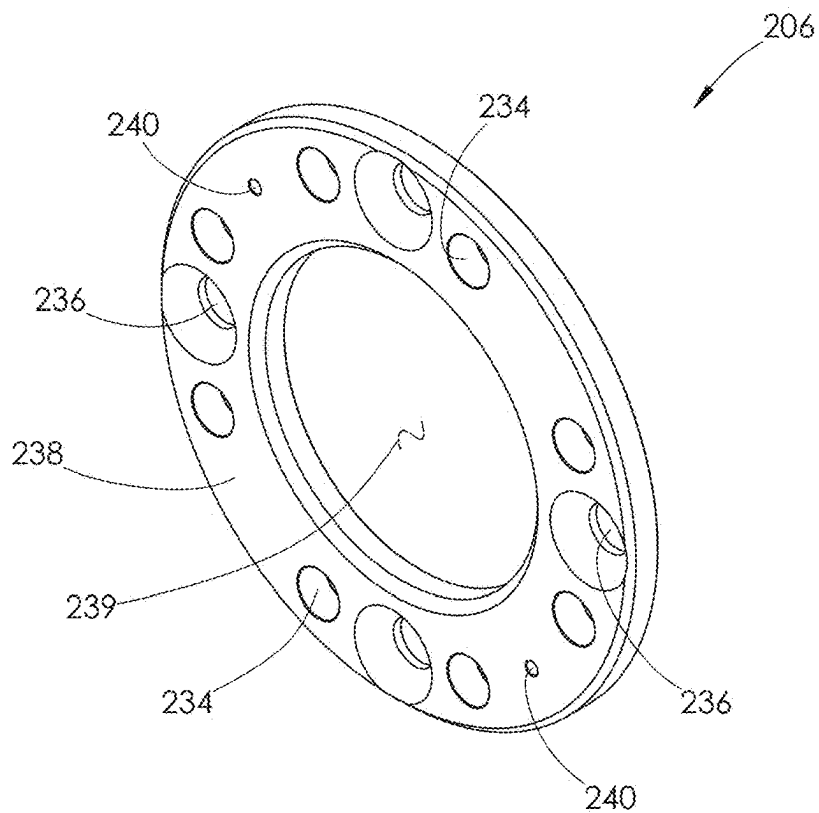


FIG. 30

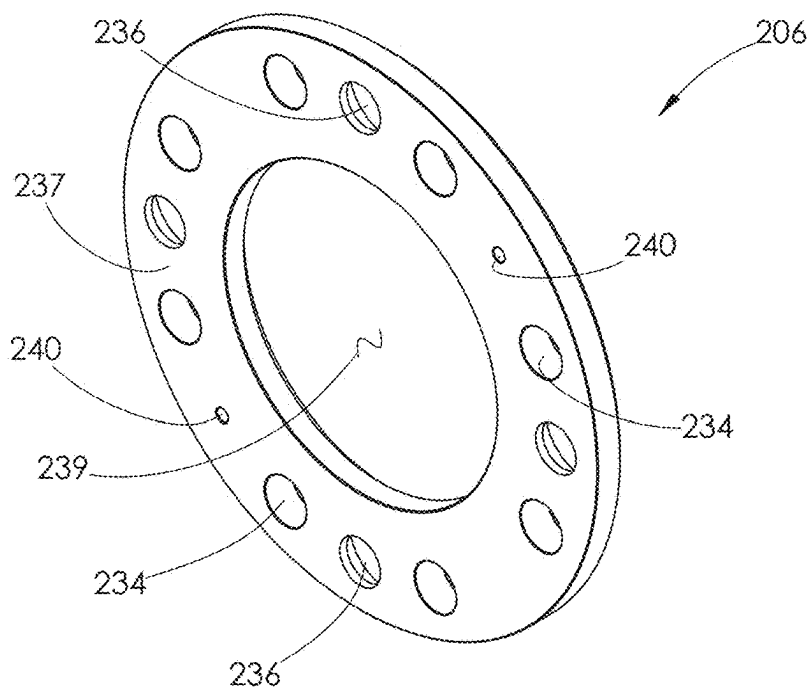


FIG. 31

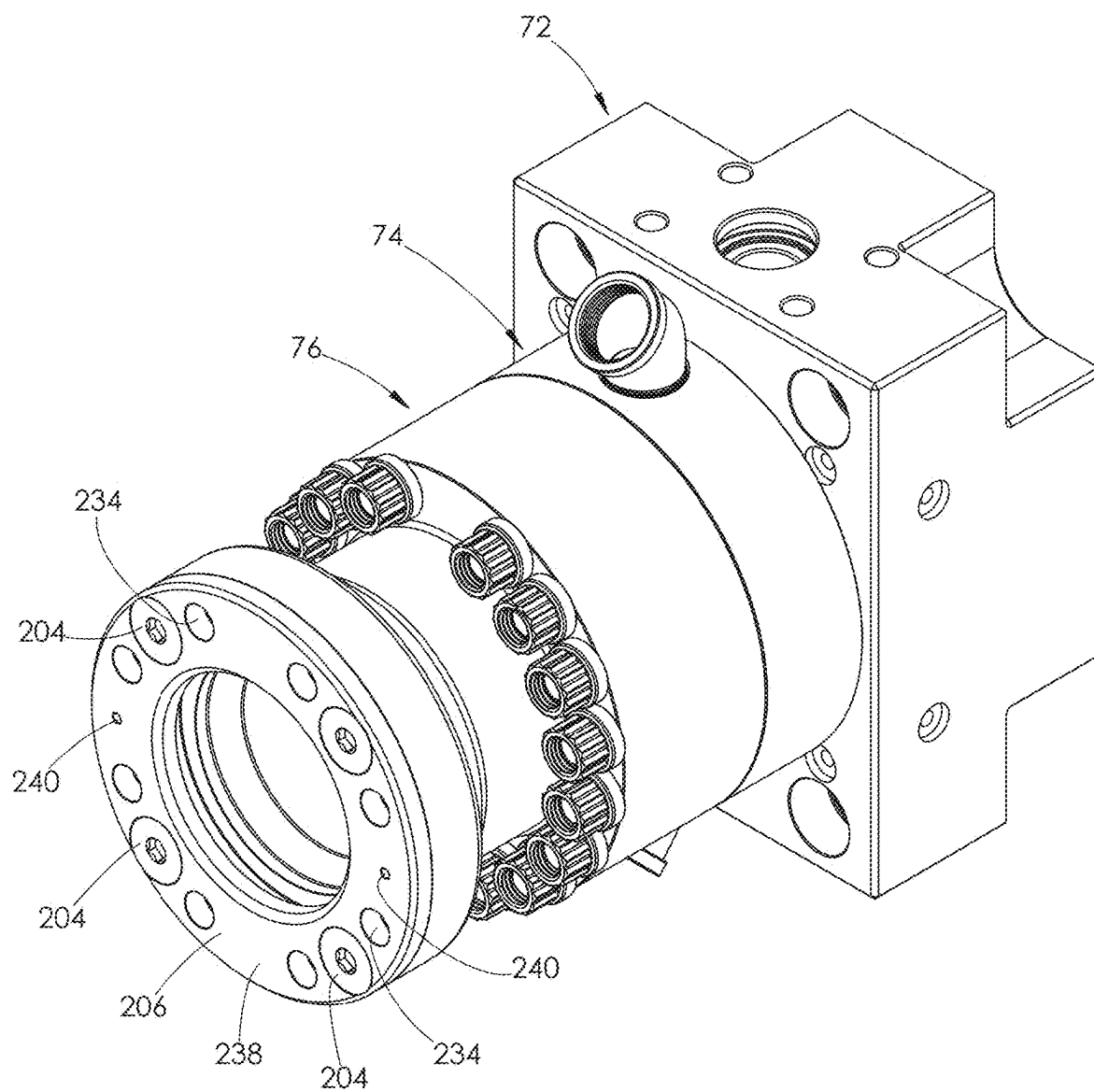
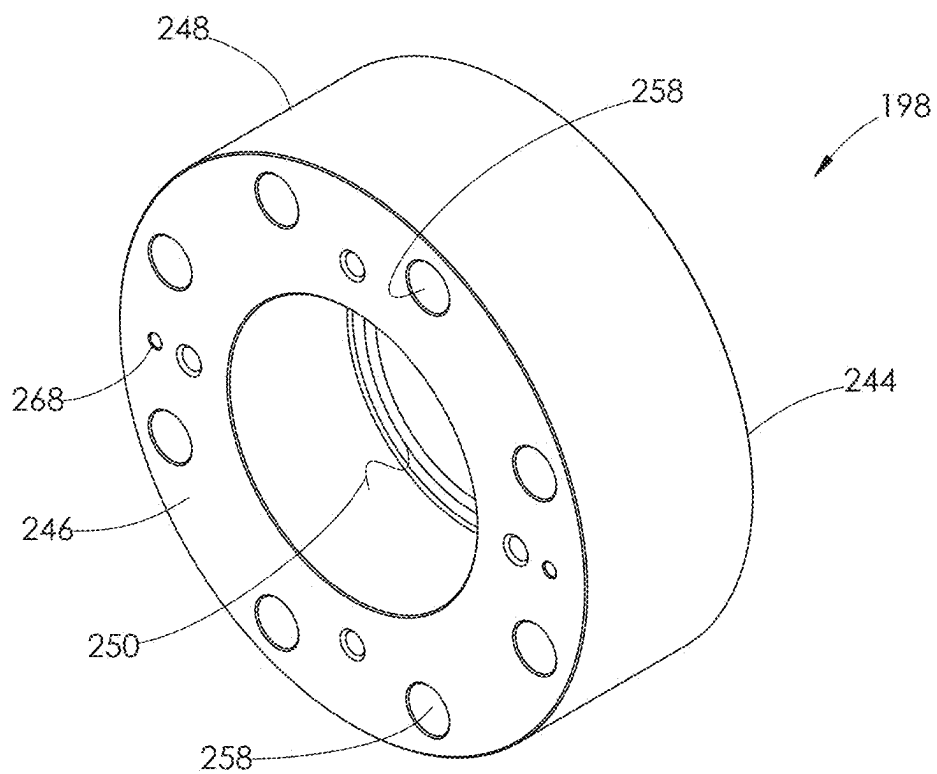
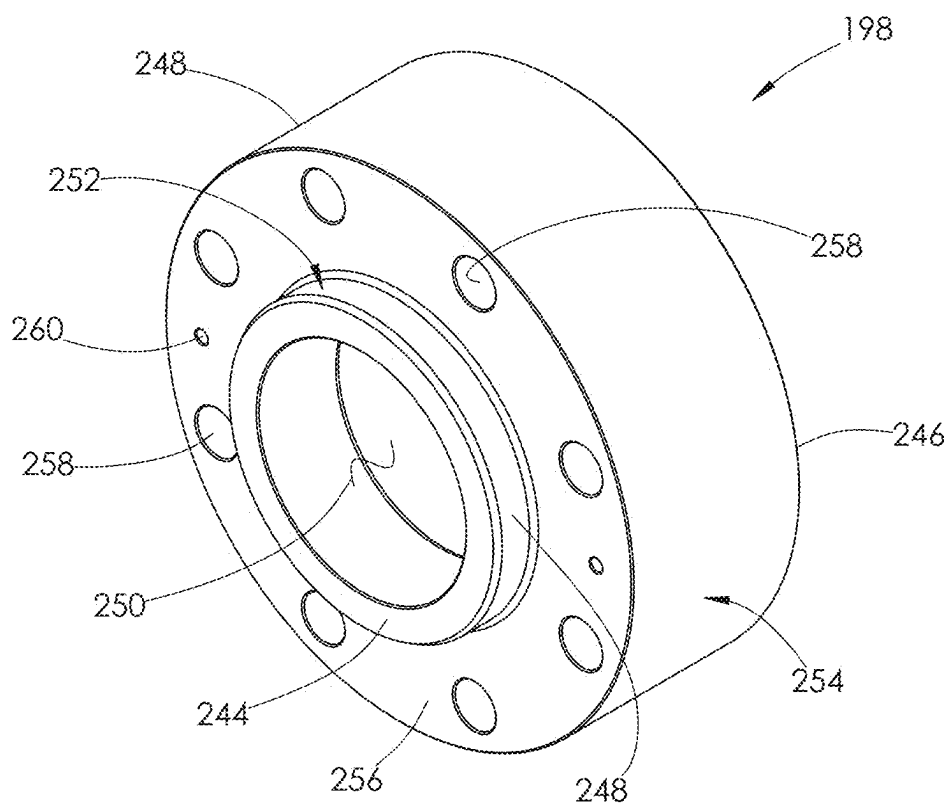


FIG. 32



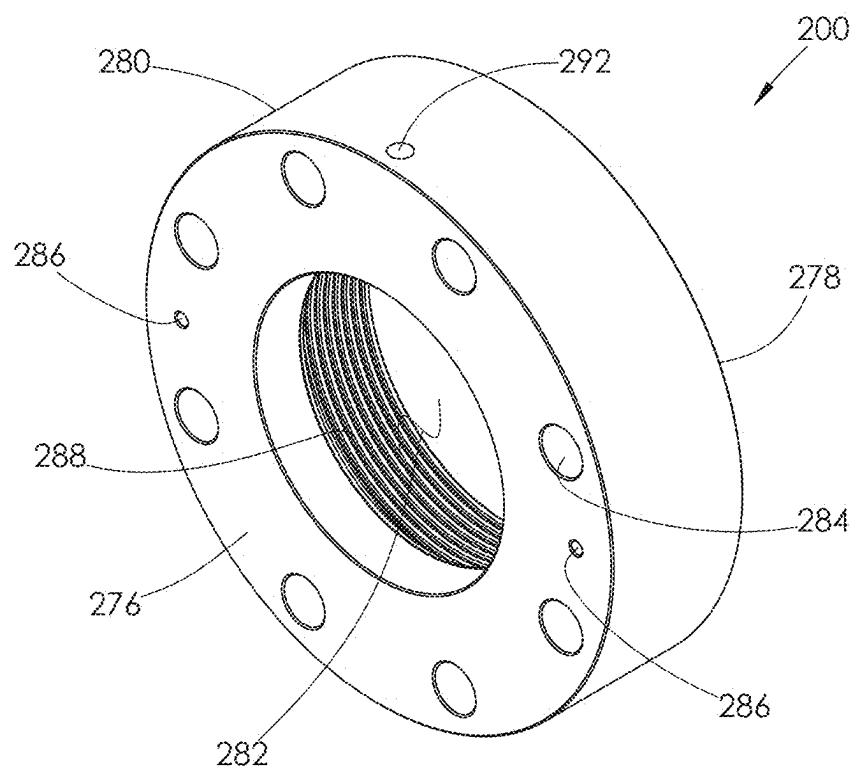


FIG. 35

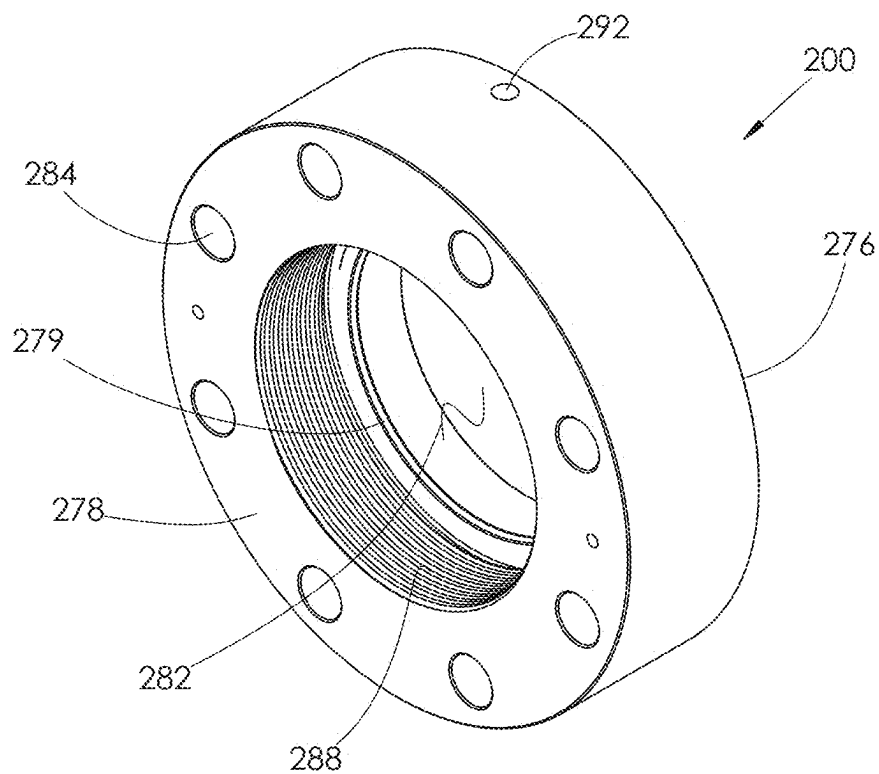


FIG. 36

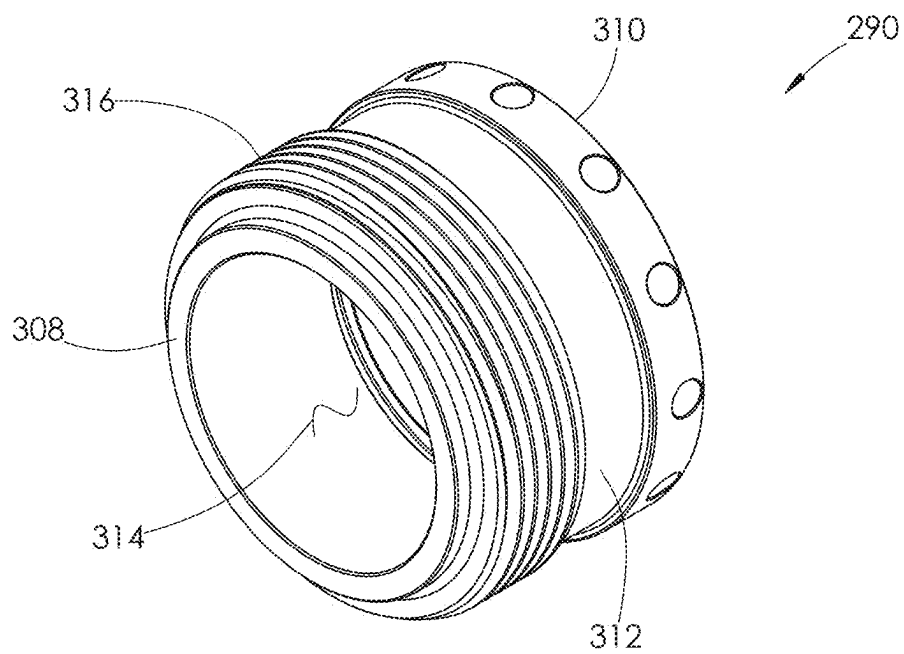


FIG. 37

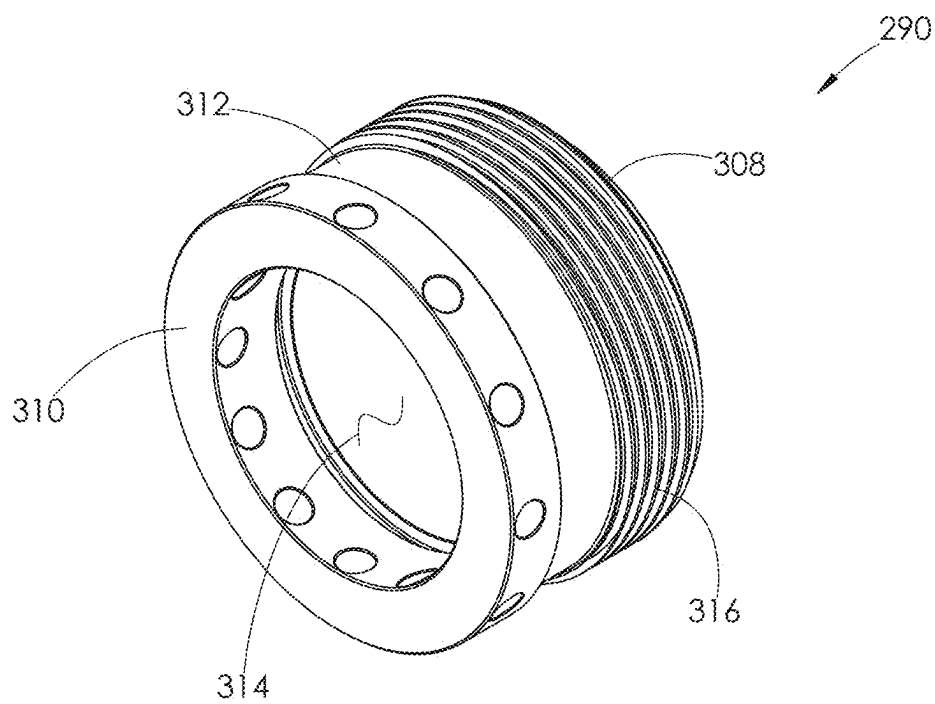


FIG. 38

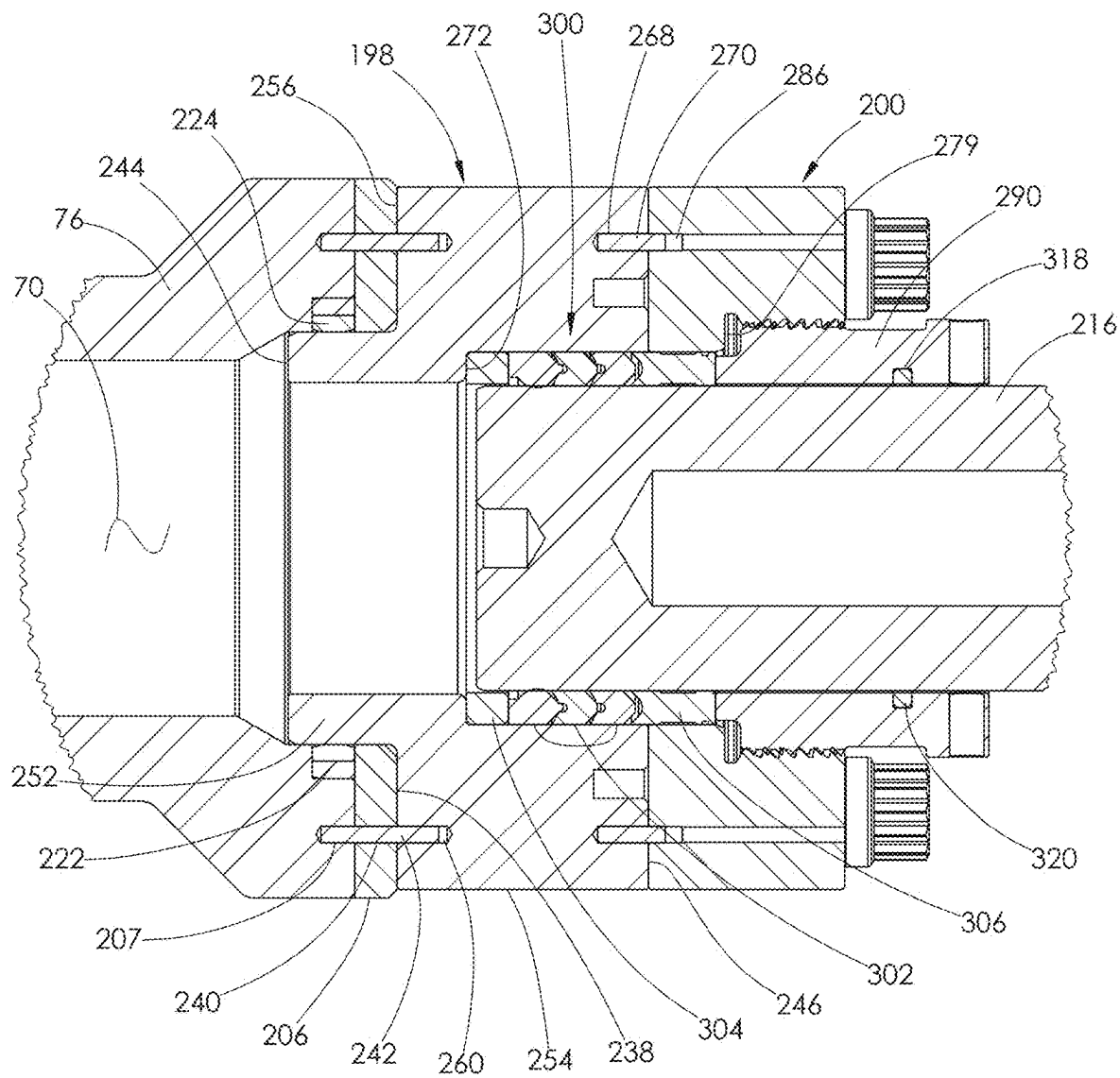


FIG. 39

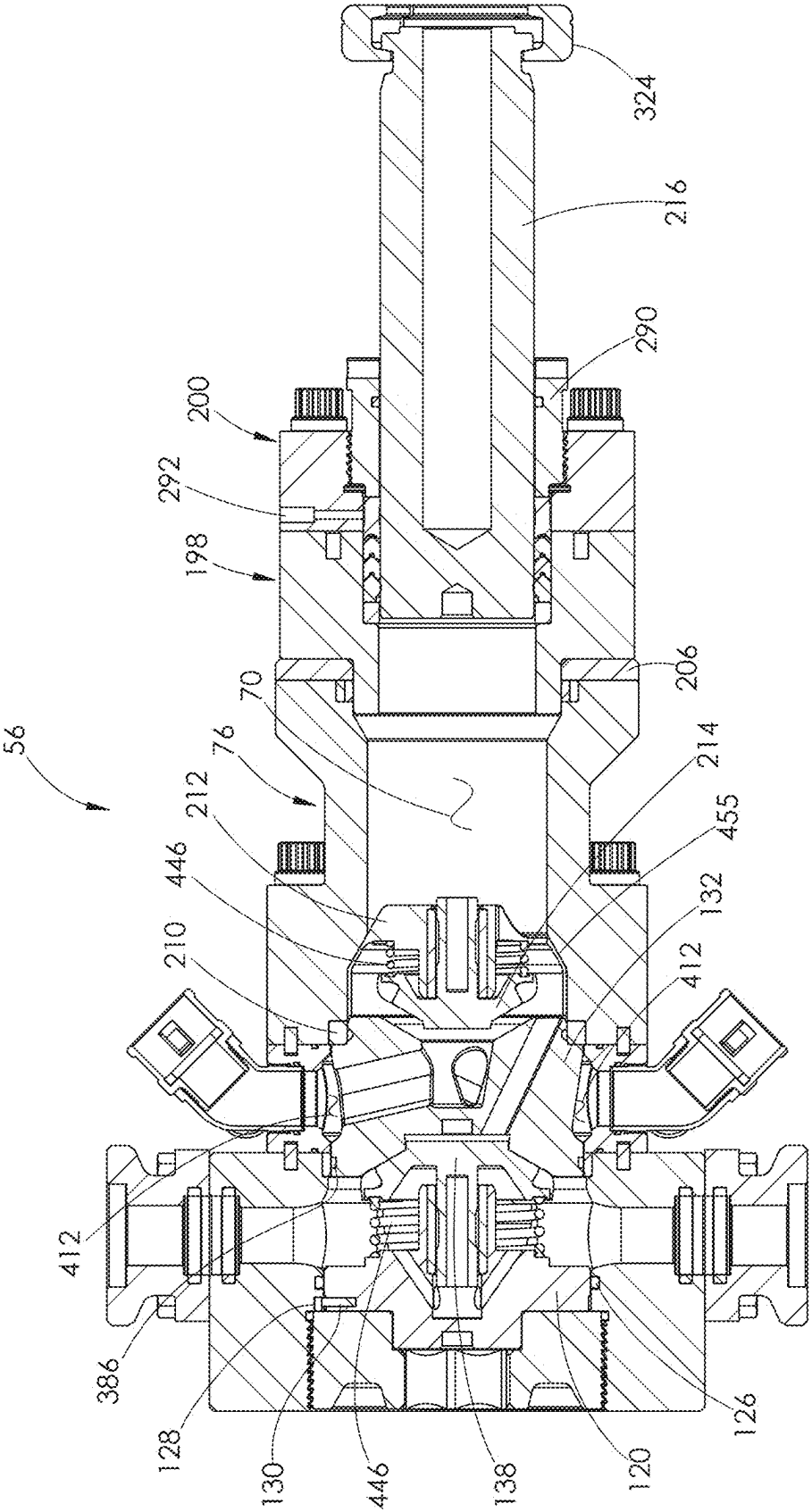
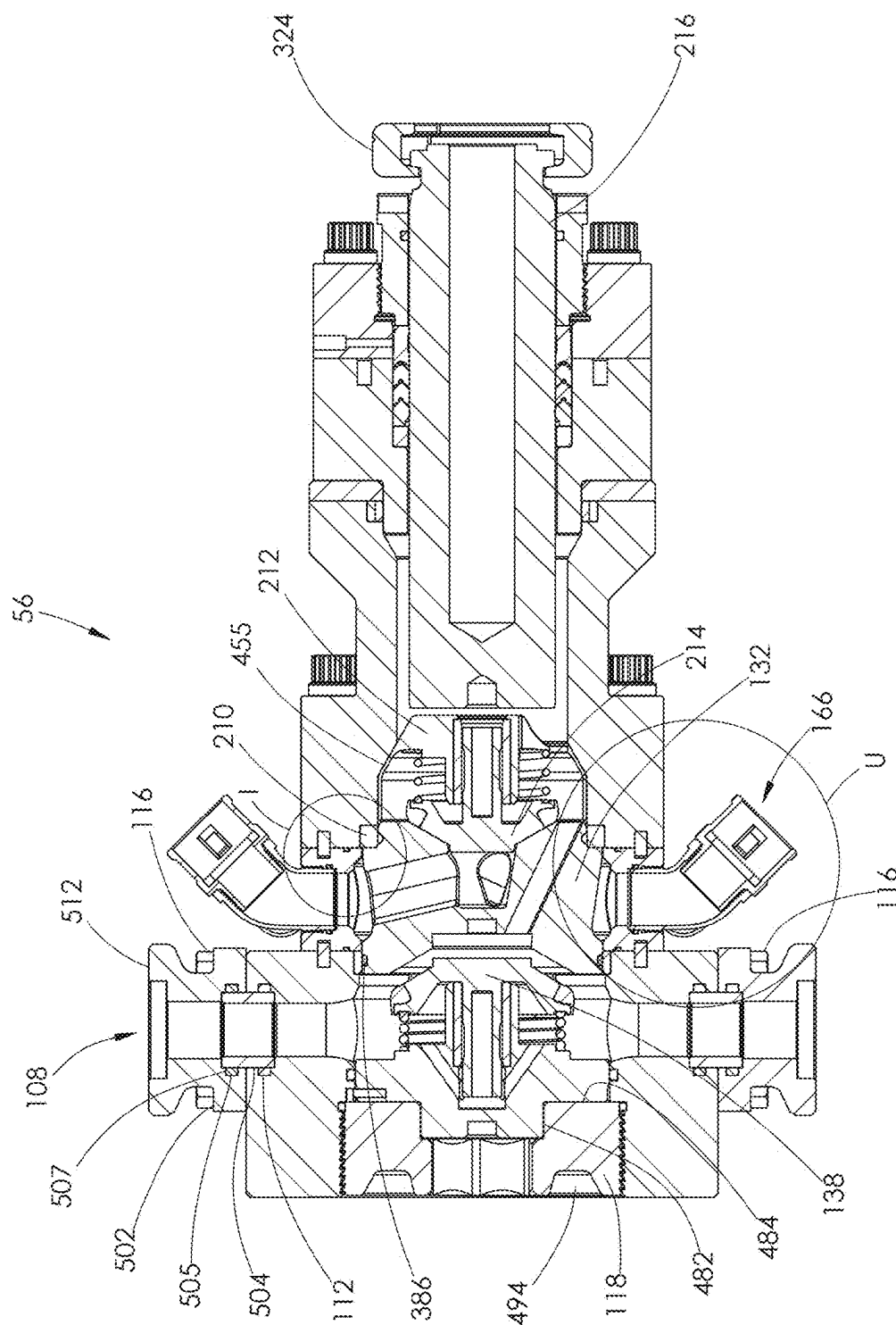


FIG. 40



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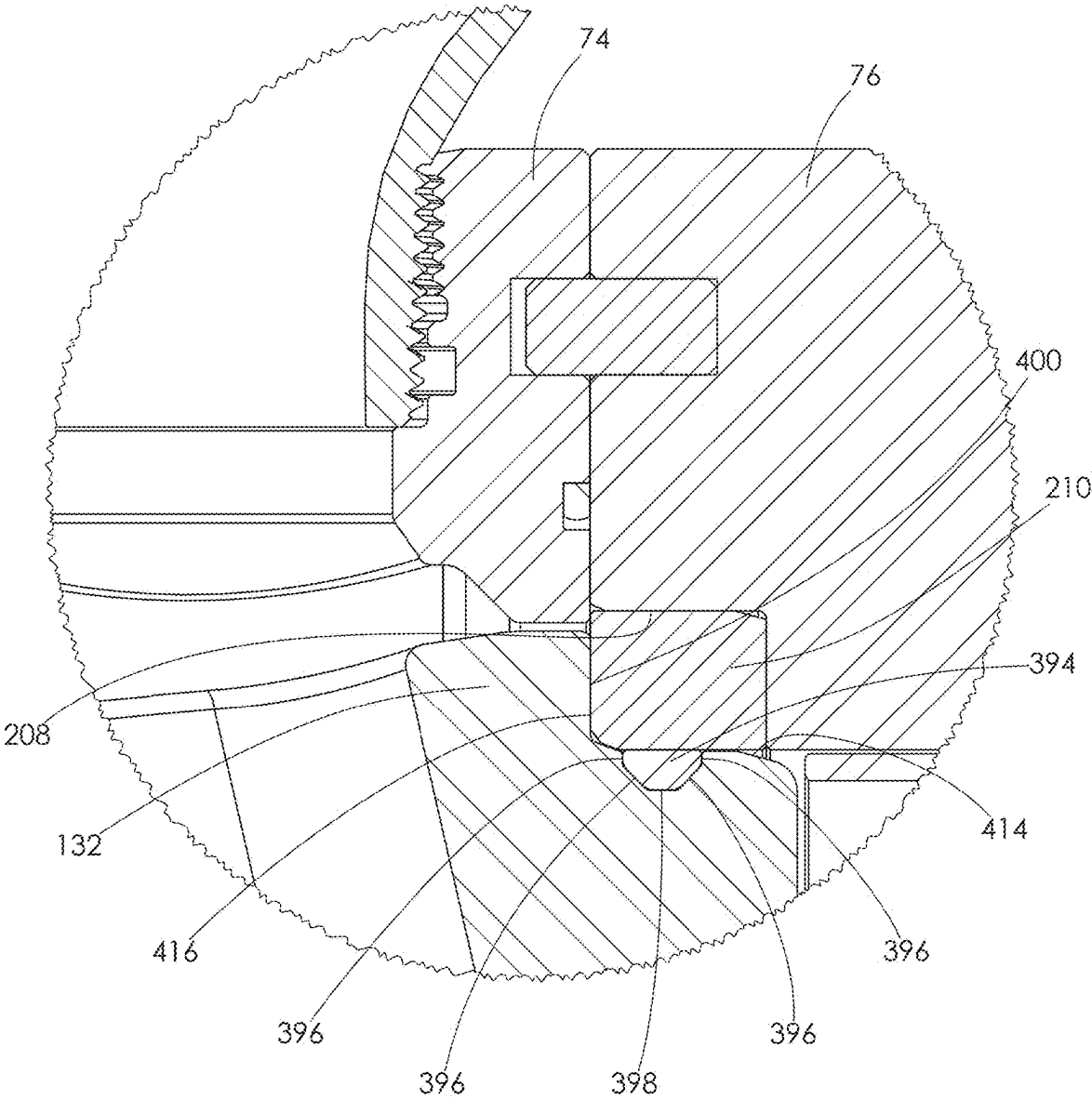


FIG. 42

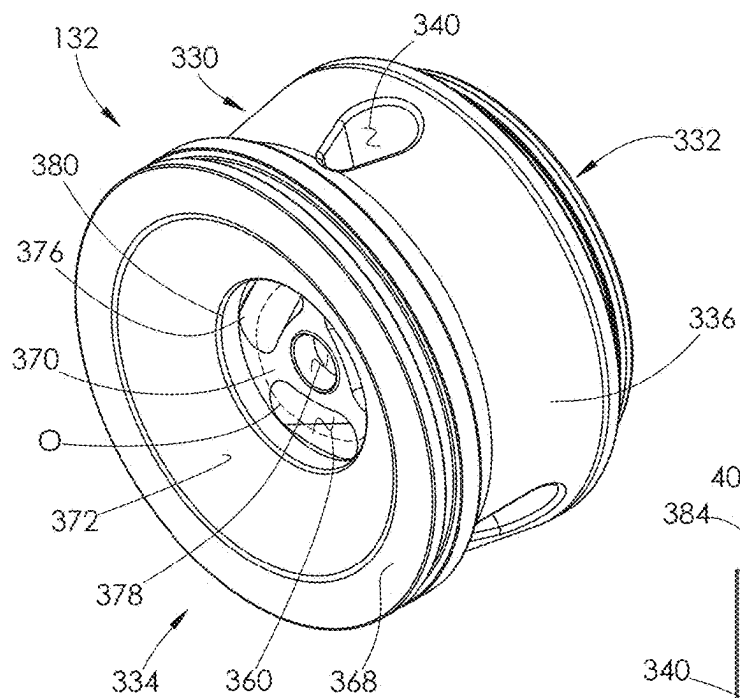


FIG. 43

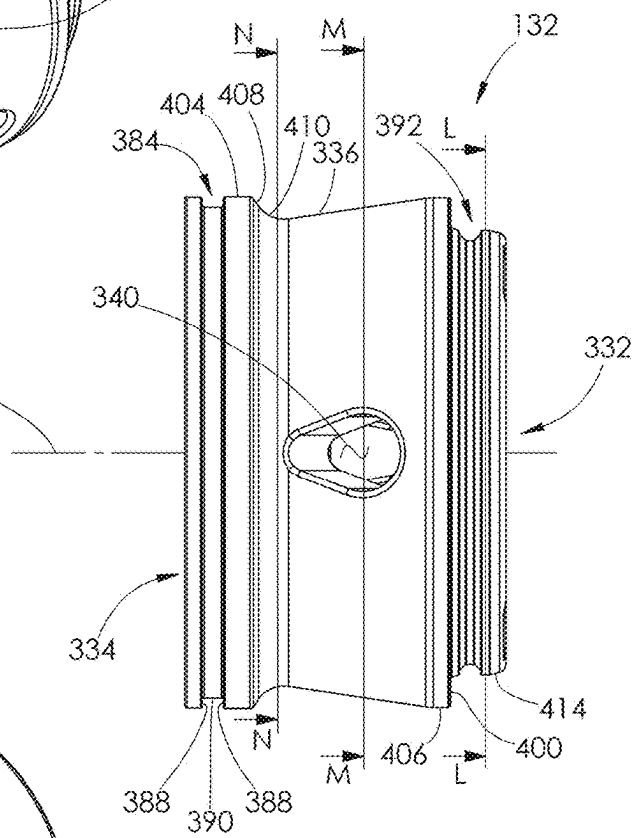


FIG. 44

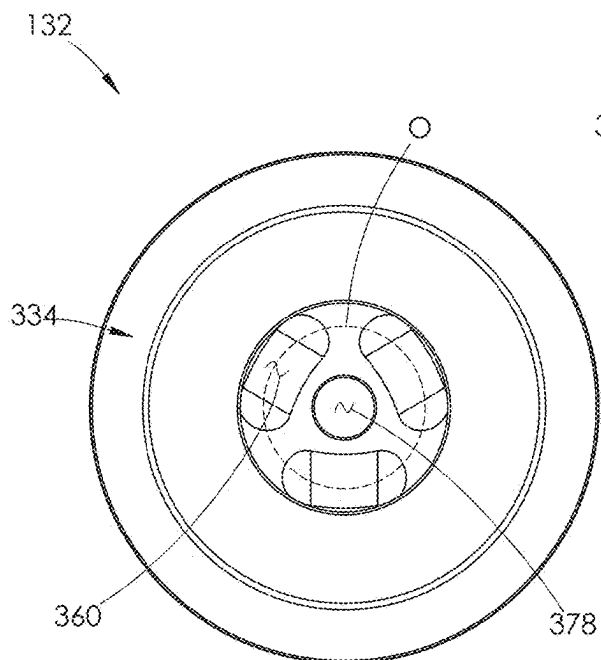


FIG. 45

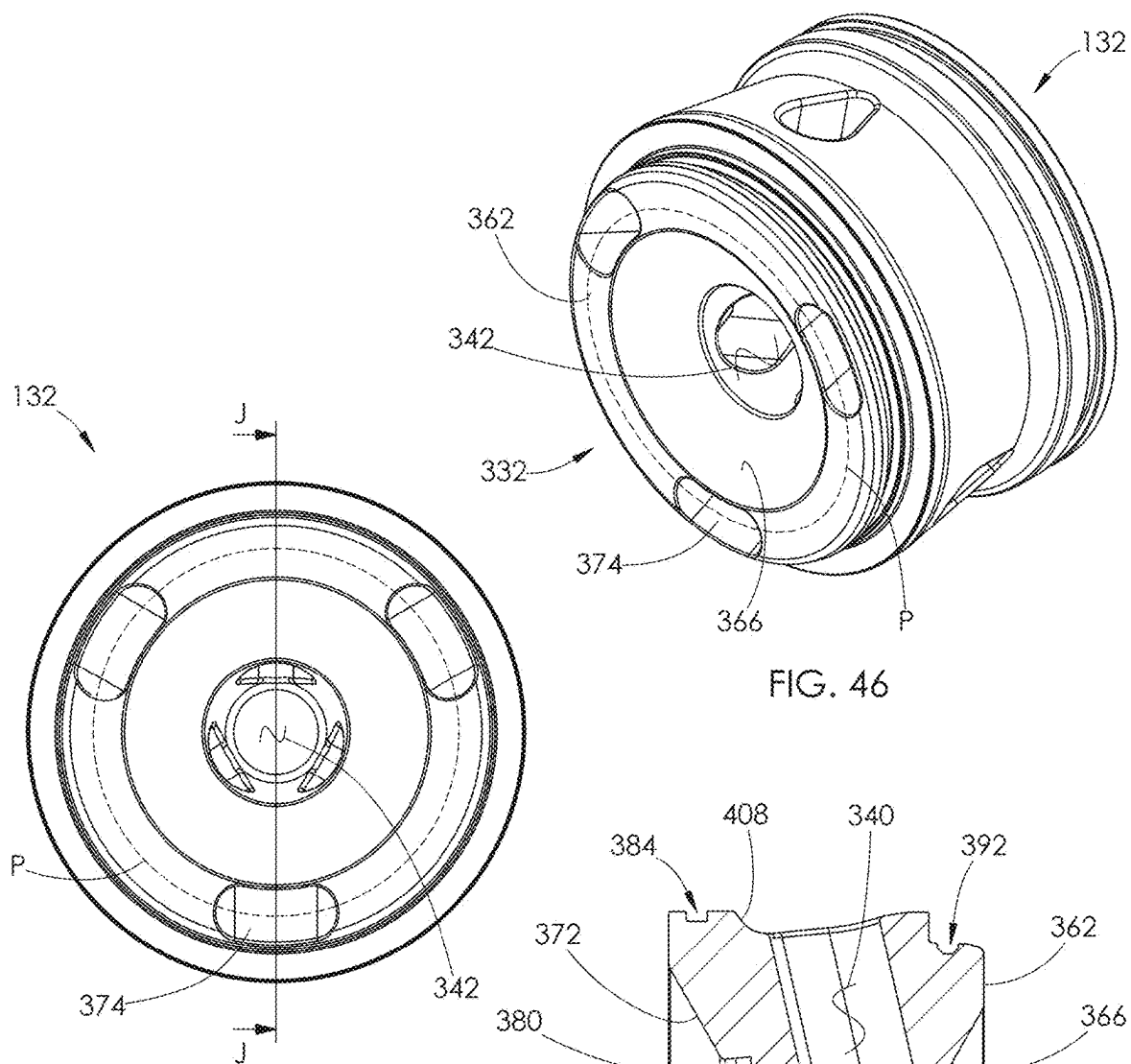


FIG. 47

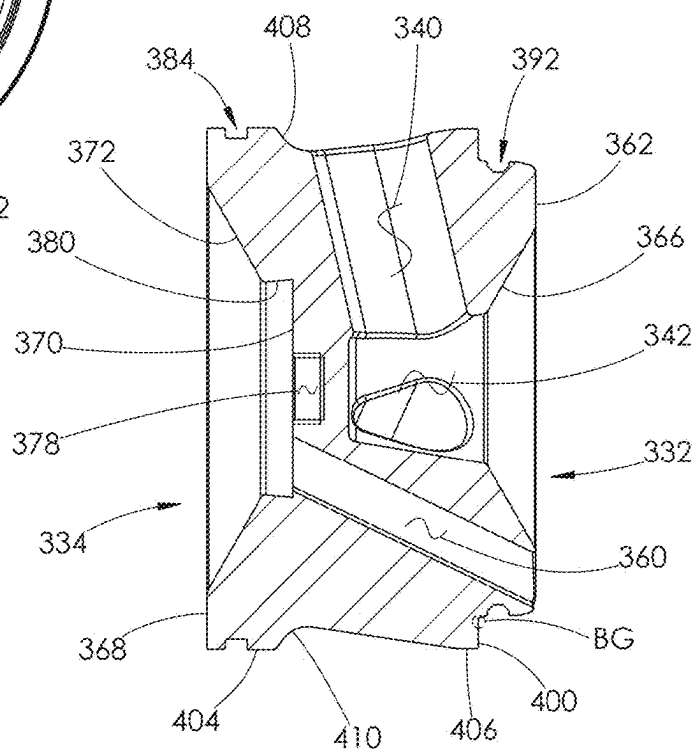


FIG. 48

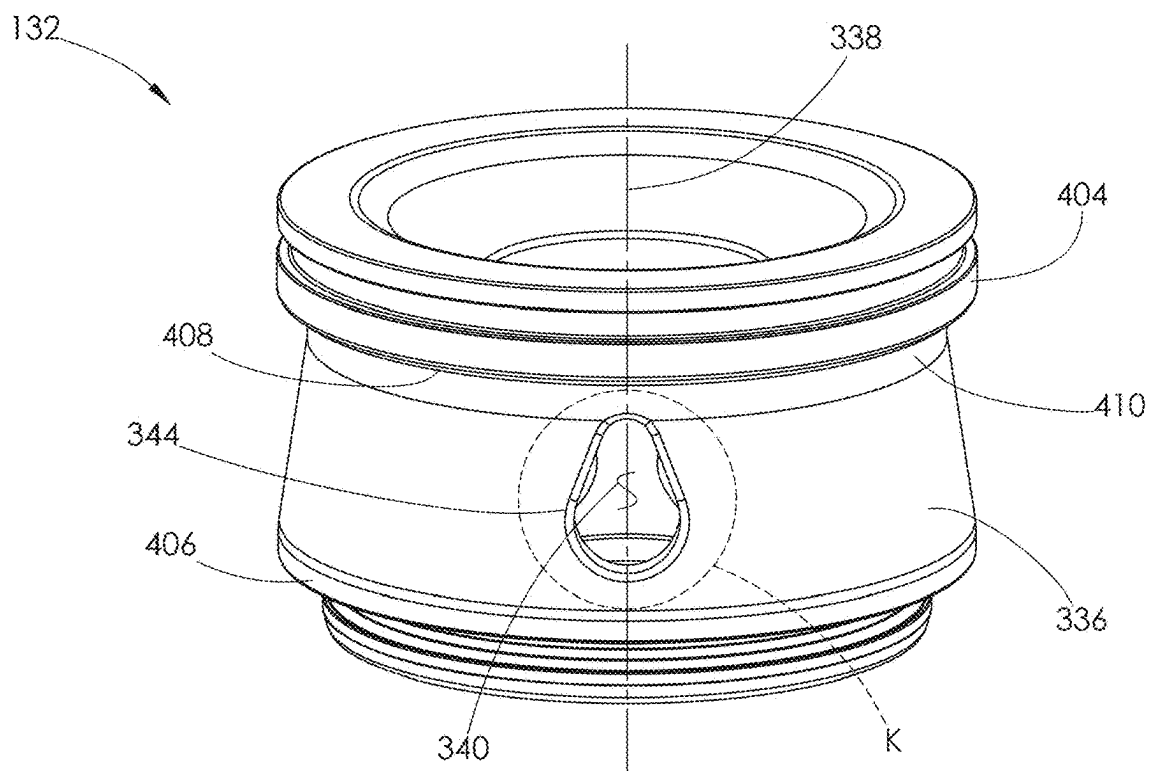


FIG. 49

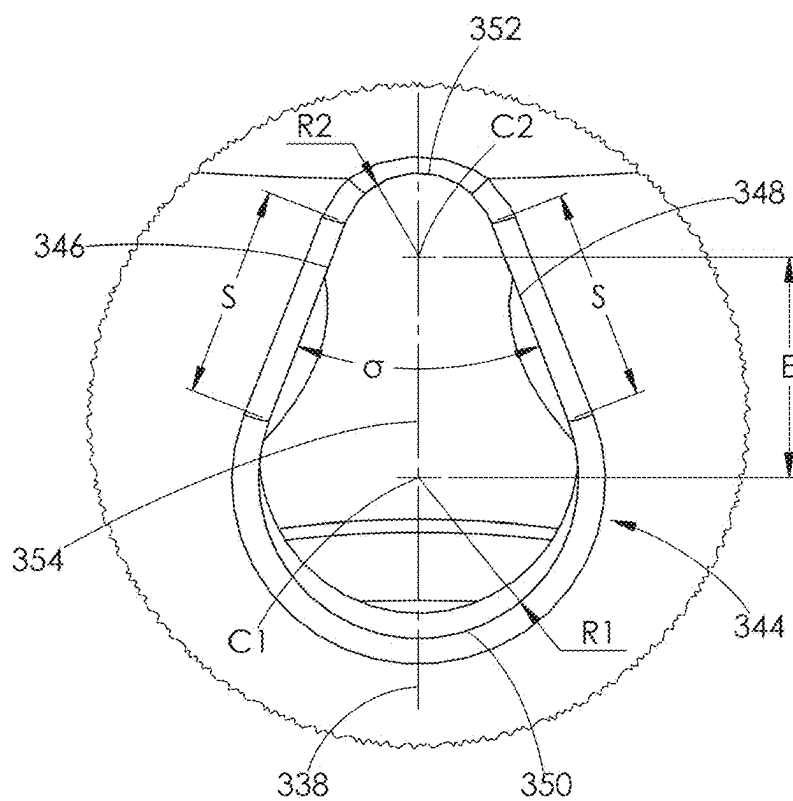


FIG. 50

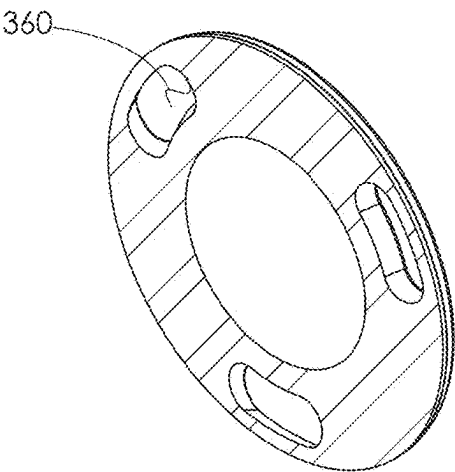


FIG. 51

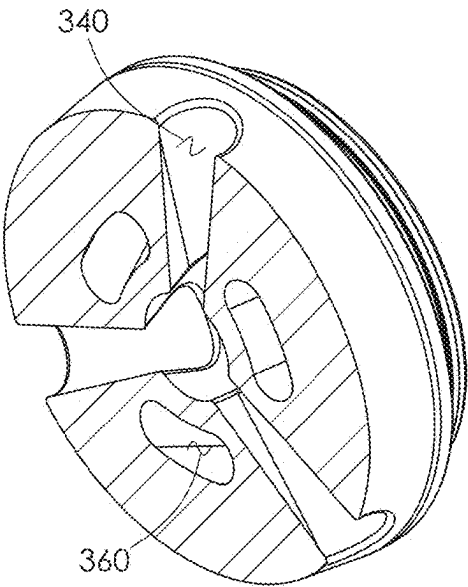


FIG. 52

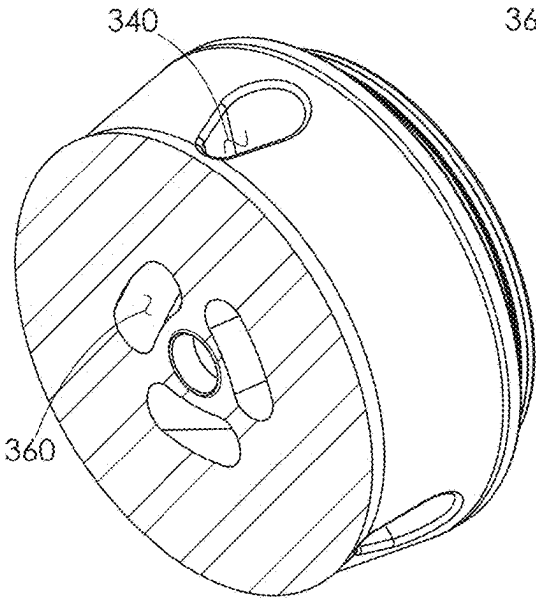


FIG. 53

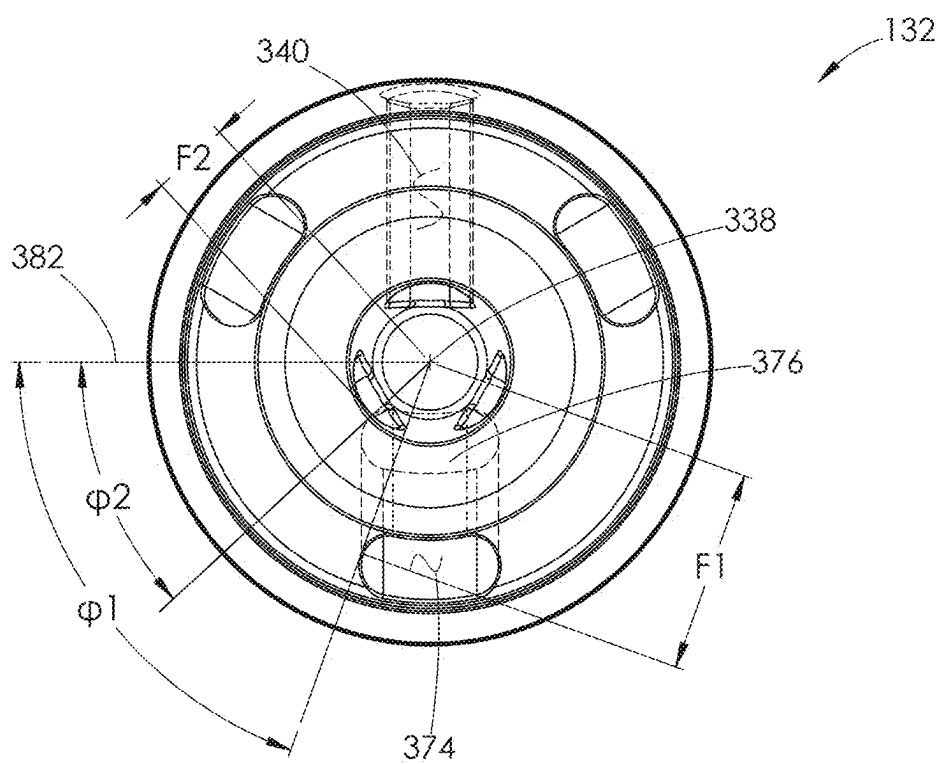


FIG. 54

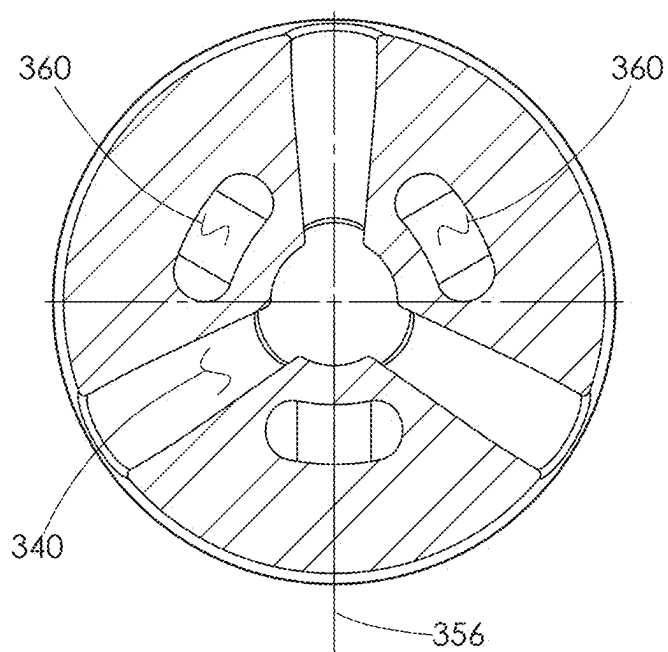


FIG. 55

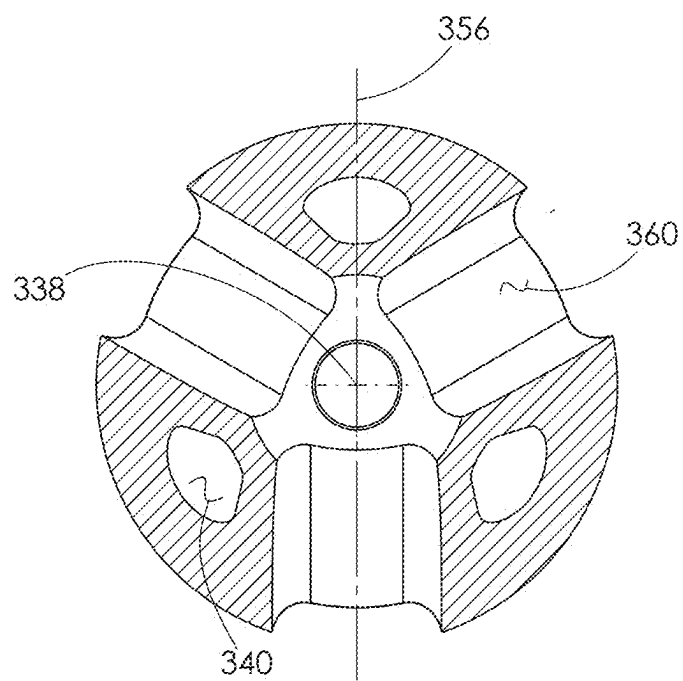


FIG. 56

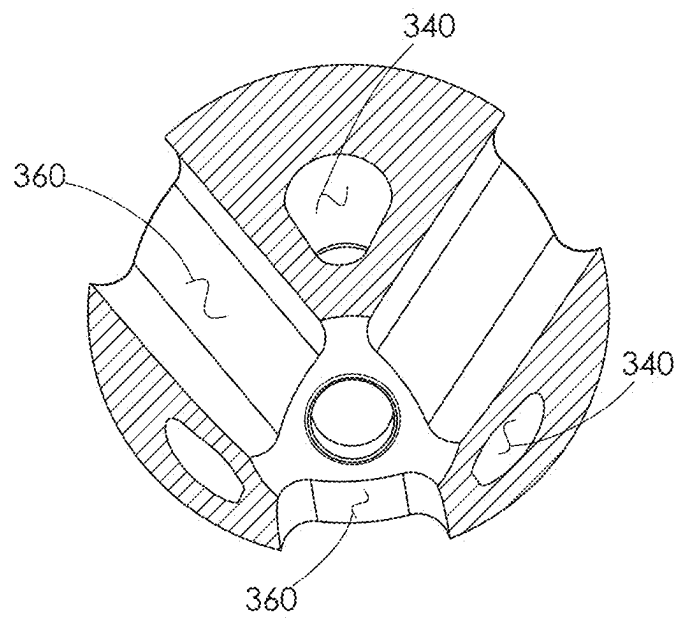


FIG. 57

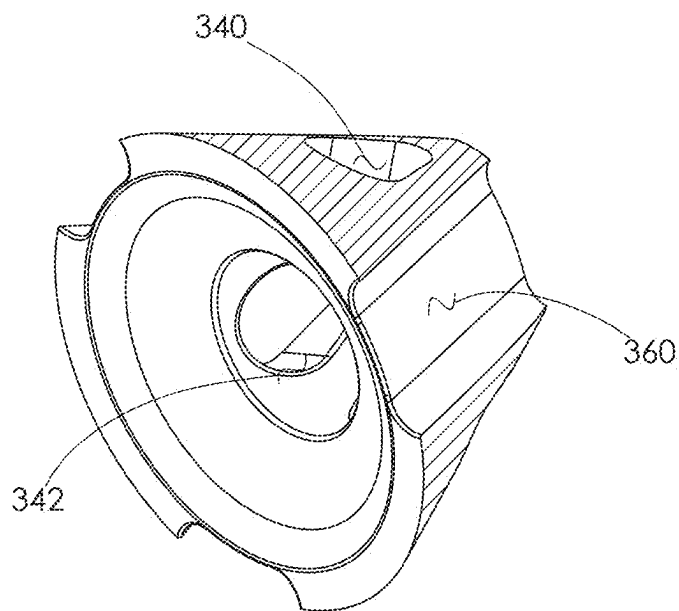


FIG. 58

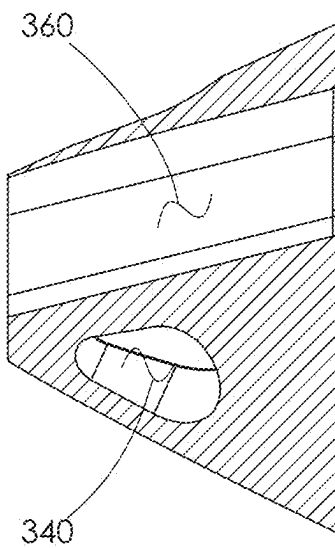


FIG. 59

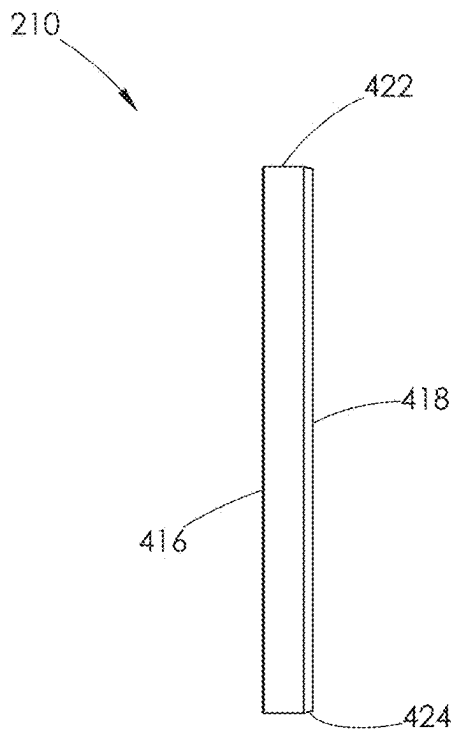


FIG. 60

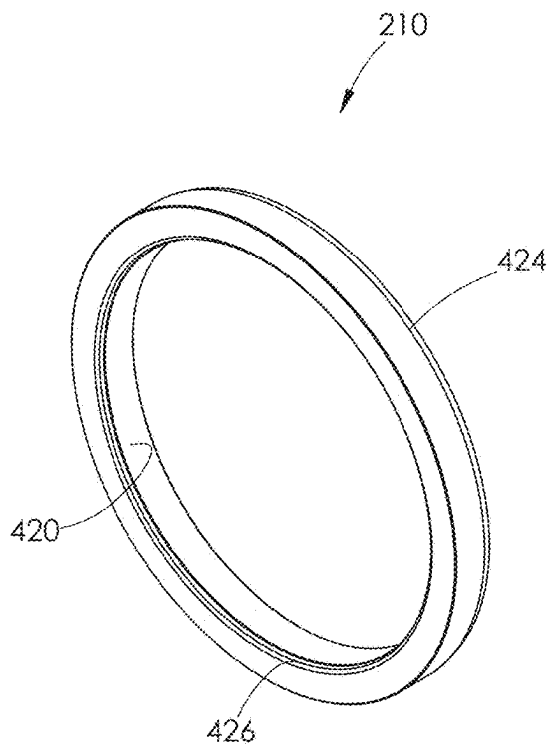


FIG. 61

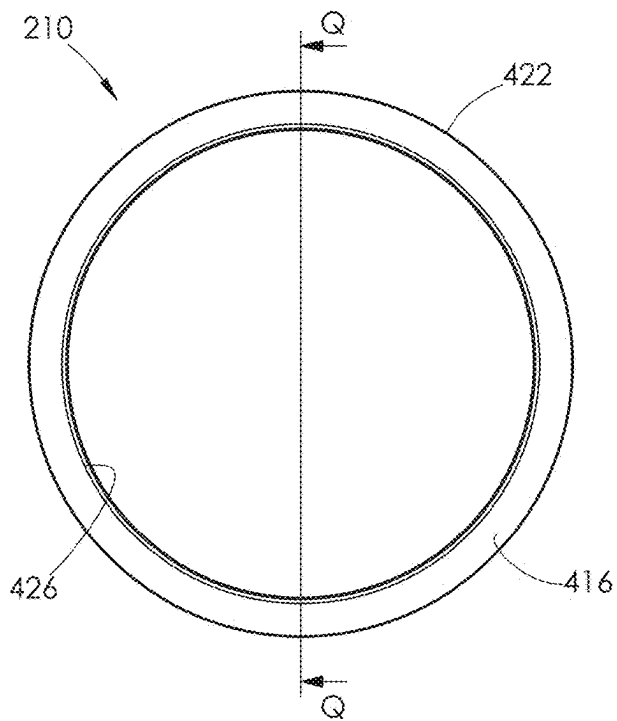


FIG. 62

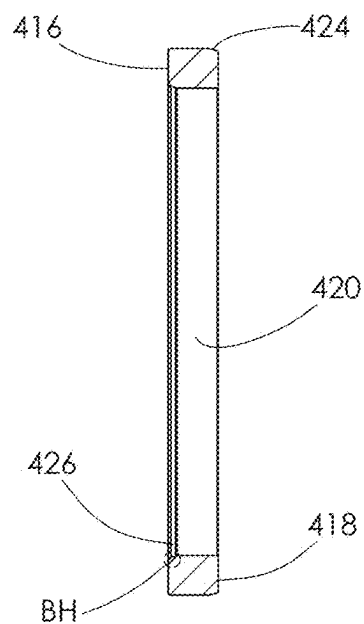


FIG. 63

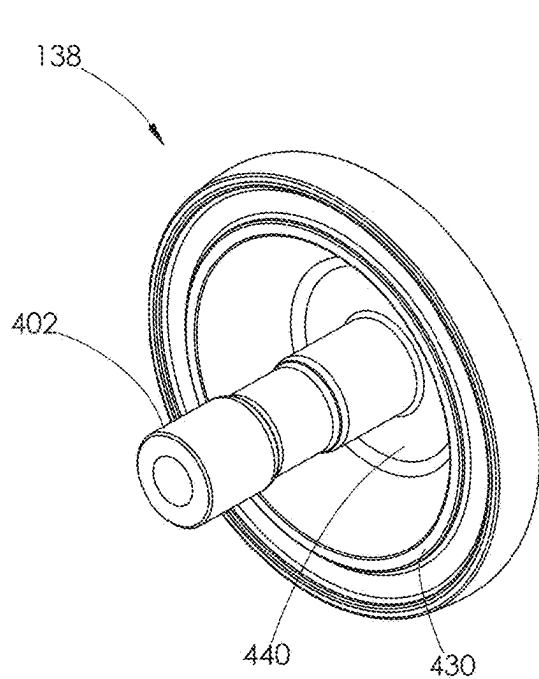


FIG. 64

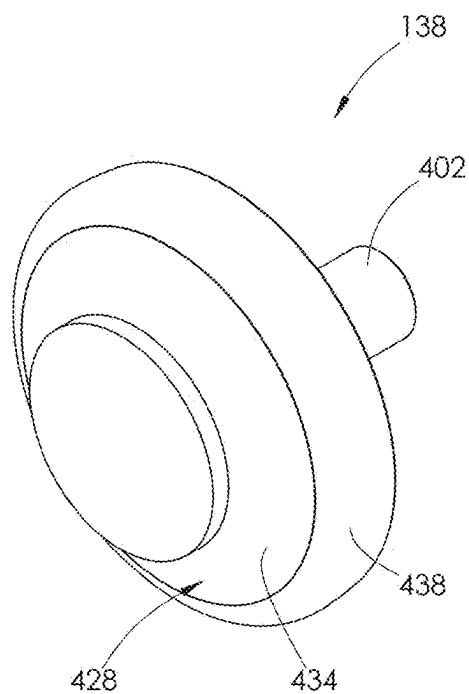


FIG. 65

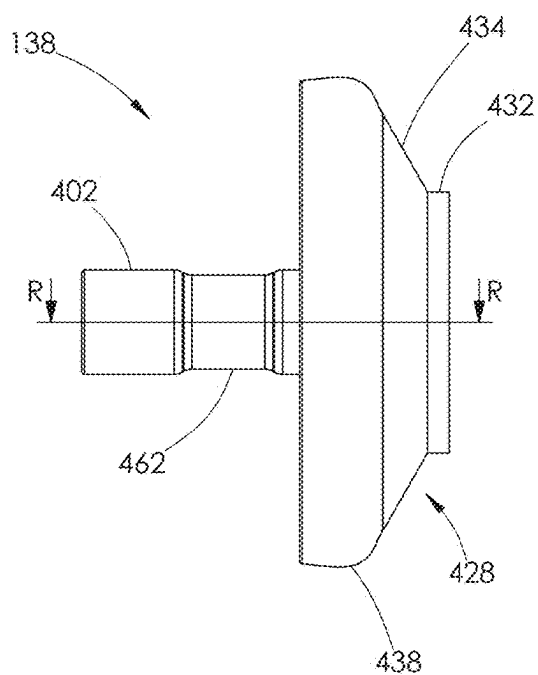


FIG. 66

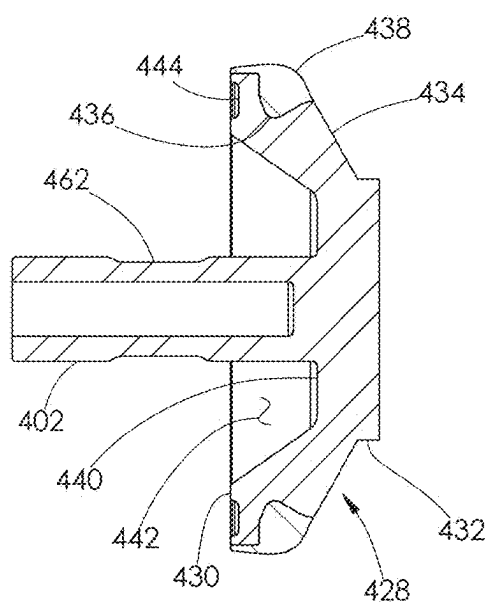


FIG. 67

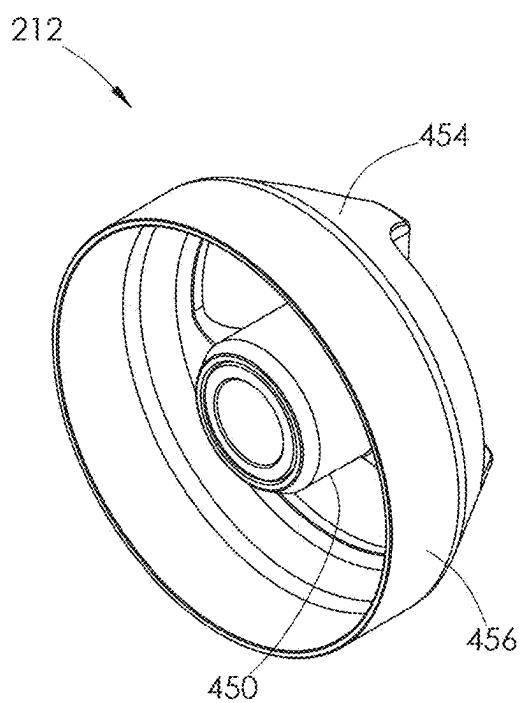


FIG. 68

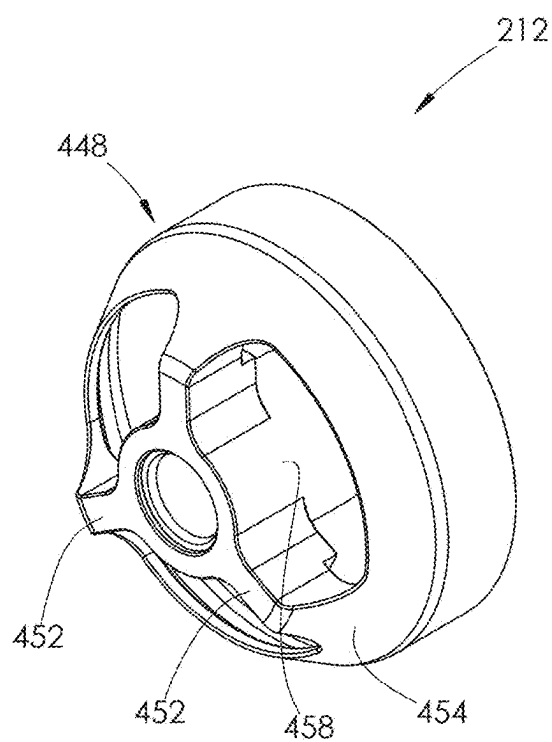


FIG. 69

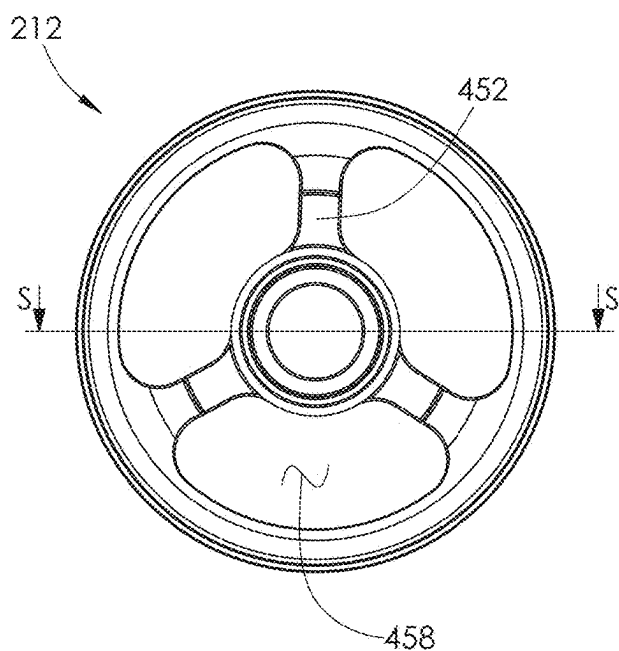


FIG. 70

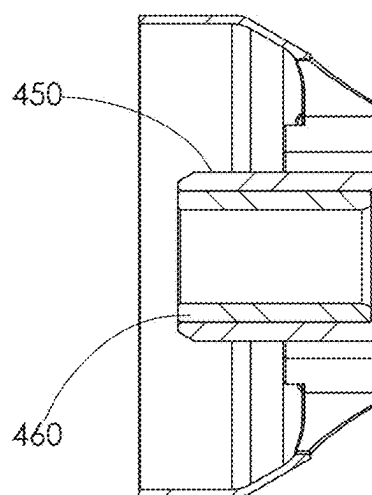


FIG. 71

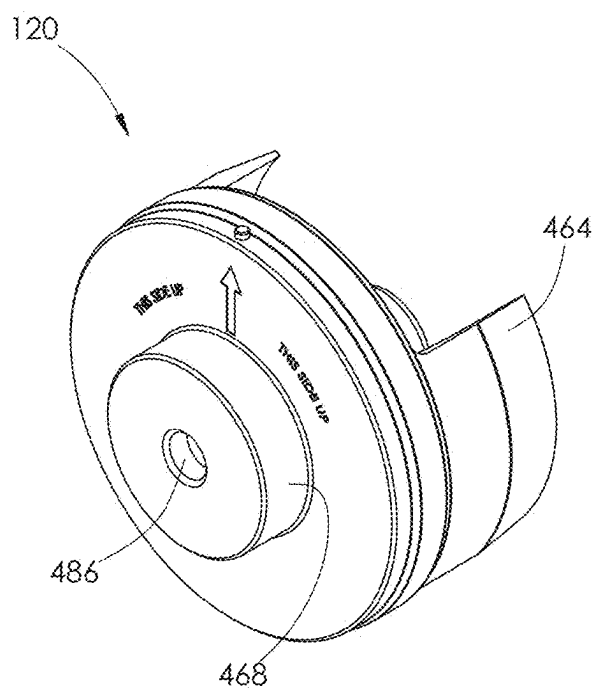


FIG. 72

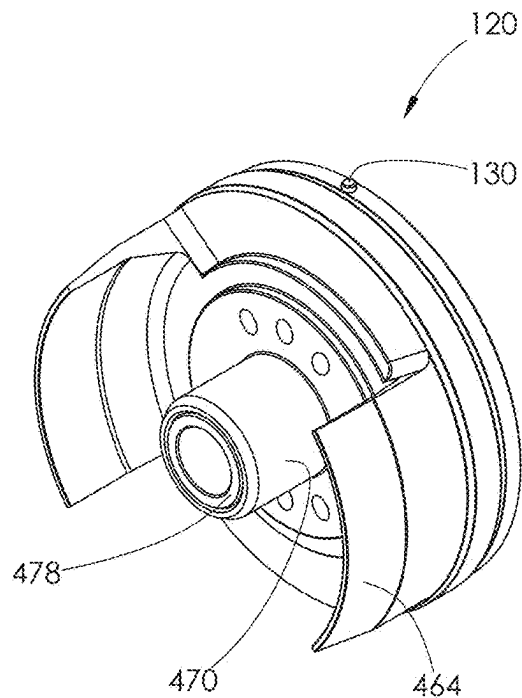


FIG. 73

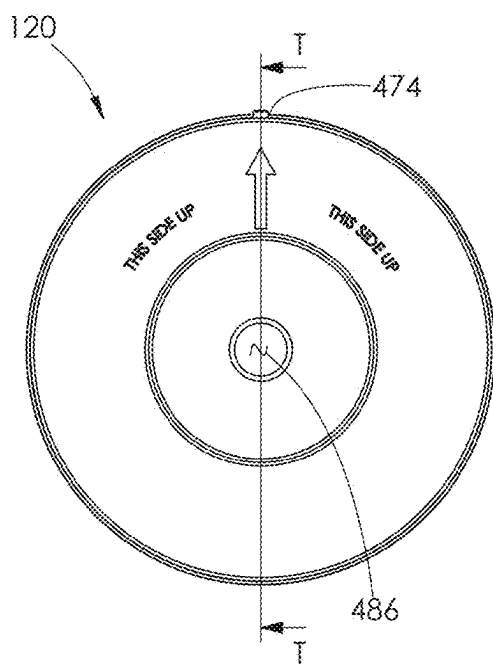


FIG. 74

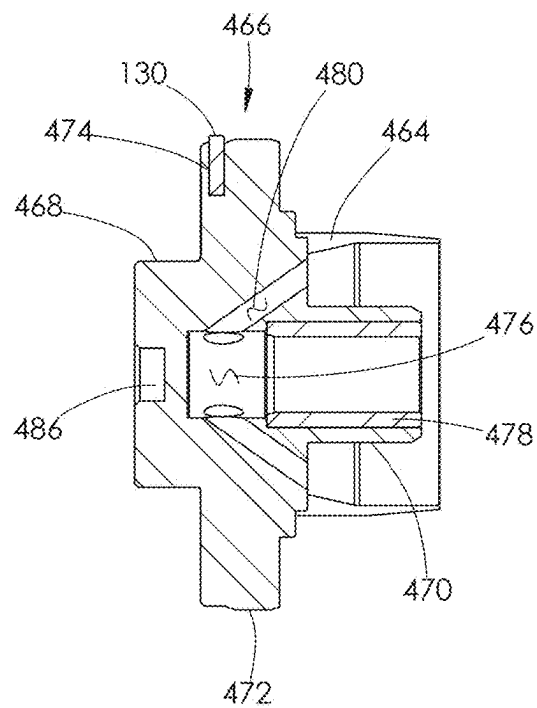


FIG. 75

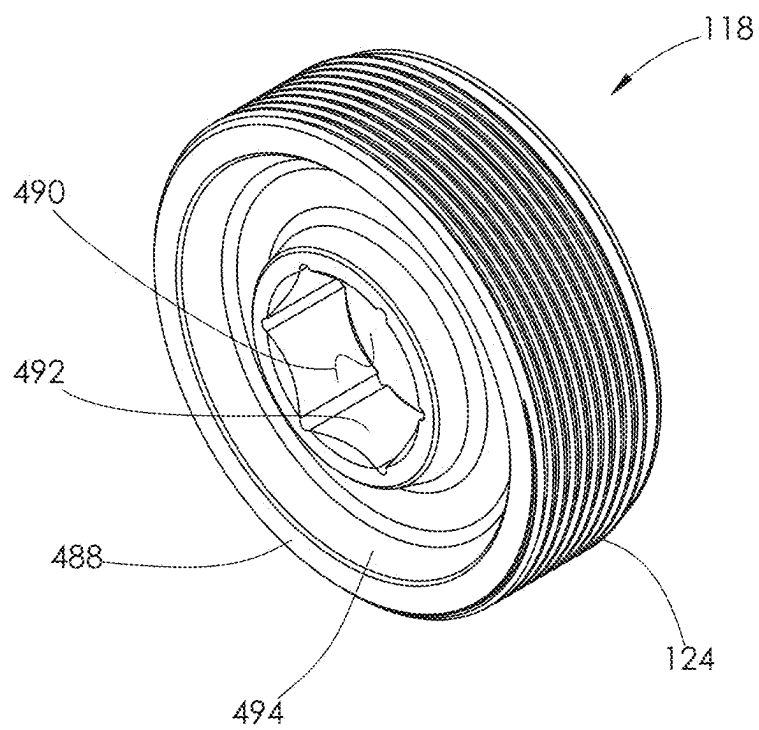


FIG. 76

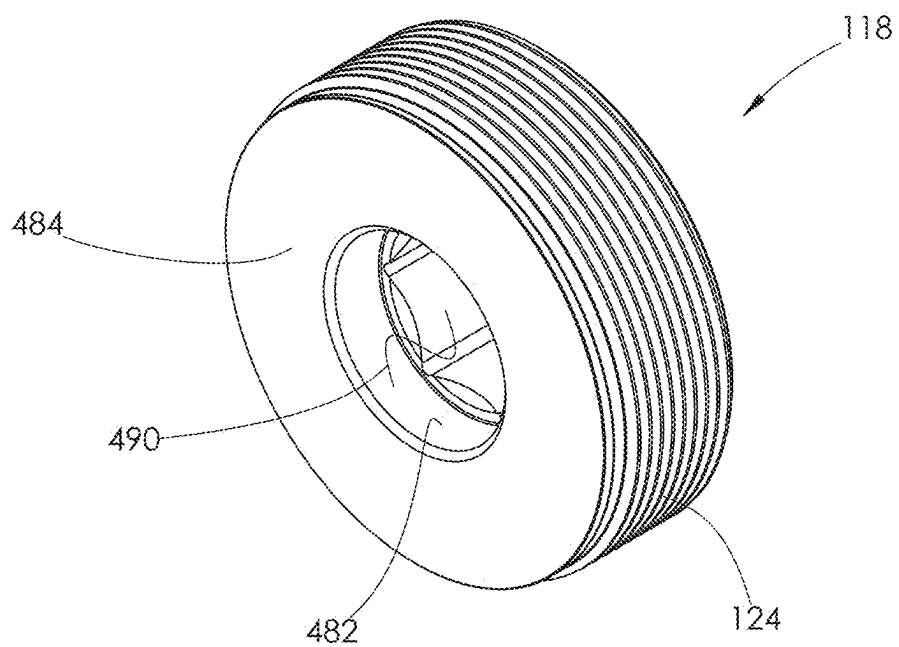


FIG. 77

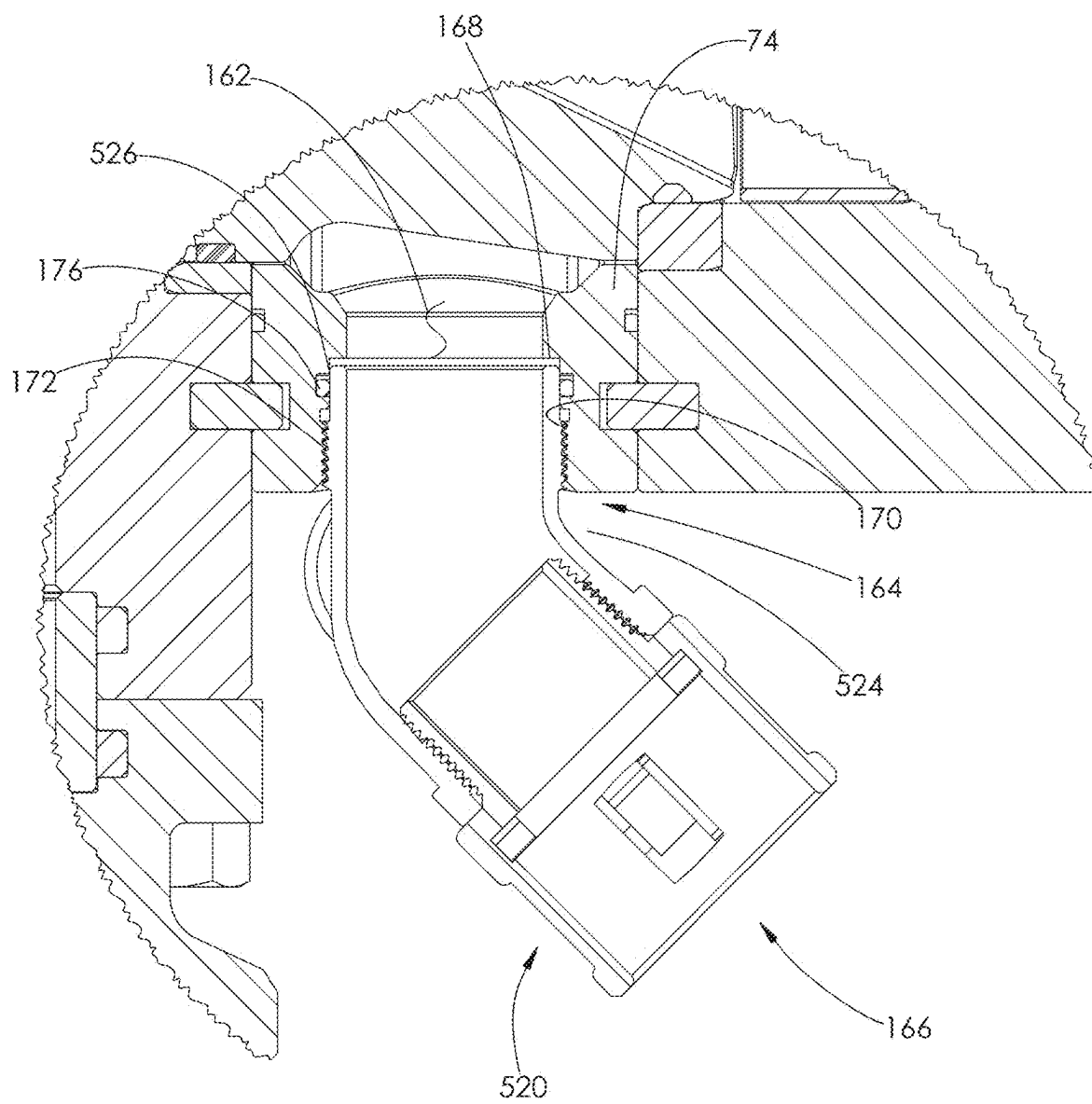


FIG. 78

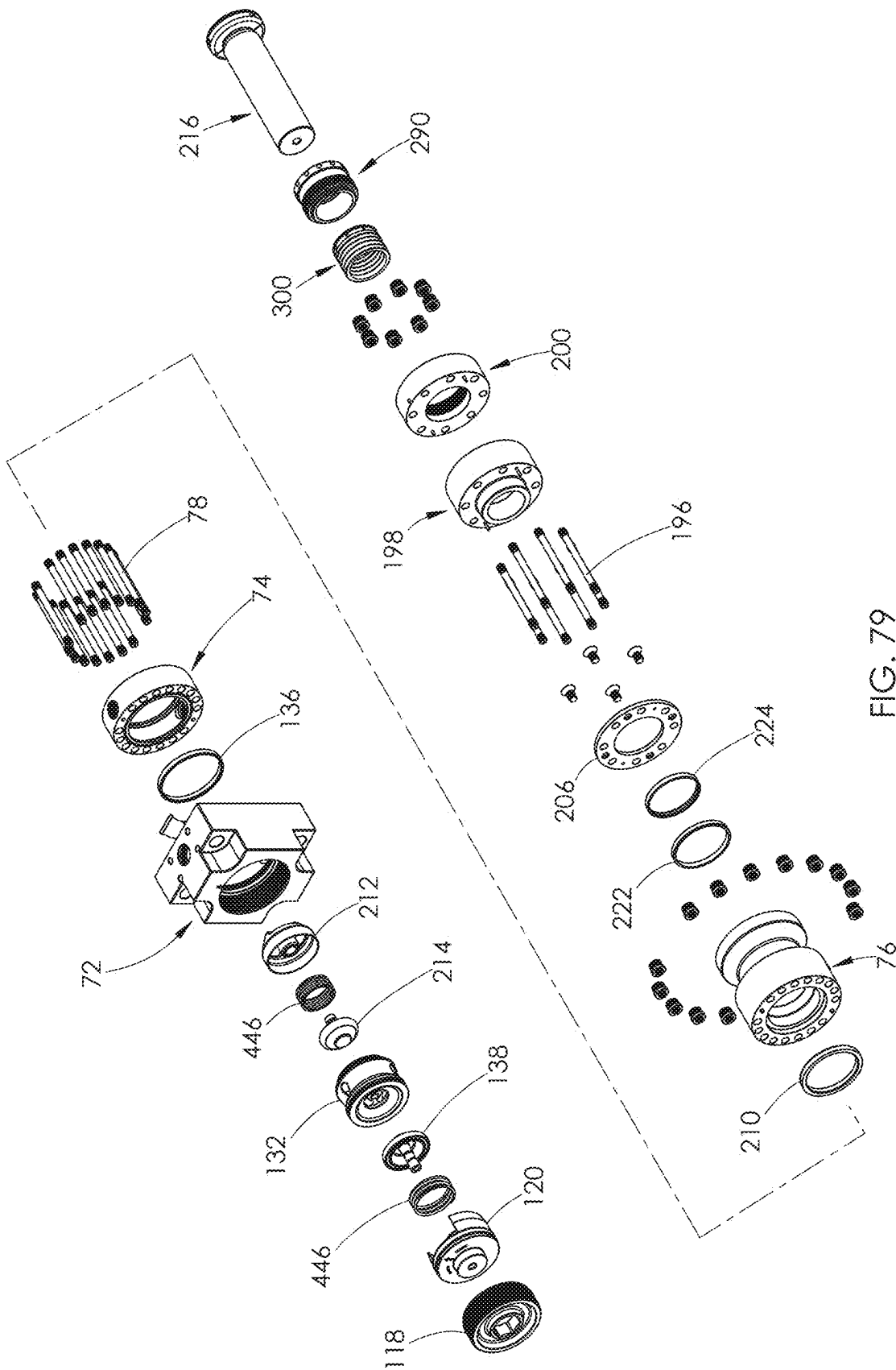


FIG. 79

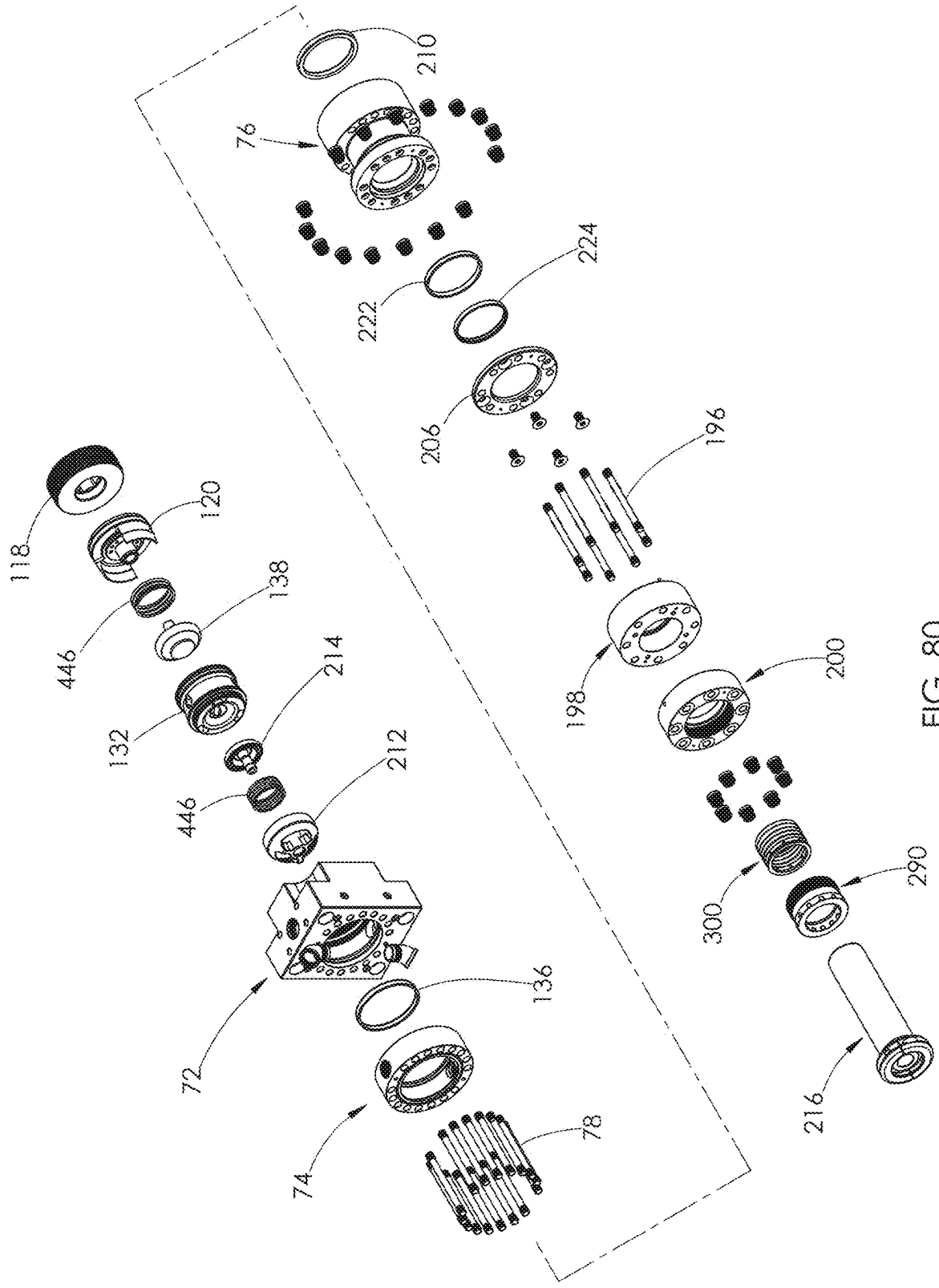


FIG. 80

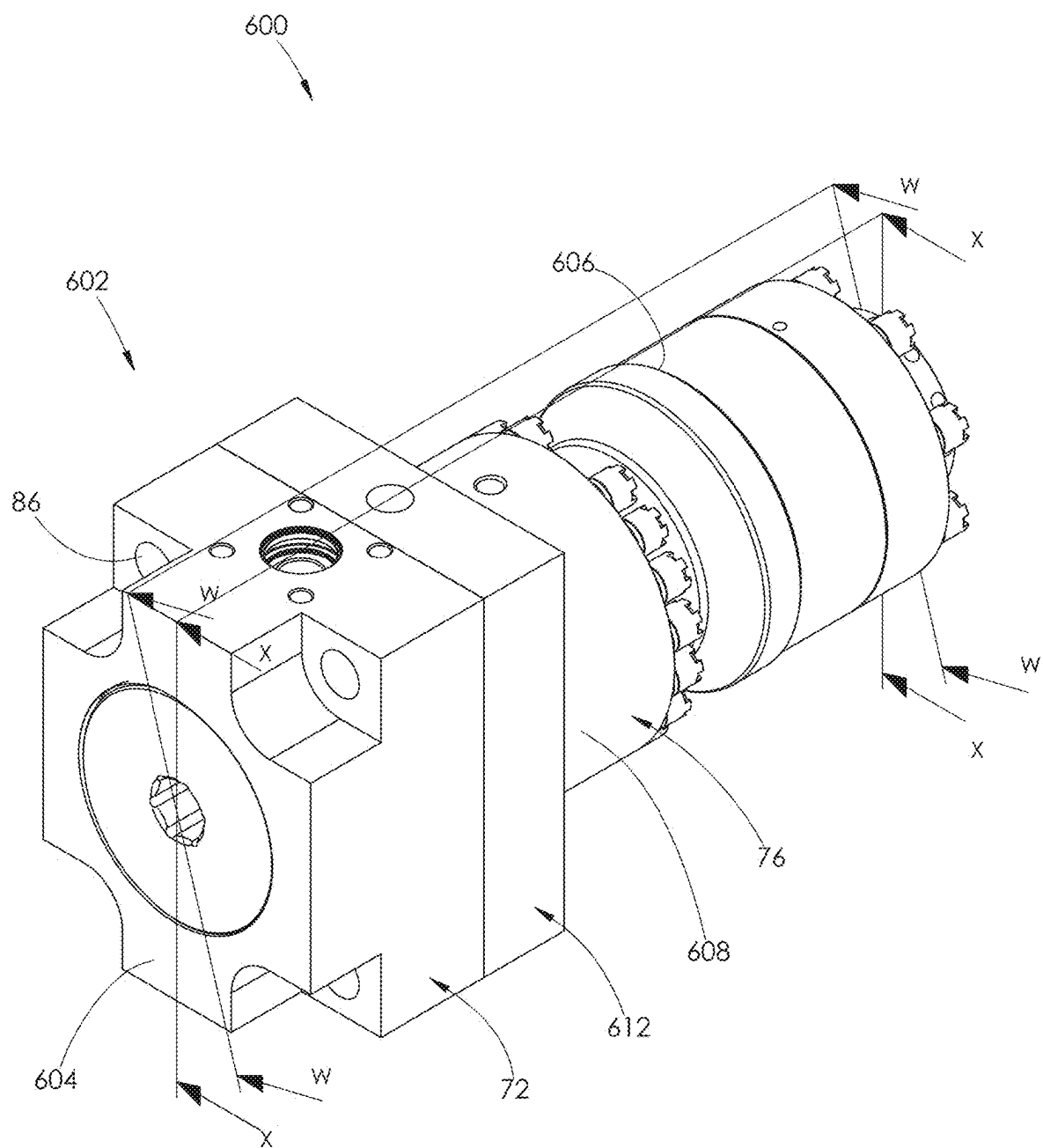


FIG. 81

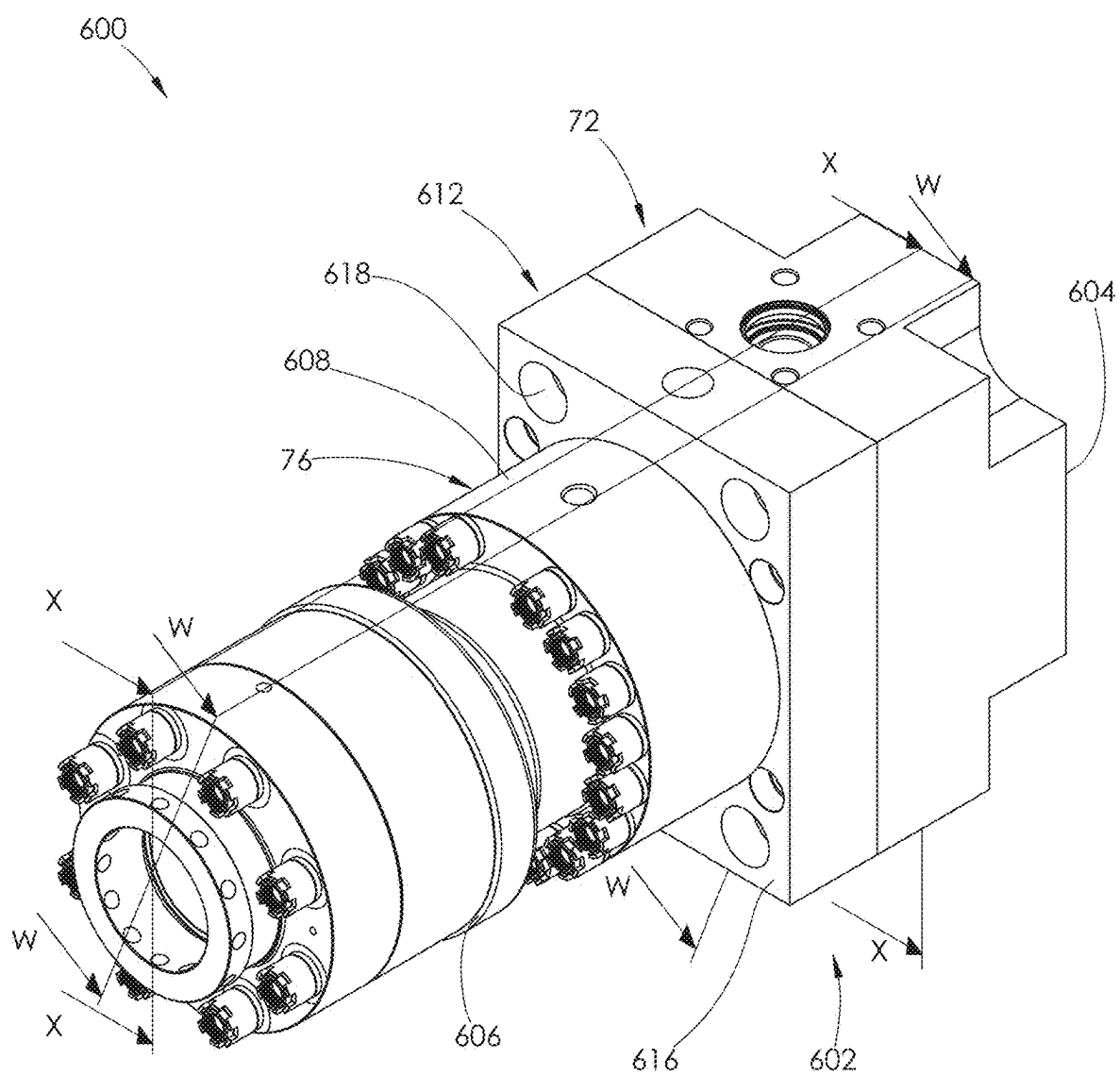


FIG. 82

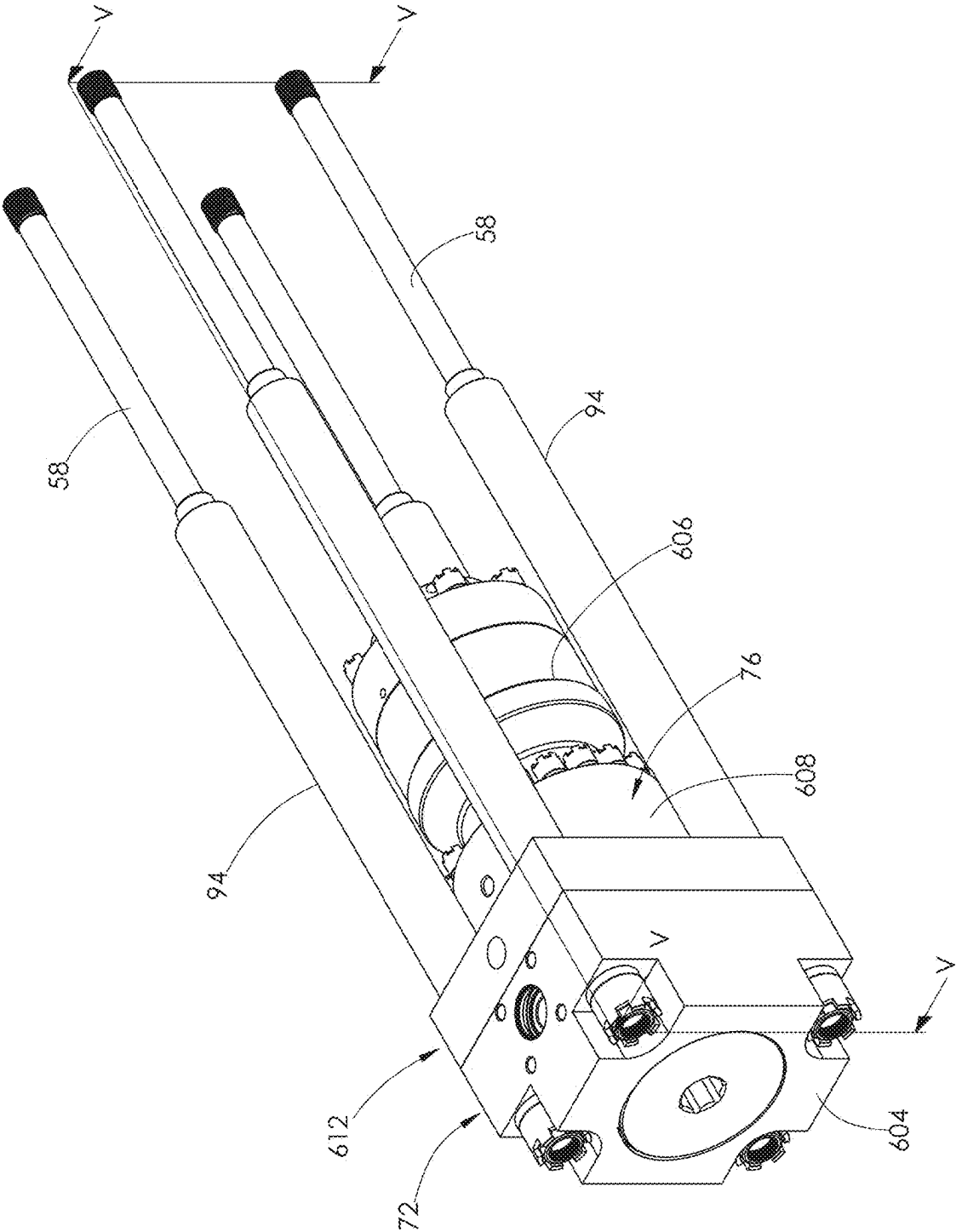


FIG. 83

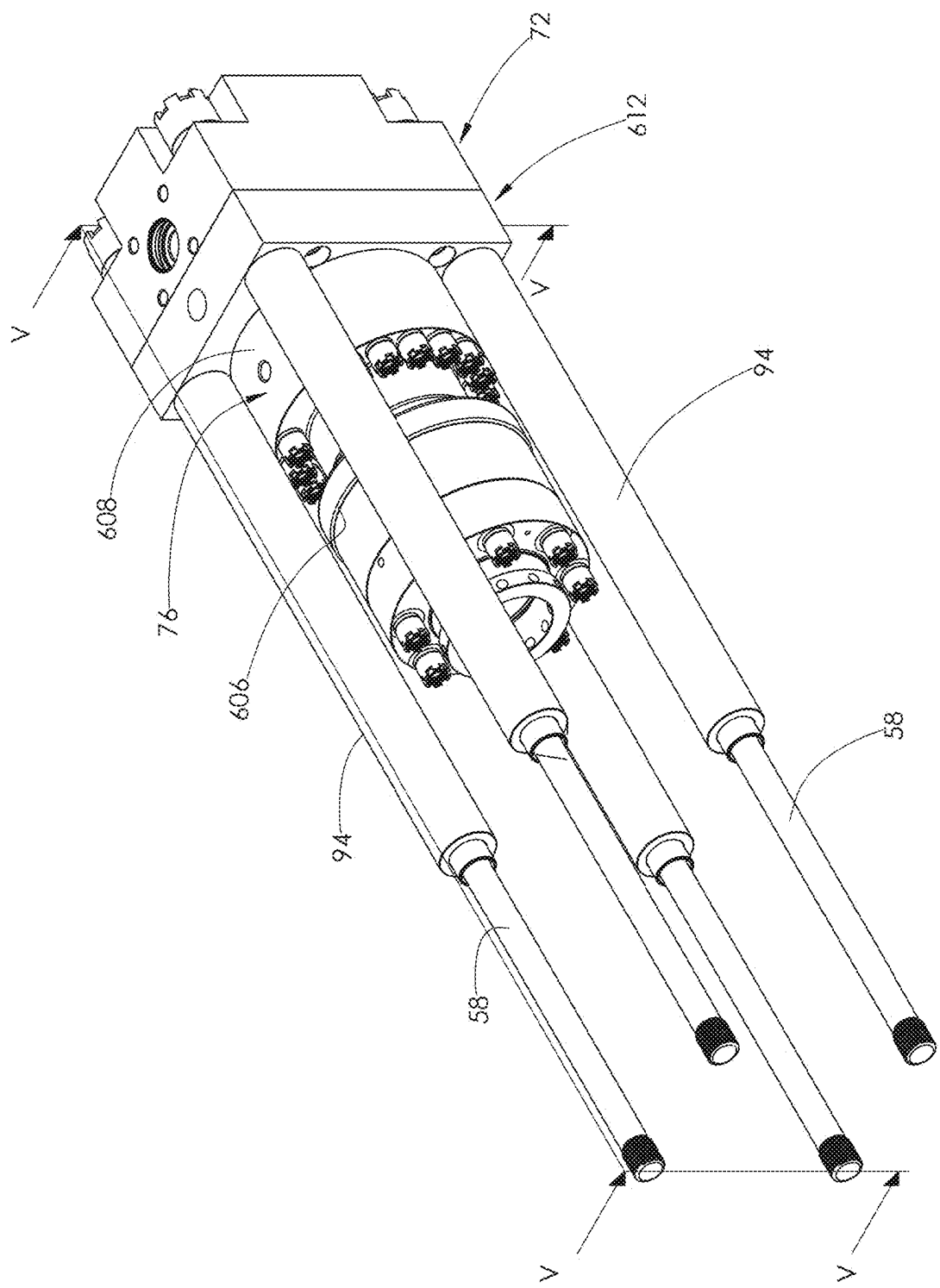


FIG. 84

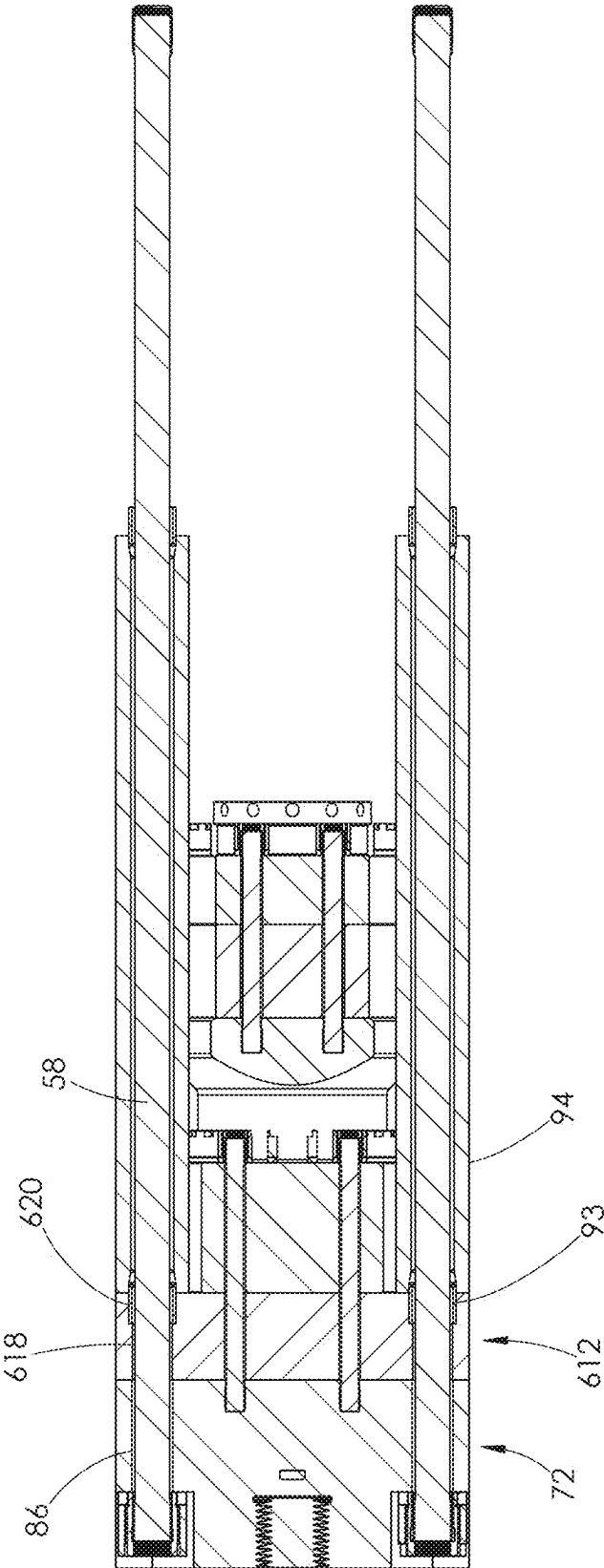


FIG. 85

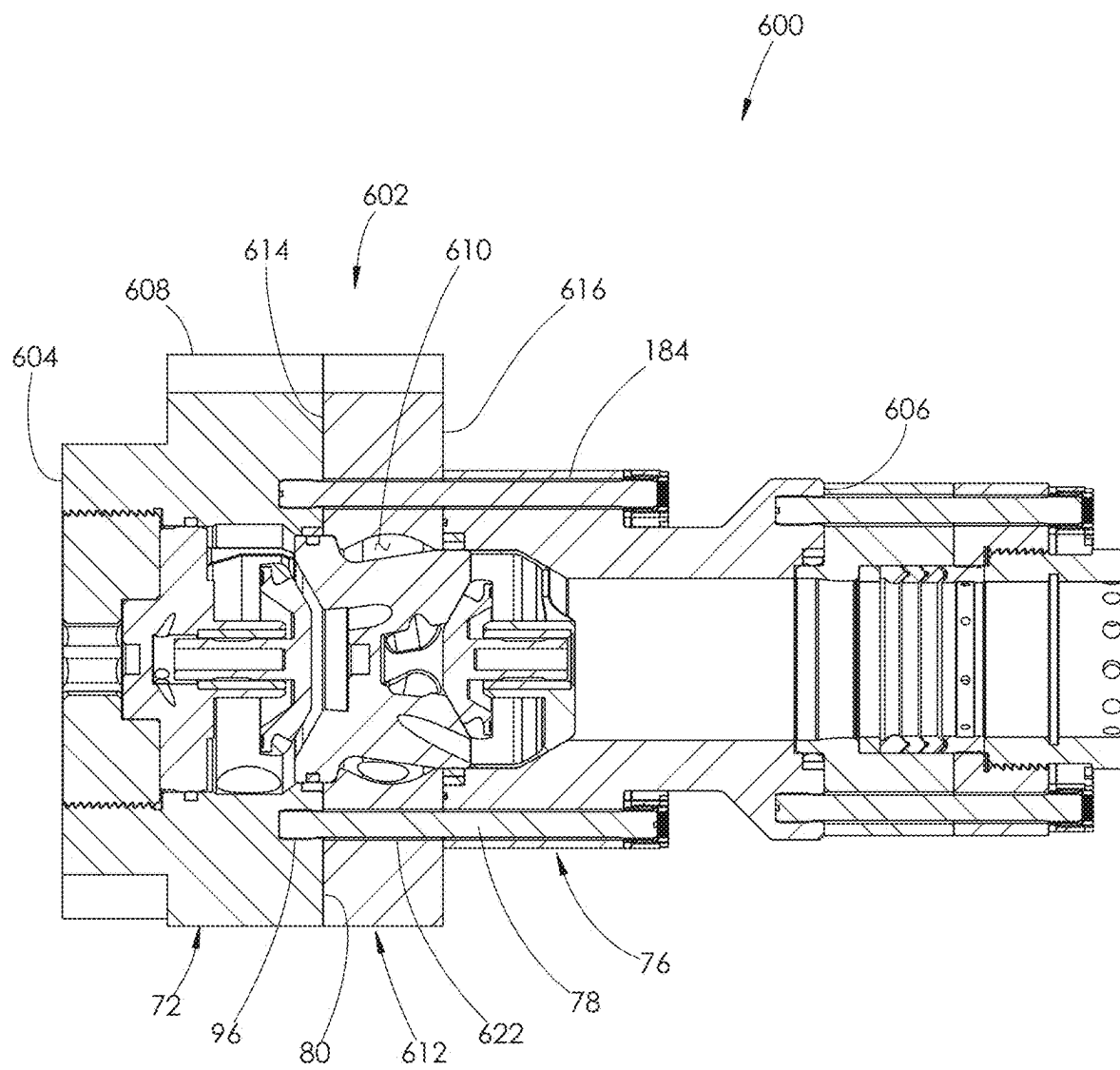


FIG. 86

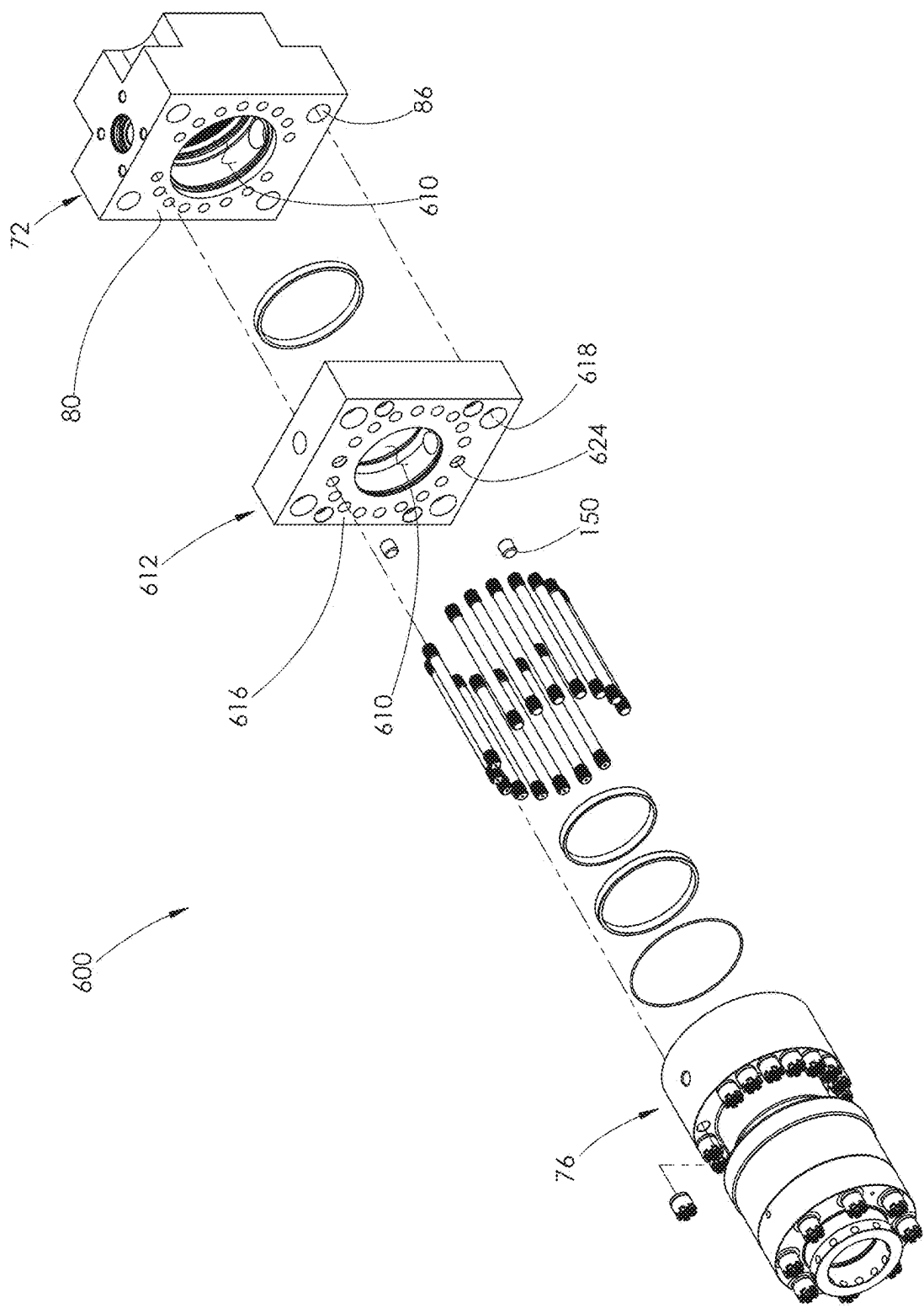


FIG. 87

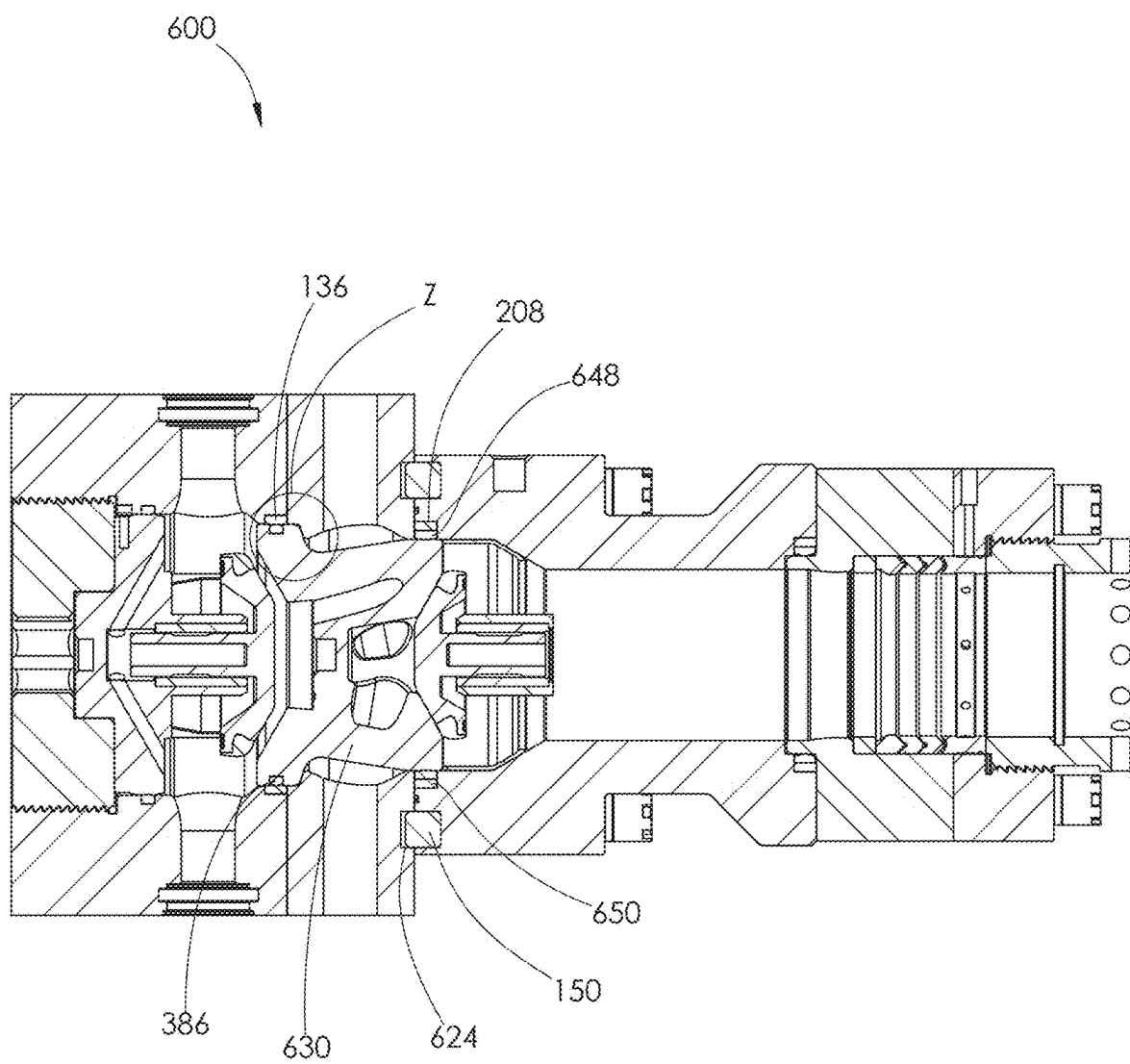


FIG. 88

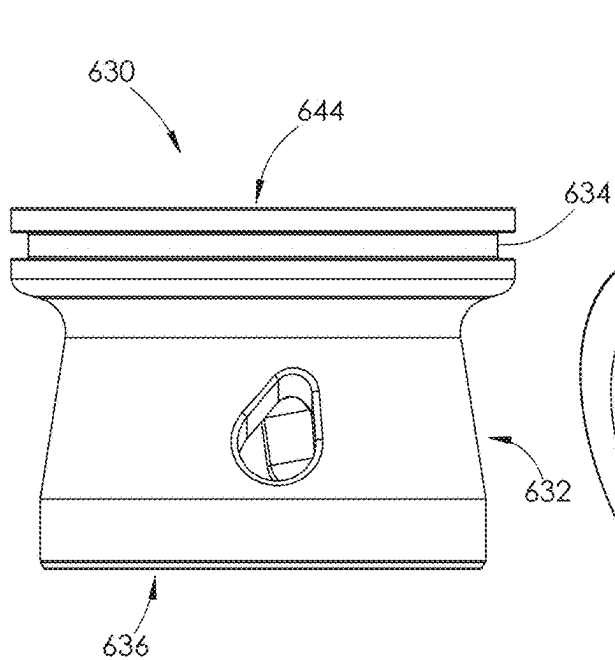


FIG. 89

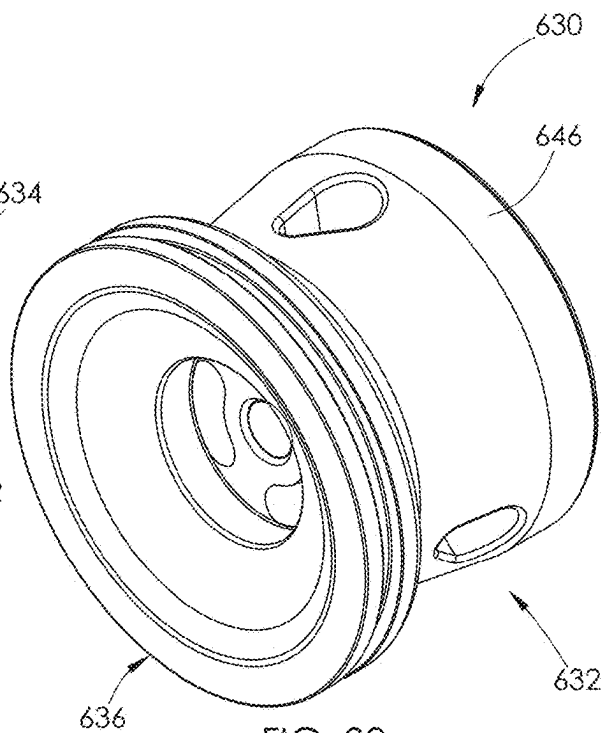


FIG. 90

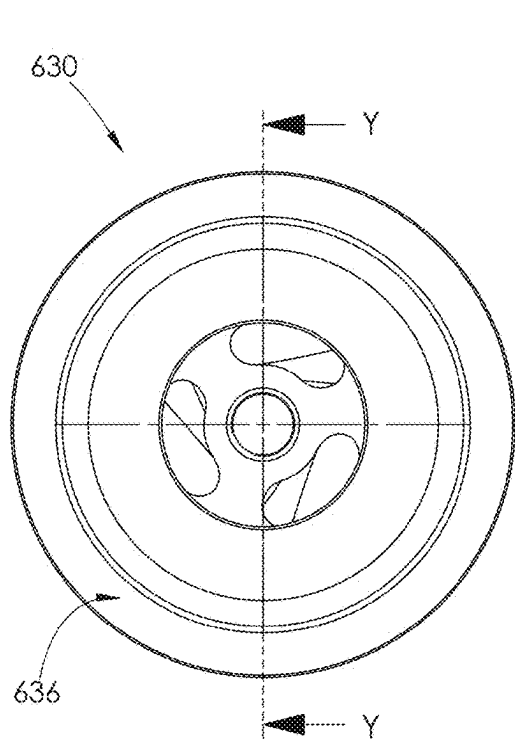


FIG. 91

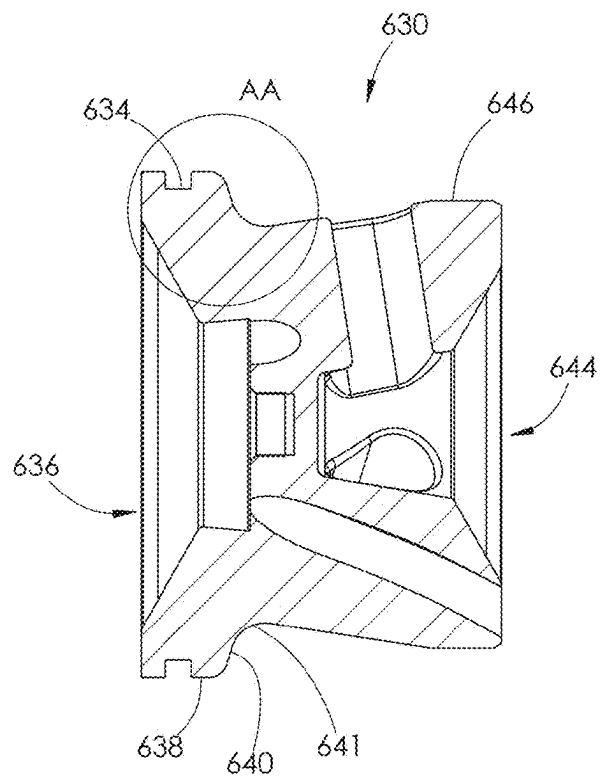


FIG. 92

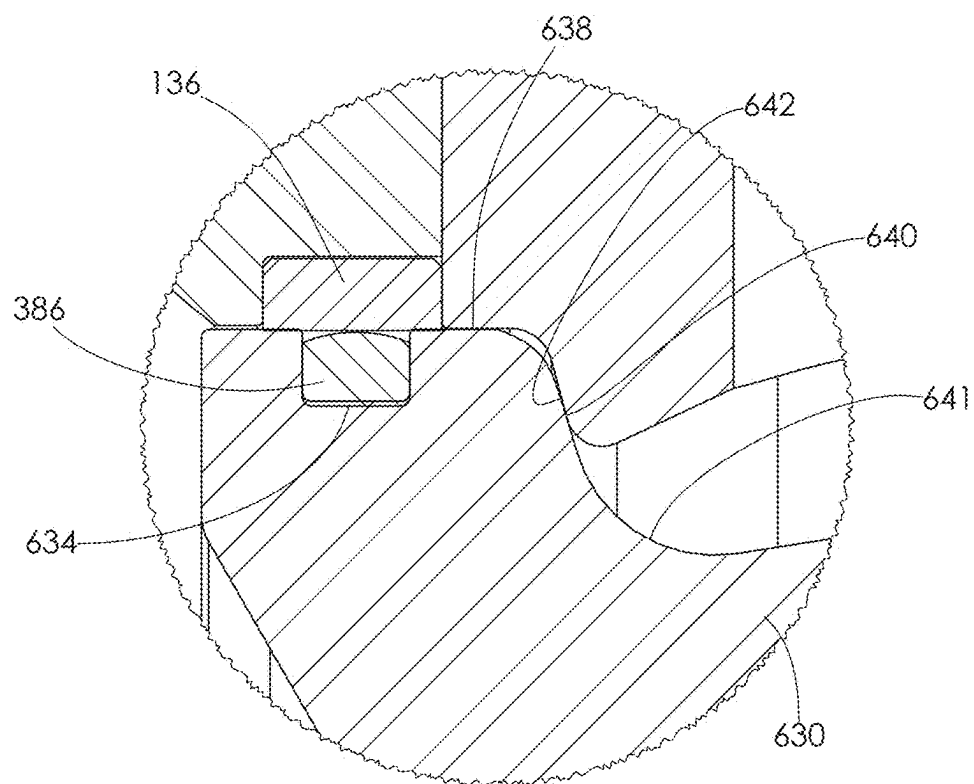


FIG. 93

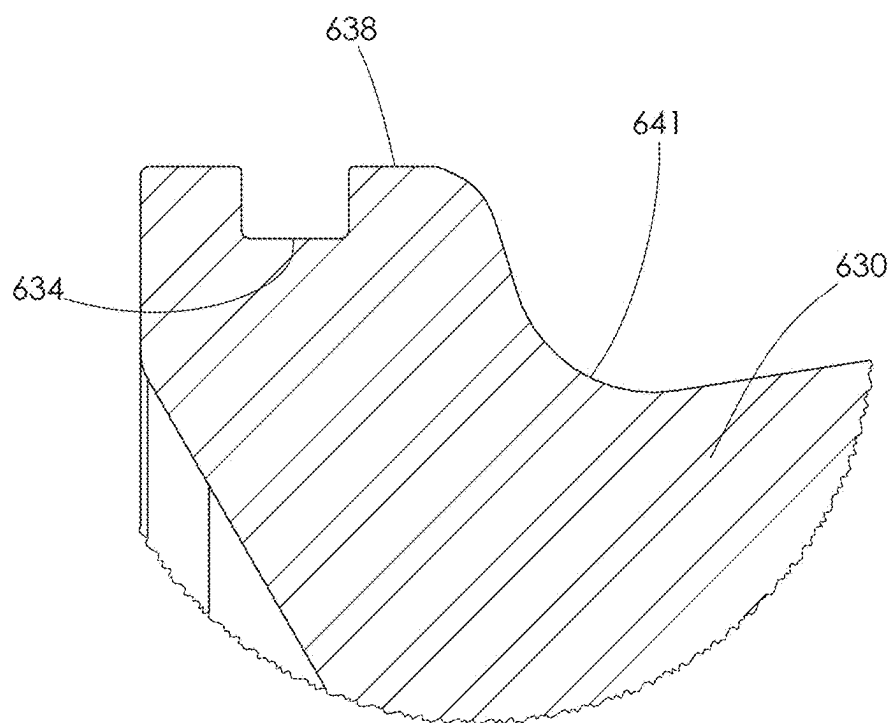


FIG. 94

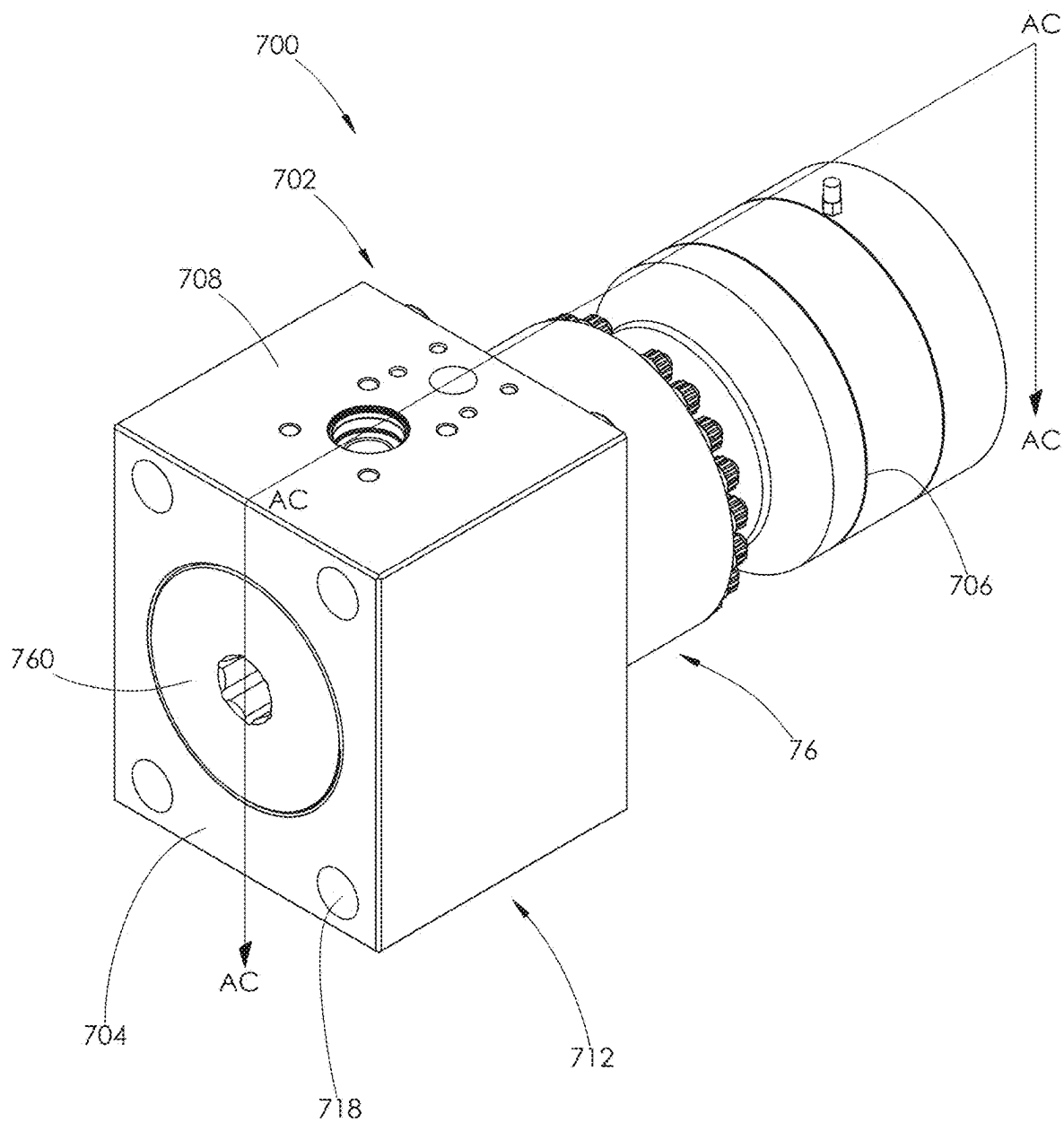


FIG. 95

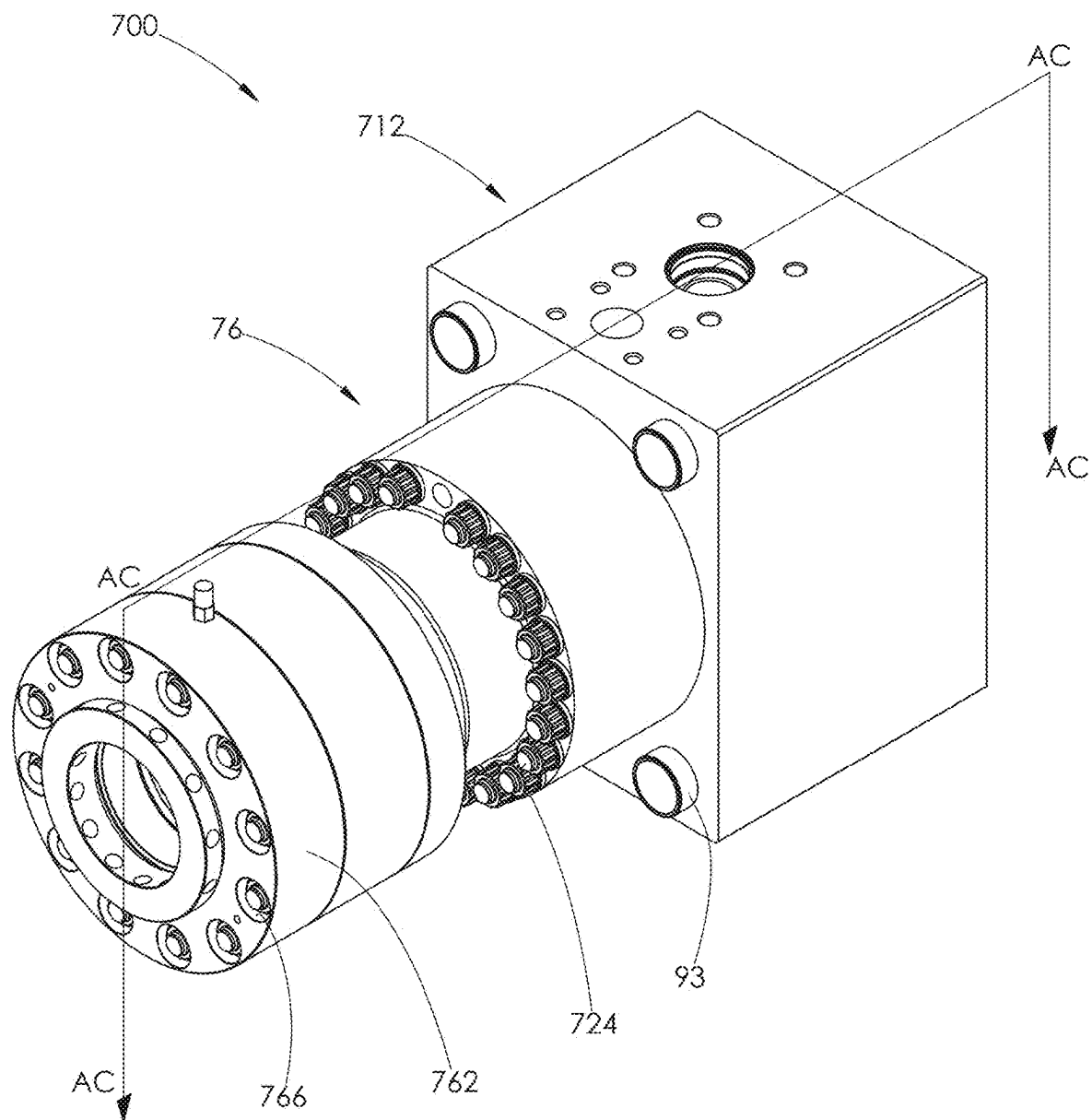


FIG. 96

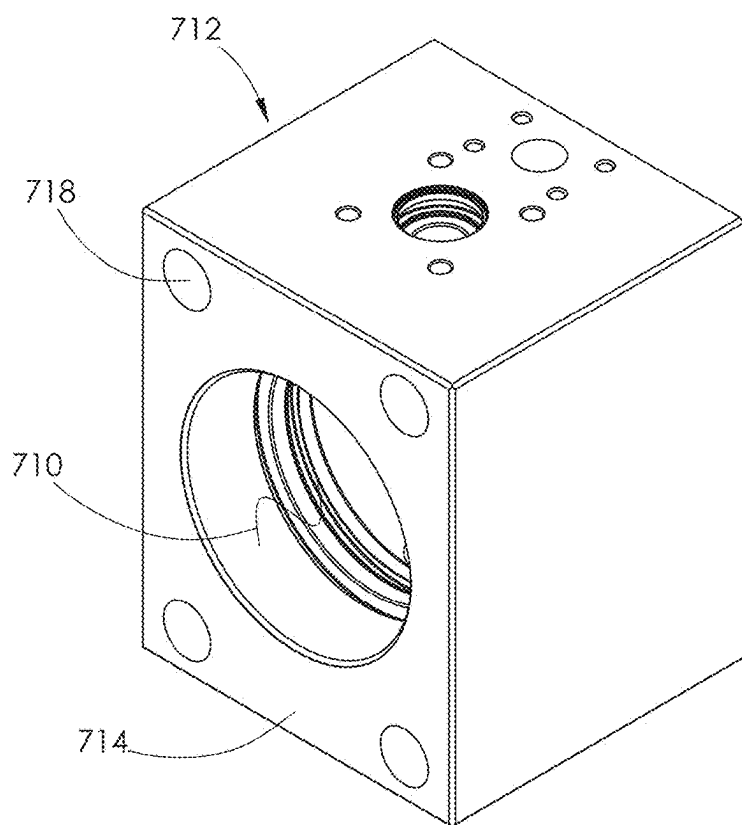


FIG. 97

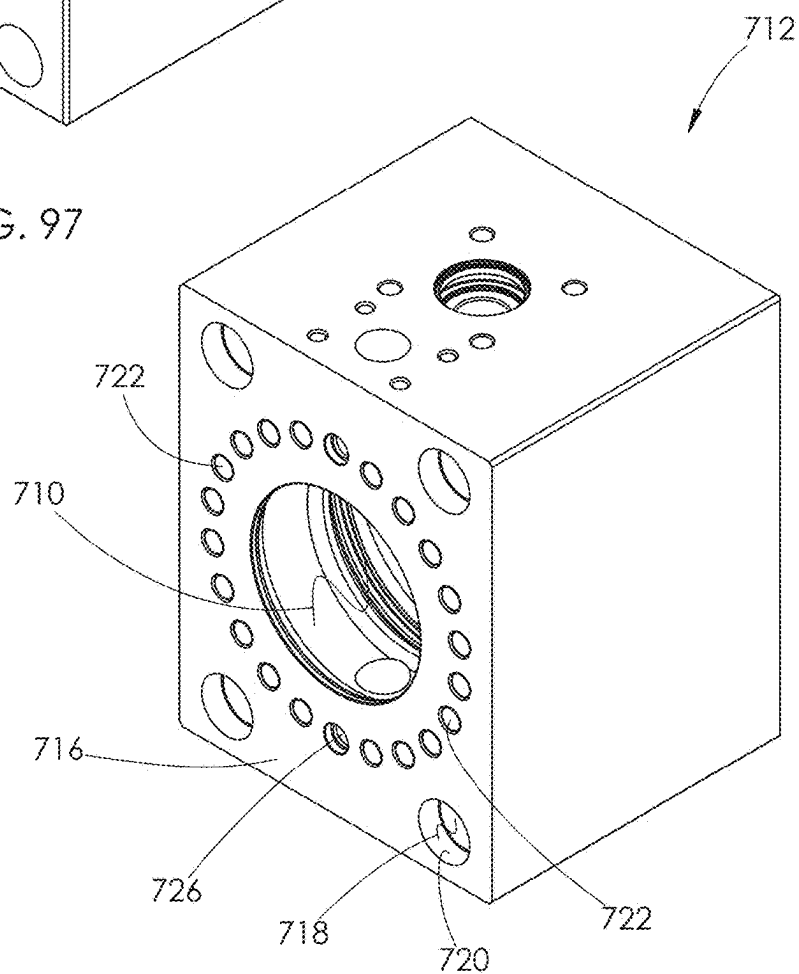
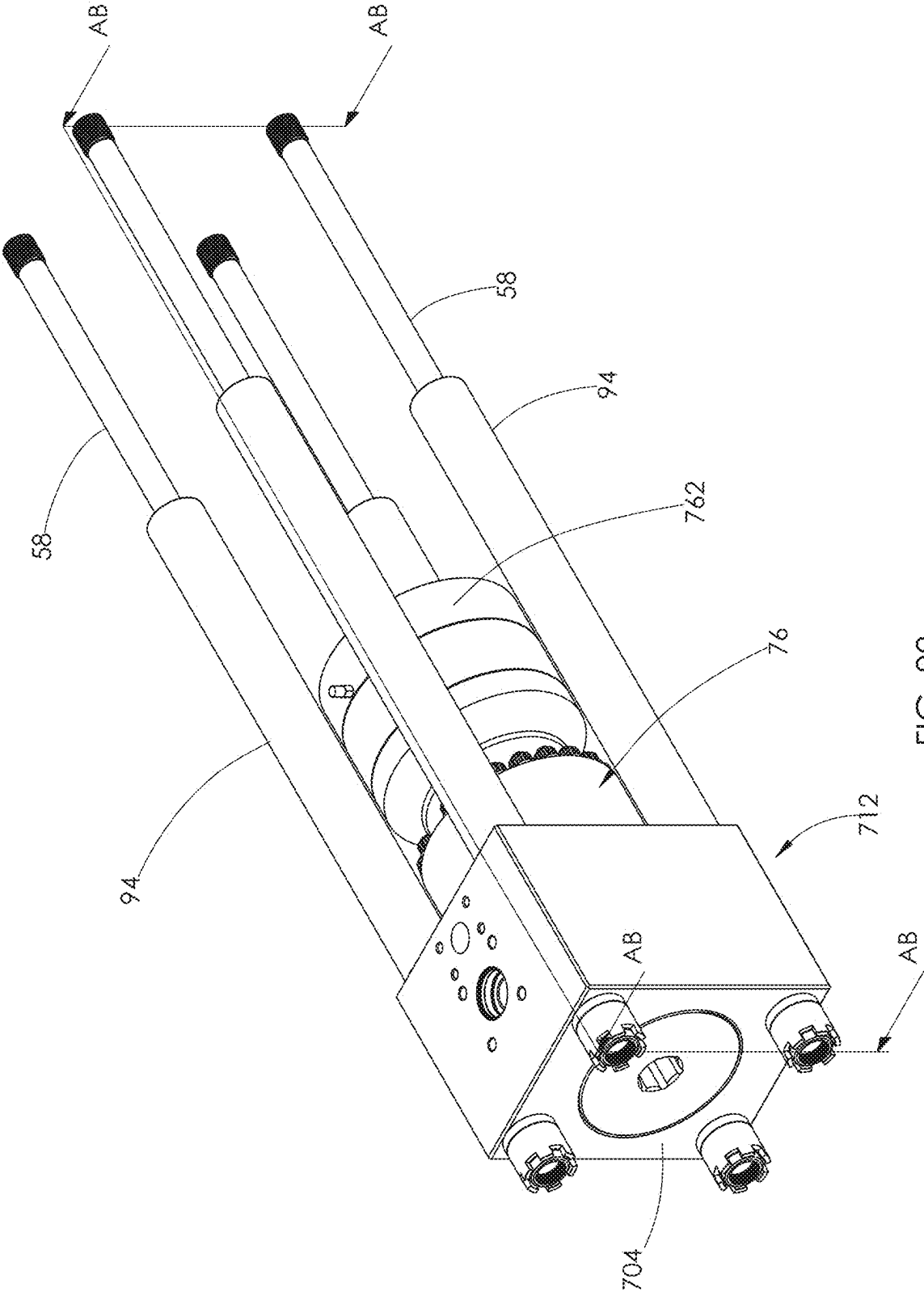


FIG. 98



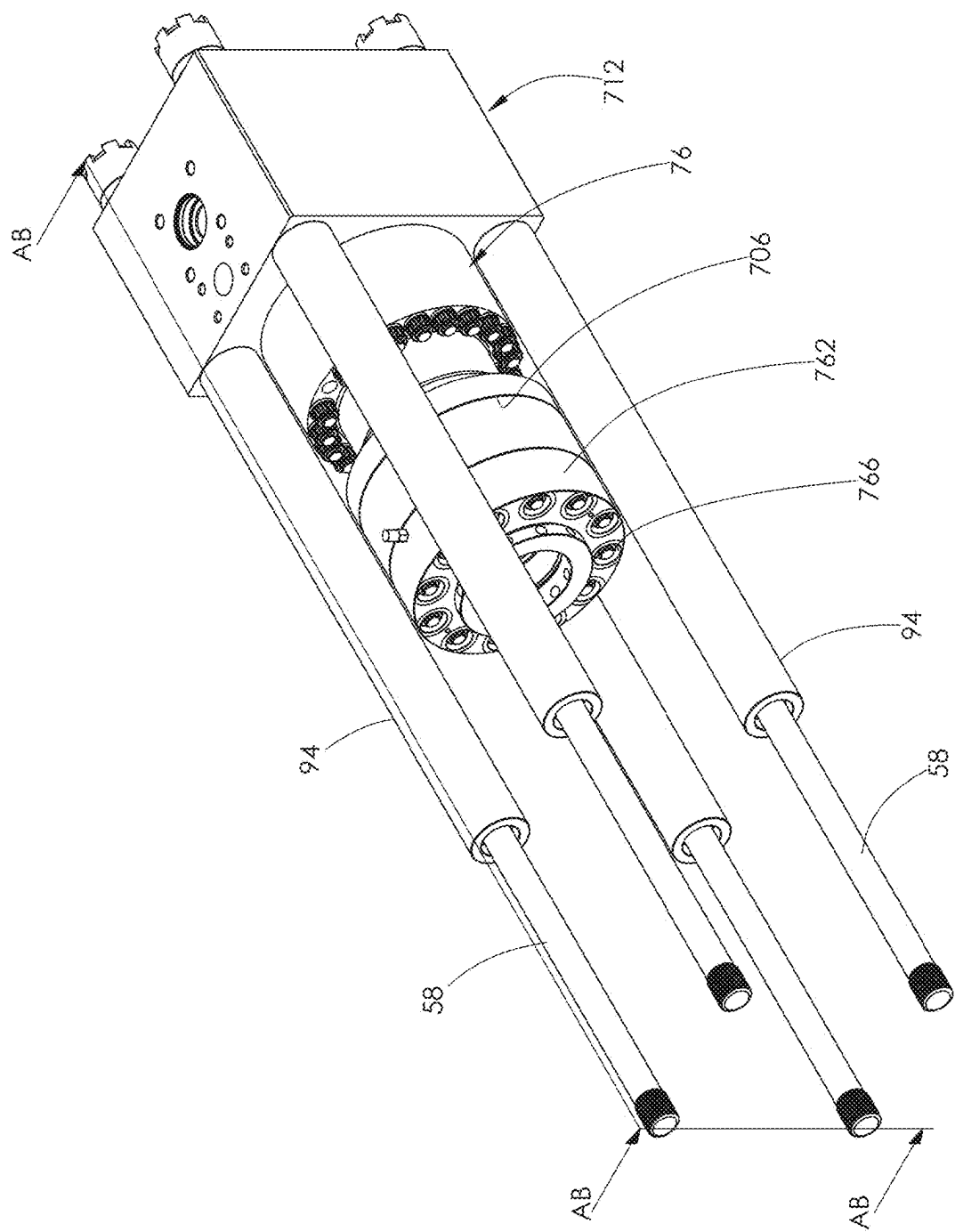


FIG. 100

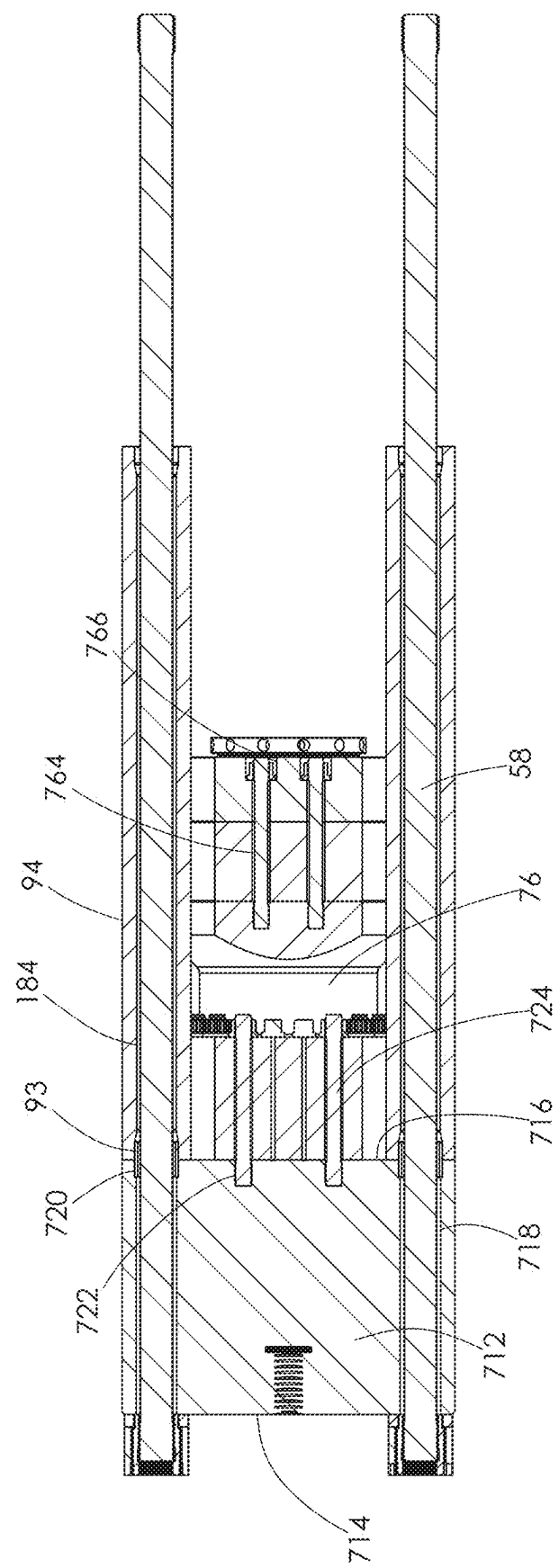


FIG. 101

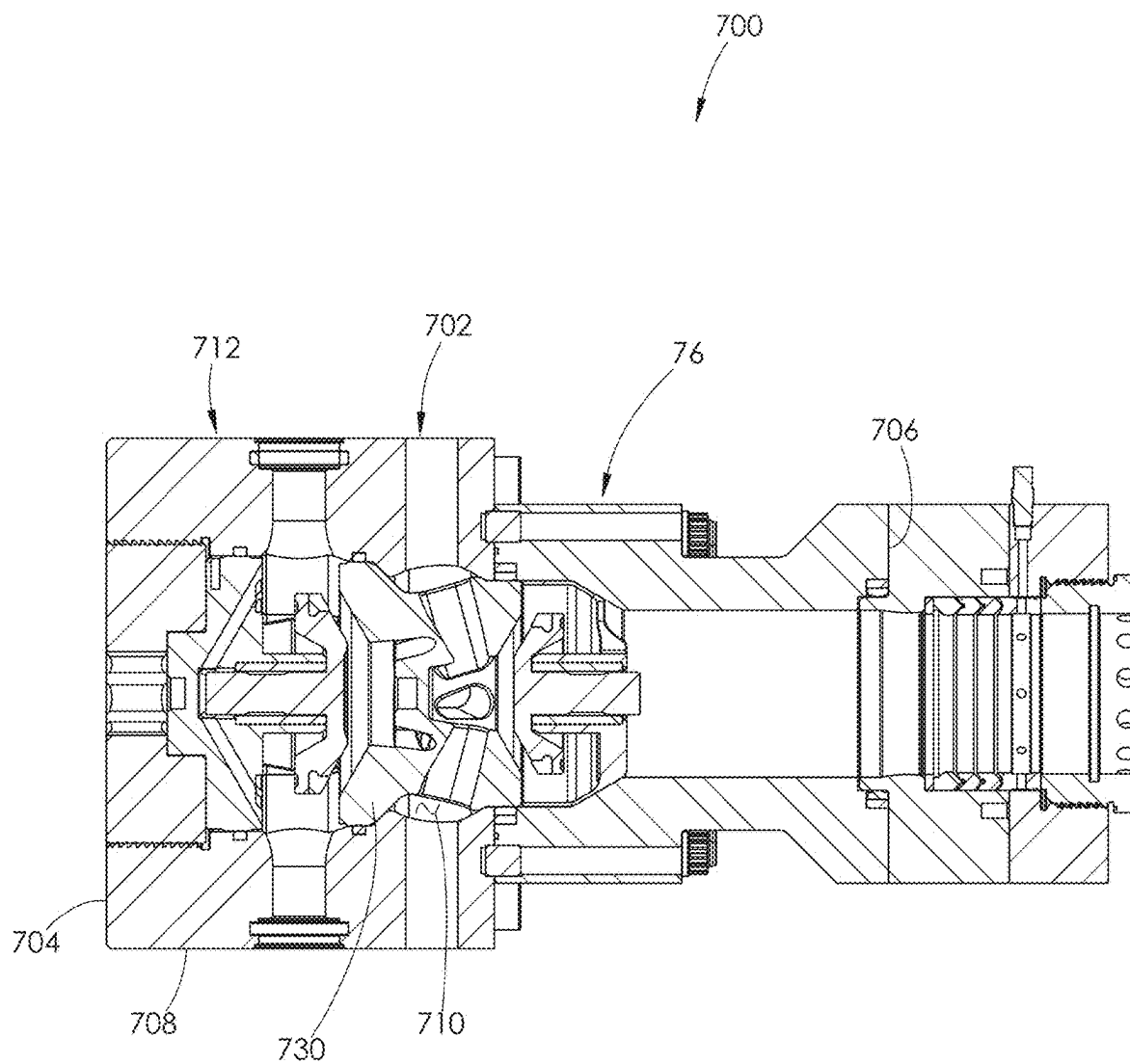


FIG. 102

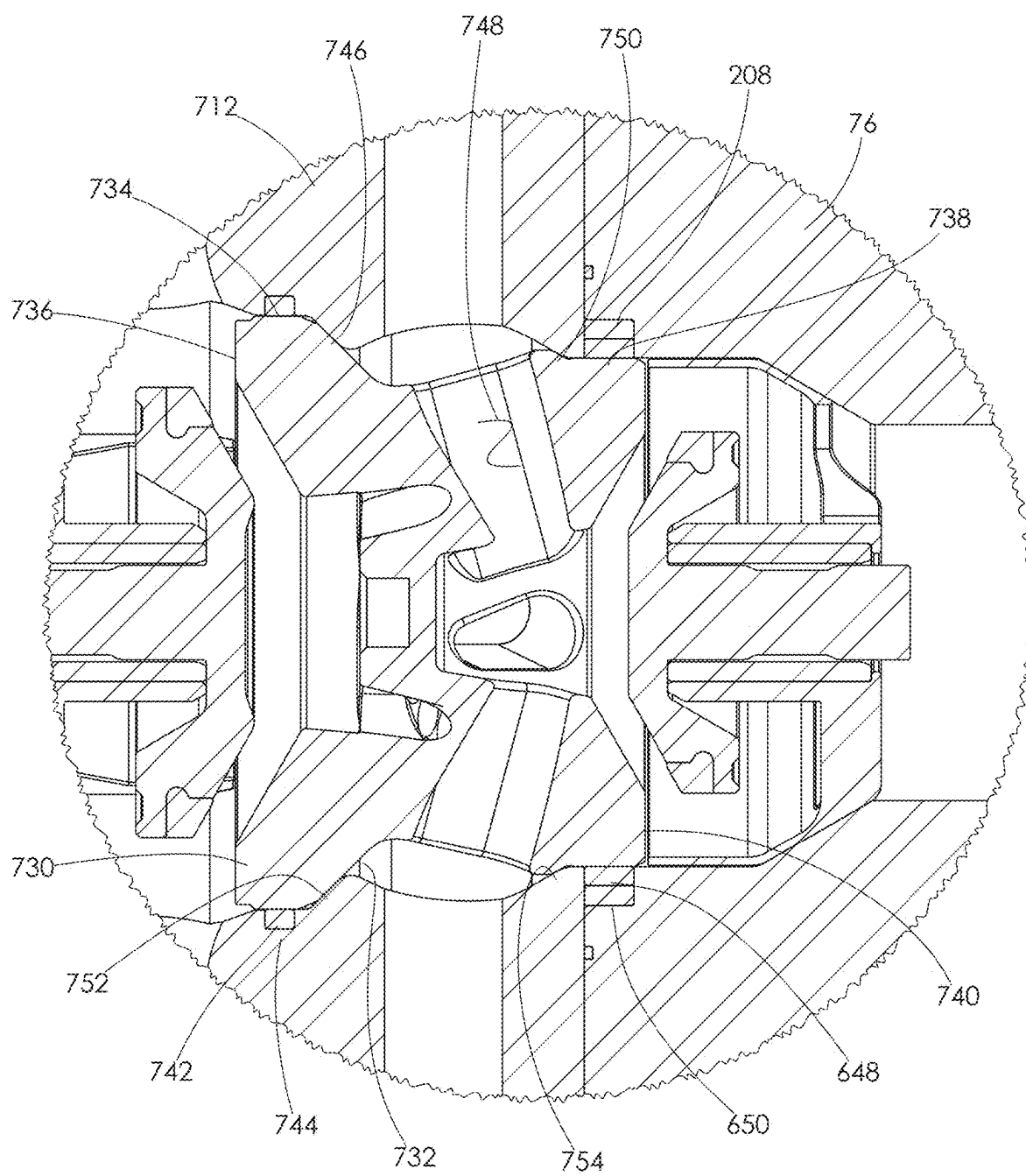


FIG. 103

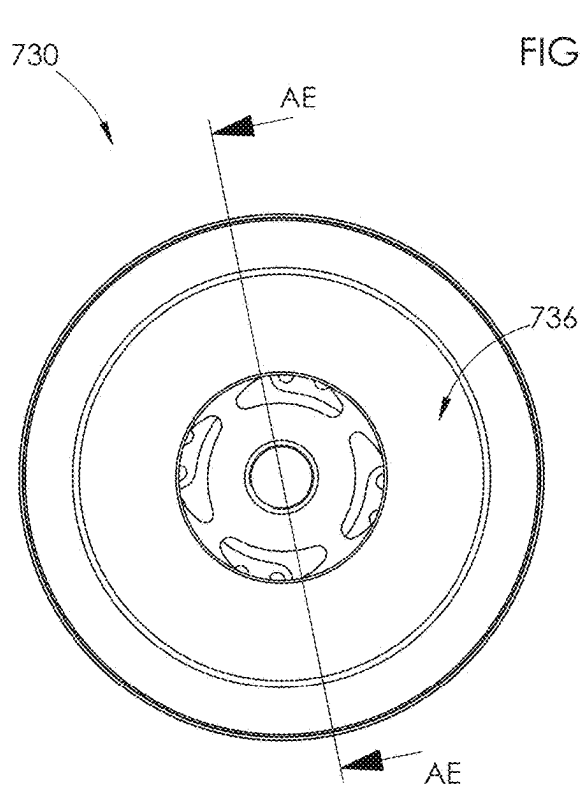
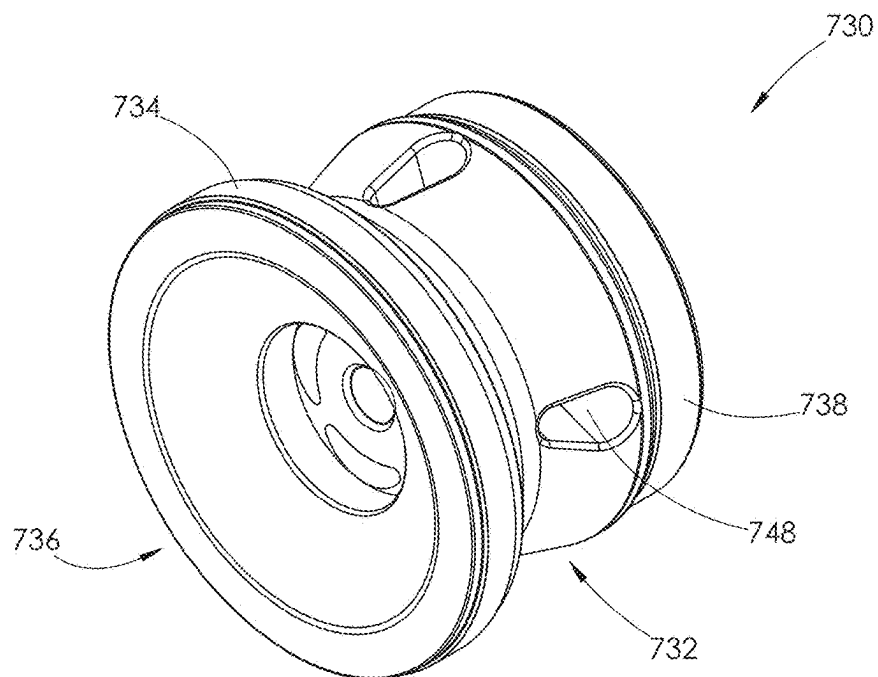


FIG. 105

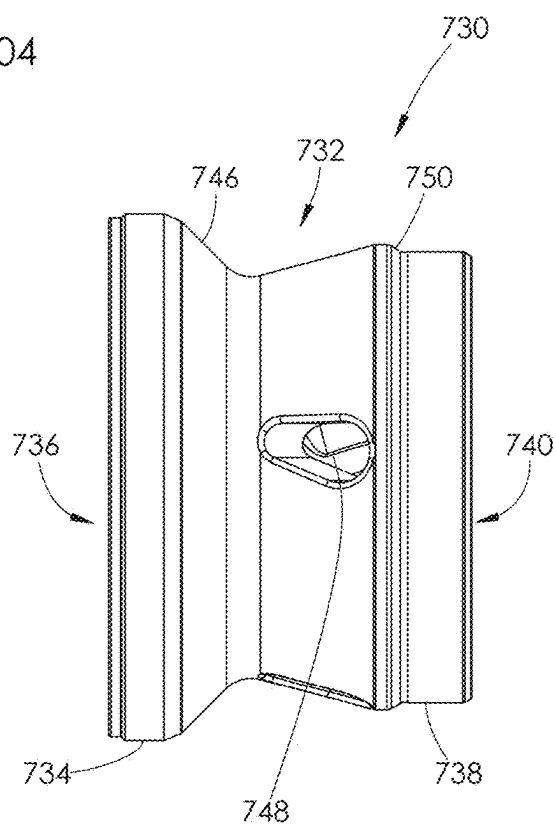
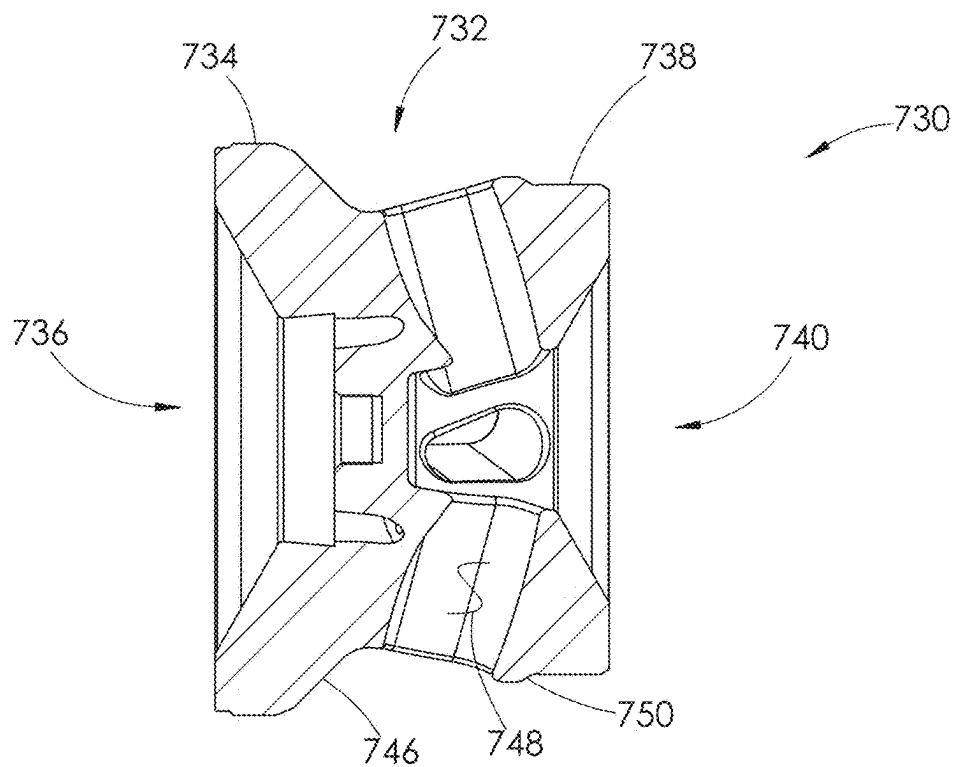
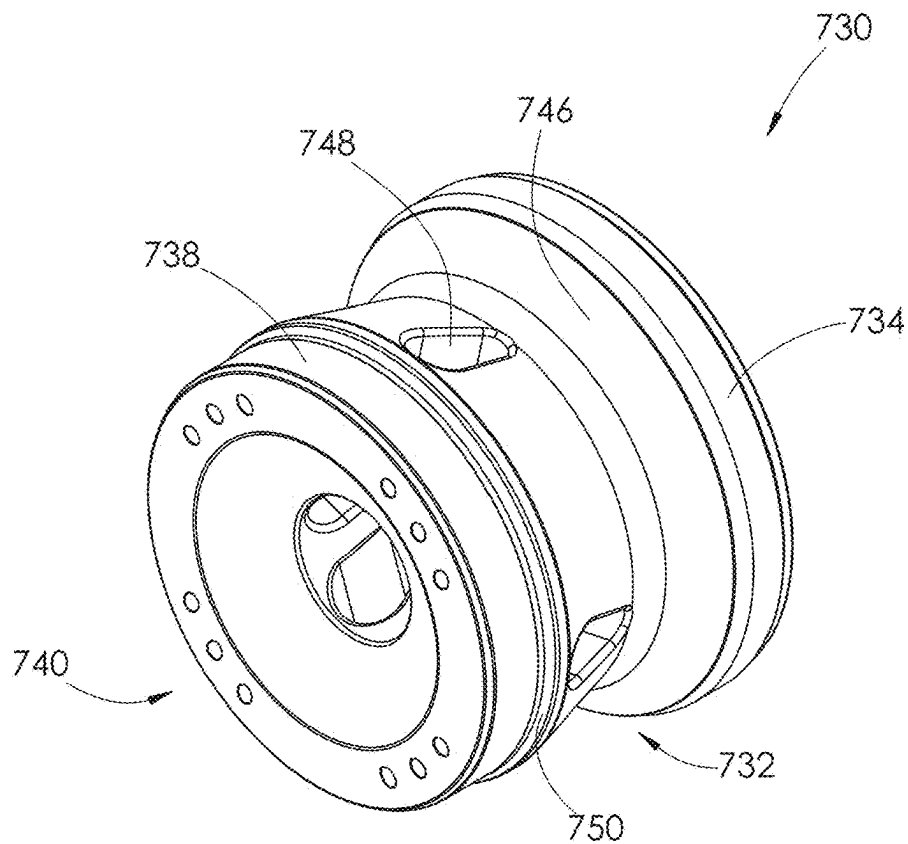


FIG. 106



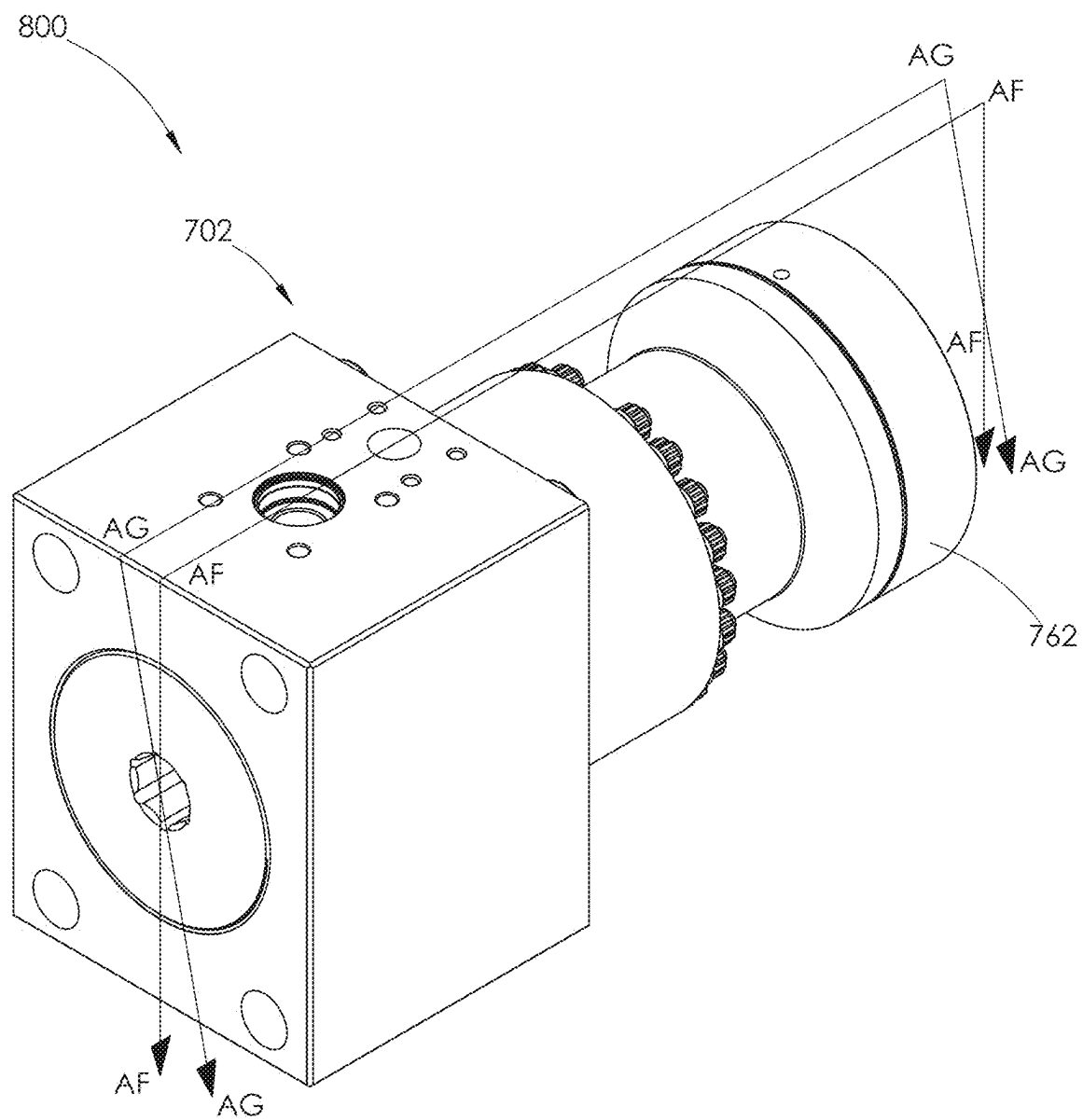


FIG. 109

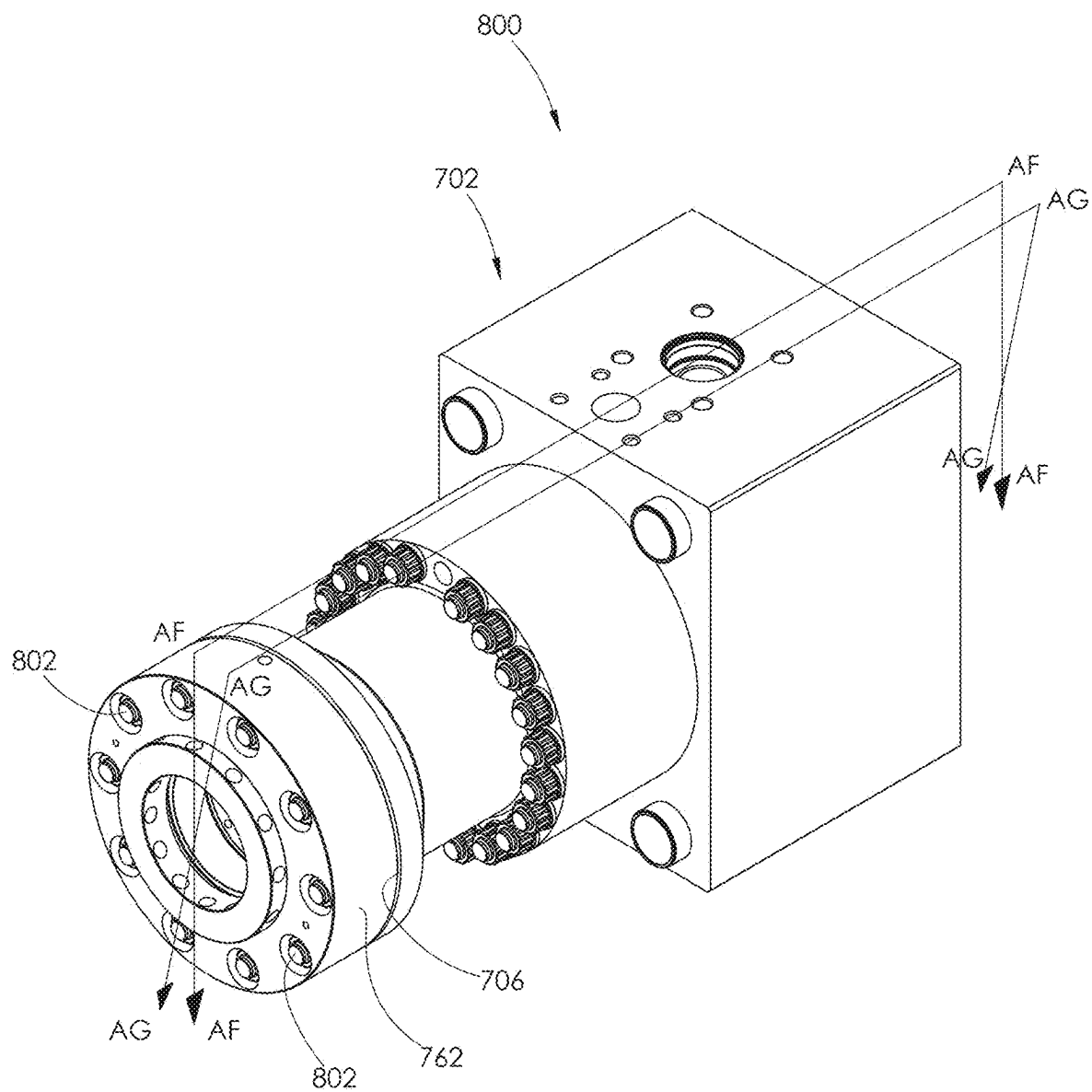


FIG. 110

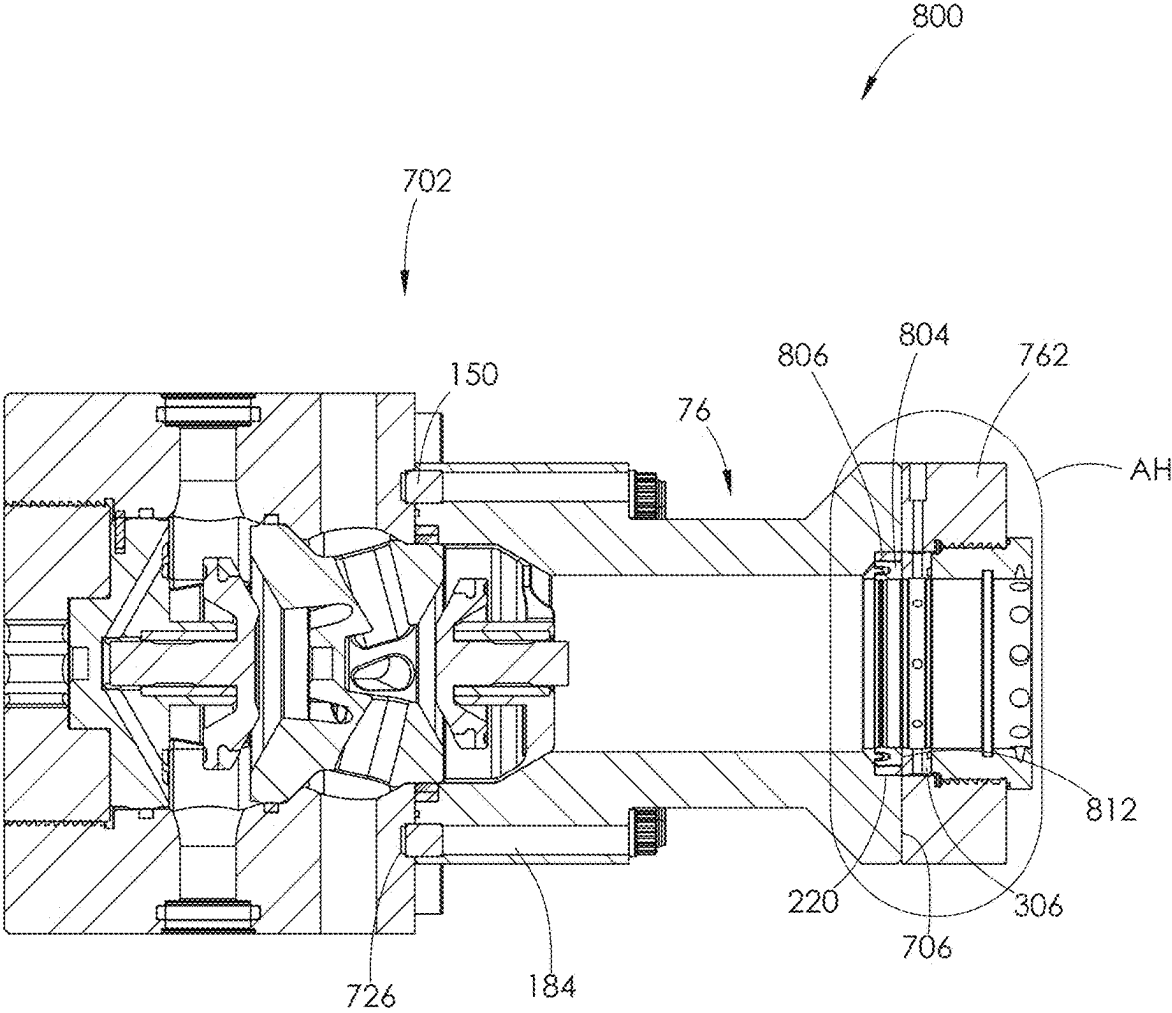


FIG. 111

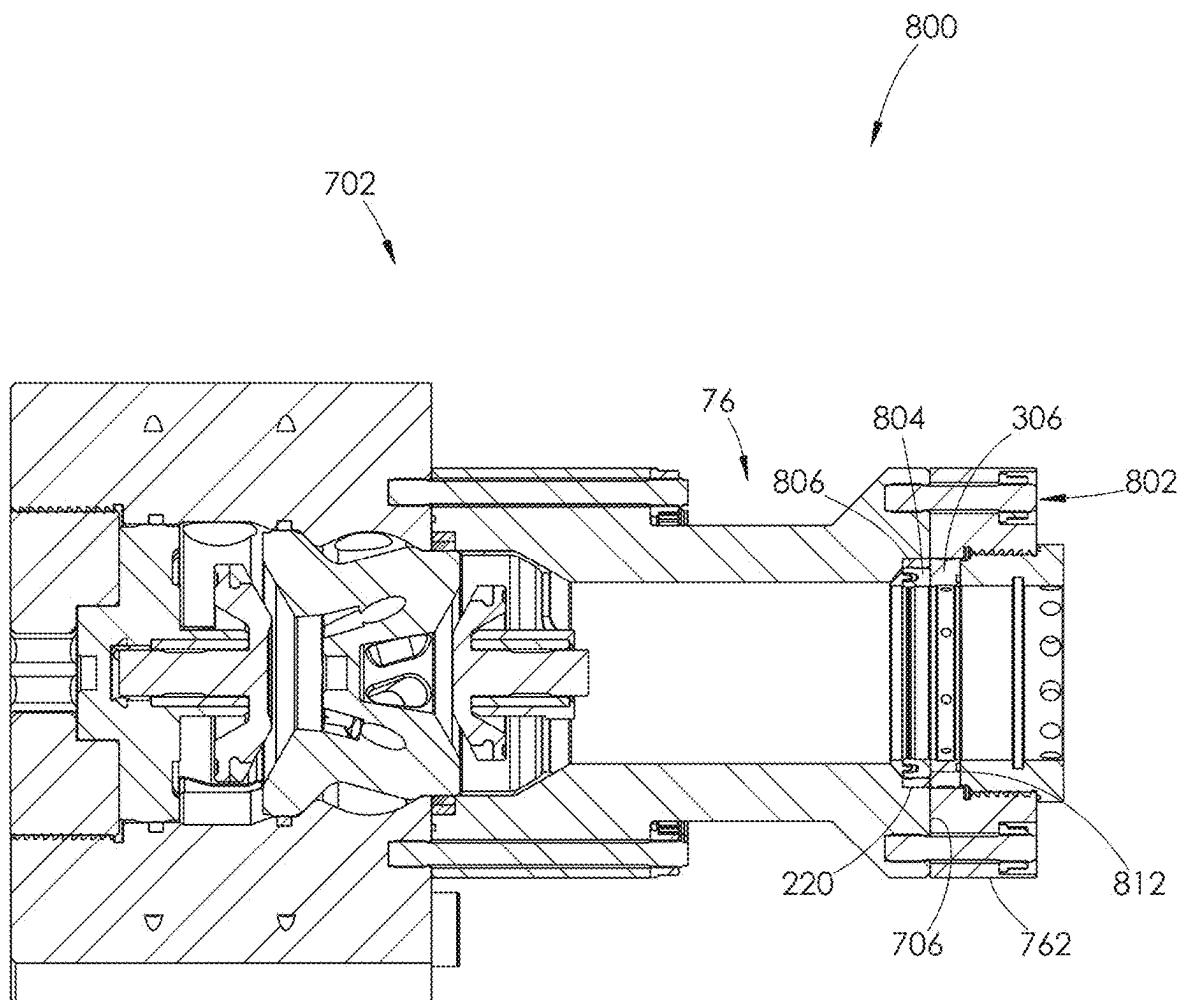


FIG. 112

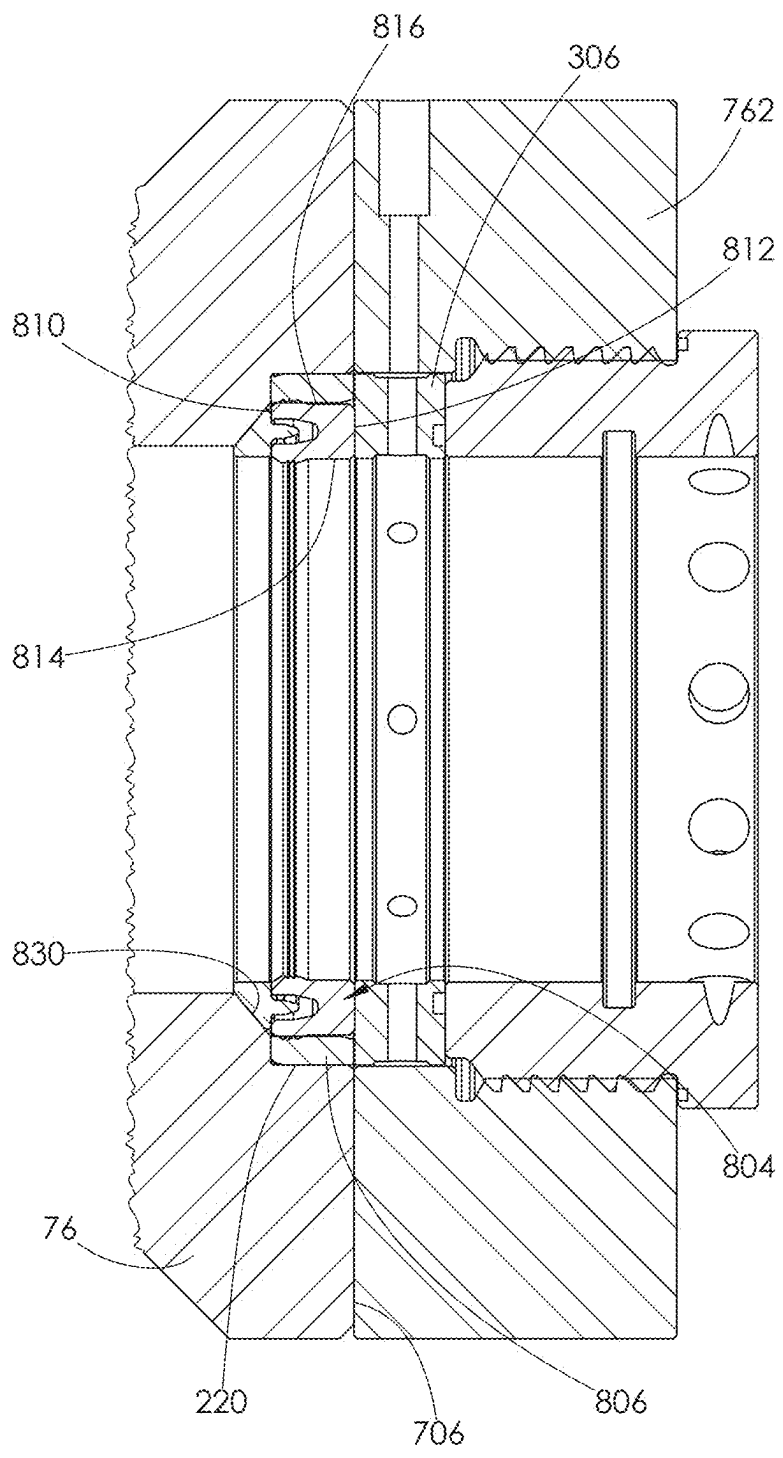


FIG. 113

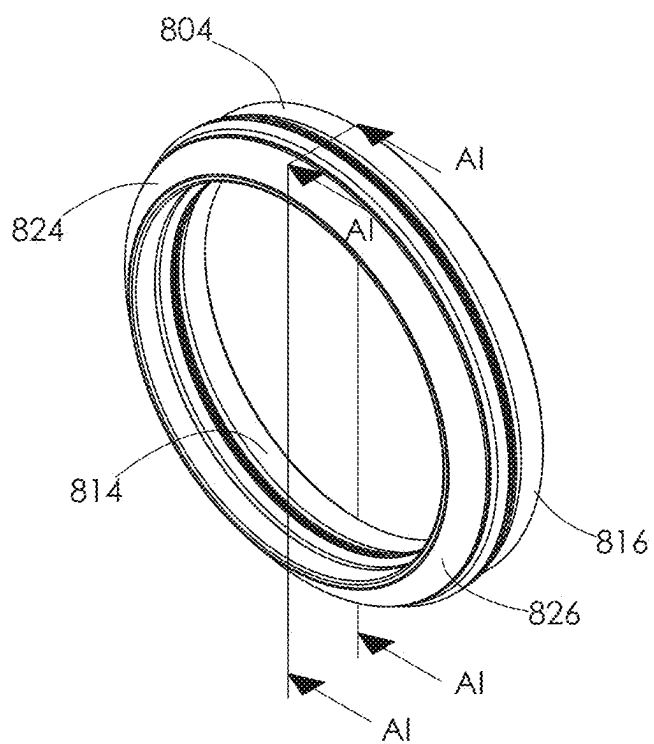


FIG. 114

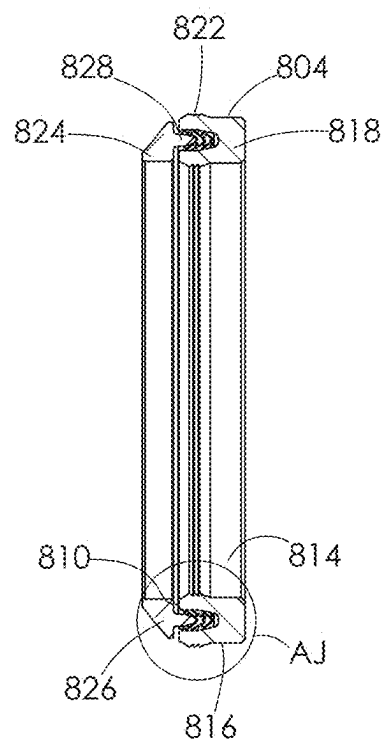


FIG. 115

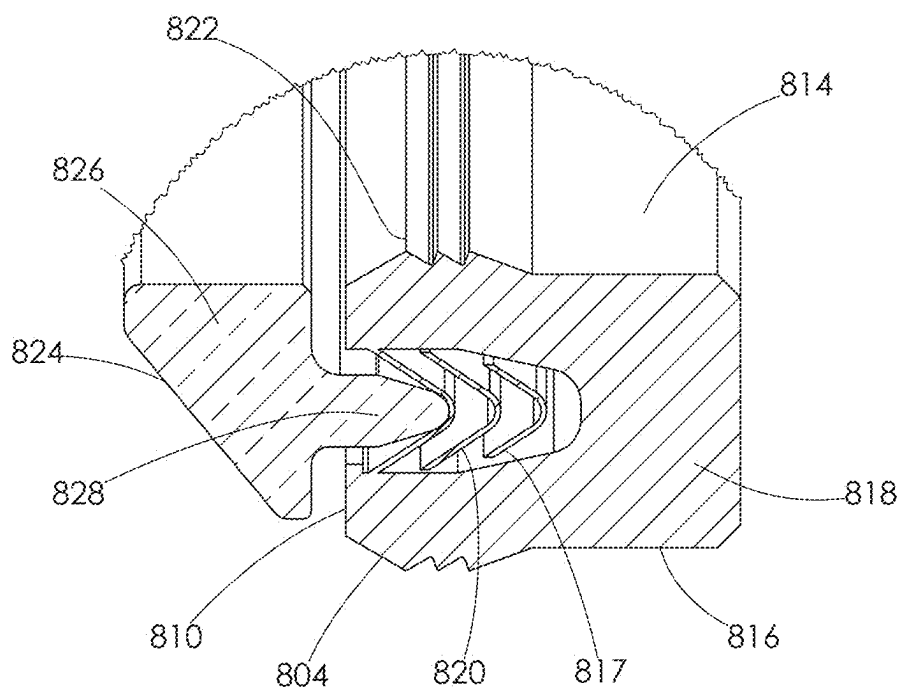


FIG. 116

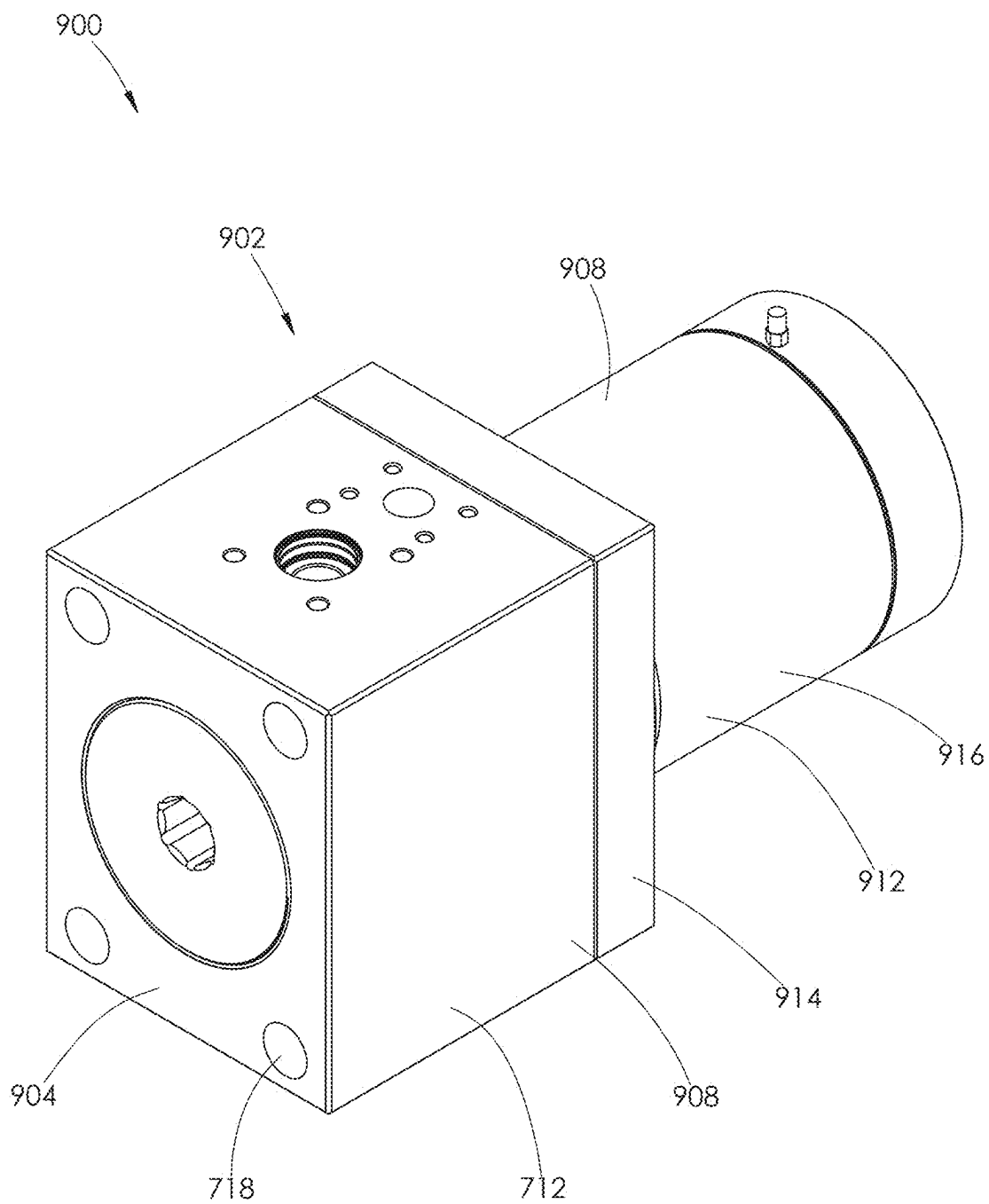


FIG. 117

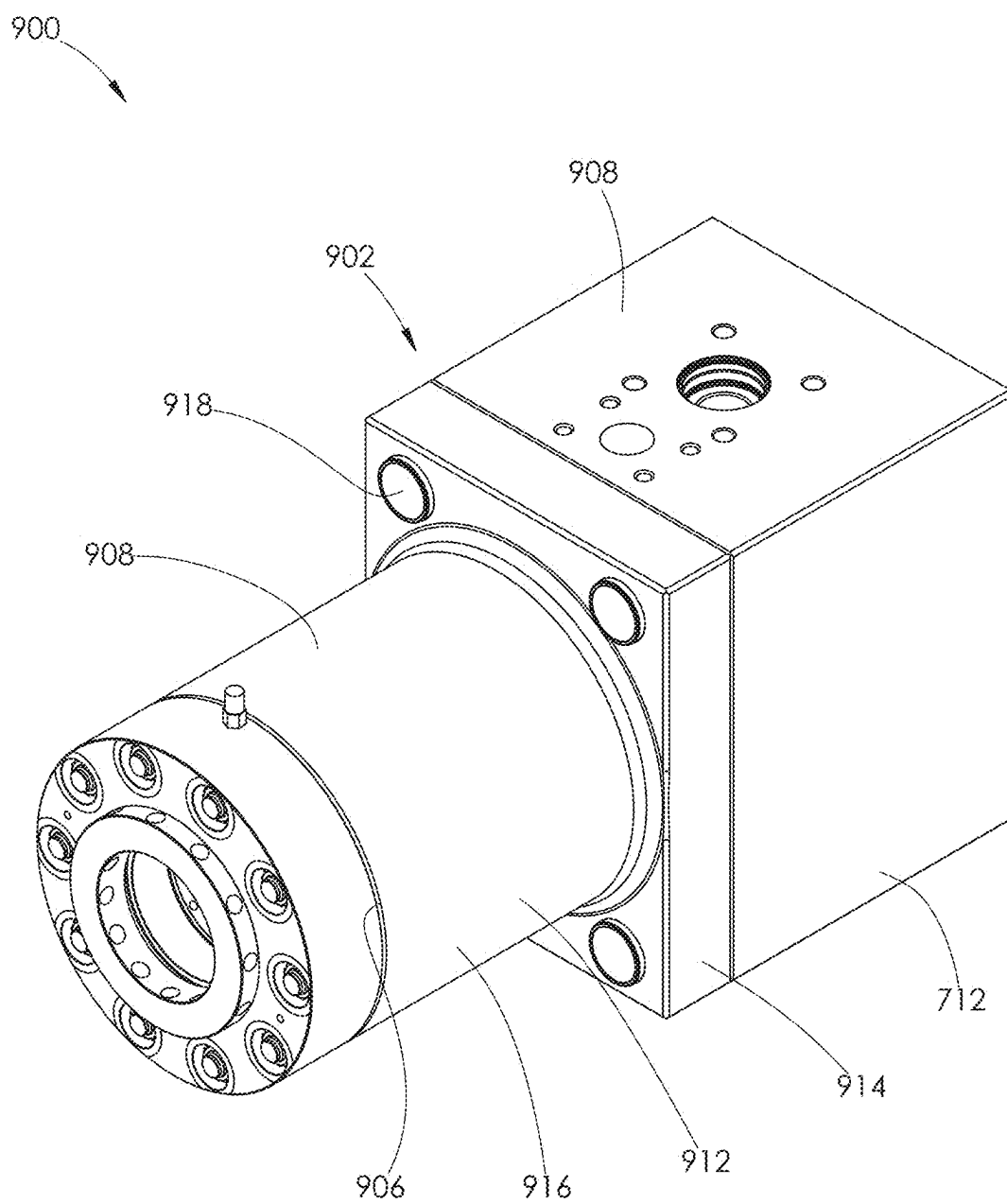


FIG. 118

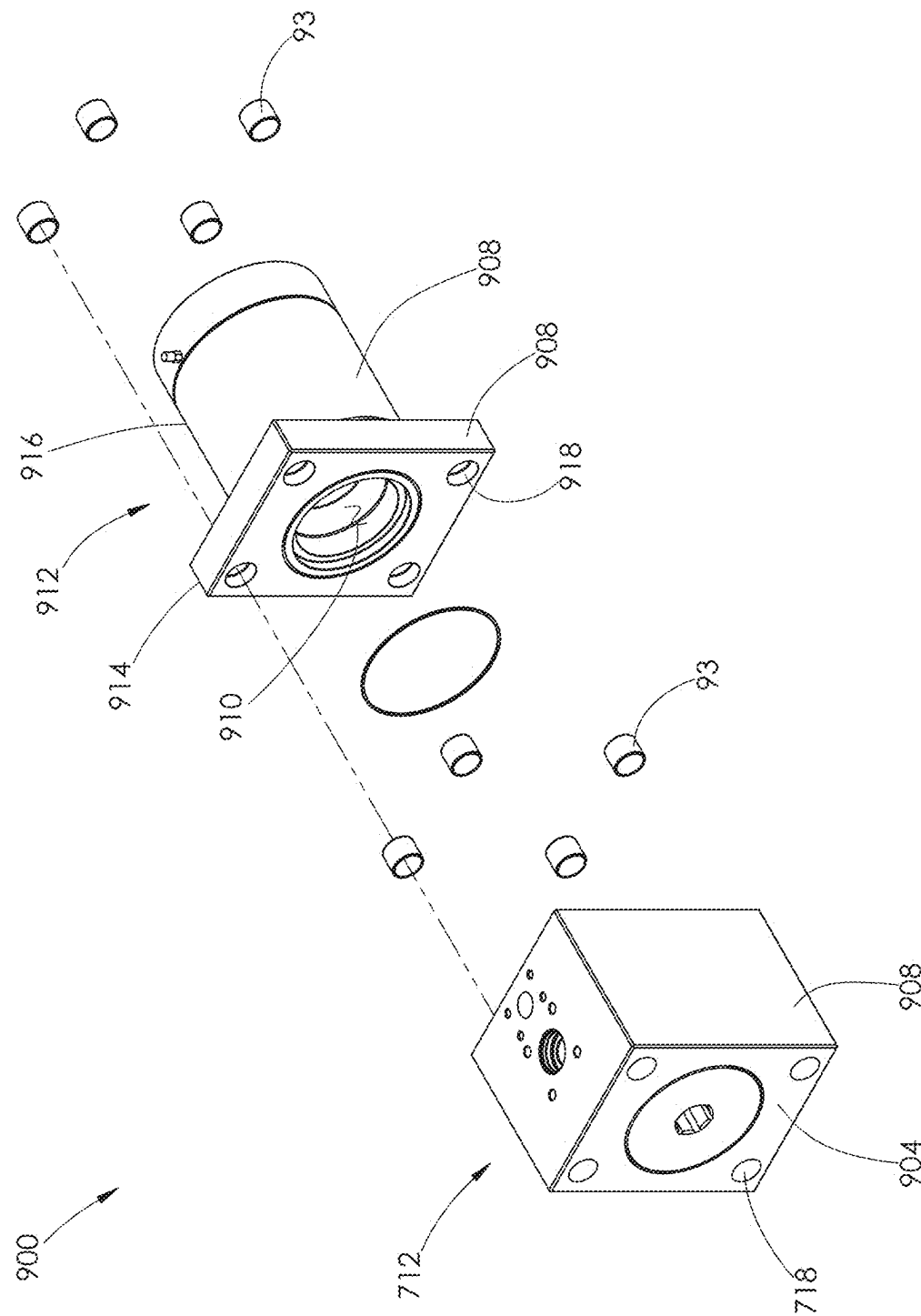


FIG. 119

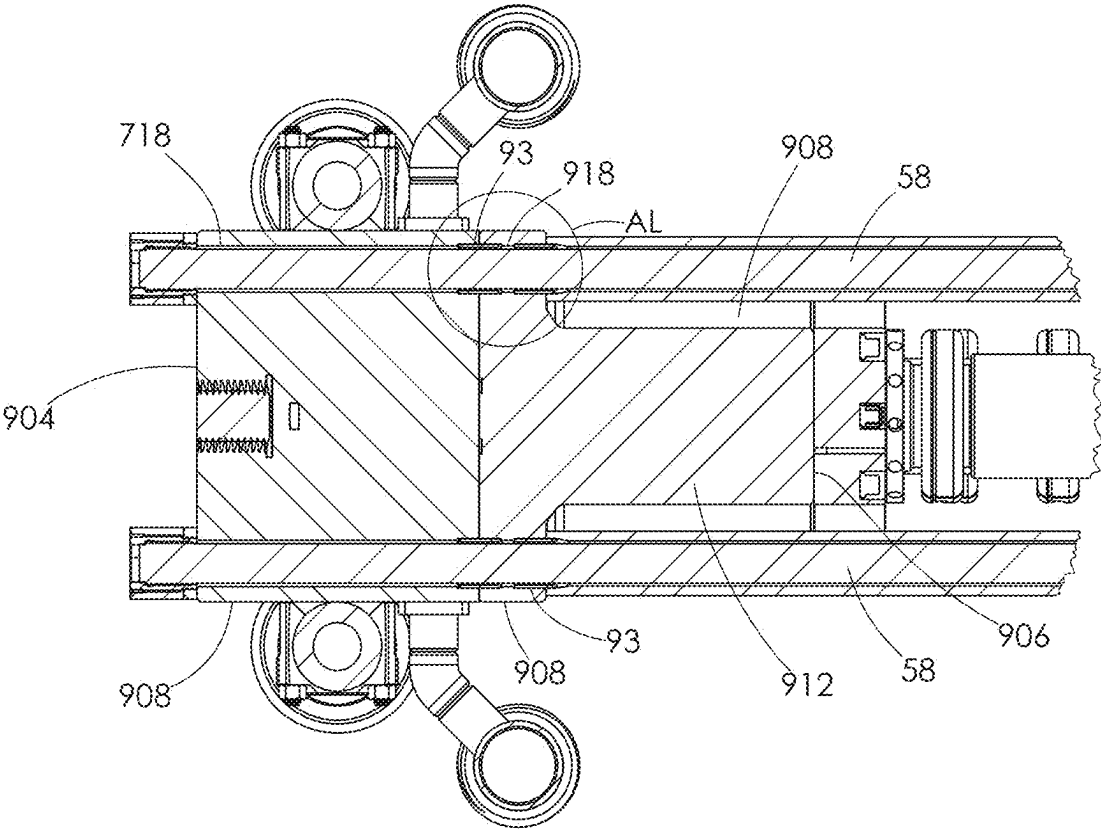


FIG. 120

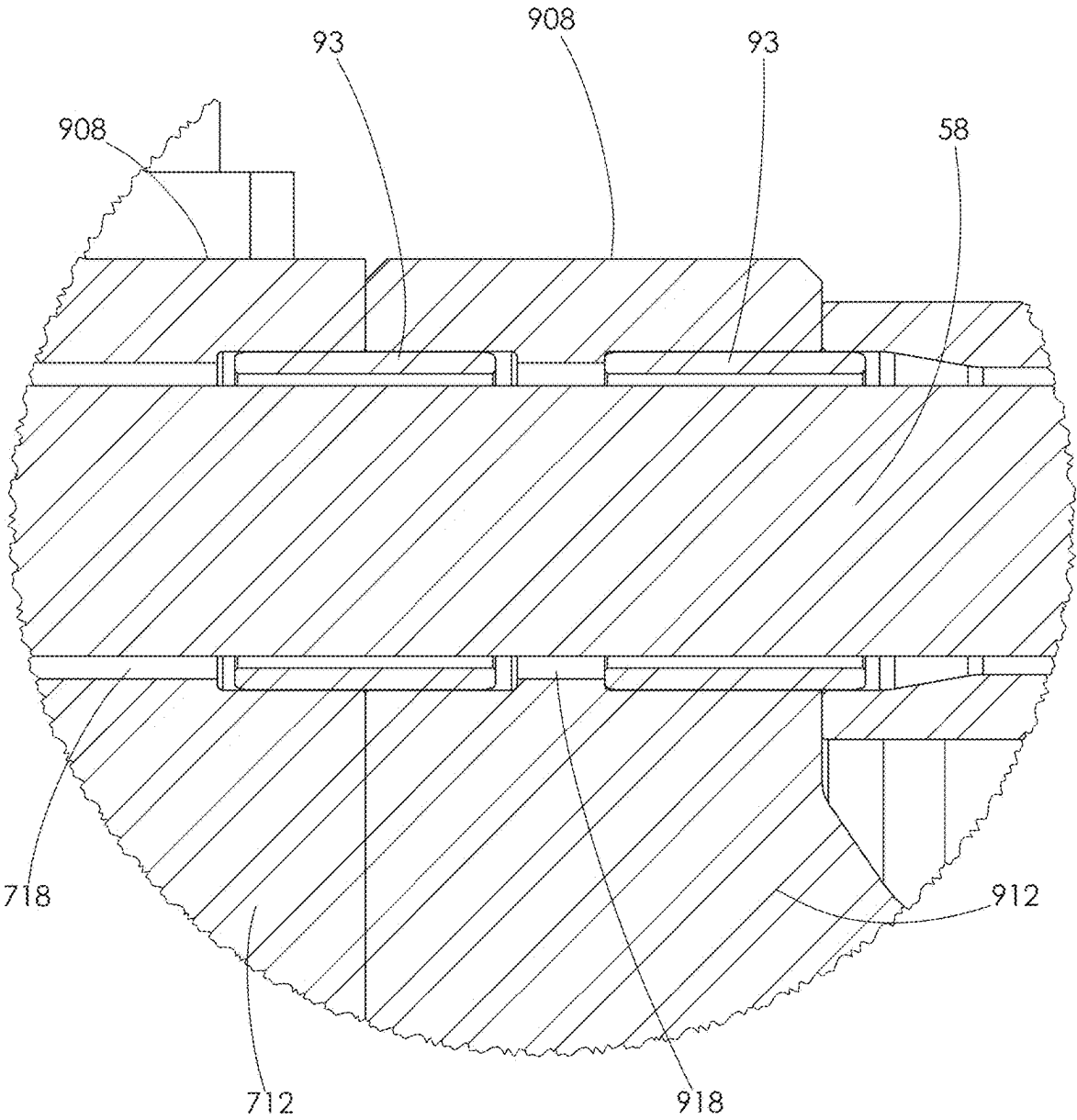


FIG. 121

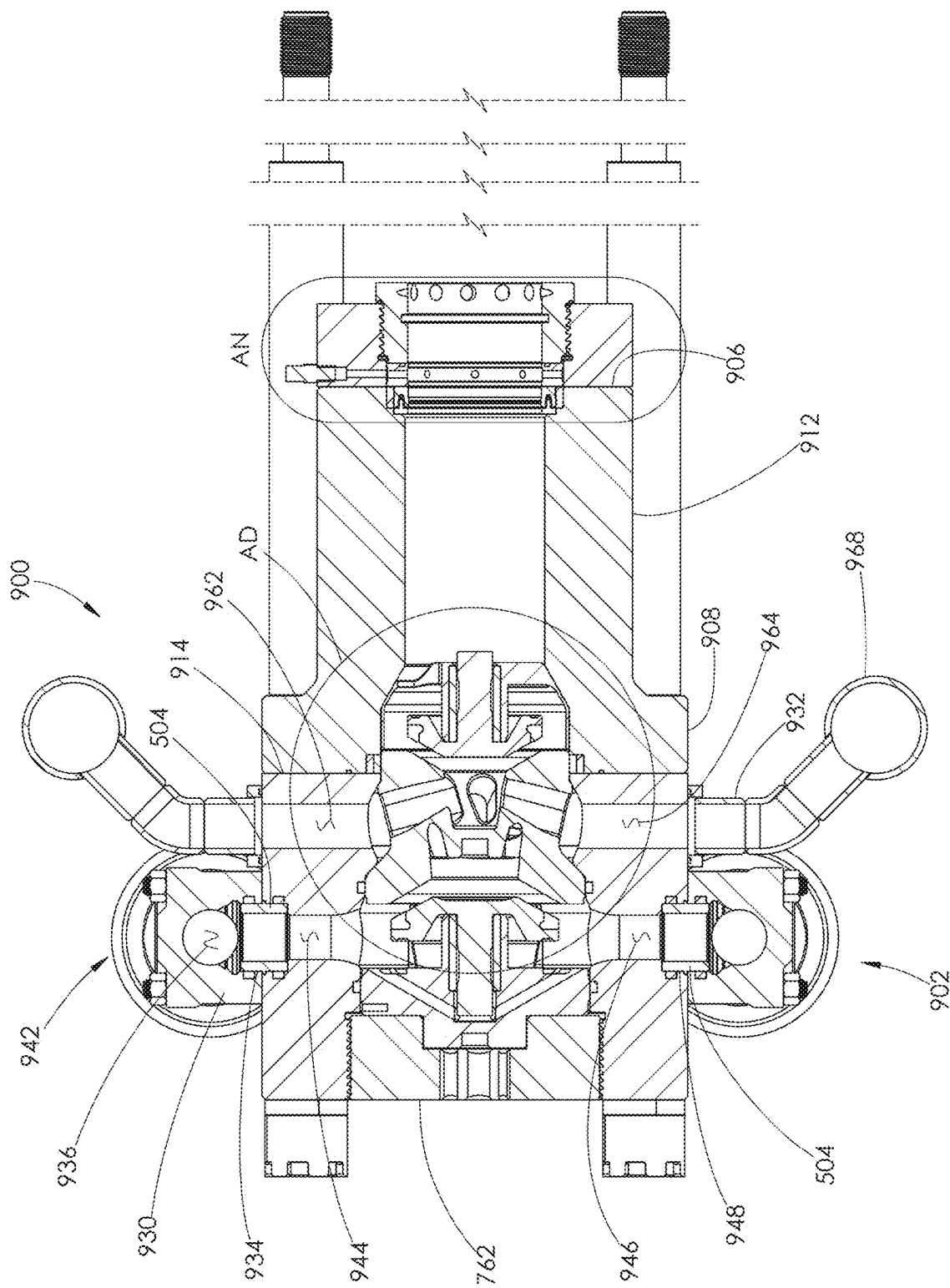


FIG. 122

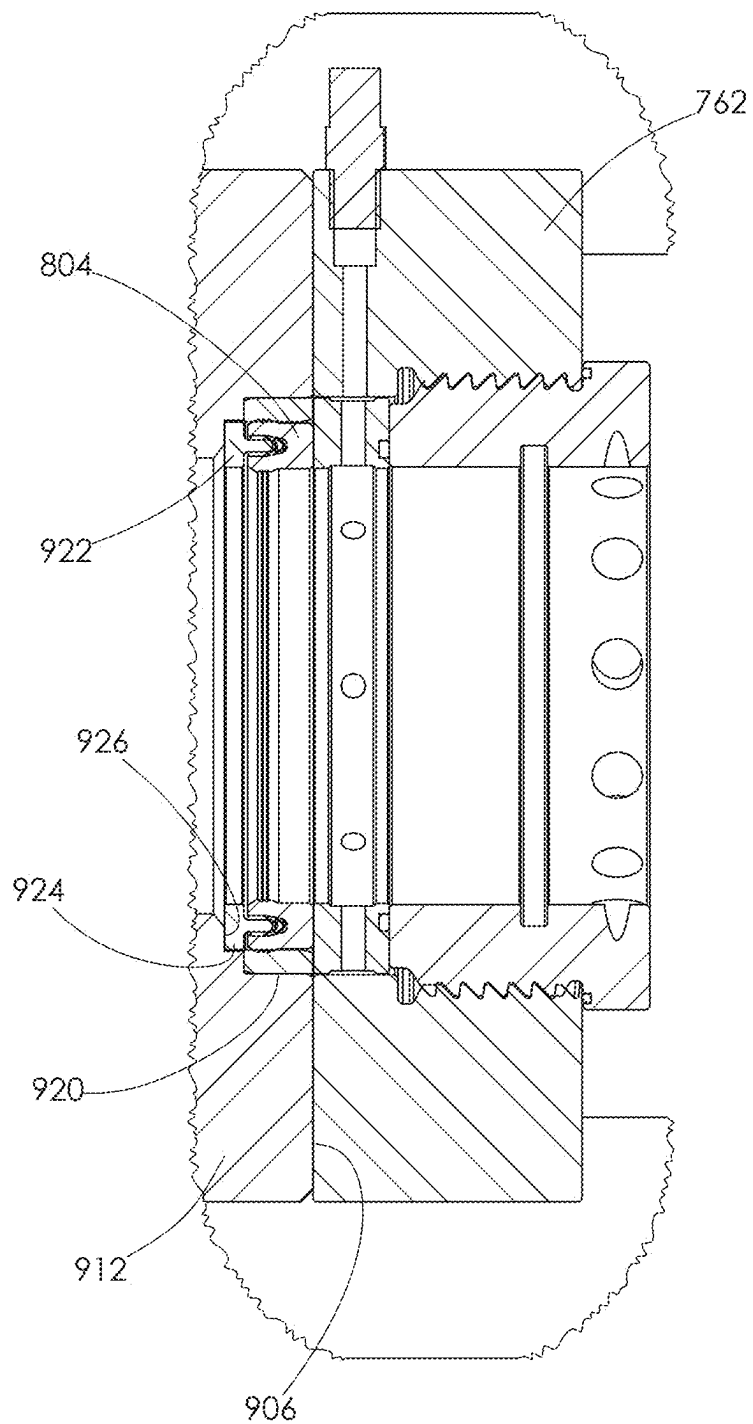


FIG. 123

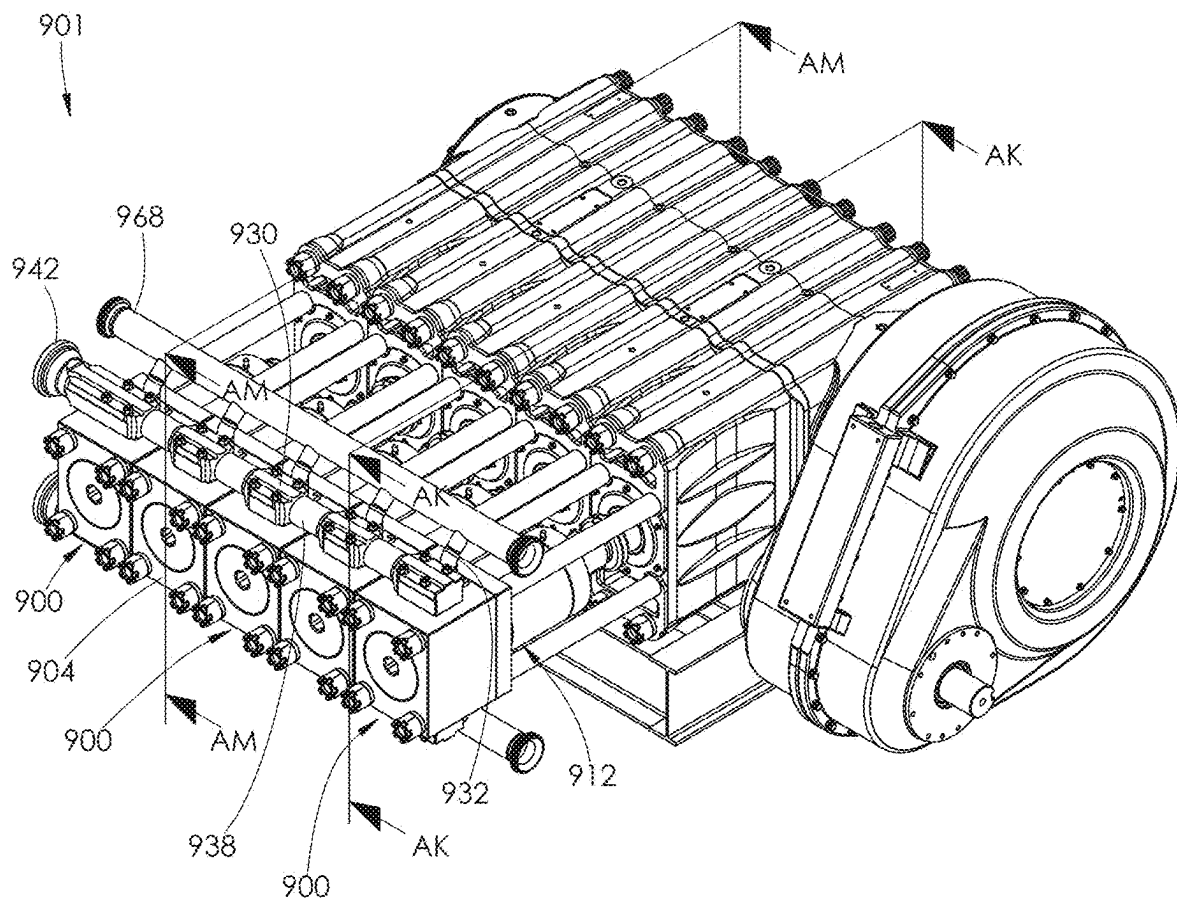
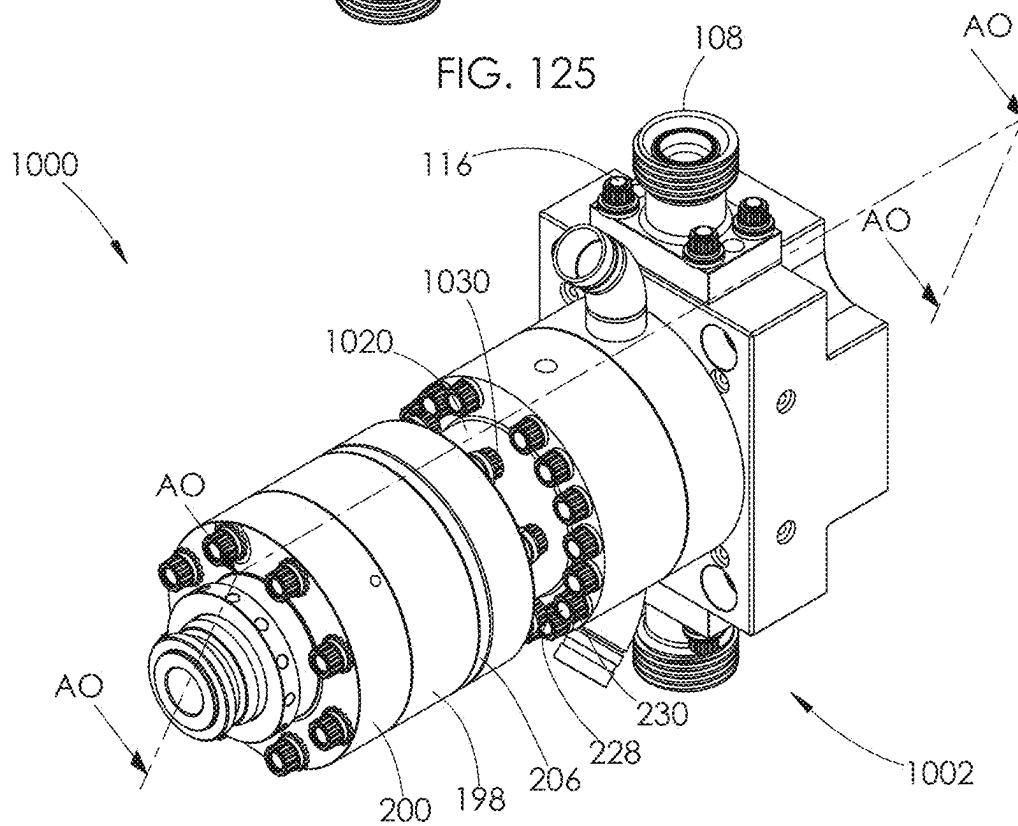
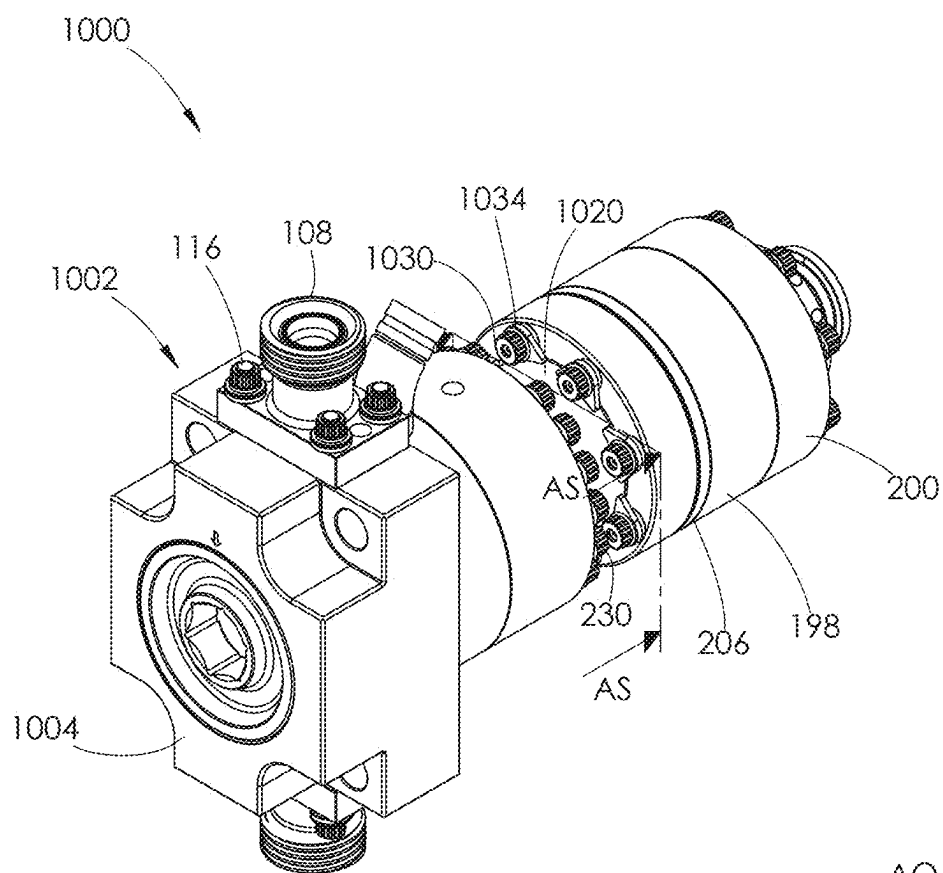


FIG. 124



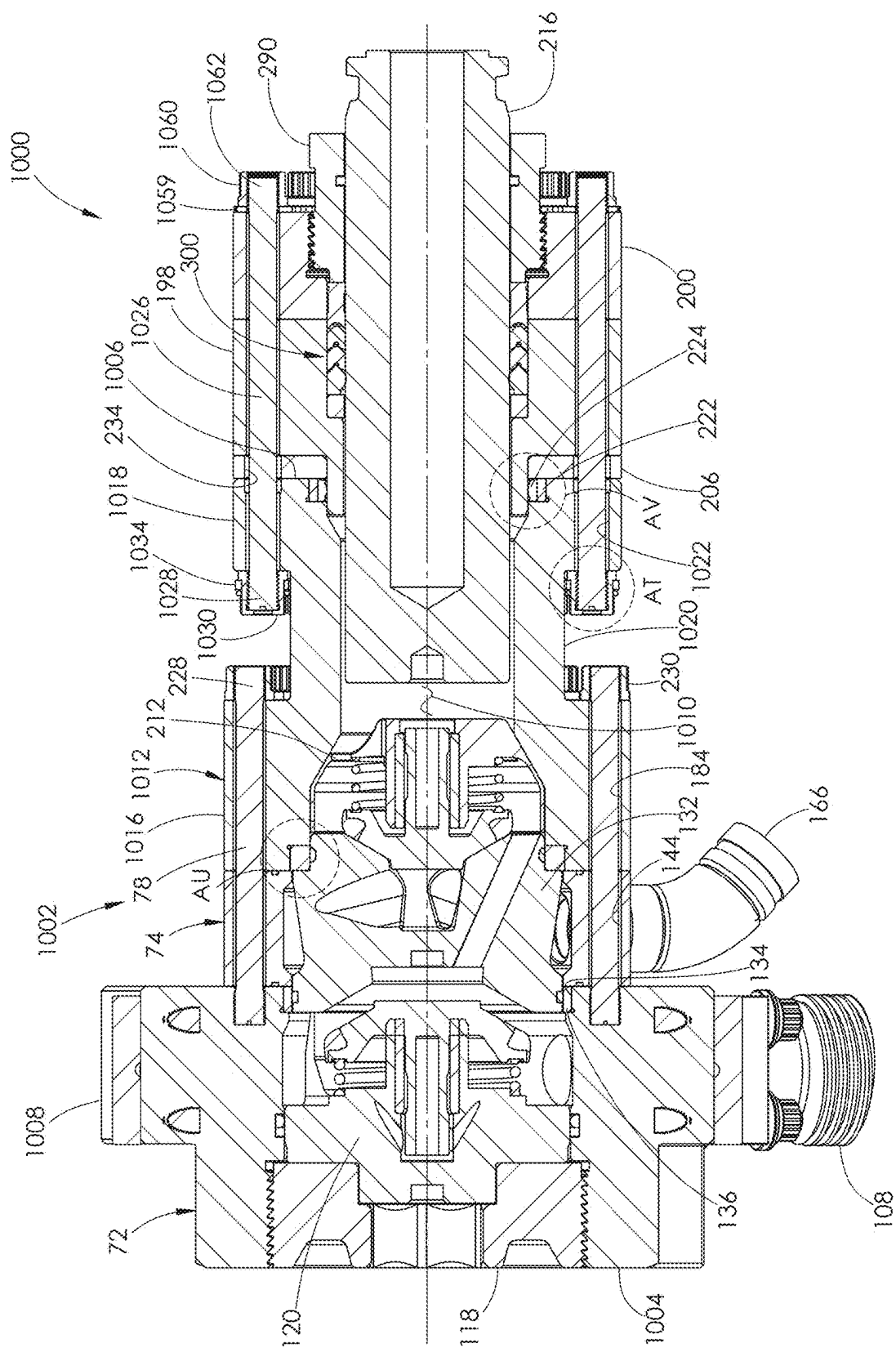


FIG. 127

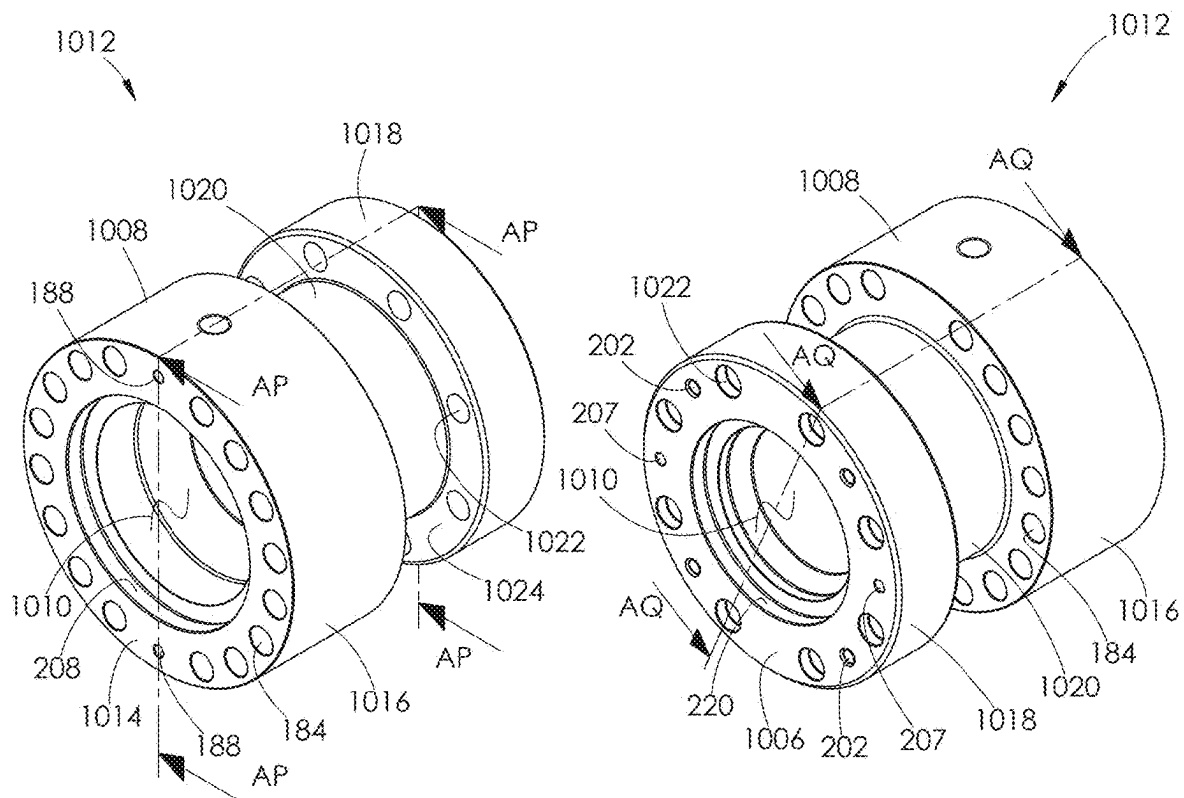


FIG. 128

FIG. 129

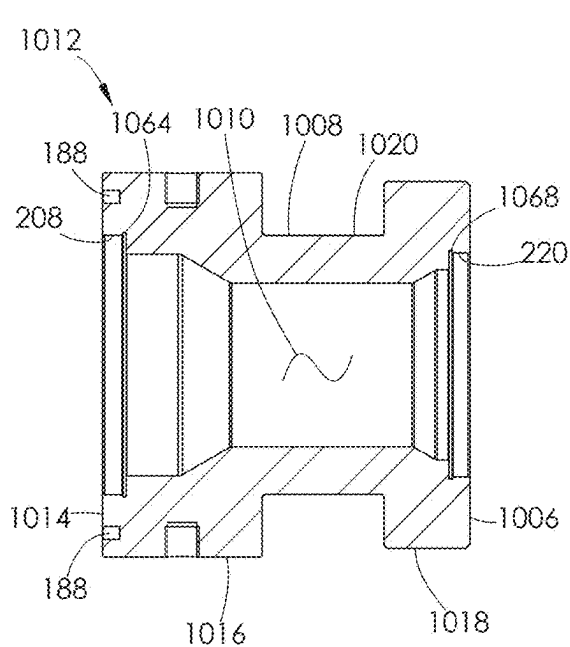


FIG. 130

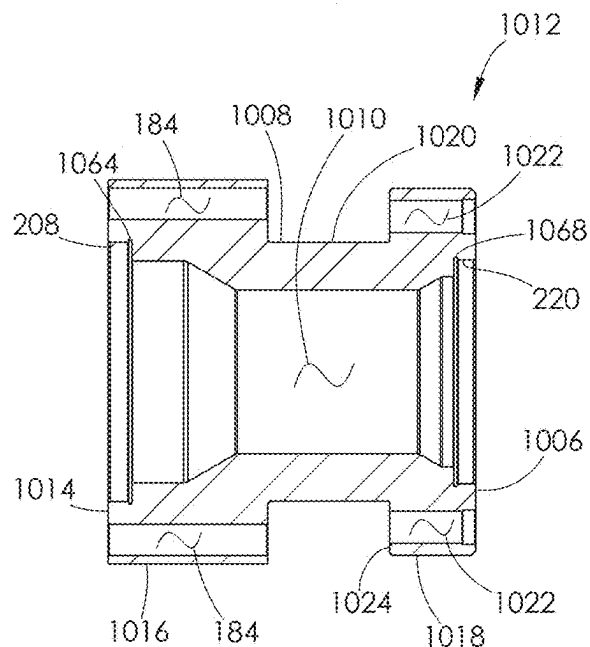


FIG. 131

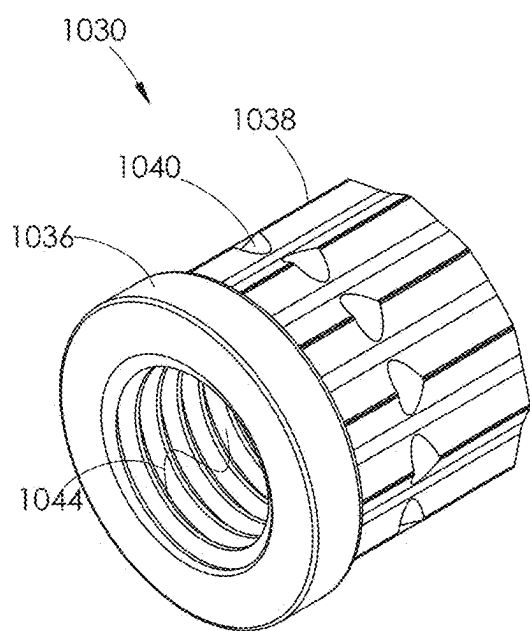


FIG. 132

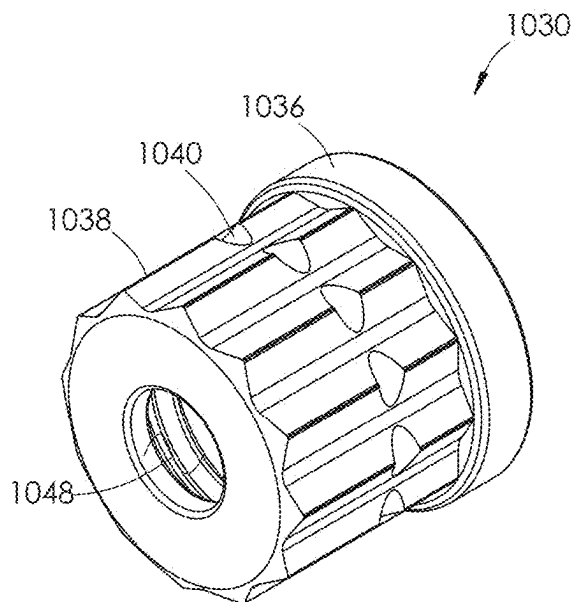


FIG. 133

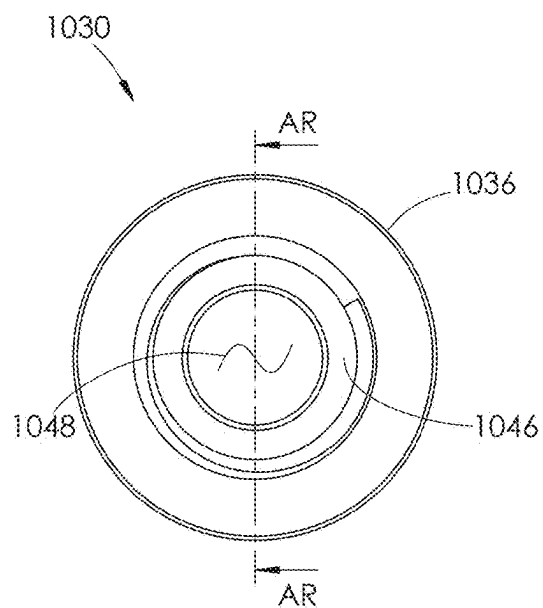


FIG. 134

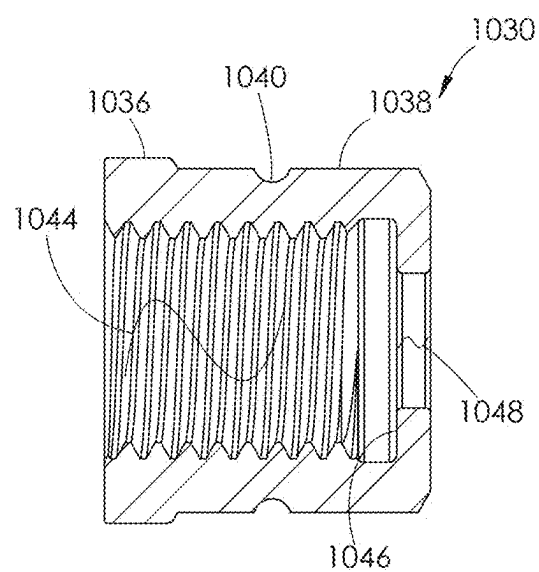


FIG. 135

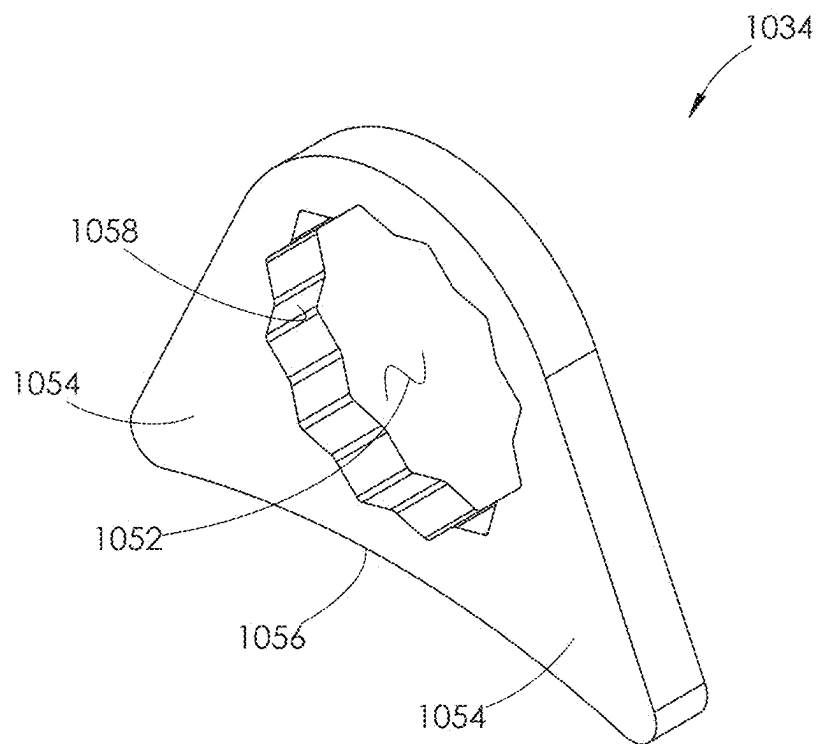


FIG. 136

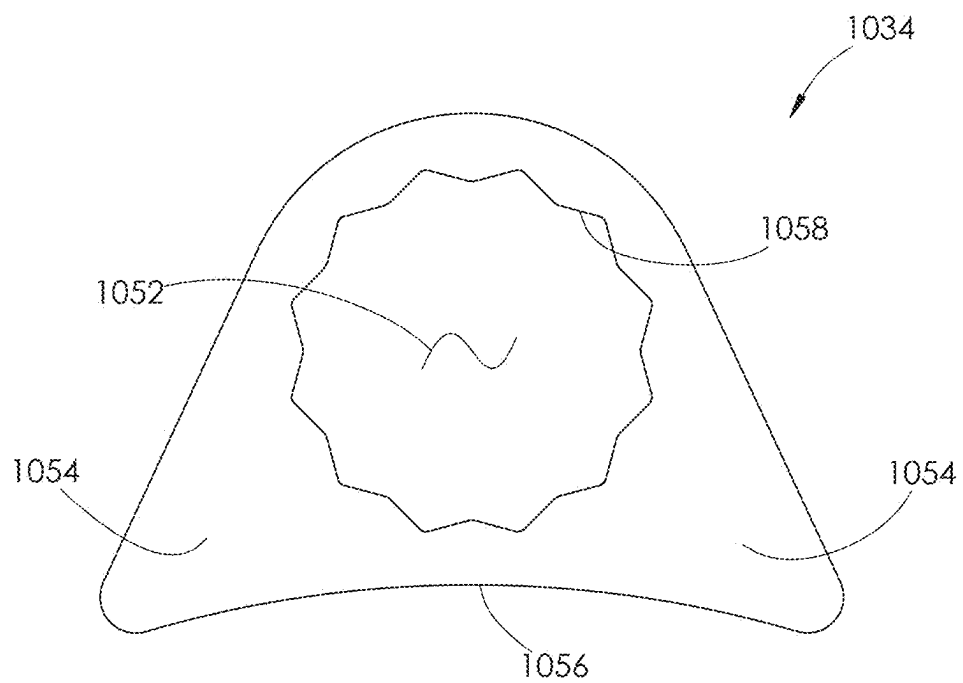


FIG. 137

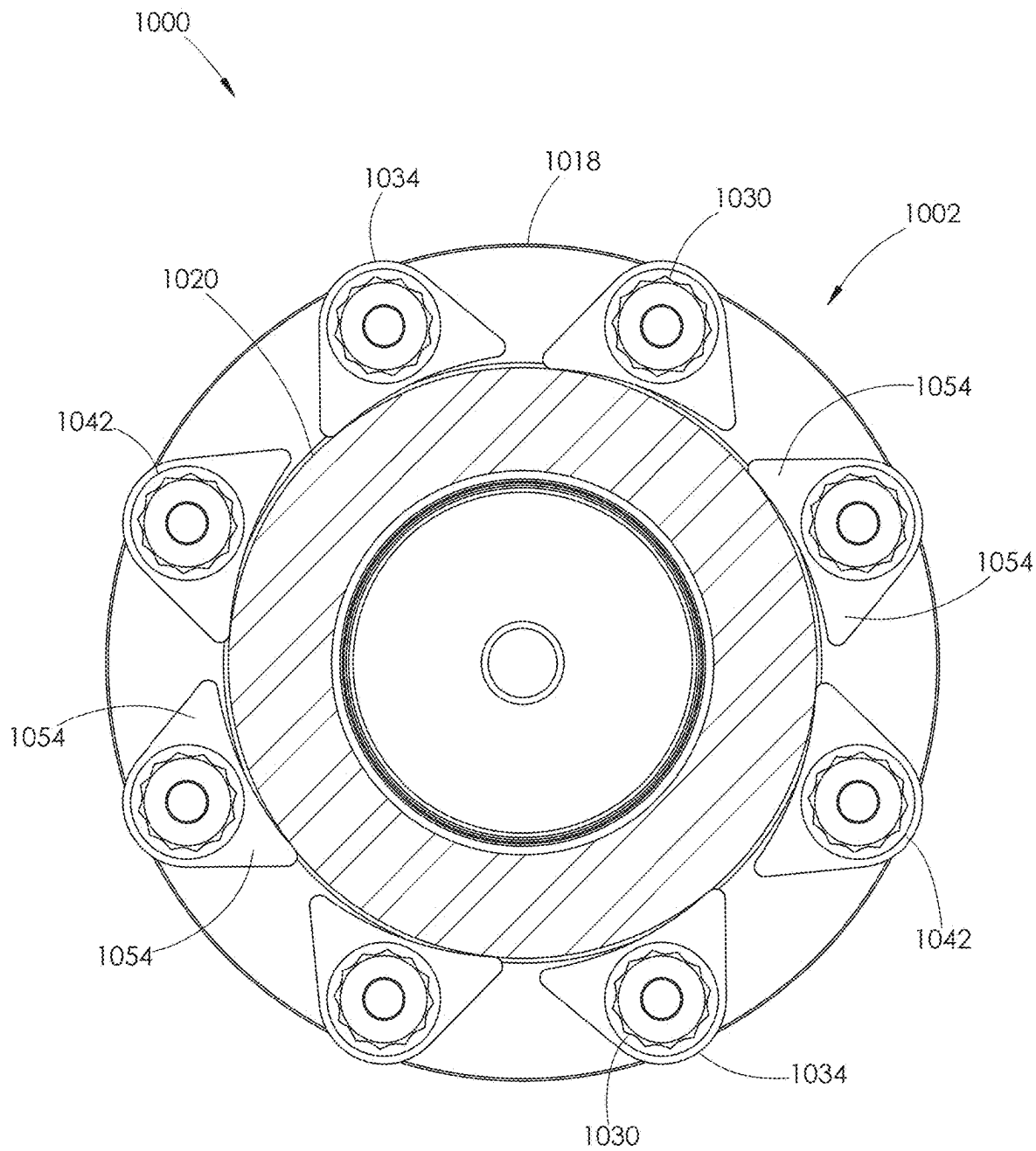


FIG. 138

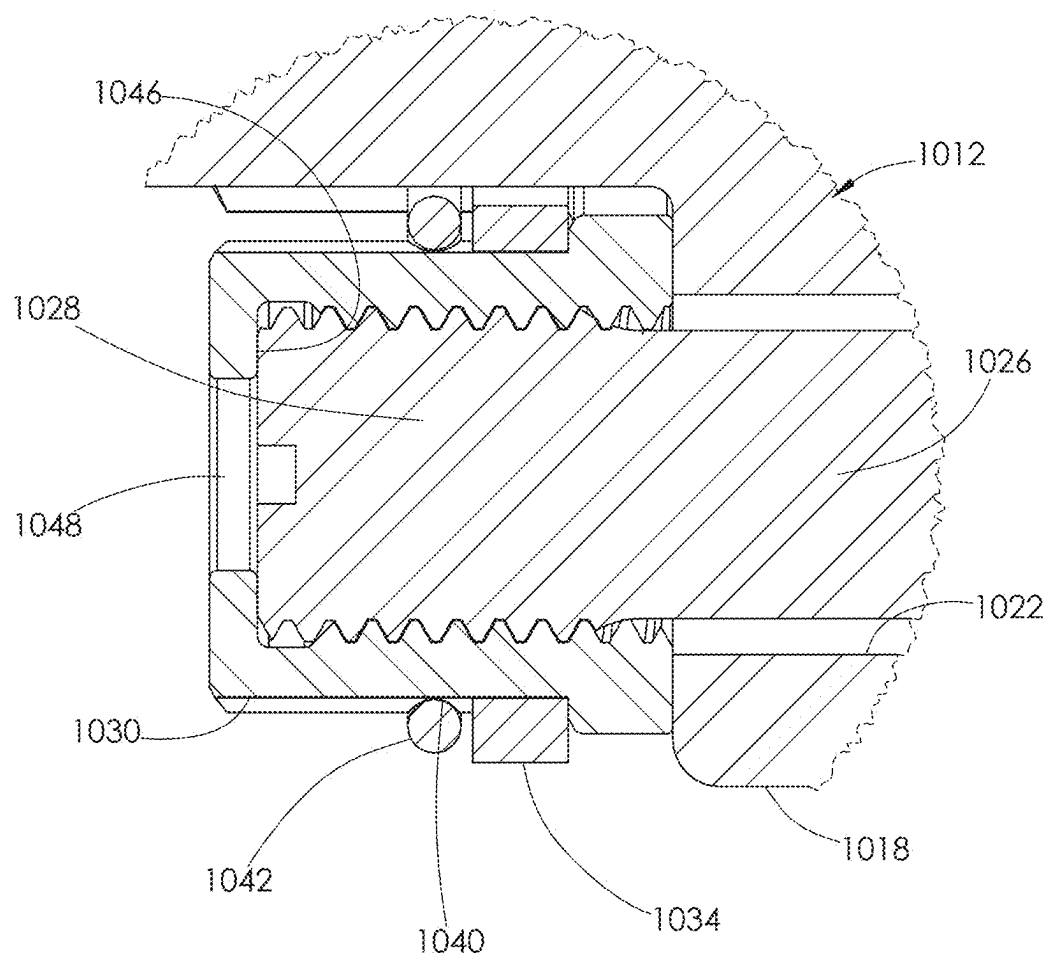


FIG. 139

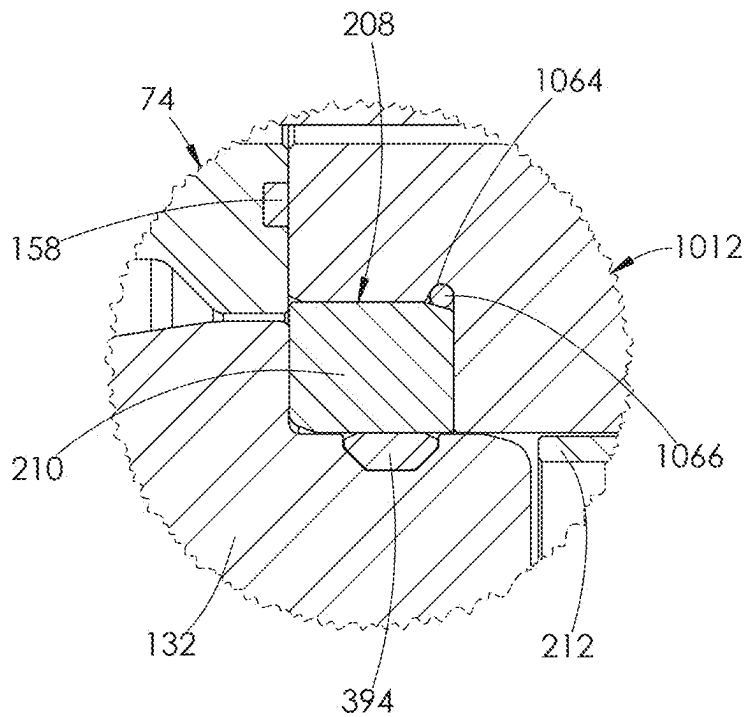


FIG. 140

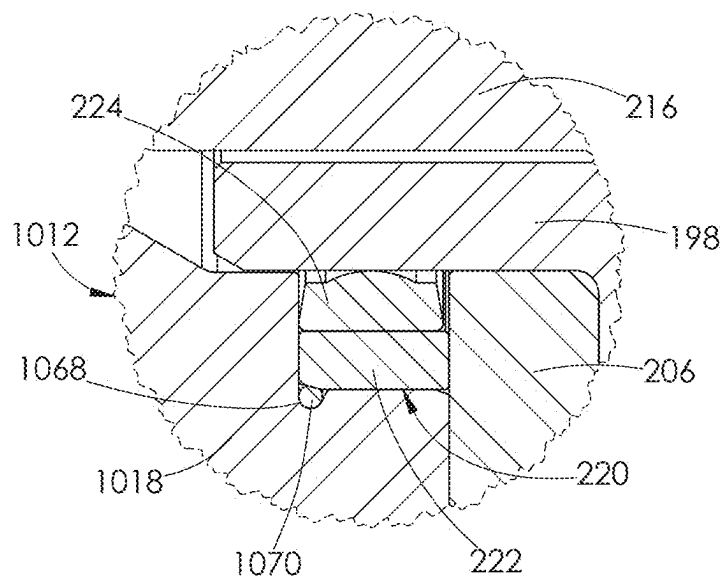


FIG. 141

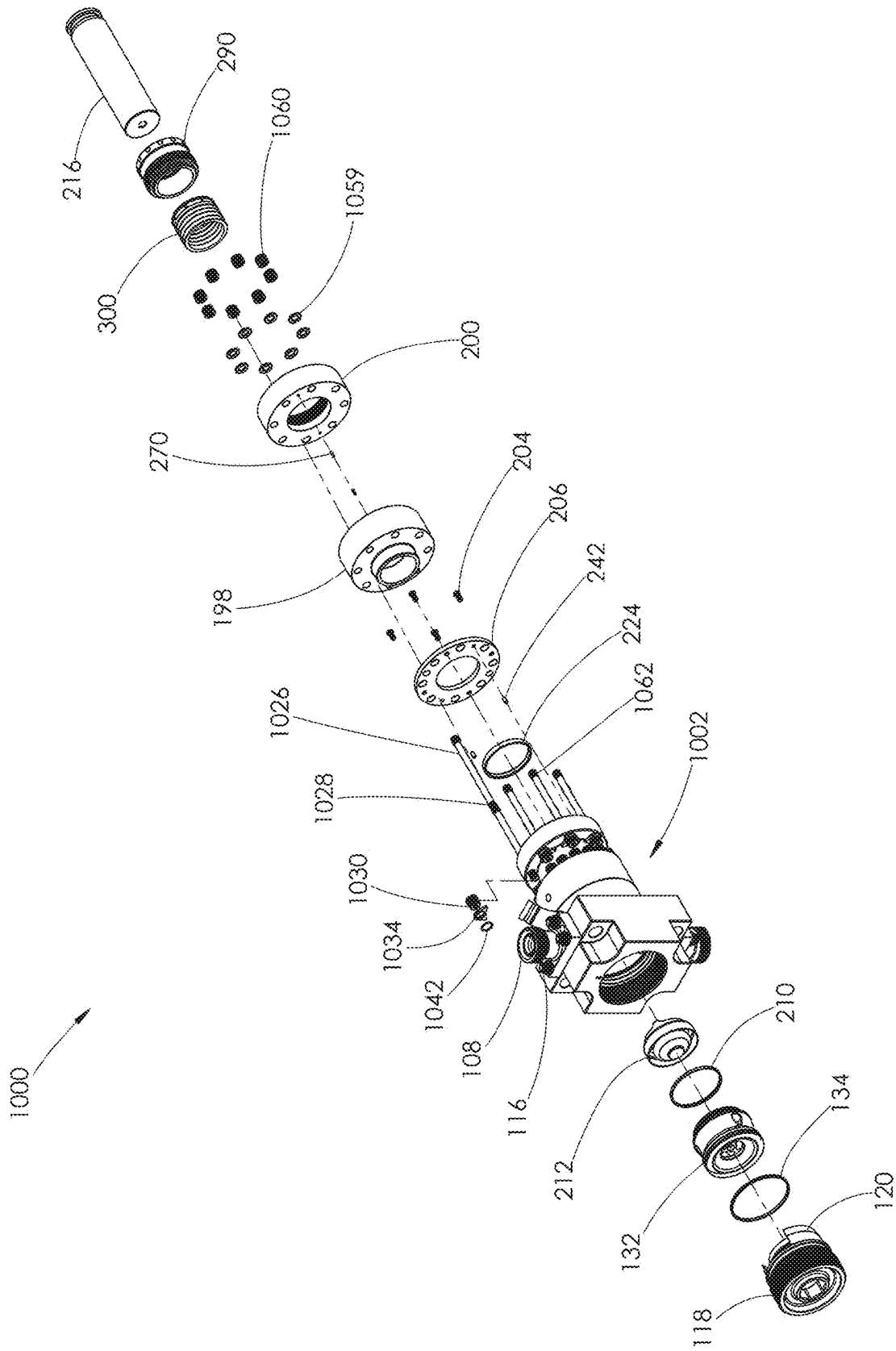


FIG. 142

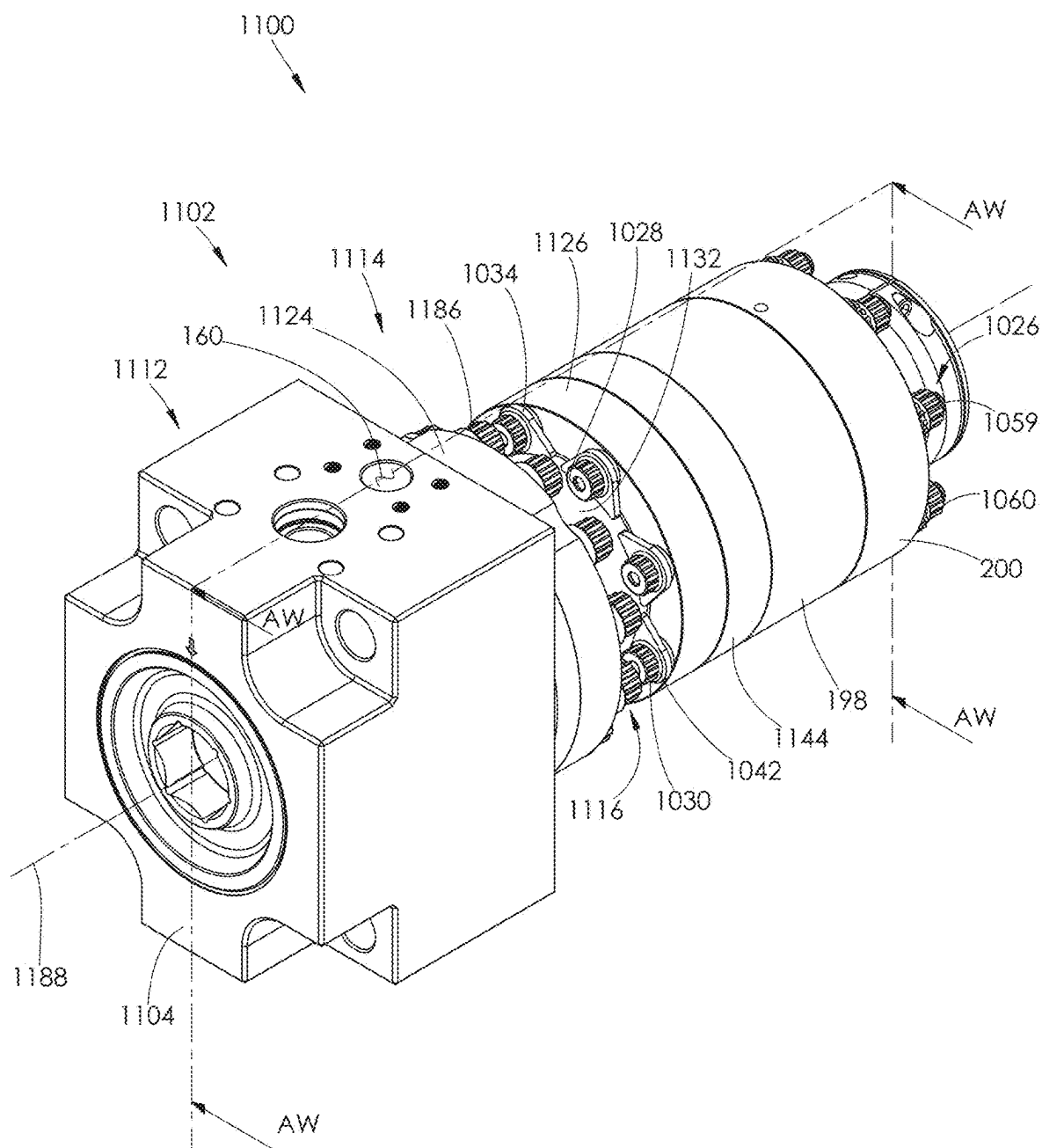


FIG. 143

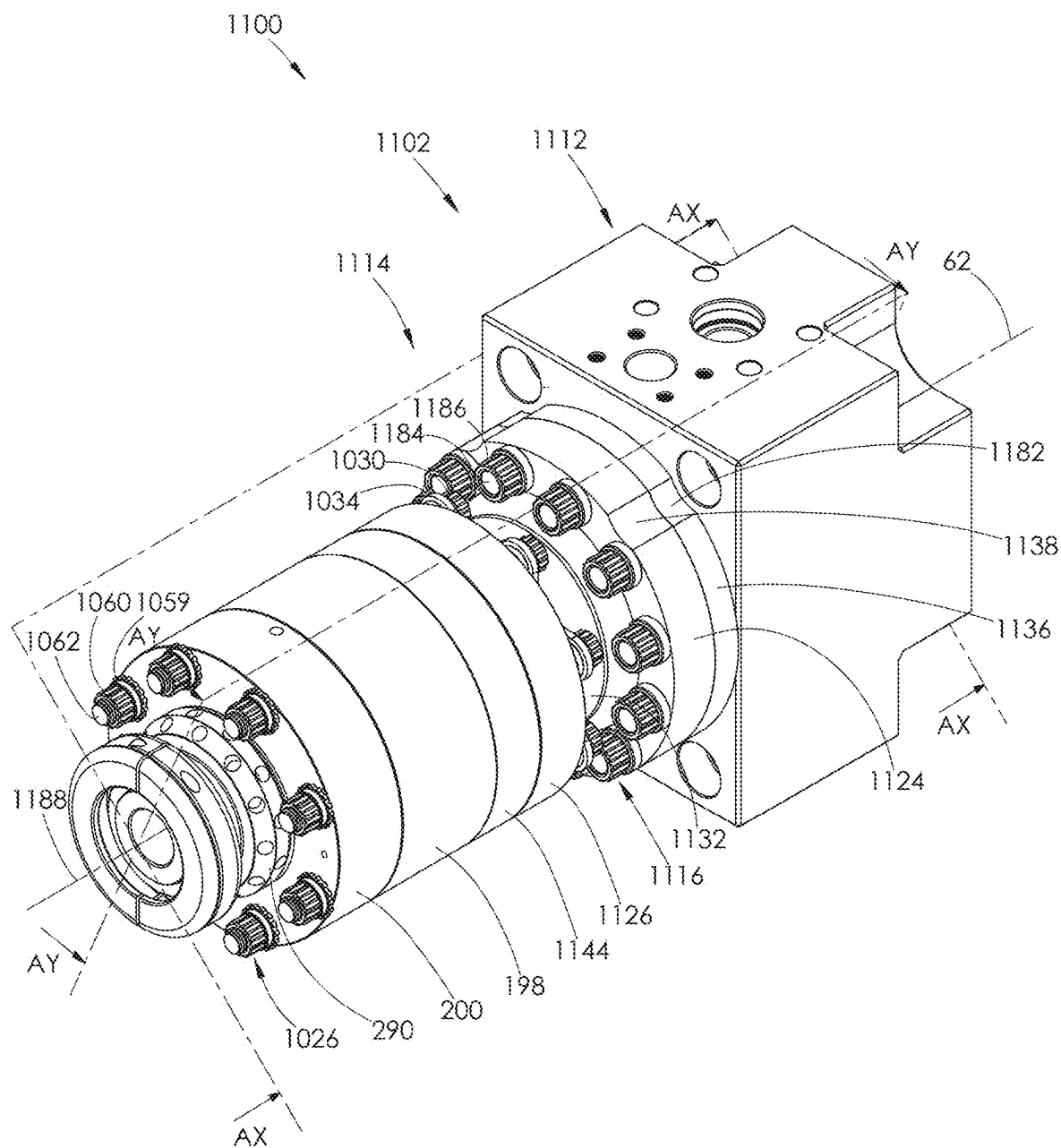


FIG. 144

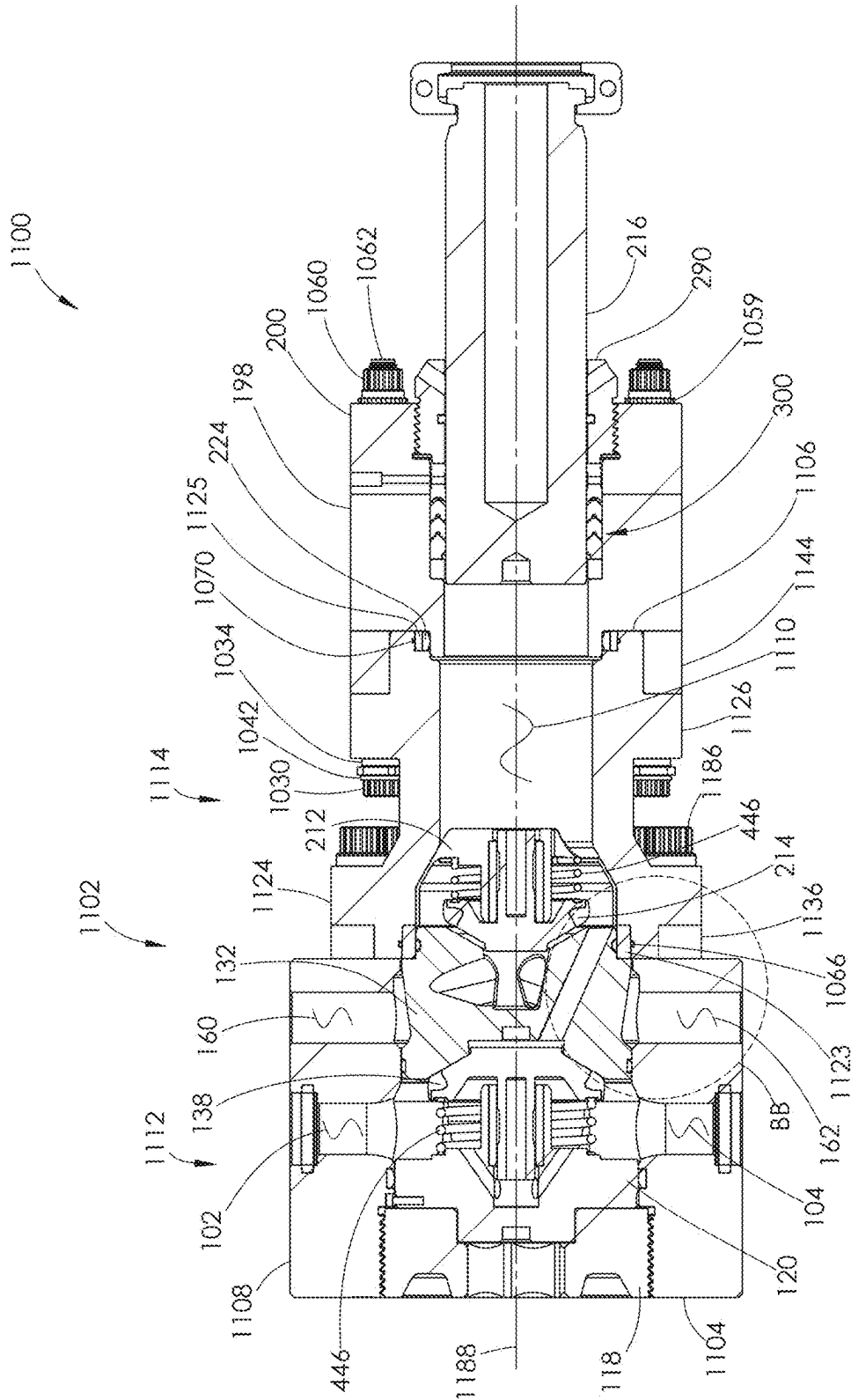


FIG. 145

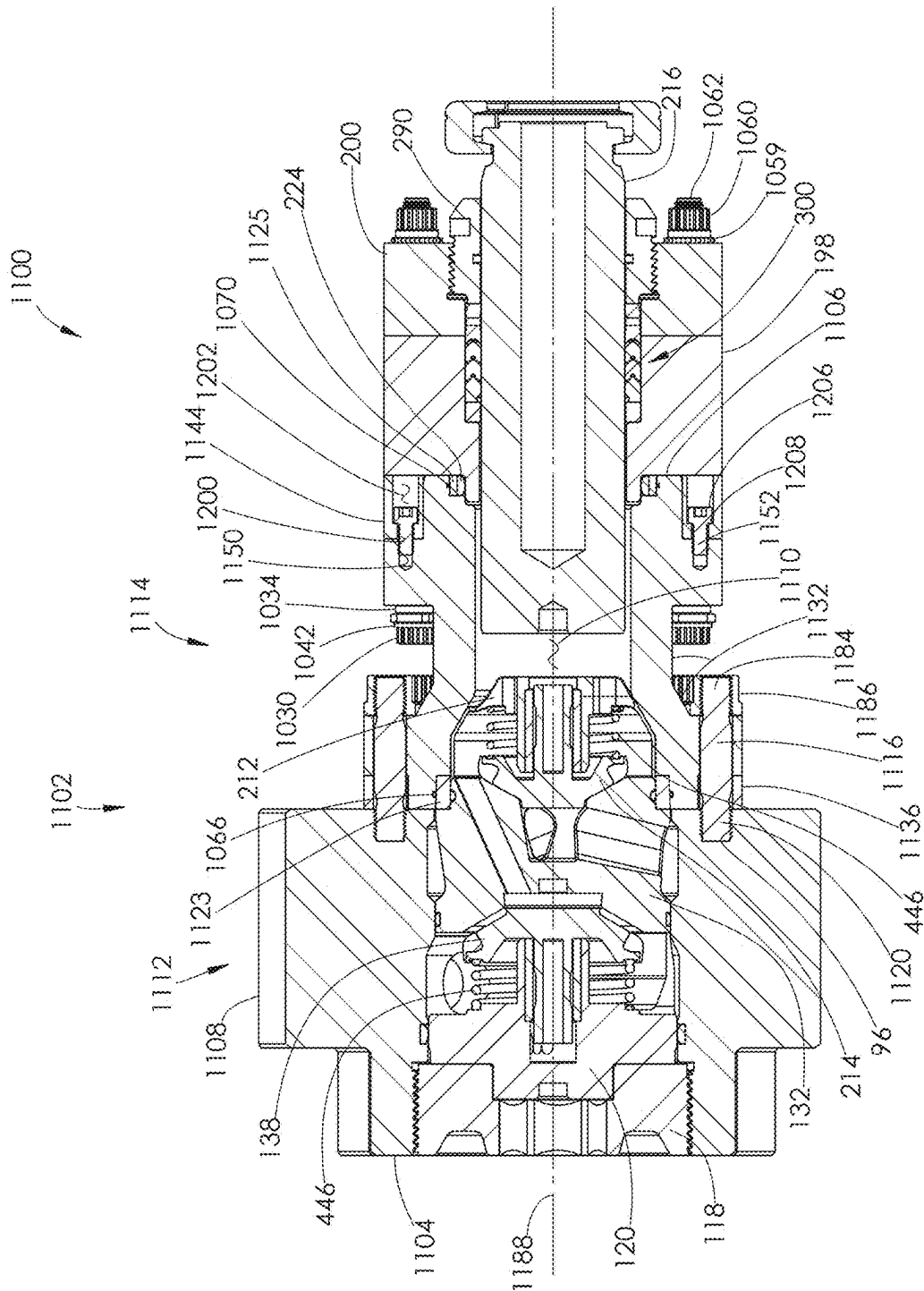


FIG. 146

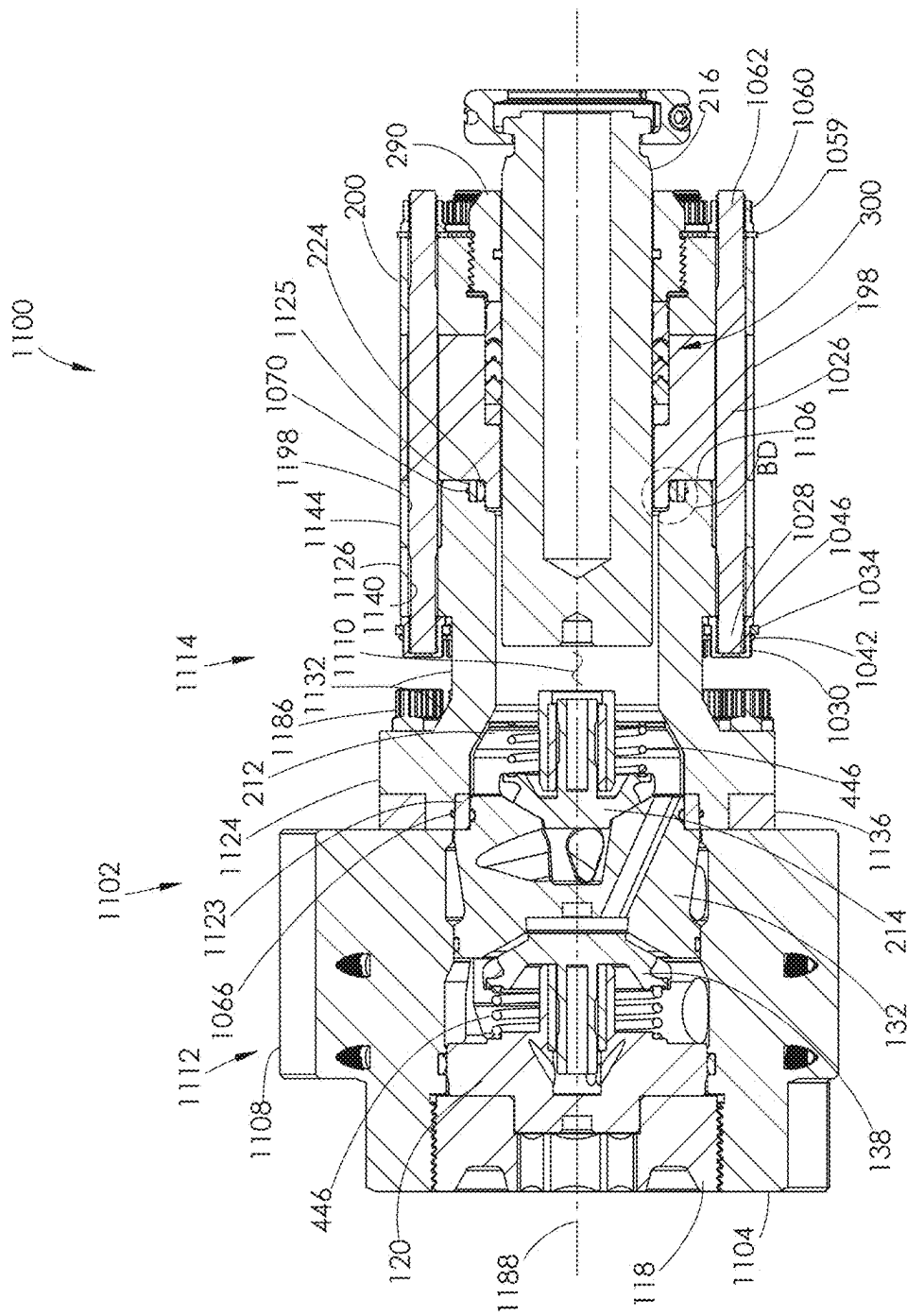


FIG. 147

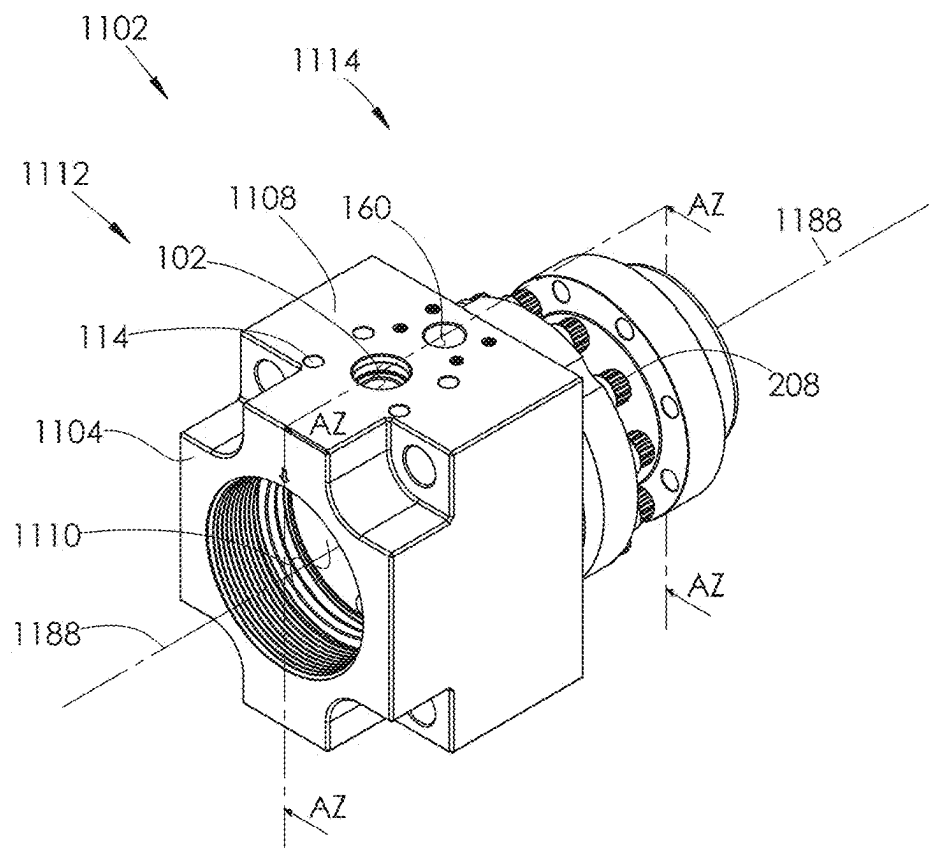


FIG. 148

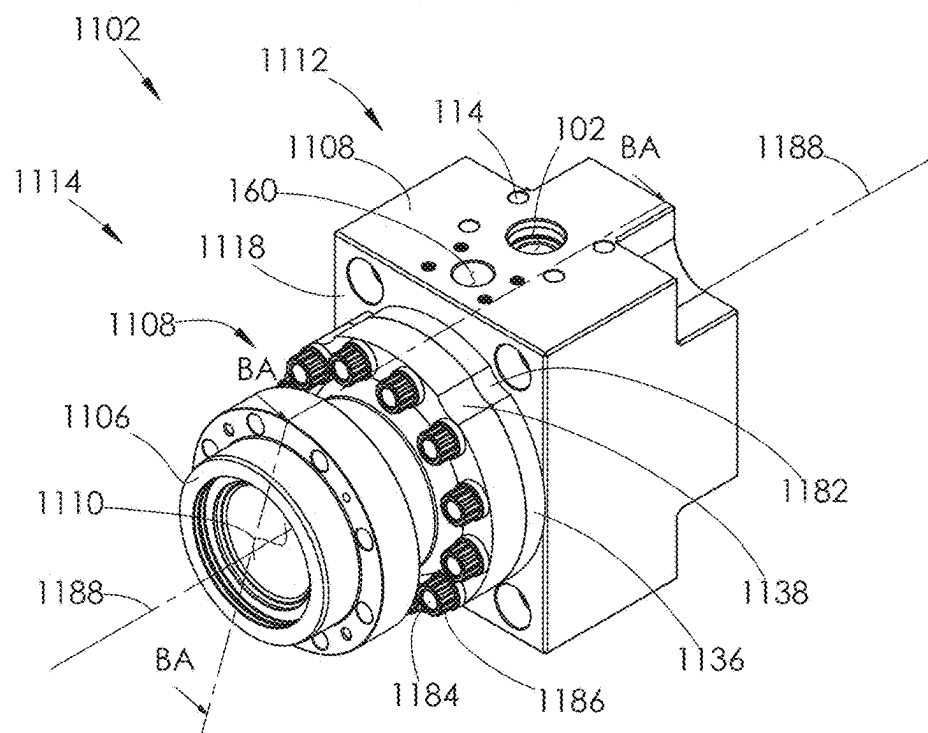


FIG. 149

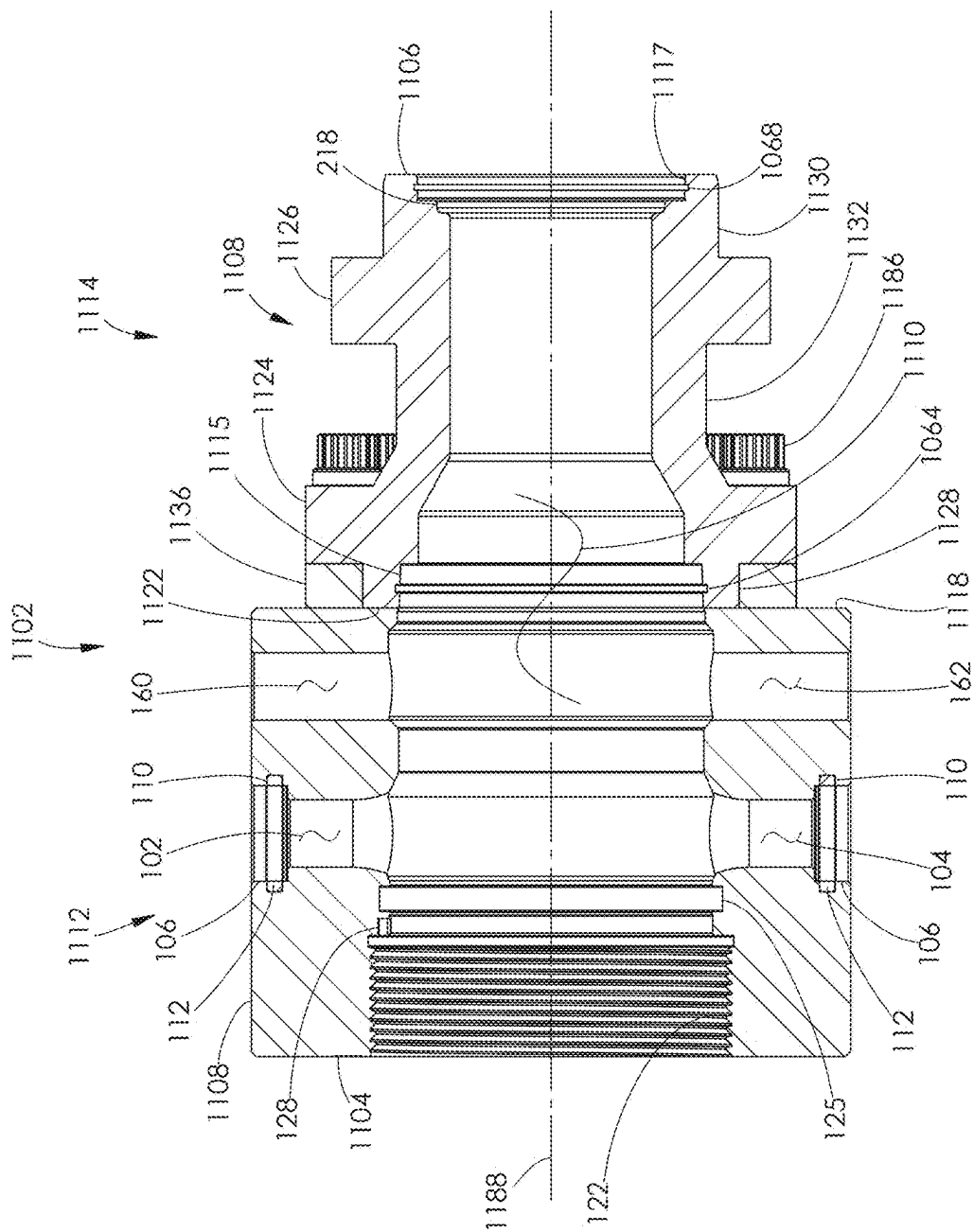


FIG. 150

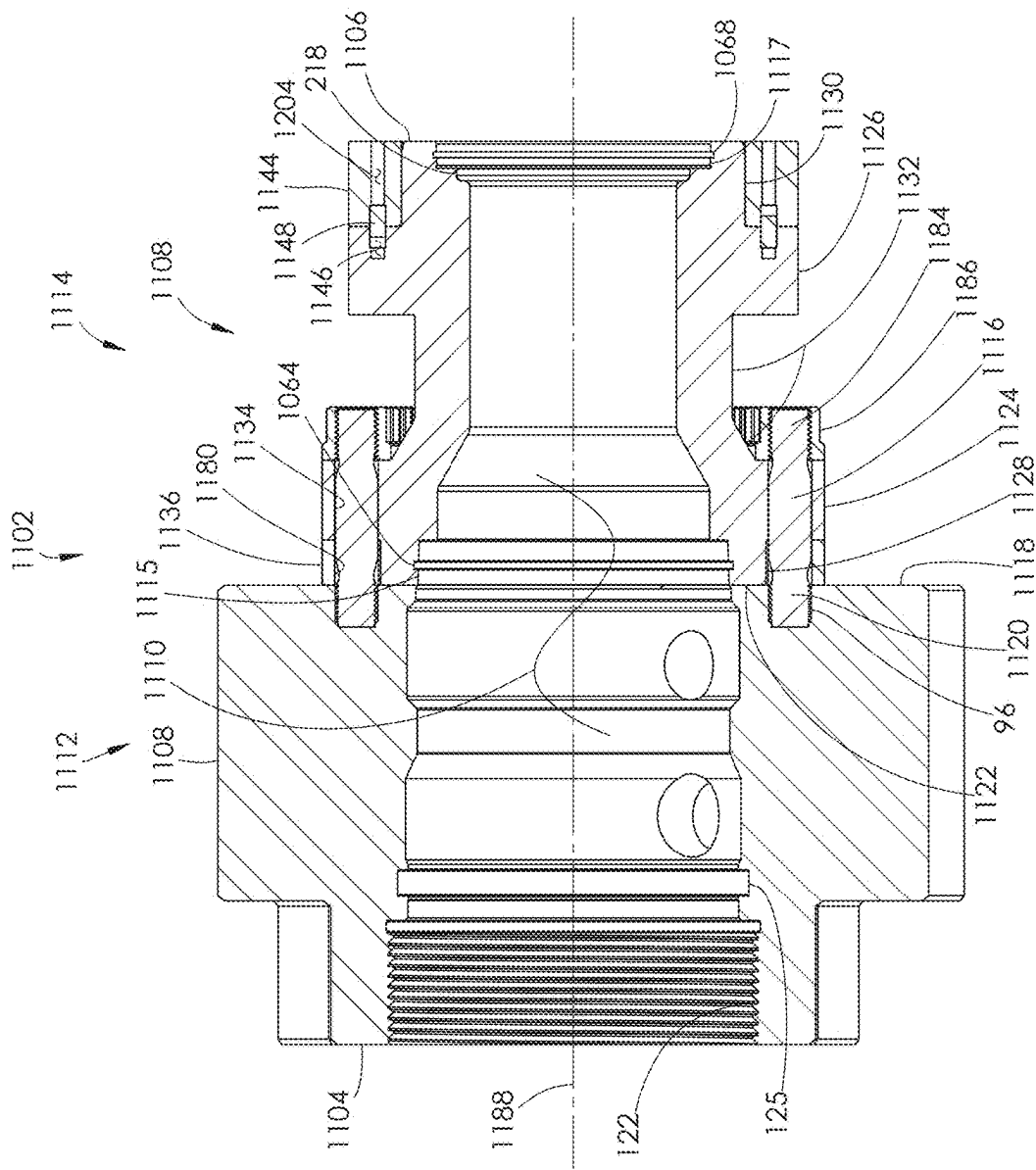


FIG. 151

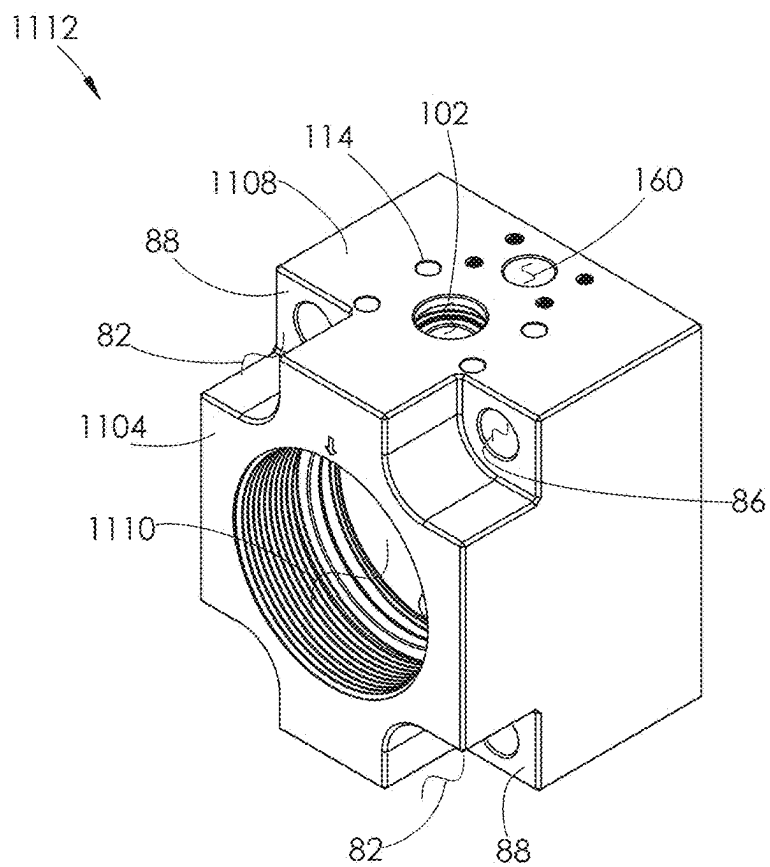


FIG. 152

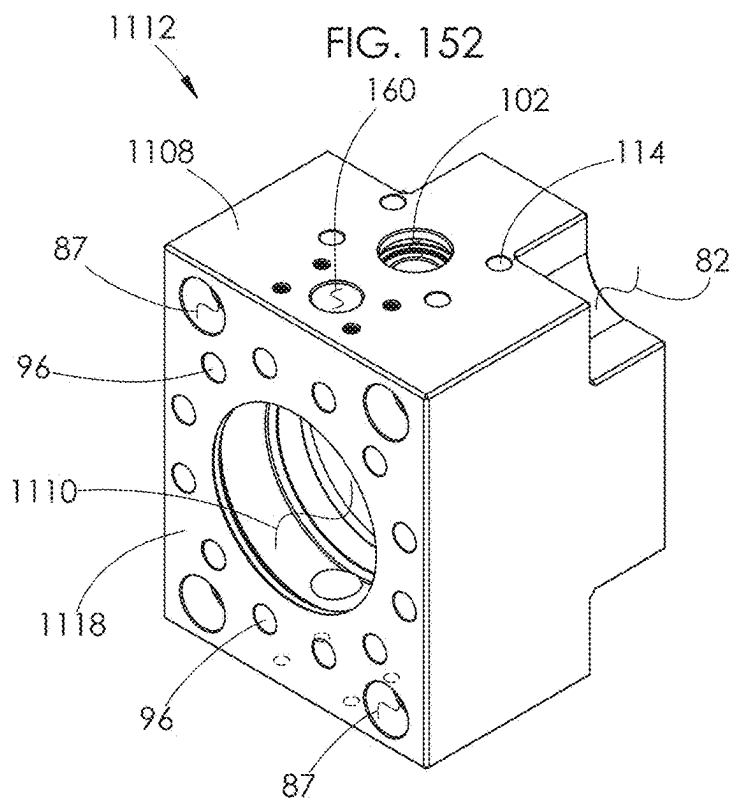


FIG. 153

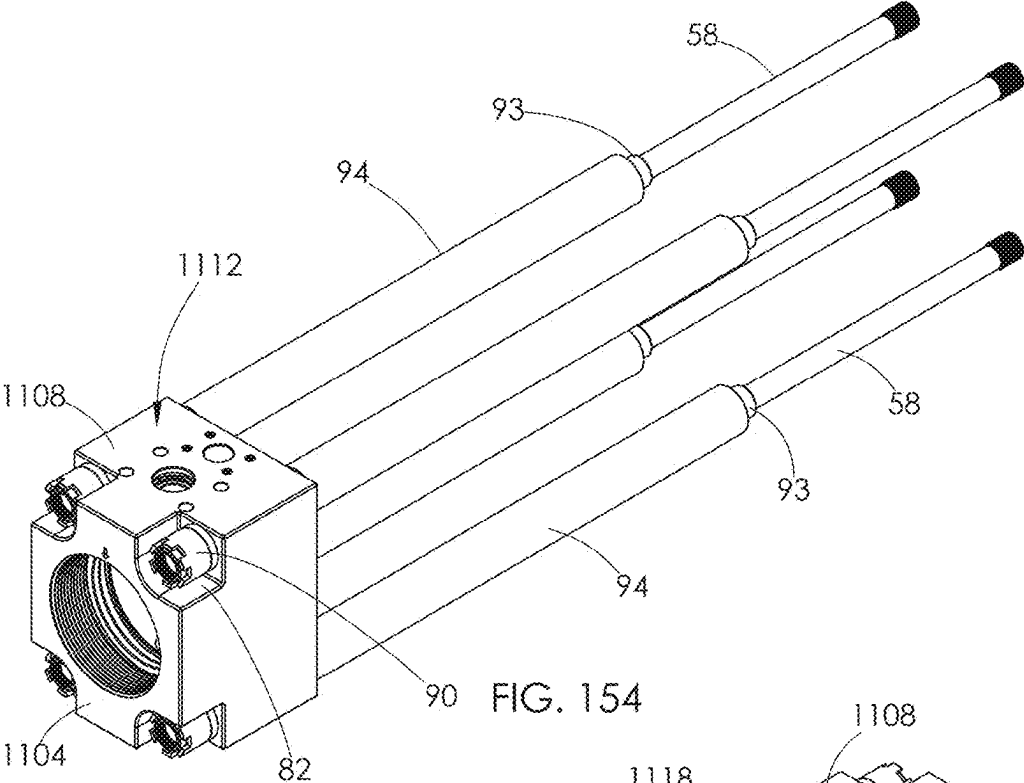


FIG. 154

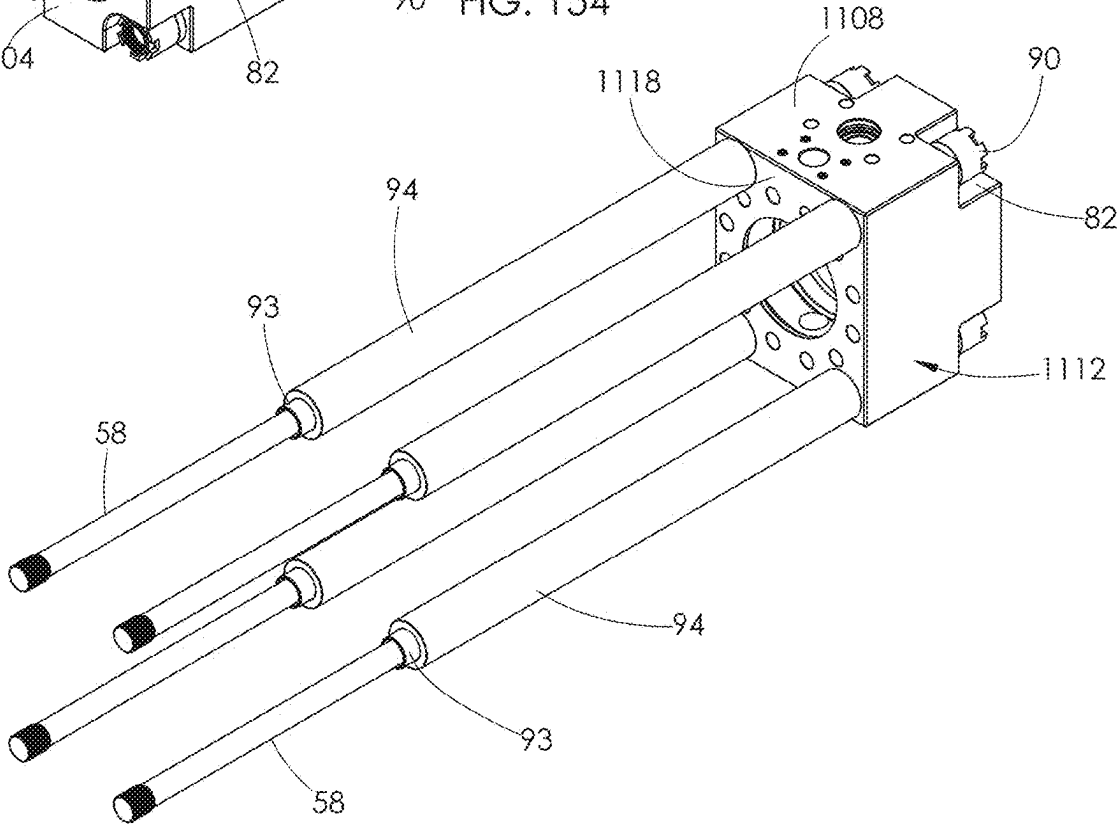


FIG. 155

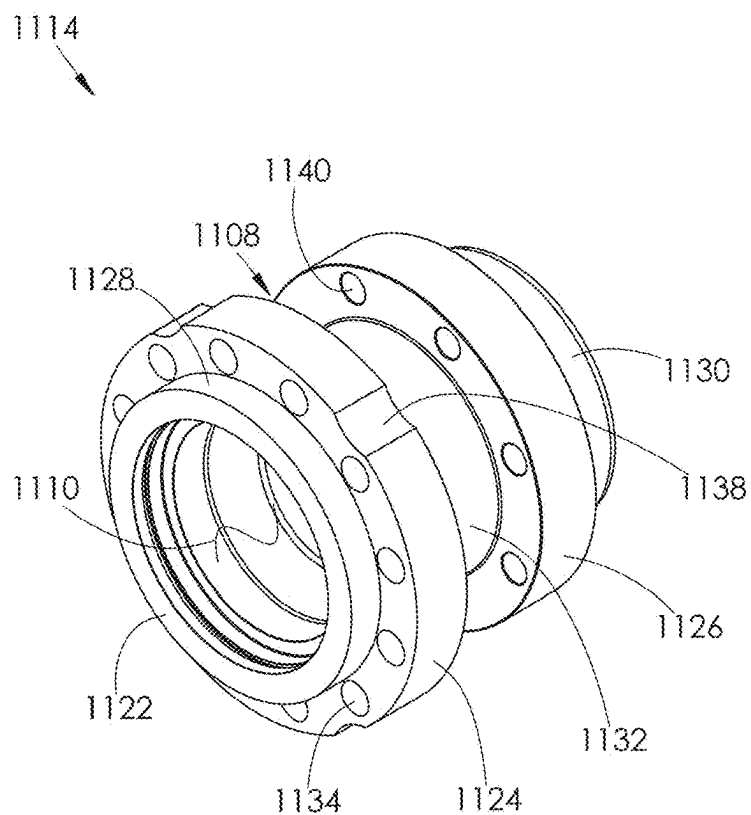


FIG. 156

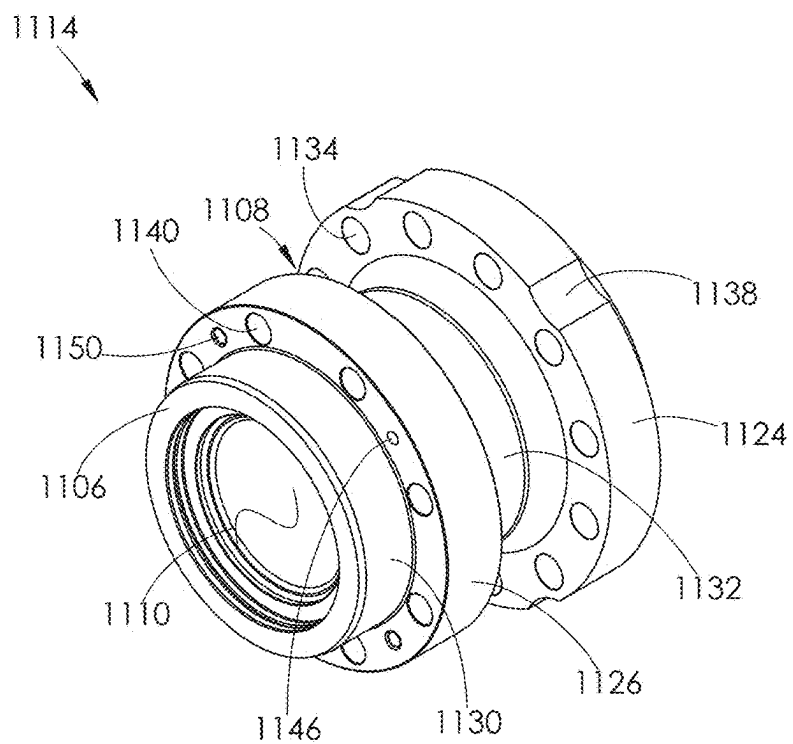


FIG. 157

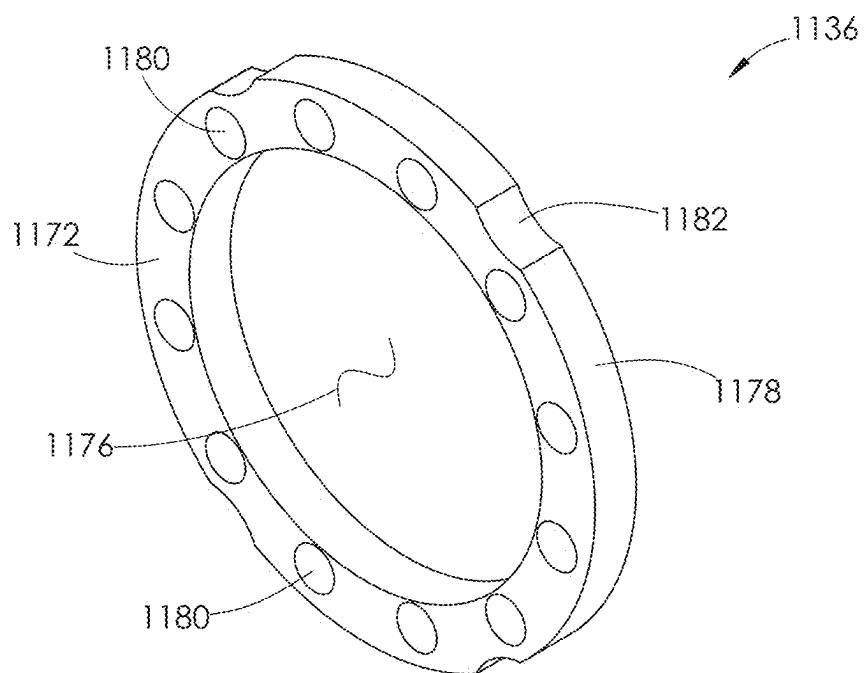


FIG. 158

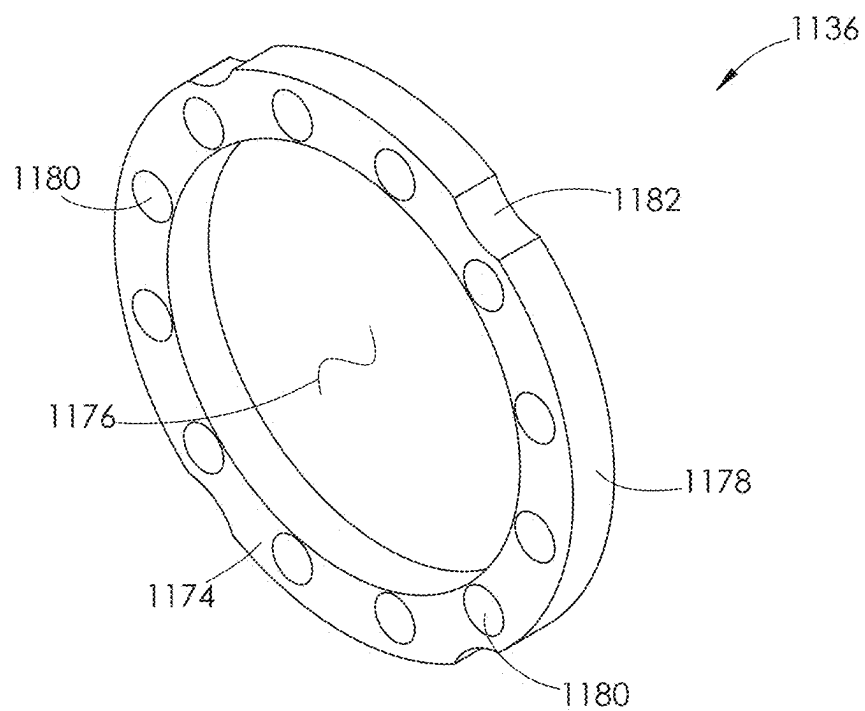


FIG. 159

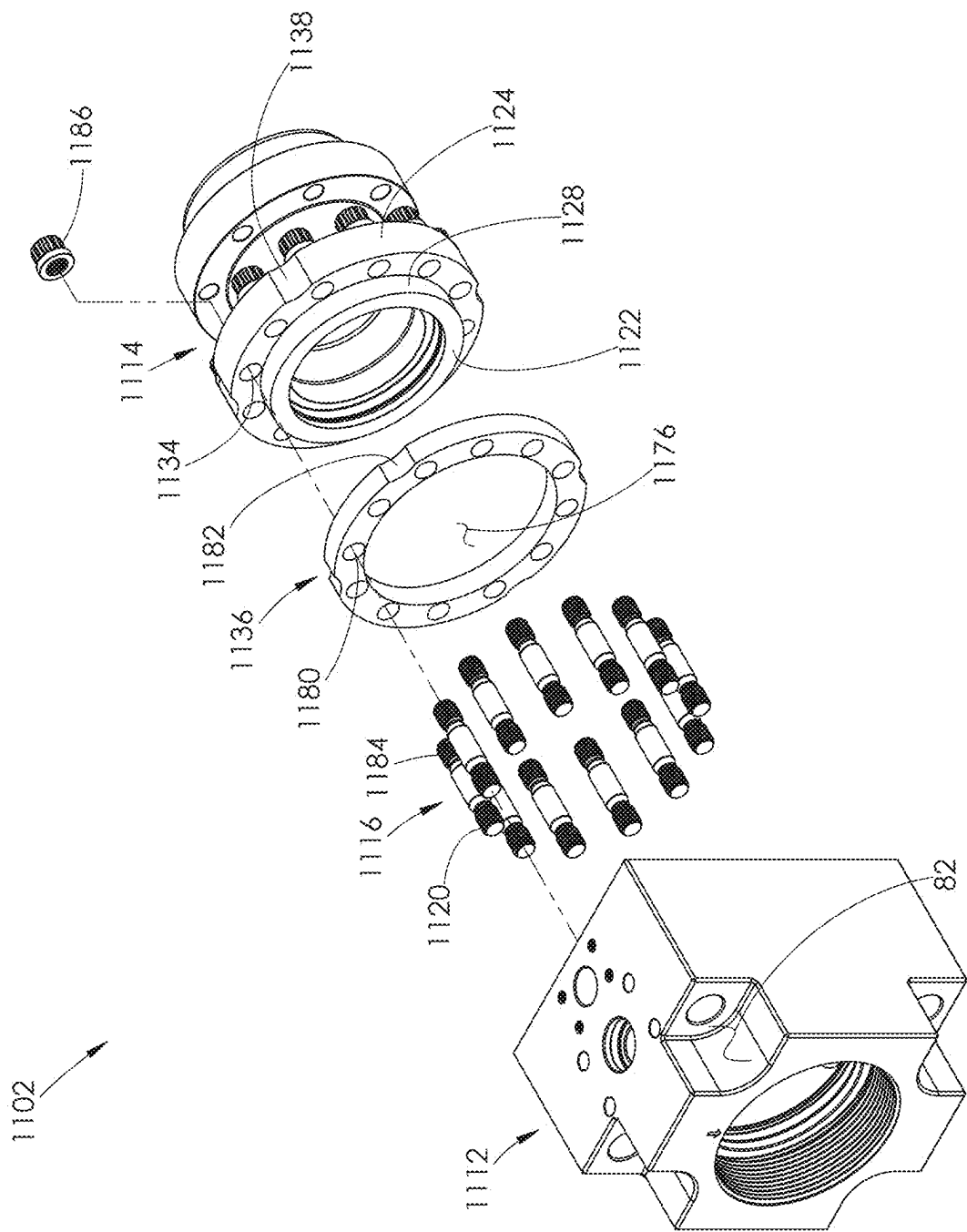


FIG. 160

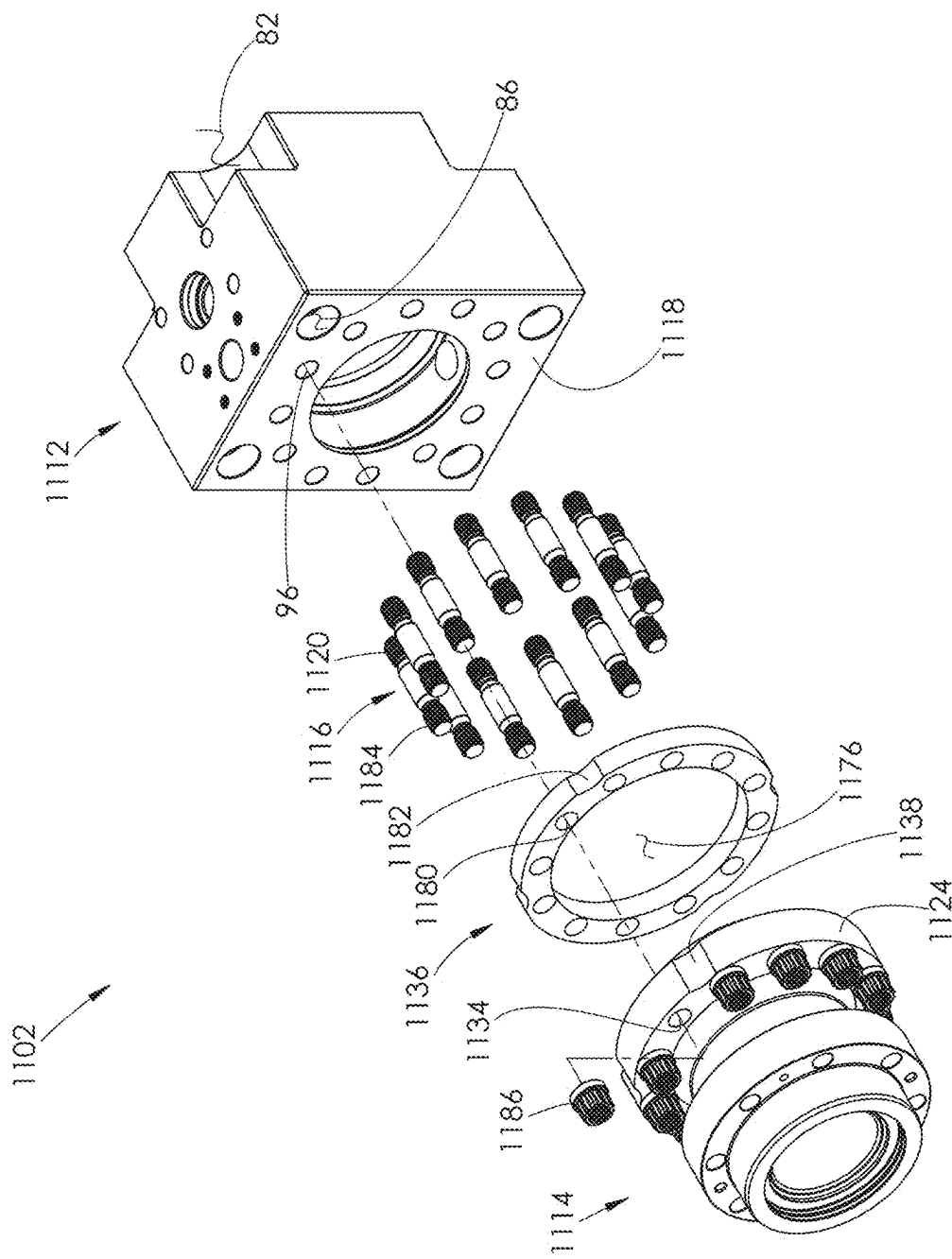


FIG. 161

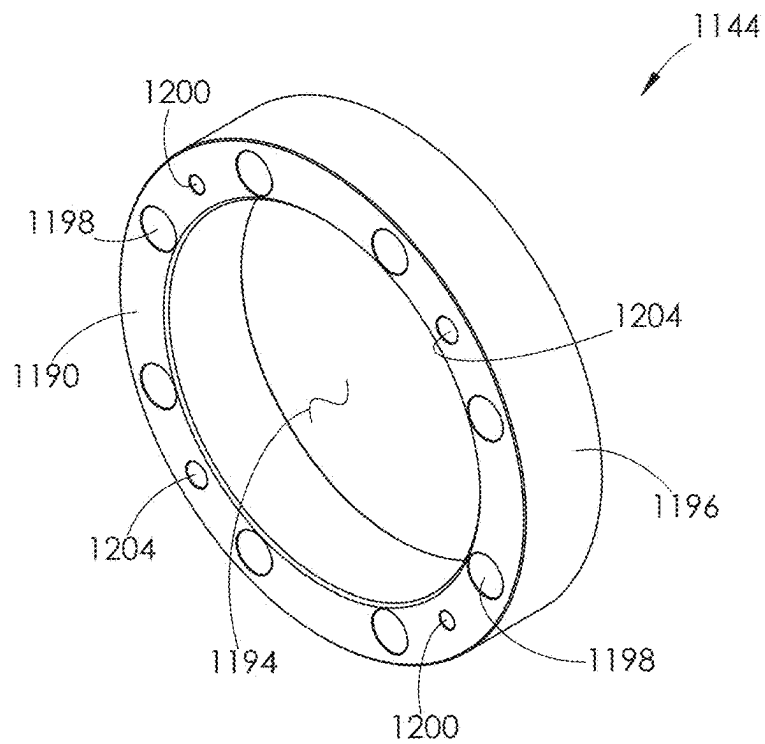


FIG. 162

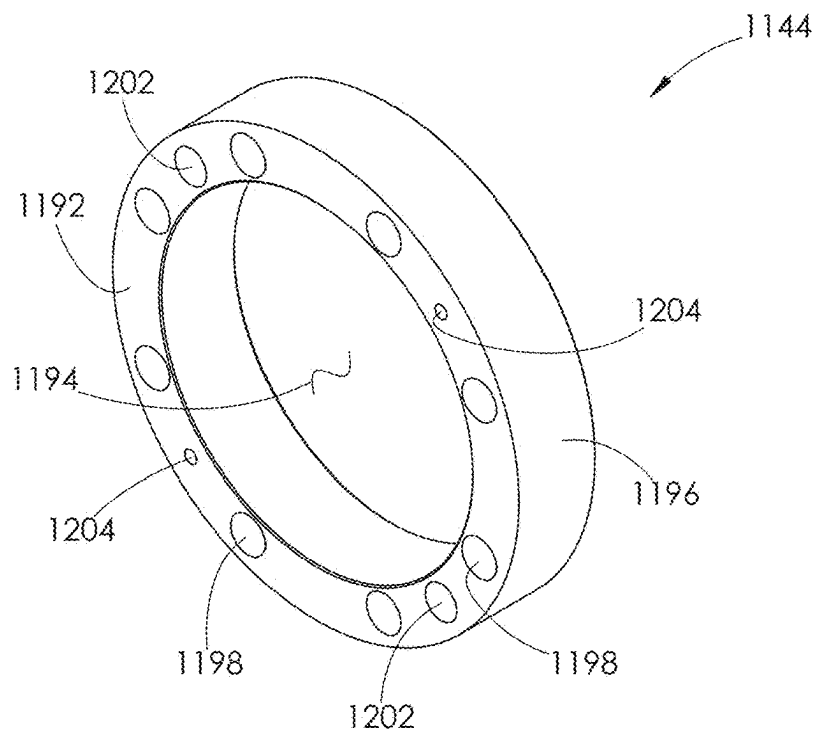


FIG. 163

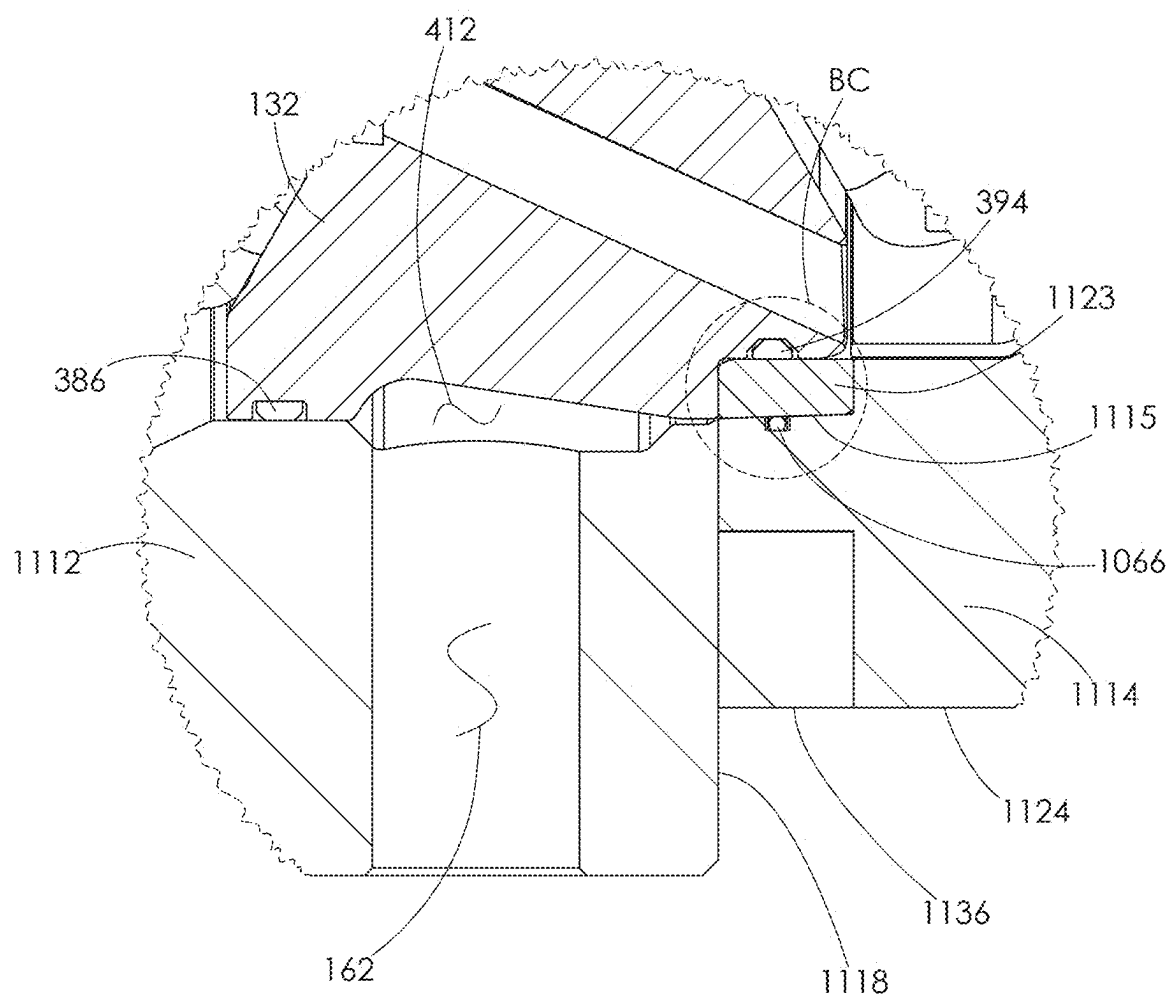


FIG. 164

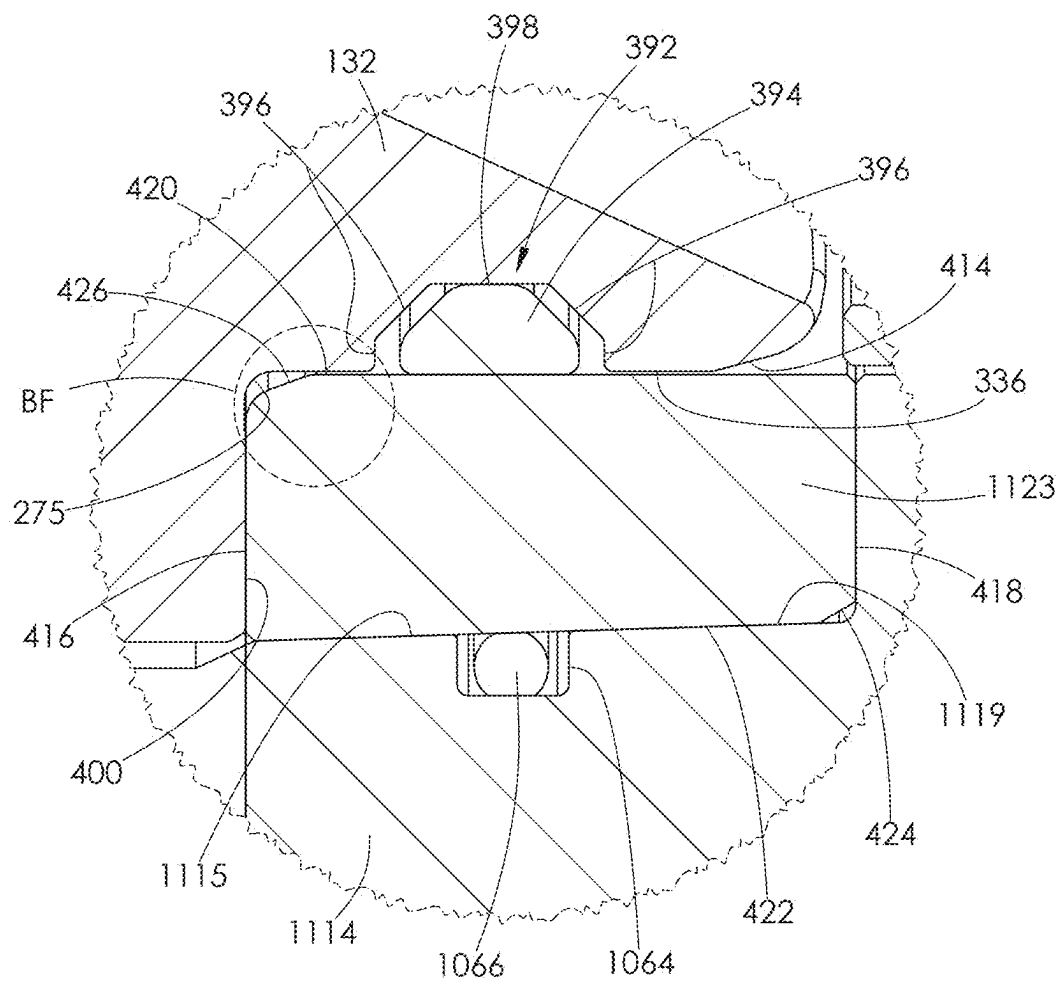


FIG. 165

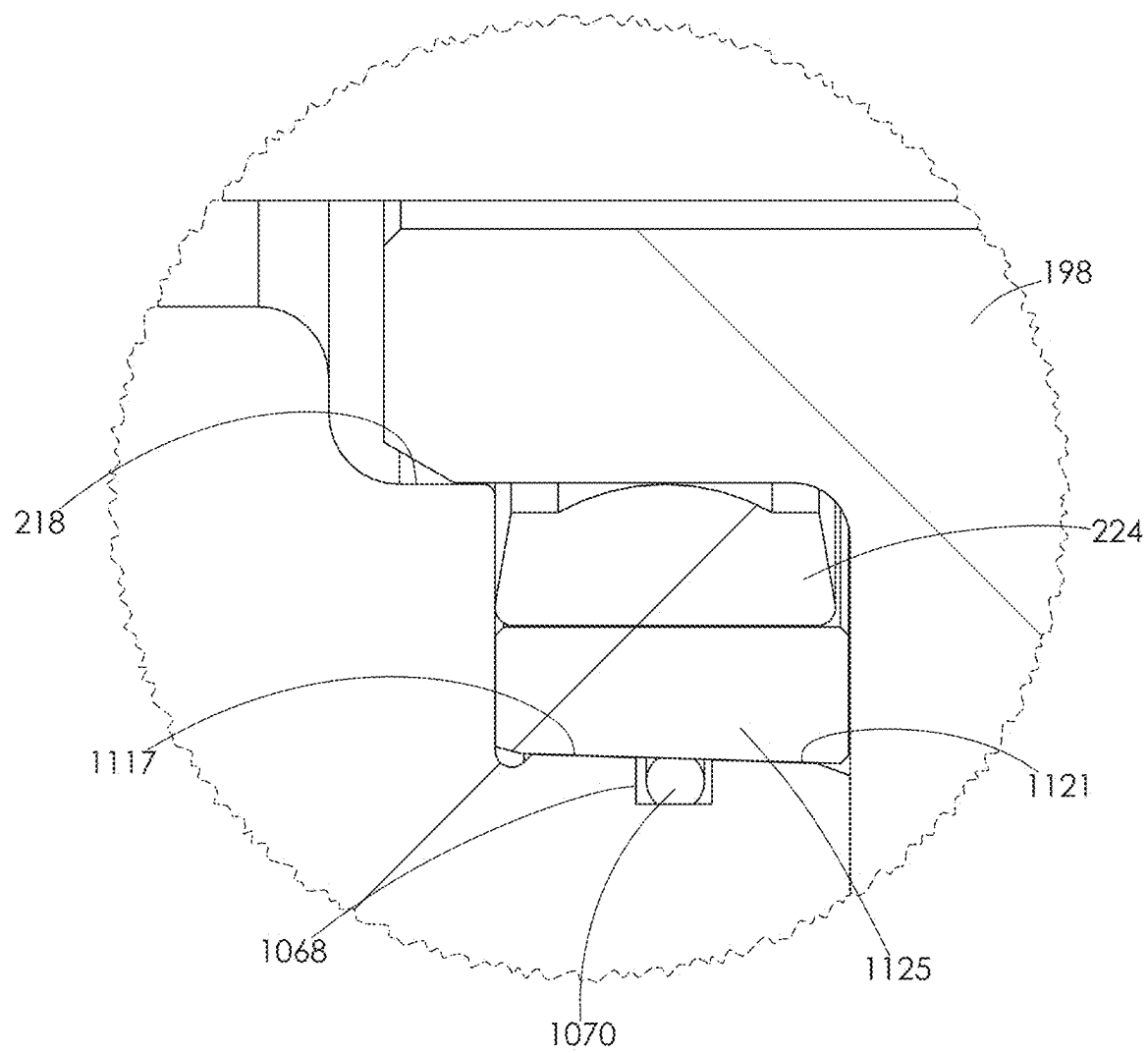


FIG. 166

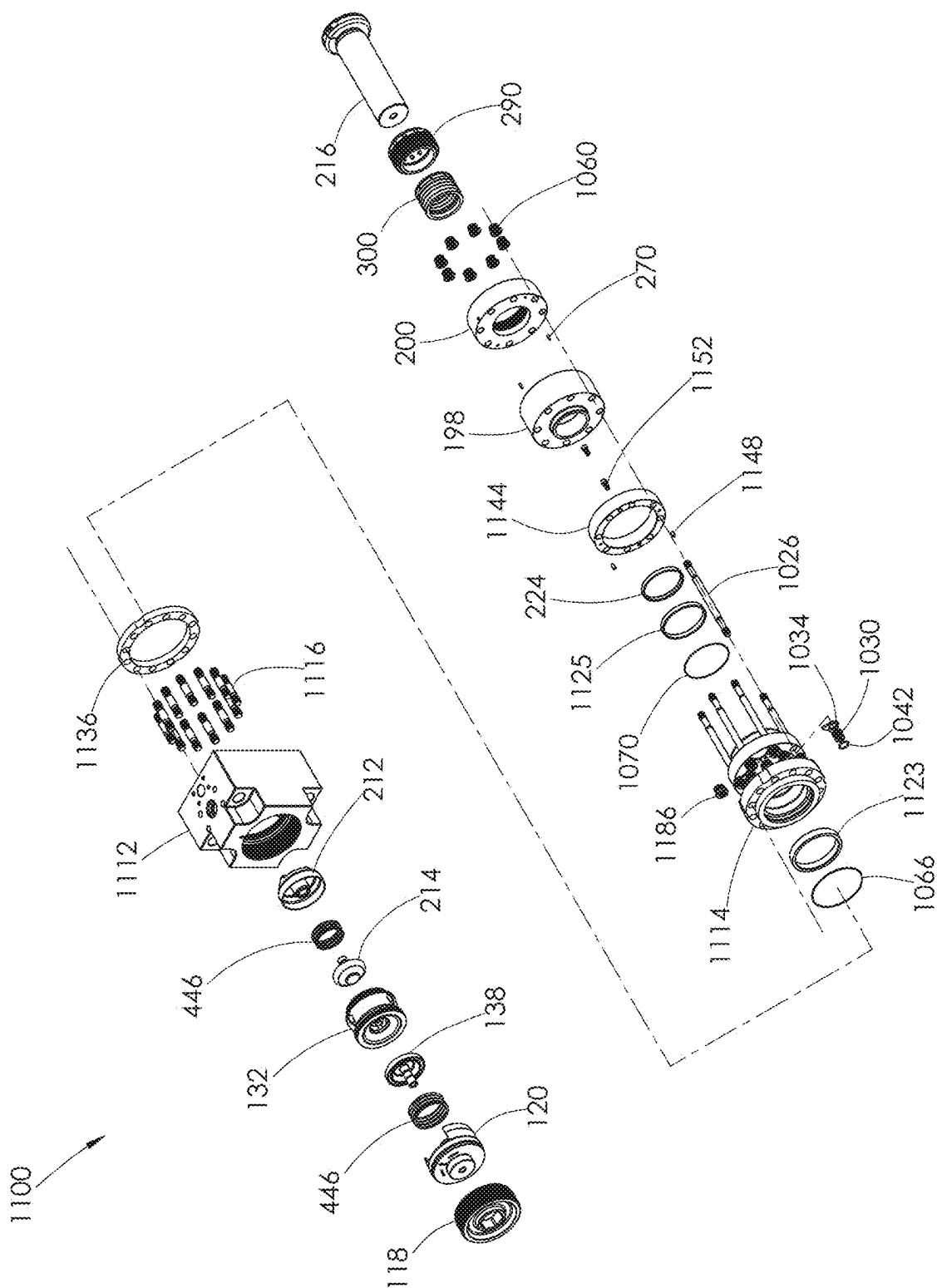


FIG. 167

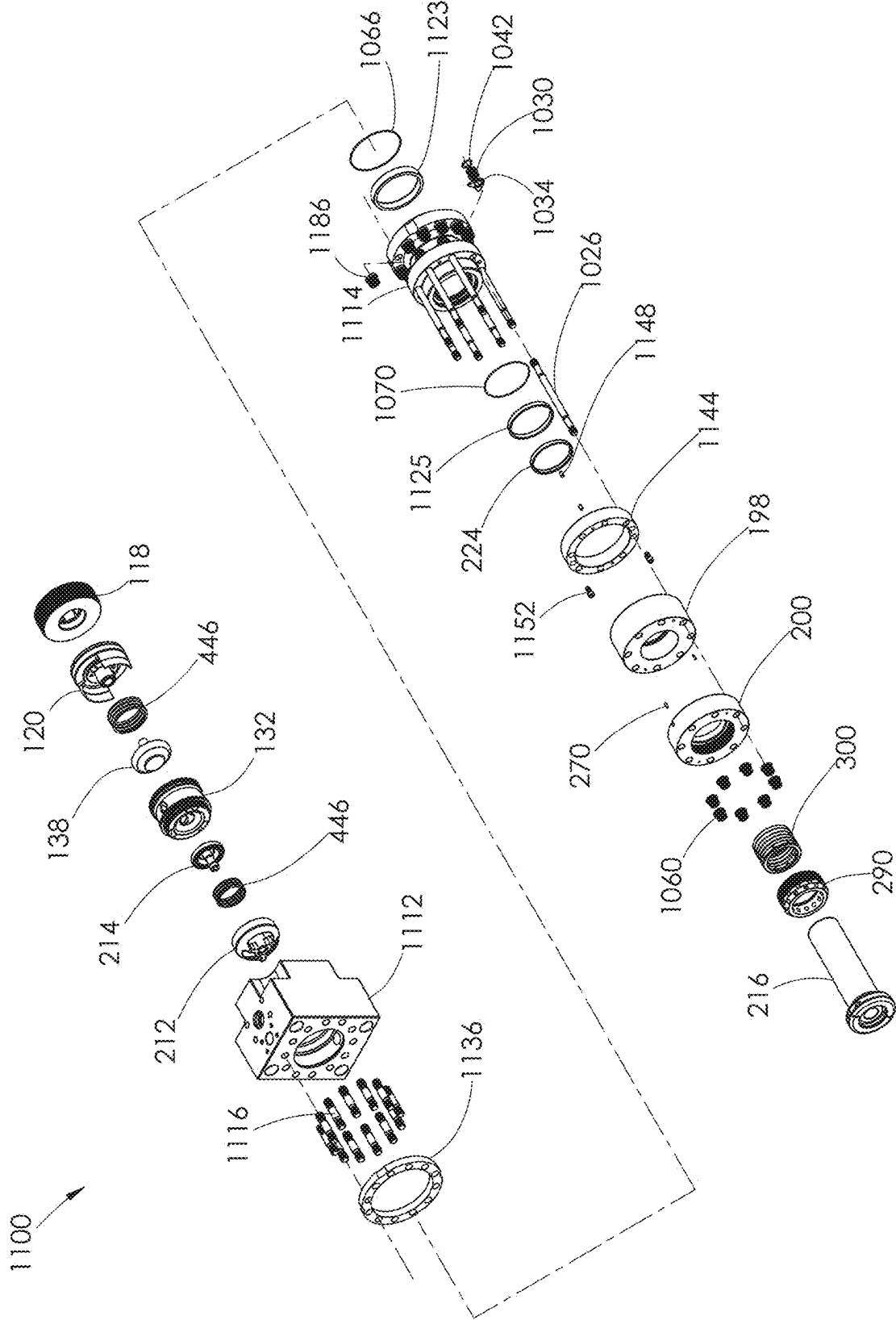


FIG. 168

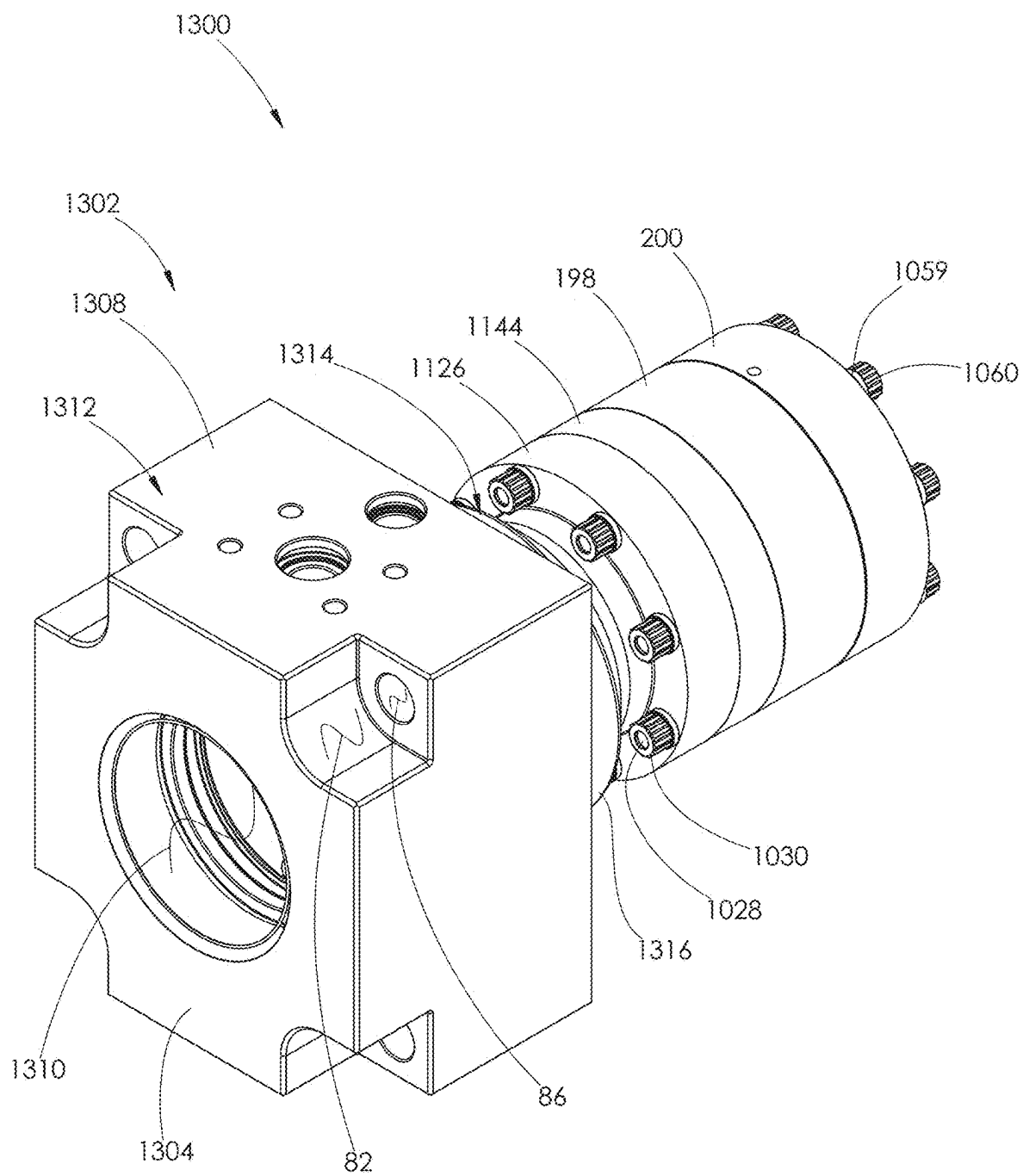


FIG. 169

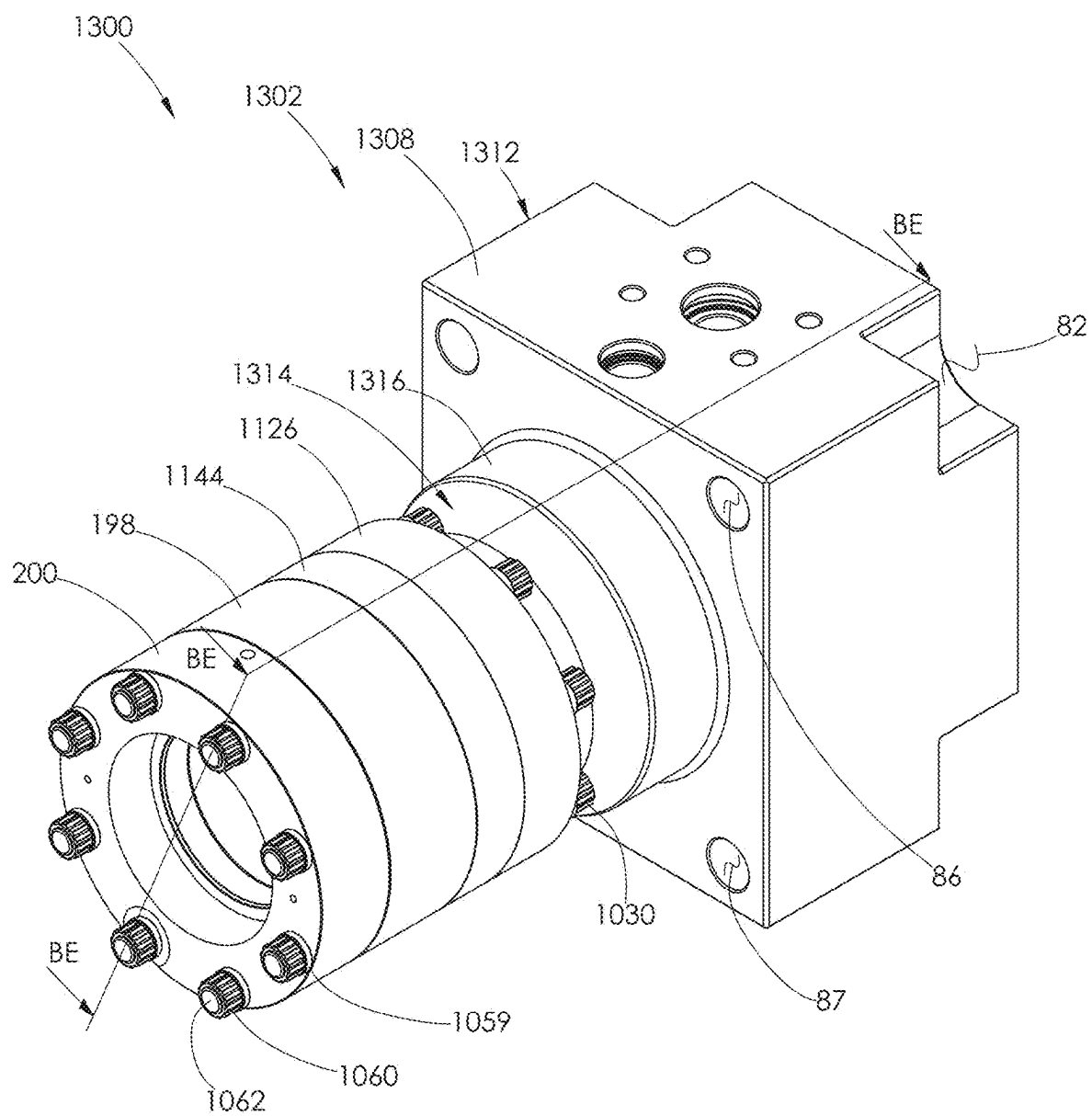


FIG. 170

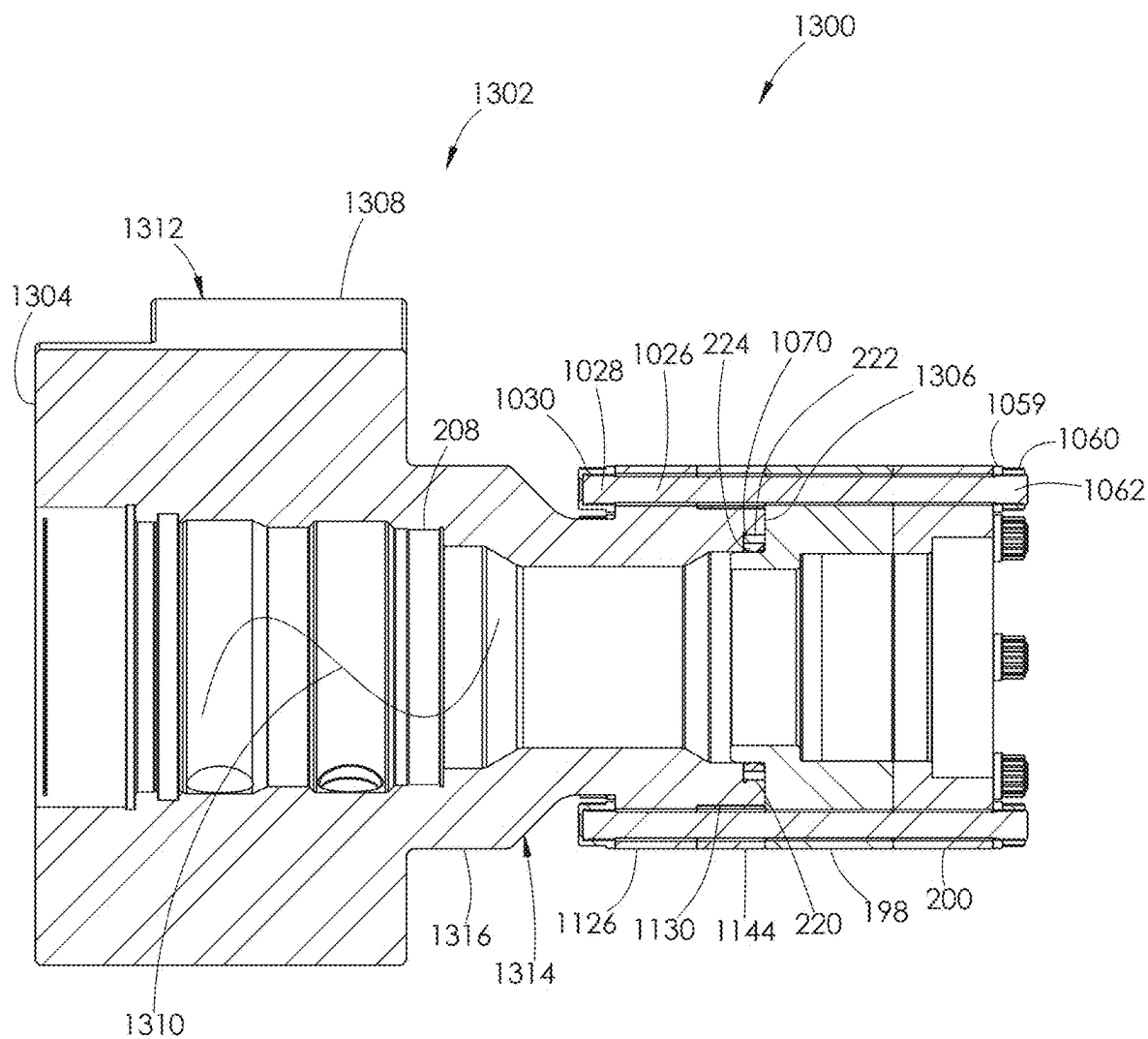


FIG. 171

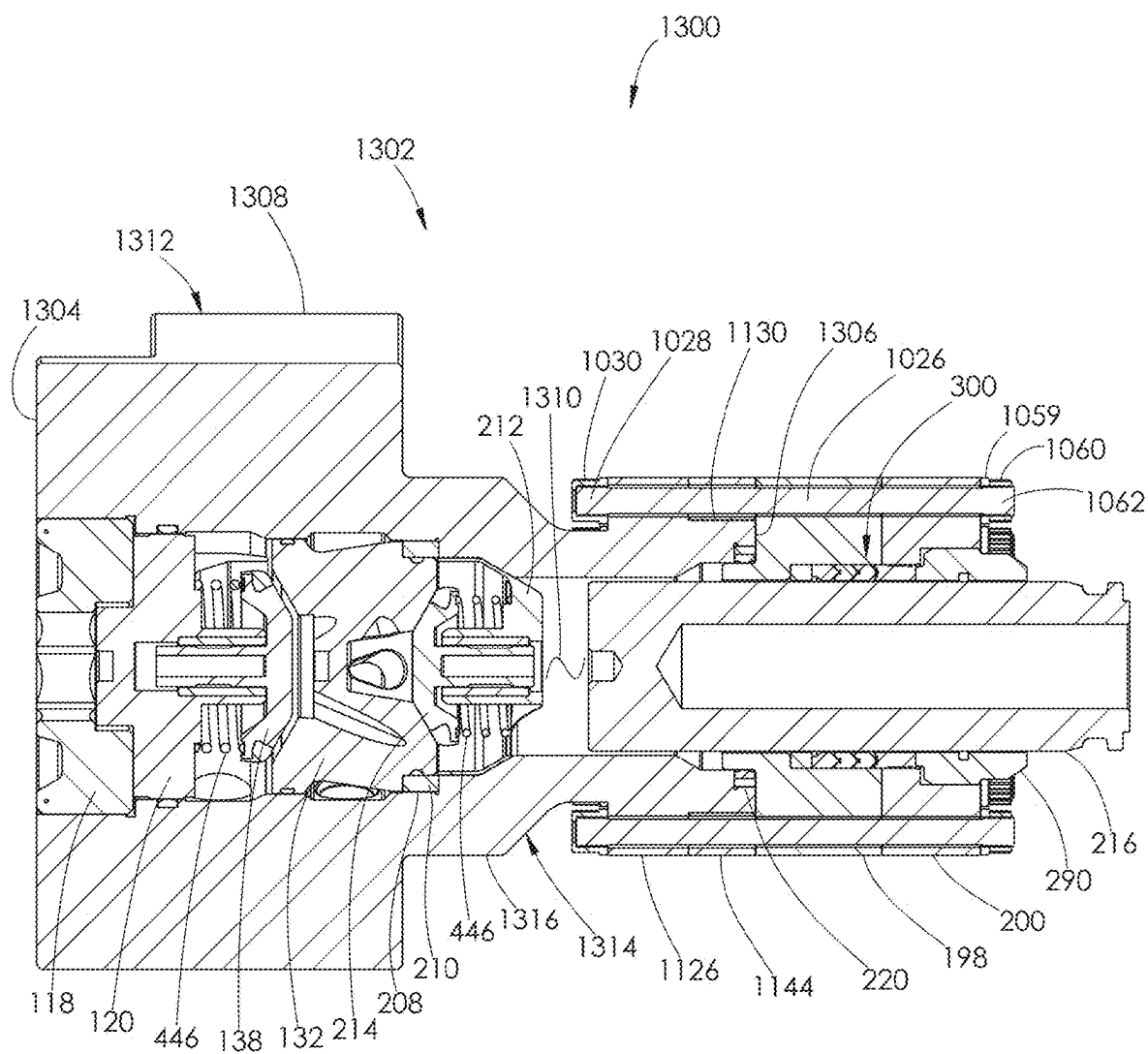


FIG. 172

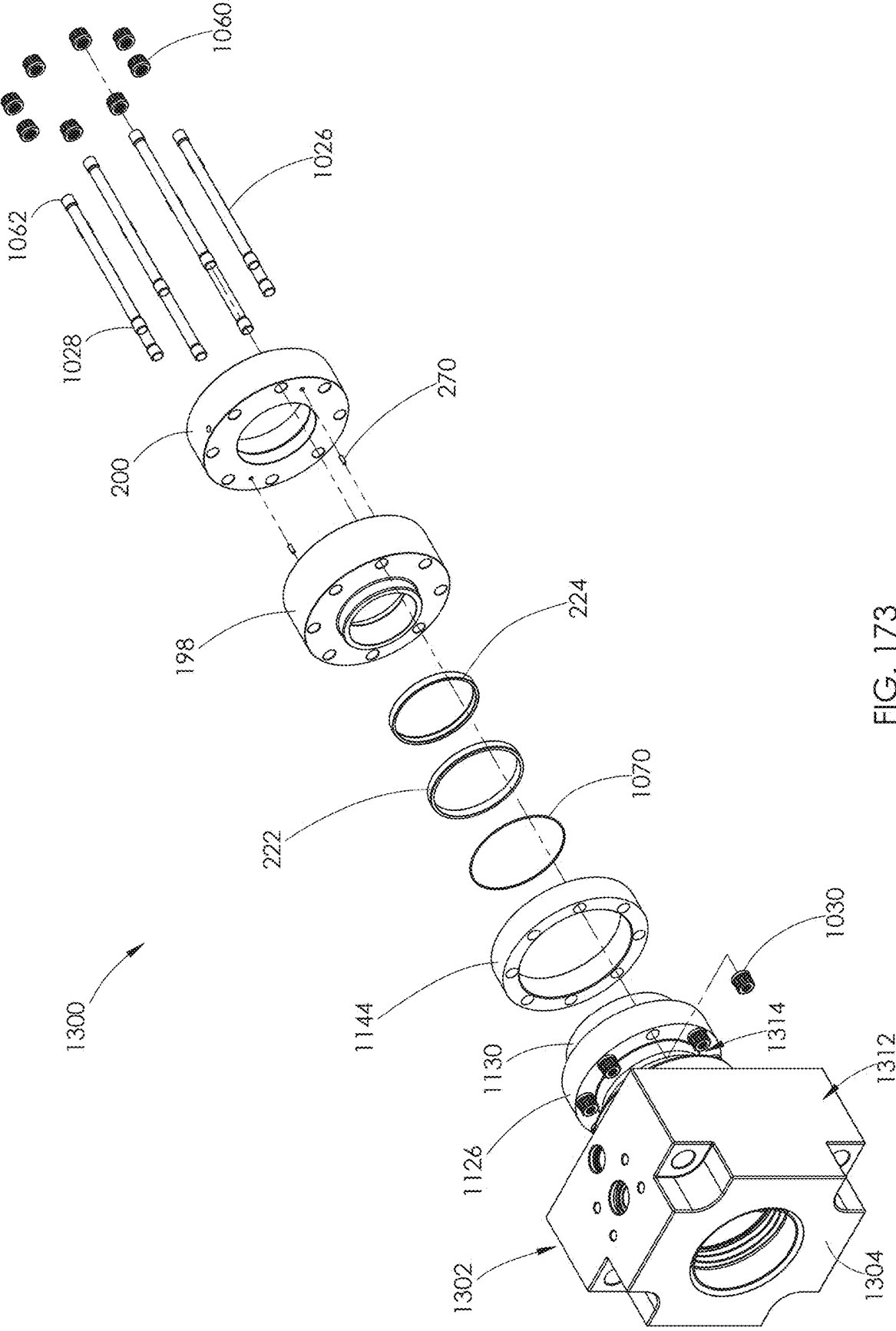


FIG. 173

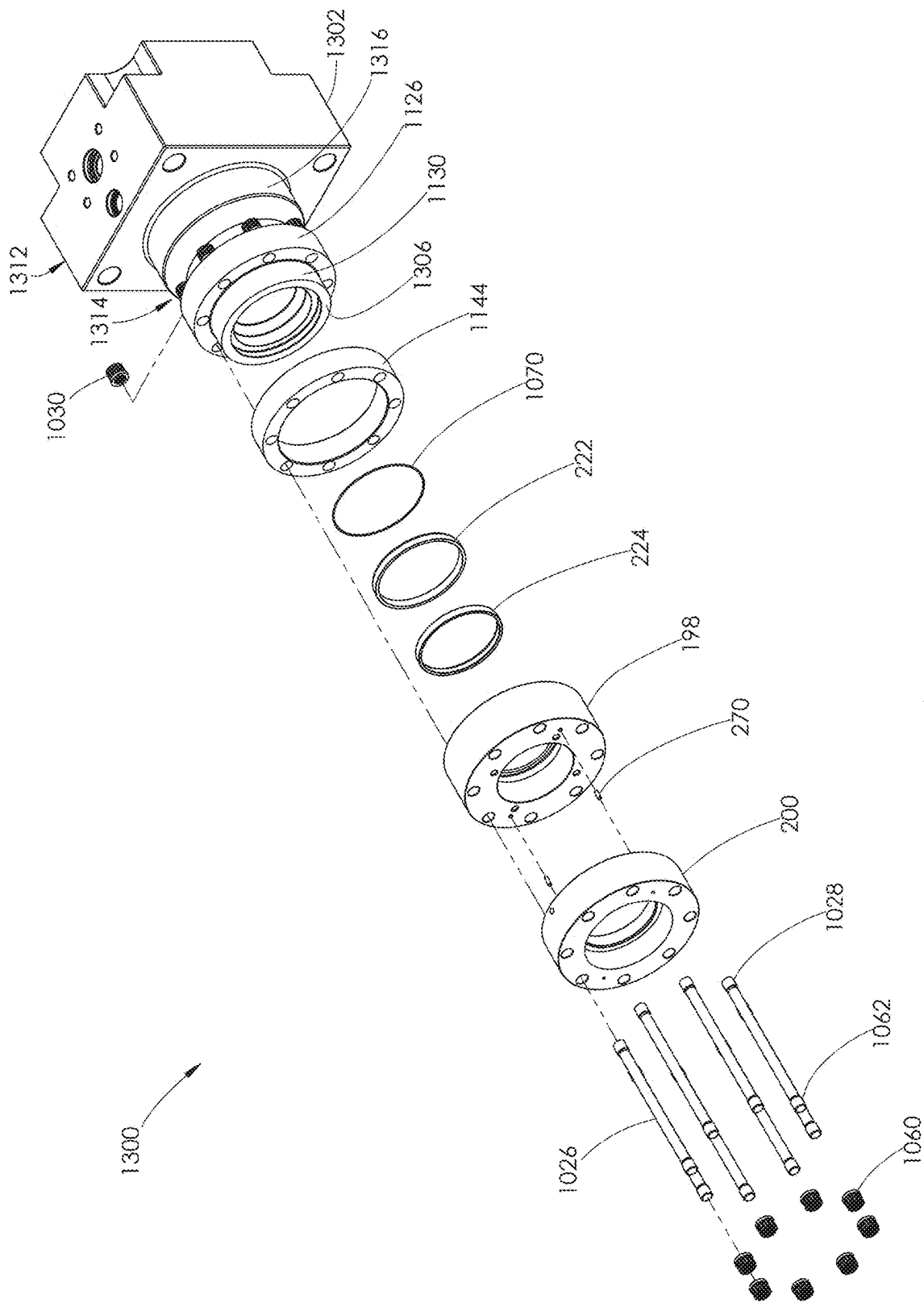


FIG. 174

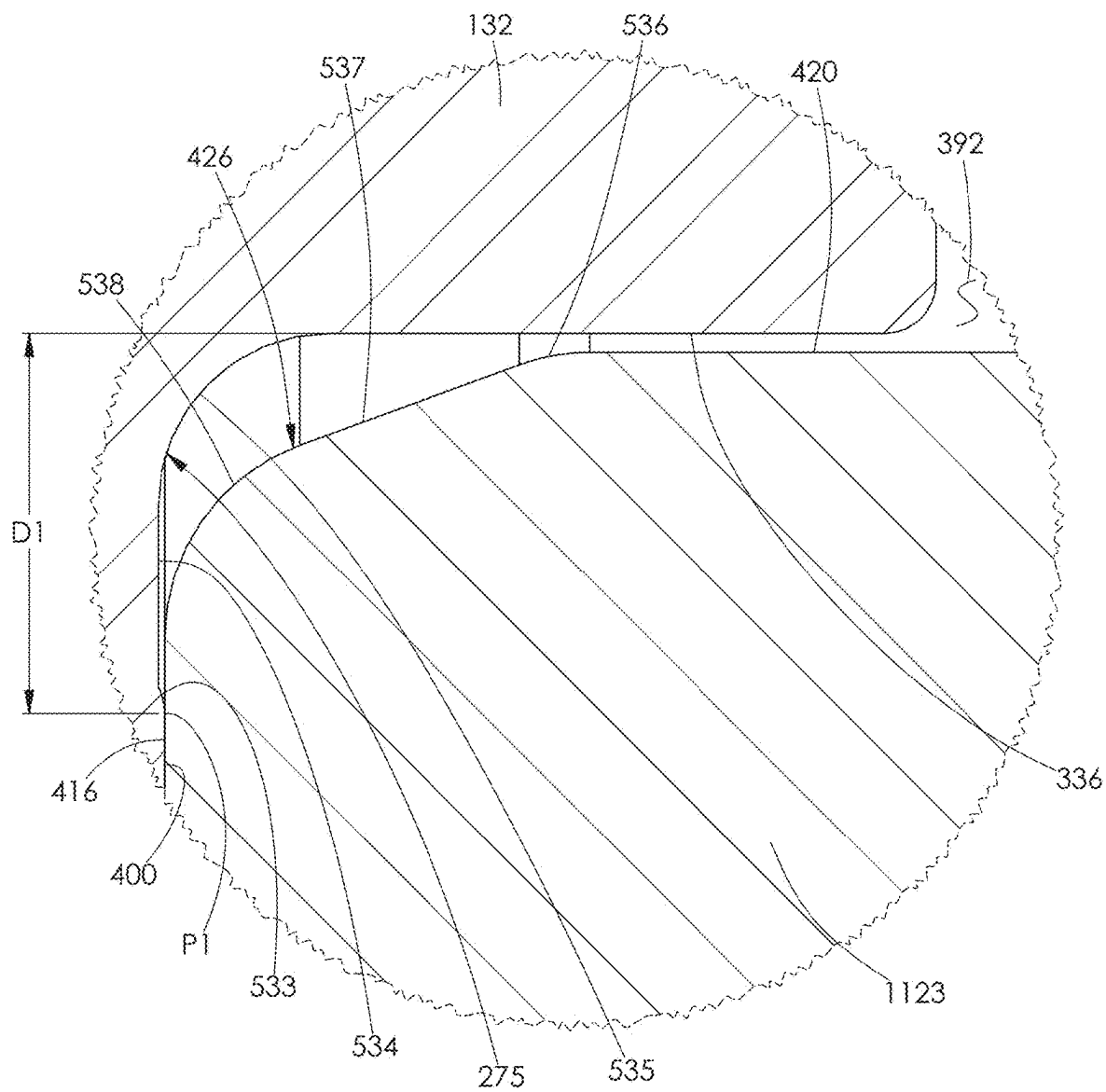


FIG. 175

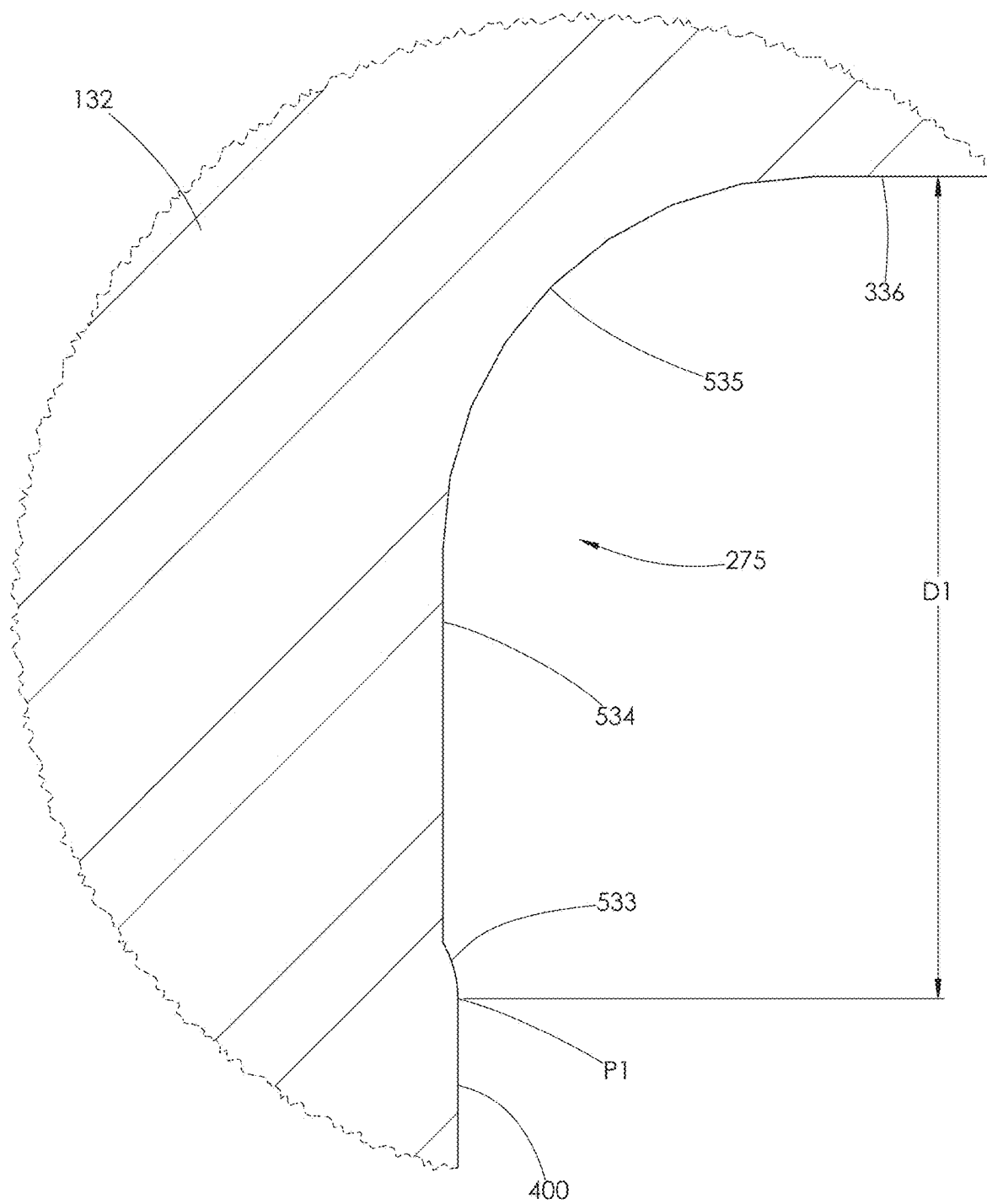


FIG. 176

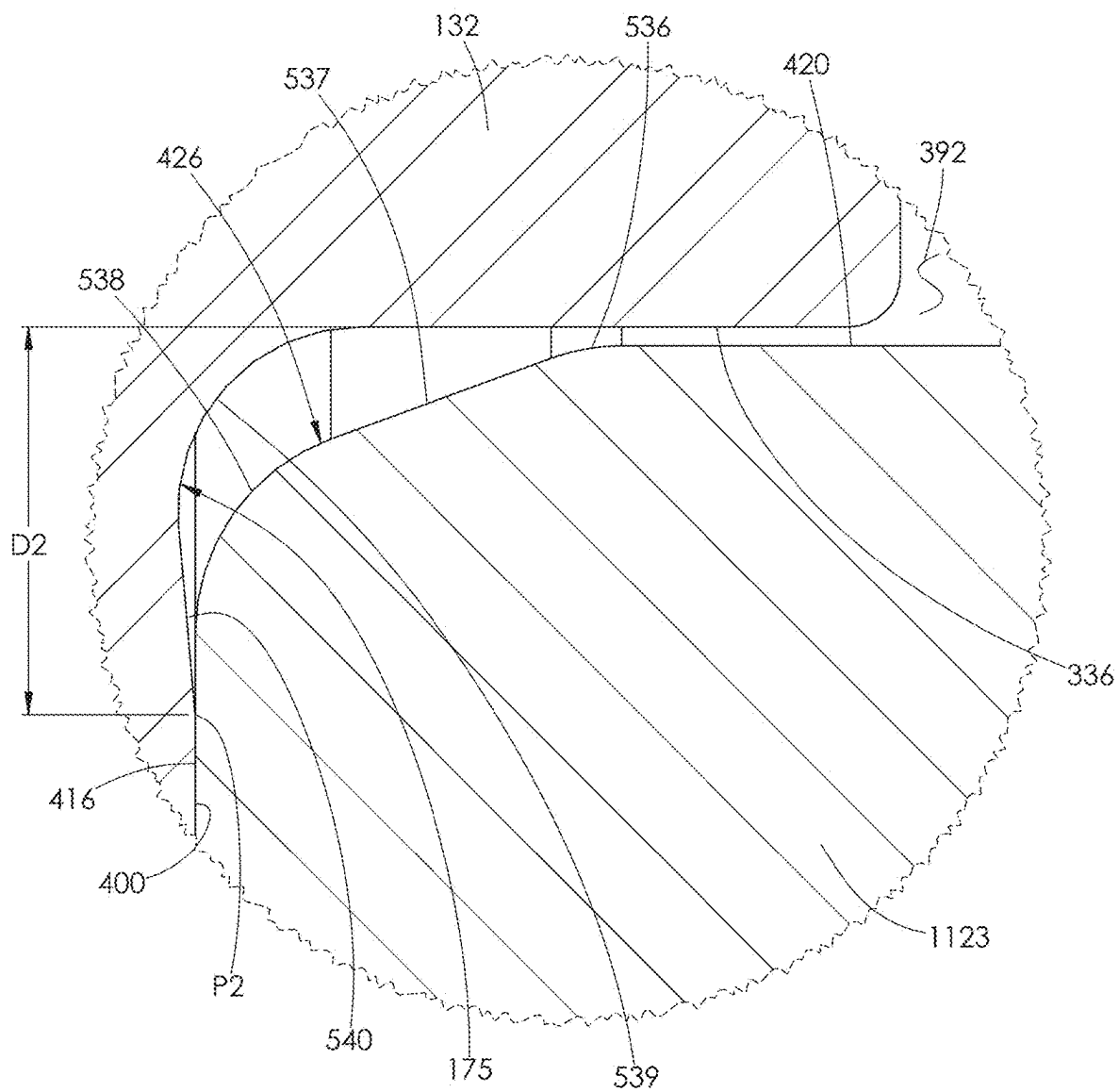


FIG. 177

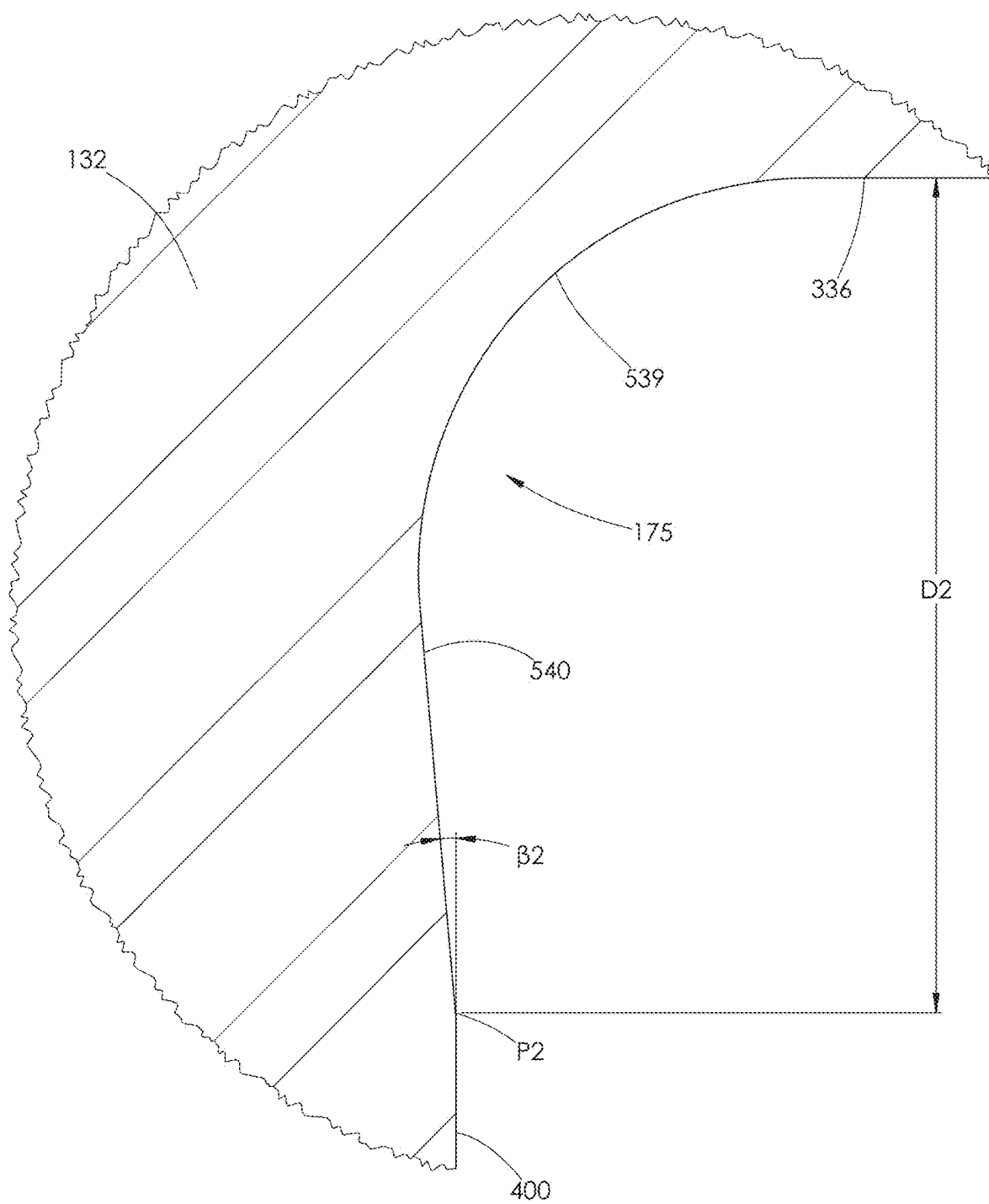


FIG. 178

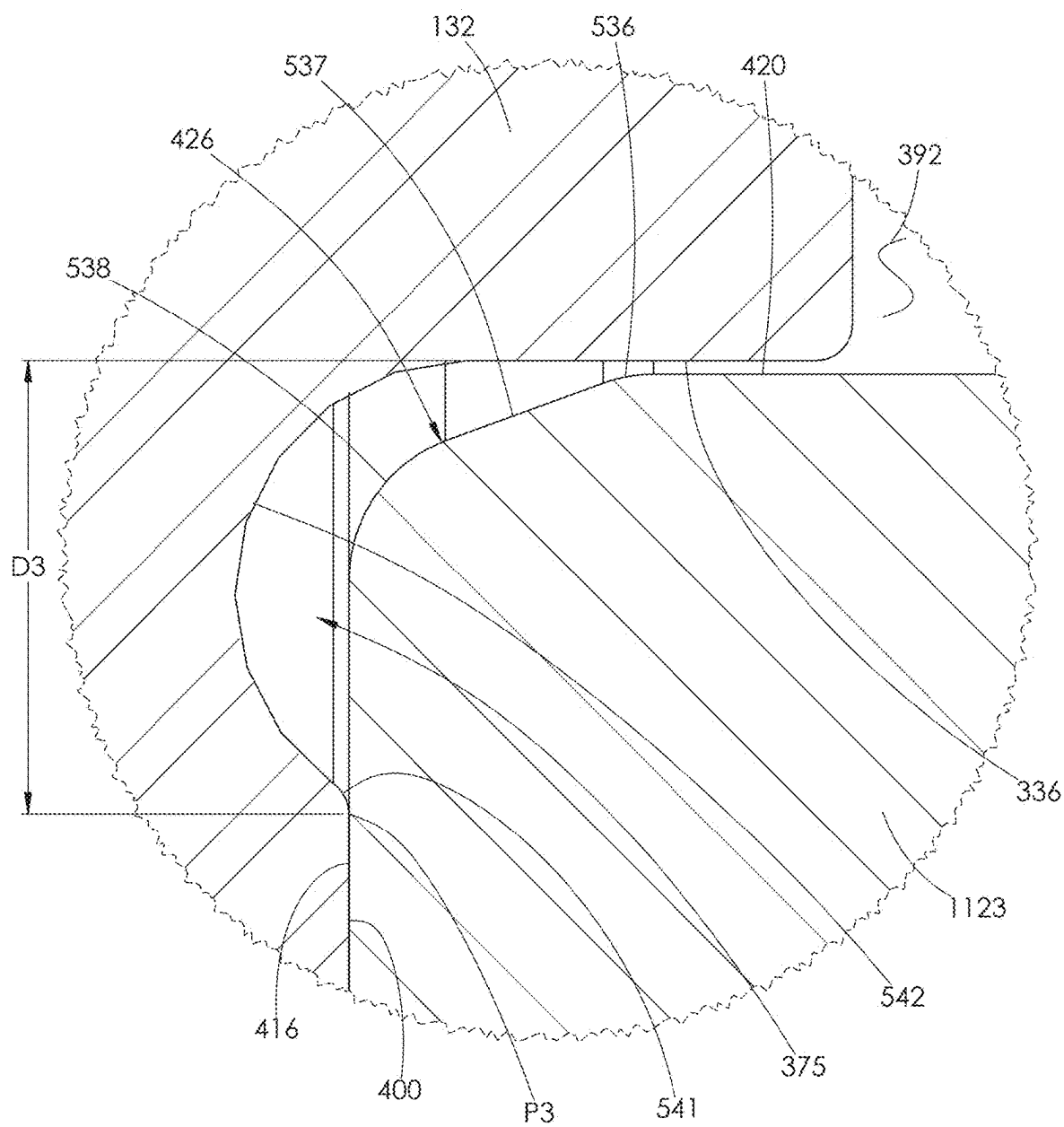


FIG. 179

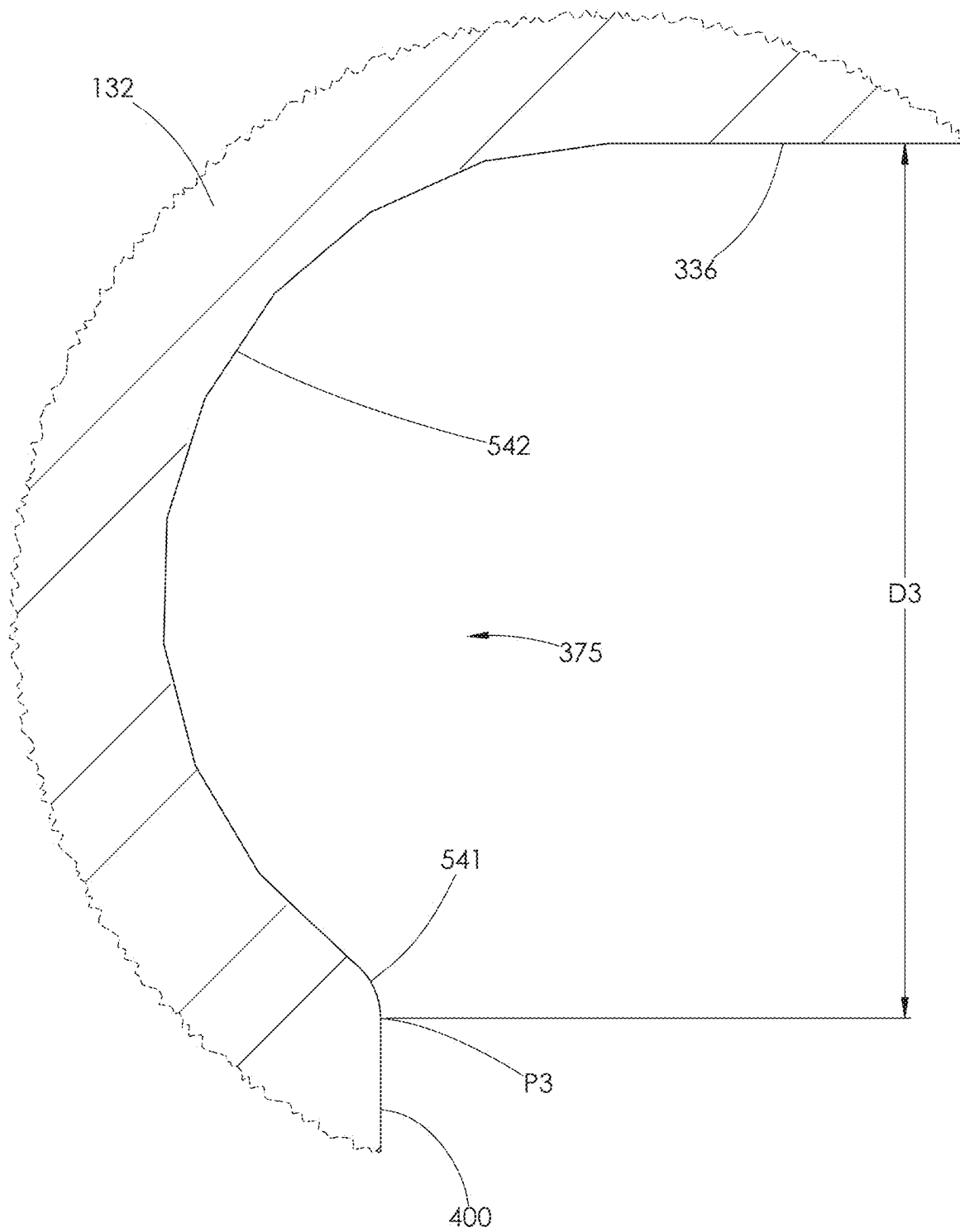


FIG. 180

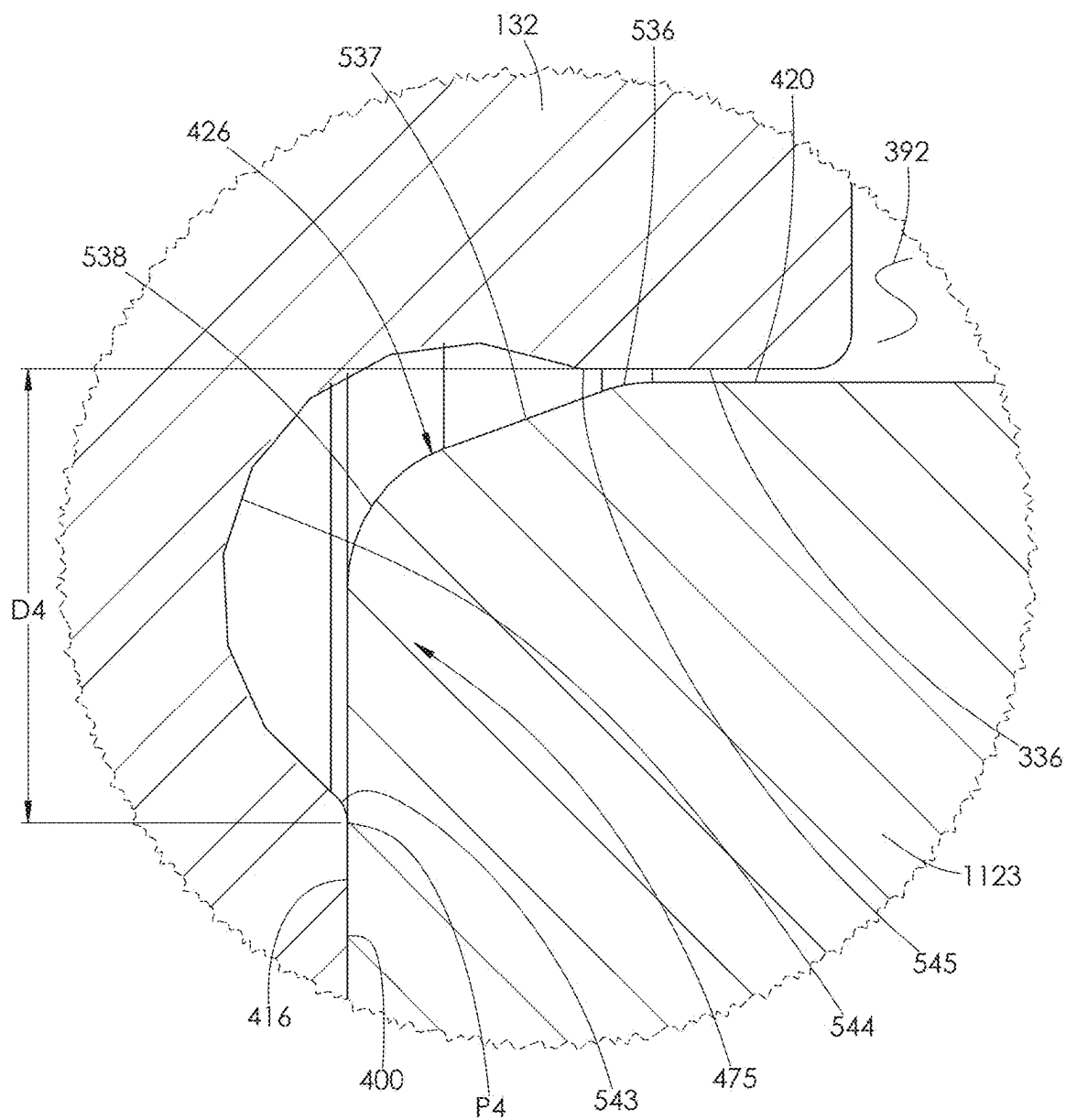


FIG. 181

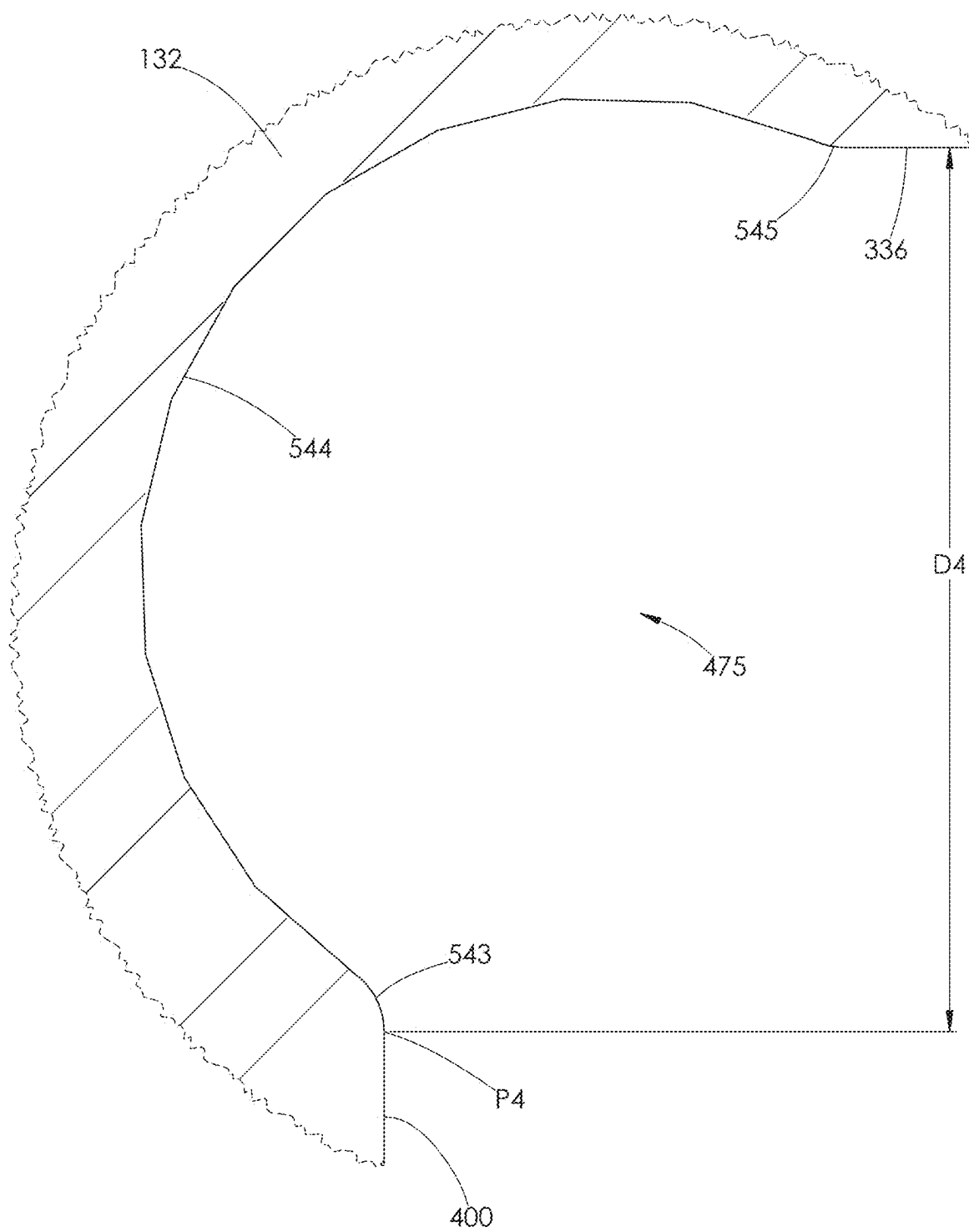


FIG. 182

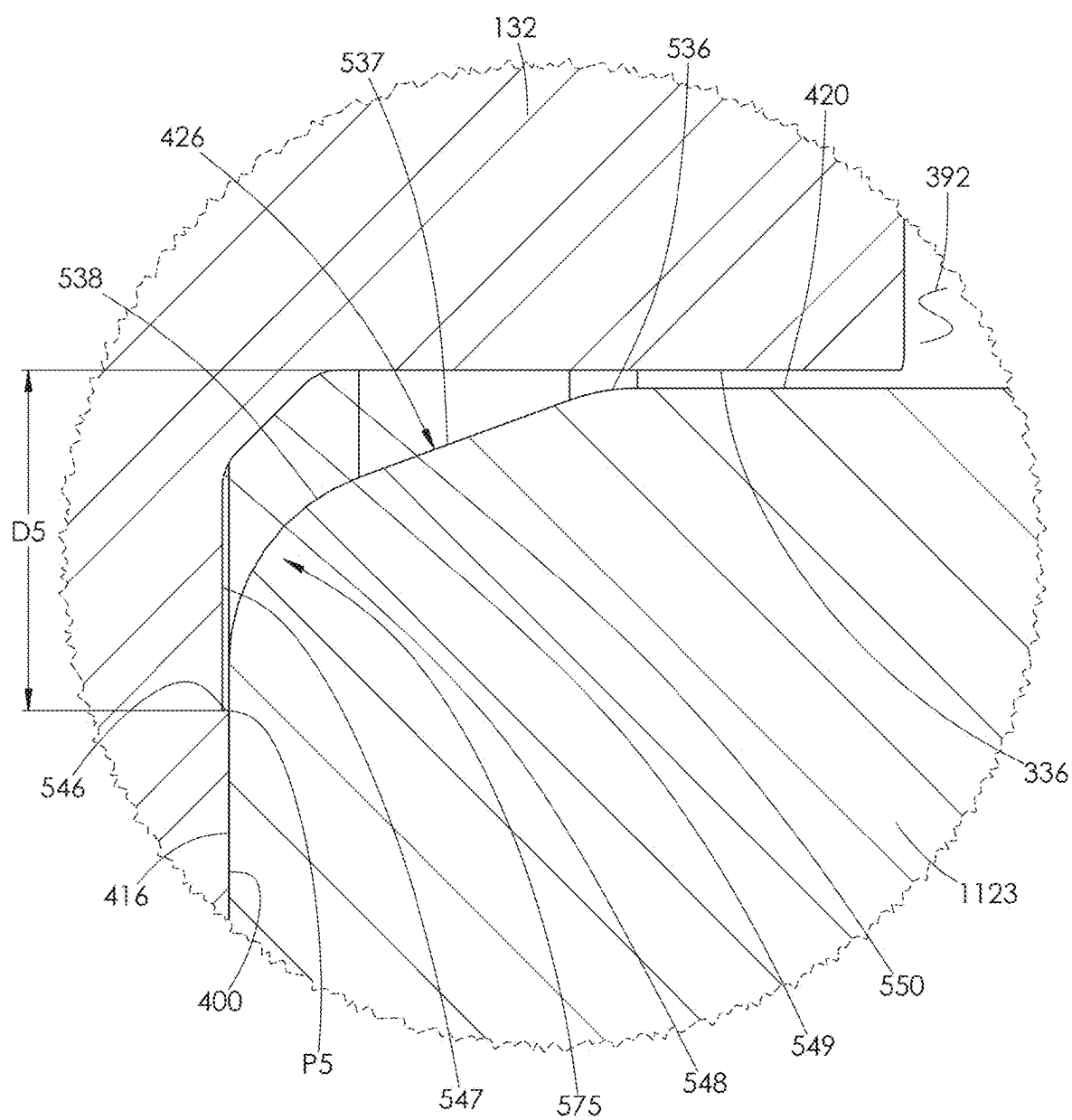


FIG. 183

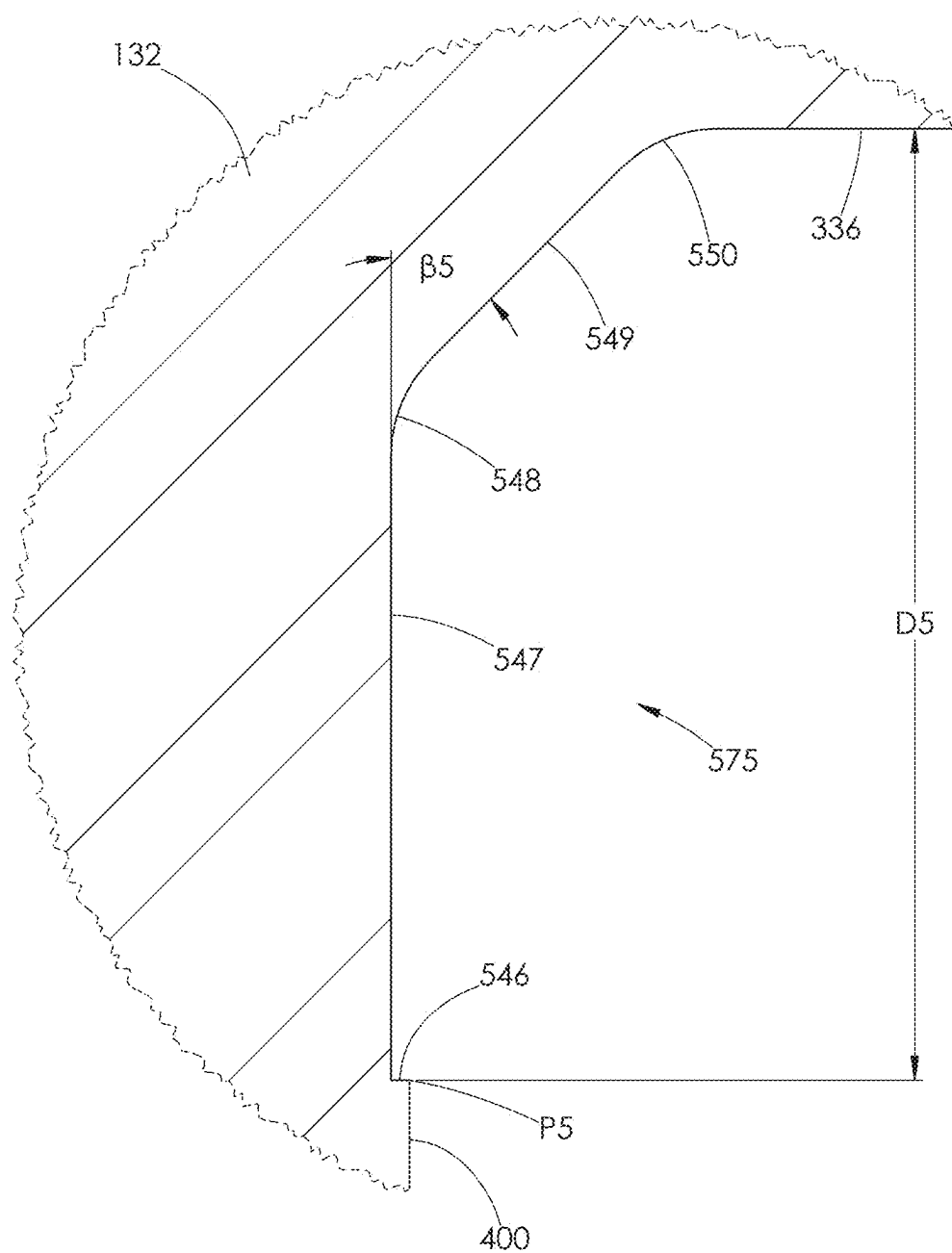


FIG. 184

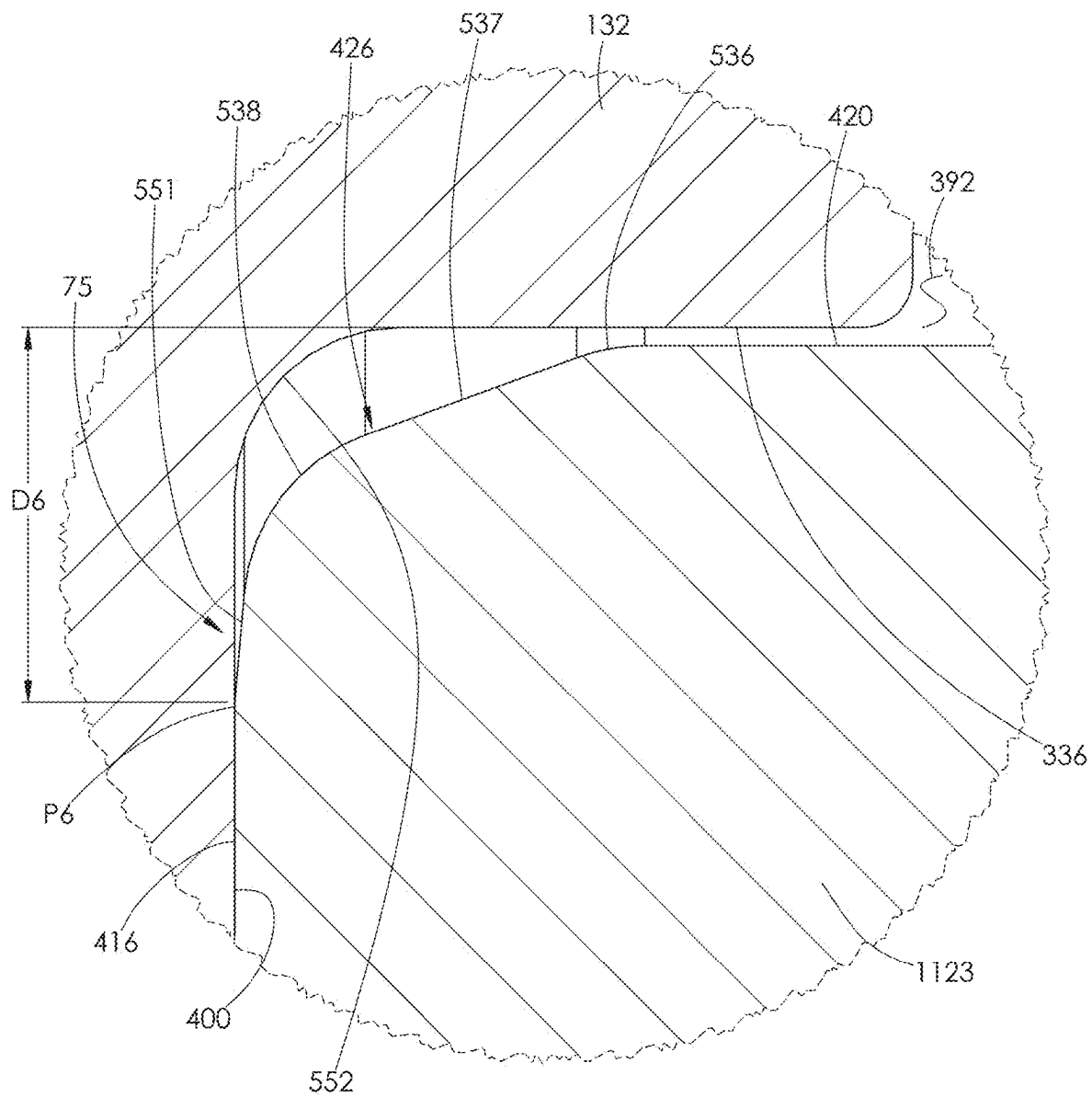


FIG. 185

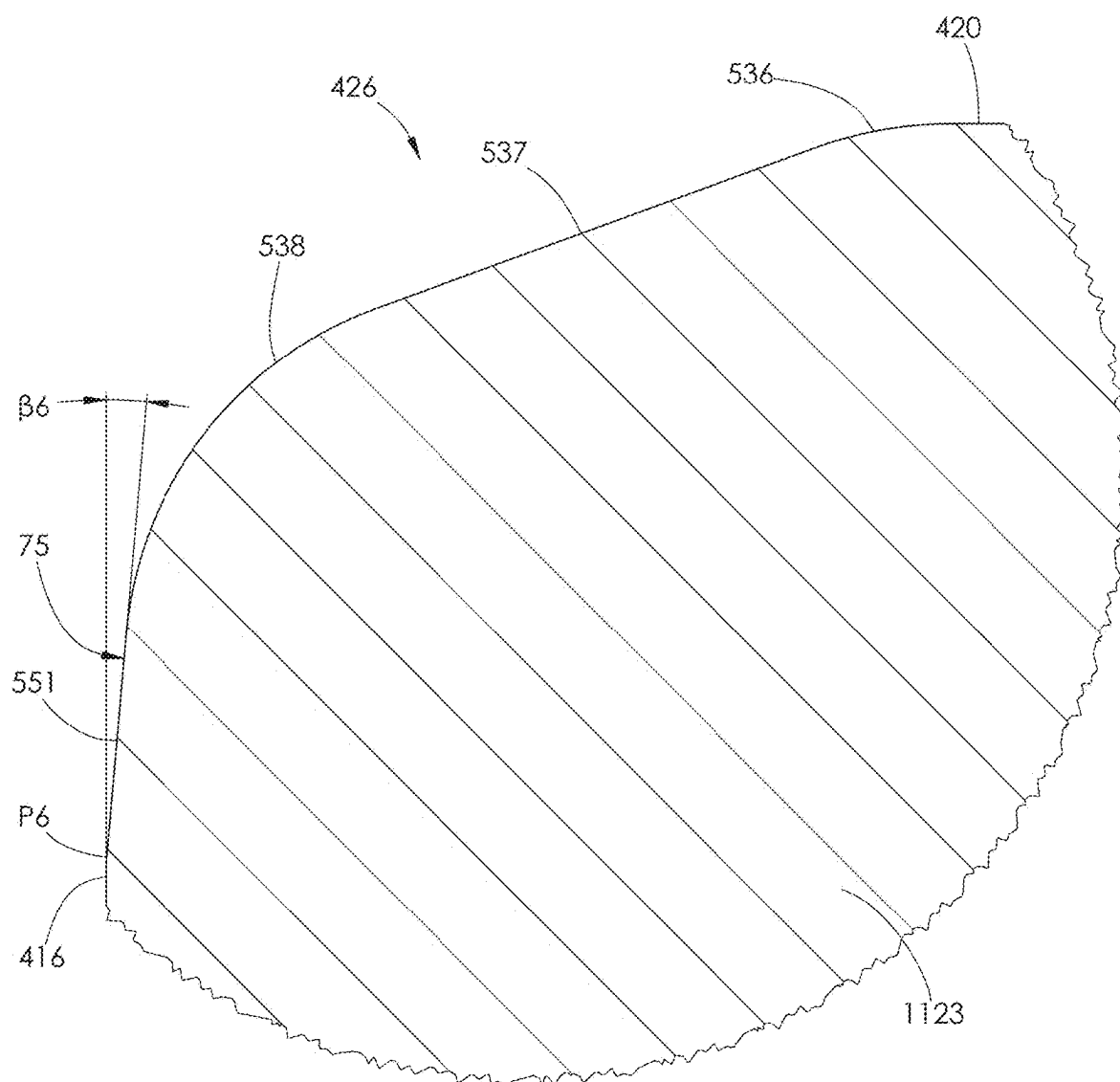


FIG. 186

FLUID END ASSEMBLY

[0001] This application is a Continuation-in-Part of U.S. patent application Ser. No. 17/550,552, now U.S. Pat. No. 11,644,018, authored by Thomas et al., and filed on Dec. 14, 2021, the entire contents of which are incorporated herein by reference (hereinafter “the ‘552 application”). This application is also a Continuation-in-Part of U.S. patent application Ser. No. 18/488,435, now U.S. Pat. No. 12,000,257, authored by Foster et al., and filed on Oct. 17, 2023, the entire contents of which are incorporated herein by reference.

[0002] This application also claims the benefit of the following U.S. provisional patent applications: Ser. No. 63/233,241, authored by Foster et al., and filed on Aug. 14, 2021; Ser. No. 63/235,251, authored by Foster et al., and filed on Aug. 20, 2021; Ser. No. 63/240,889, authored by Foster et al., and filed on Sep. 4, 2021; Ser. No. 63/246,099, authored by Foster et al., and filed on Sep. 20, 2021; Ser. No. 63/301,524, authored by Foster et al., and filed on Jan. 21, 2022; Ser. No. 63/304,070, authored by Foster et al., and filed on Jan. 28, 2022; Ser. No. 63/310,269, authored by Foster et al., and filed on Feb. 15, 2022; Ser. No. 63/312,541, authored by Foster et al., and filed on Feb. 22, 2022; Ser. No. 63/416,644, authored by Foster et al., and filed on Oct. 17, 2022; Ser. No. 63/422,637, authored by Foster et al., and filed on Nov. 4, 2022; Ser. No. 63/506,222, authored by Foster et al., and filed on Jun. 5, 2023; and Ser. No. 63/508,577, authored by Foster et al., and filed on Jun. 16, 2023. The entire contents of all of the above listed provisional patent applications are incorporated herein by reference.

BACKGROUND

[0003] Various industrial applications may require the delivery of high volumes of highly pressurized fluids. For example, hydraulic fracturing (commonly referred to as “fracking”) is a well stimulation technique used in oil and gas production, in which highly pressurized fluid is injected into a cased wellbore. As shown for example in FIG. 1, the pressured fluid flows through perforations 10 in a casing 12 and creates fractures 14 in deep rock formations 16. Pressurized fluid is delivered to the casing 12 through a wellhead 18 supported on the ground surface 20. Sand or other small particles (commonly referred to as “proppants”) are normally delivered with the fluid into the rock formations 16. The proppants help hold the fractures 14 open after the fluid is withdrawn. The resulting fractures 14 facilitate the extraction of oil, gas, brine, or other fluid trapped within the rock formations 16.

[0004] Fluid ends are devices used in conjunction with a power source to pressurize the fluid used during hydraulic fracturing operations. A single fracking operation may require the use of two or more fluid ends at one time. For example, six fluid ends 22 are shown operating at a wellsite 24 in FIG. 2. Each of the fluid ends 22 is attached to a power end 26 in a one-to-one relationship. The power end 26 serves as an engine or motor for the fluid end 22. Together, the fluid end 22 and power end 26 function as a high pressure pump.

[0005] Continuing with FIG. 2, a single fluid end 22 and its corresponding power end 26 are typically positioned on a truck bed 28 at the wellsite 24 so that they may be easily moved, as needed. The fluid and proppant mixture to be pressurized is normally held in large tanks 30 at the wellsite

24. An intake piping system 32 delivers the fluid and proppant mixture from the tanks 30 to each fluid end 22. A discharge piping system 33 transfers the pressurized fluid from each fluid end 22 to the wellhead 18, where it is delivered into the casing 12 shown in FIG. 1.

[0006] Fluid ends operate under notoriously extreme conditions, enduring the same pressures, vibrations, and abrasives that are needed to fracture the deep rock formations 16, shown in FIG. 1. Fluid ends may operate at pressures of 5,000-15,000 pounds per square inch (psi) or greater. Fluid used in hydraulic fracturing operations is typically pumped through the fluid end at a pressure of at least 8,000 psi, and more typically between 10,000 and 15,000 psi. However, the pressure may reach up to 22,500 psi. The power end used with the fluid end typically has a power output of at least 2,250 horsepower during hydraulic fracturing operations. A single fluid end typically produces a fluid volume of about 400 gallons, or 10 barrels, per minute during a fracking operation. A single fluid end may operate in flow ranges from 170 to 630 gallons per minute, or approximately 4 to 15 barrels per minute. When a plurality of fluid ends are used together, the fluid ends collectively may deliver as much as 4,200 gallons per minute or 100 barrels per minute to the wellbore.

[0007] In contrast, mud pumps known in the art typically operate at a pressure of less than 8,000 psi. Mud pumps are used to deliver drilling mud to a rotating drill bit within the wellbore during drilling operations. Thus, the drilling mud does not need to have as high of fluid pressure as fracking fluid. A fluid end does not pump drilling mud. A power end used with mud pumps typically has a power output of less than 2,250 horsepower. Mud pumps generally produce a fluid volume of about 150-600 gallons per minute, depending on the size of pump used.

[0008] In further contrast, a fluid jetting pump known in the art typically operates at pressures of 30,000-90,000 psi. Jet pumps are used to deliver a highly concentrated stream of fluid to a desired area. Jet pumps typically deliver fluid through a wand. Fluid ends do not deliver fluid through a wand. Unlike fluid ends, jet pumps are not used in concert with a plurality of other jet pumps. Rather, only a single jet pump is used to pressurize fluid. A power end used with a jet pump typically has a power output of about 1,000 horsepower. Jet pumps generally produce a fluid volume of about 10 gallons per minute.

[0009] High operational pressures may cause a fluid end to expand or crack. Such a structural failure may lead to fluid leakage, which leaves the fluid end unable to produce and maintain adequate fluid pressures. Moreover, if proppants are included in the pressurized fluid, those proppants may cause erosion at weak points within the fluid end, resulting in additional failures.

[0010] It is not uncommon for conventional fluid ends to experience failure after only several hundred operating hours. Yet, a single fracking operation may require as many as fifty (50) hours of fluid end operation. Thus, a traditional fluid end may require replacement after use on as few as two fracking jobs. There is a need in the industry for a fluid end configured to avoid or significantly delay the structures or conditions that cause wear or failures within a fluid end.

BRIEF DESCRIPTION OF THE DRAWINGS

First Embodiment

[0011] FIG. 1 is an illustration of the underground environment of a hydraulic fracturing operation.

[0012] FIG. 2 illustrates above-ground equipment used in a hydraulic fracturing operation.

[0013] FIG. 3 is a front perspective view of one embodiment of a high pressure pump disclosed herein.

[0014] FIG. 4 is a front perspective view of the fluid end assembly shown in FIG. 3 attached to a plurality of stay rods.

[0015] FIG. 5 is a rear perspective view of the fluid end assembly and stay rods shown in FIG. 4.

[0016] FIG. 6 is a front elevational view of the fluid end assembly and stay rods shown in FIG. 4.

[0017] FIG. 7 is a front perspective view of the fluid end assembly shown in FIGS. 3 and 4.

[0018] FIG. 8 is a rear perspective view of the fluid end assembly shown in FIG. 7.

[0019] FIG. 9 is a cross-sectional view of one of the fluid end sections making up the fluid end assembly shown in FIG. 6, taken along lines A-A.

[0020] FIG. 10 is a front perspective view of one of the housings used with one of the fluid end sections shown in FIG. 7.

[0021] FIG. 11 is a rear perspective view of the housing shown in FIG. 10.

[0022] FIG. 12 is a front elevational view of the housing shown in FIG. 10.

[0023] FIG. 13 is a rear elevational view of the housing shown in FIG. 10.

[0024] FIG. 14 is a cross-sectional view of the housing shown in FIG. 13, taken along line B-B.

[0025] FIG. 15 is a cross-sectional view of the housing shown in FIG. 13, taken along line C-C.

[0026] FIG. 16 is a cross-sectional view of the housing shown in FIG. 13, taken along line D-D.

[0027] FIG. 17 is a front perspective view of the first section of the housing shown in FIG. 10.

[0028] FIG. 18 is a rear perspective view of the first section shown in FIG. 17.

[0029] FIG. 19 is a front perspective view of the first section shown in FIG. 17, but the first section has a plurality of stay rods attached thereto.

[0030] FIG. 20 is a rear perspective view of the first section and stay rods shown in FIG. 19.

[0031] FIG. 21 is a cross-sectional view of the first section and stay rods shown in FIG. 19, taken along line E-E.

[0032] FIG. 22 is an enlarged view of area F shown in FIG. 21.

[0033] FIG. 23 is a front perspective view of a second section of the housing shown in FIG. 10.

[0034] FIG. 24 is a rear perspective view of the second section shown in FIG. 23.

[0035] FIG. 25 is a front perspective view of the third section of the housing shown in FIG. 10.

[0036] FIG. 26 is a rear perspective view of the third section shown in FIG. 25.

[0037] FIG. 27 is a front perspective and exploded view of the housing shown in FIG. 10.

[0038] FIG. 28 is a rear perspective and exploded view of the housing shown in FIG. 10.

[0039] FIG. 29 is a cross-sectional view of the fluid end assembly shown in FIG. 6, taken along line G-G.

[0040] FIG. 30 is a rear perspective view of the retention plate shown in FIGS. 9 and 29.

[0041] FIG. 31 is a front perspective view of the retention plate shown in FIG. 30.

[0042] FIG. 32 is a rear perspective view of the retention plate shown in FIG. 30 attached to the housing shown in FIG. 10.

[0043] FIG. 33 is a front perspective view of the stuffing box shown in FIGS. 9 and 29.

[0044] FIG. 34 is a rear perspective view of the stuffing box shown in FIG. 33.

[0045] FIG. 35 is a front perspective view of the rear retainer shown in FIGS. 9 and 29.

[0046] FIG. 36 is a rear perspective view of the rear retainer shown in FIG. 35.

[0047] FIG. 37 is a front perspective view of the packing nut shown in FIGS. 9 and 29.

[0048] FIG. 38 is a rear perspective view of the packing nut shown in FIG. 37.

[0049] FIG. 39 is an enlarged cross-sectional view of the components attached to a rear surface of the housing in FIG. 9.

[0050] FIG. 40 is the cross-sectional view of the fluid end assembly shown in FIG. 9. The plunger is retracted, and the discharge valve is shown in a closed position.

[0051] FIG. 41 is the cross-sectional view of the fluid end assembly shown in FIG. 40, but the plunger is extended within the fluid end section and the suction valve is in a closed position.

[0052] FIG. 42 is an enlarged view of area I shown in FIG. 41.

[0053] FIG. 43 is a front perspective view of the fluid routing plug shown in FIGS. 40 and 41.

[0054] FIG. 44 is a top plan view of the fluid routing plug shown in FIG. 43.

[0055] FIG. 45 is a front elevational view of the fluid routing plug shown in FIG. 43.

[0056] FIG. 46 is a rear perspective view of the fluid routing plug shown in FIG. 43.

[0057] FIG. 47 is a rear elevational view of the fluid routing plug shown in FIG. 43.

[0058] FIG. 48 is a cross-sectional view of the fluid routing plug shown in FIG. 47, taken along line J-J.

[0059] FIG. 49 is a top perspective view of the fluid routing plug shown in FIG. 43.

[0060] FIG. 50 is an enlarged view of area K shown in FIG. 49.

[0061] FIG. 51 is a perspective cross-sectional view of the fluid routing plug shown in FIG. 44, taken along line L-L.

[0062] FIG. 52 is a perspective cross-sectional view of the fluid routing plug shown in FIG. 44, taken along line M-M.

[0063] FIG. 53 is a perspective cross-sectional view of the fluid routing plug shown in FIG. 44, taken along line N-N.

[0064] FIG. 54 is a rear elevational view of the fluid routing plug shown in FIG. 43, but one of the suction passages and one of the discharge passages are shown in phantom.

[0065] FIG. 55 is a cross-sectional view of the fluid routing plug shown in FIG. 44, taken along line M-M.

[0066] FIG. 56 is a front elevational and conical-sectional view of the fluid routing plug shown in FIG. 43. The conical-section is taken from line O in FIGS. 43 and 45 to line P in FIGS. 46 and 47.

[0067] FIG. 57 is a front perspective and conical-sectional view of the fluid routing plug shown in FIG. 56.

[0068] FIG. 58 is a rear perspective and conical-sectional view of the fluid routing plug shown in FIG. 56.

[0069] FIG. 59 is a side elevational and conical-sectional view of the fluid routing plug shown in FIG. 56.

[0070] FIG. 60 is a side elevational view of the hardened insert shown in FIGS. 40 and 41.

[0071] FIG. 61 is a front perspective view of the hardened insert shown in FIG. 60.

[0072] FIG. 62 is a front elevational view of the hardened insert shown in FIG. 60.

[0073] FIG. 63 is a cross-sectional view of the hardened insert shown in FIG. 62, taken along line Q-Q.

[0074] FIG. 64 is a front perspective view of the discharge valve shown in FIGS. 40 and 41.

[0075] FIG. 65 is a rear perspective view of the discharge valve shown in FIG. 64.

[0076] FIG. 66 is a side elevational view of the discharge valve shown in FIG. 64.

[0077] FIG. 67 is a cross-sectional view of the discharge valve shown in FIG. 66, taken along line R-R.

[0078] FIG. 68 is a front perspective view of the suction valve guide shown in FIGS. 40 and 41.

[0079] FIG. 69 is a rear perspective view of the suction valve guide shown in FIG. 68.

[0080] FIG. 70 is a front elevational view of the suction valve guide shown in FIG. 68.

[0081] FIG. 71 is a cross-sectional view of the suction valve guide shown in FIG. 70, taken along line S-S.

[0082] FIG. 72 is a front perspective view of the discharge plug shown in FIGS. 40 and 41.

[0083] FIG. 73 is a rear perspective view of the discharge plug shown in FIG. 72.

[0084] FIG. 74 is a front elevational view of the discharge plug shown in FIG. 72.

[0085] FIG. 75 is a cross-sectional view of the discharge plug shown in FIG. 74, taken along line T-T.

[0086] FIG. 76 is a front perspective view of the front retainer shown in FIGS. 40 and 41.

[0087] FIG. 77 is a rear perspective view of the front retainer shown in FIG. 76.

[0088] FIG. 78 is an enlarged view of area U shown in FIG. 41.

[0089] FIG. 79 is a front perspective and exploded view of the fluid end section shown in FIGS. 9, 29, 40, and 41.

[0090] FIG. 80 is a rear perspective and exploded view of the fluid end section shown in FIGS. 9, 29, 40, and 41.

Alternative Embodiments

[0091] FIG. 81 is a front perspective view of another embodiment of a fluid end section.

[0092] FIG. 82 is a rear perspective view of the fluid end section shown in FIG. 81.

[0093] FIG. 83 is a front perspective view of the fluid end section shown in FIG. 81 having a plurality of stay rods attached thereto.

[0094] FIG. 84 is a rear perspective view of the fluid end section and stay rods shown in FIG. 83.

[0095] FIG. 85 is a cross-sectional view of the fluid end section and stay rods shown in FIGS. 83 and 84, taken along line V-V.

[0096] FIG. 86 is a cross-sectional view of the fluid end section shown in FIGS. 81 and 82, taken along line W-W.

[0097] FIG. 87 is a rear perspective and partially exploded view of the fluid end section shown in FIG. 81.

[0098] FIG. 88 is a cross-sectional view of the fluid end section shown in FIGS. 81 and 82, taken along line X-X.

[0099] FIG. 89 is a top plan view of another embodiment of a fluid routing plug. The fluid routing plug is shown installed within the fluid end section in FIGS. 86 and 88.

[0100] FIG. 90 is a front perspective view of the fluid routing plug shown in FIG. 89.

[0101] FIG. 91 is a front elevational view of the fluid routing plug shown in FIG. 89.

[0102] FIG. 92 is a cross-sectional view of the fluid routing plug shown in FIG. 91, taken along line Y-Y.

[0103] FIG. 93 is an enlarged view of area Z shown in FIG. 88.

[0104] FIG. 94 is an enlarged view of area AA shown in FIG. 92.

[0105] FIG. 95 is a front perspective view of another embodiment of a fluid end section.

[0106] FIG. 96 is a rear perspective view of the fluid end section shown in FIG. 95.

[0107] FIG. 97 is a front perspective view of the integral section of the housing shown in FIG. 95.

[0108] FIG. 98 is a rear perspective view of the integral section shown in FIG. 97.

[0109] FIG. 99 is a front perspective view of the fluid end section shown in FIG. 95 having a plurality of stay rods attached thereto.

[0110] FIG. 100 is a rear perspective view of the fluid end section and stay rods shown in FIG. 99.

[0111] FIG. 101 is a cross-sectional view of the fluid end section and stay rods shown in FIGS. 99 and 100, taken along lines AB-AB.

[0112] FIG. 102 is a cross-sectional view of the fluid end section shown in FIGS. 95 and 96, taken along lines AC-AC.

[0113] FIG. 103 is an enlarged view of area AD shown in FIG. 122.

[0114] FIG. 104 is a front perspective view of another embodiment of a fluid routing plug. The fluid routing plug is shown installed within the fluid end section in FIG. 103.

[0115] FIG. 105 is a front elevational view of the fluid routing plug shown in FIG. 104.

[0116] FIG. 106 is a side elevational view of the fluid routing plug shown in FIG. 104.

[0117] FIG. 107 is a rear perspective view of the fluid routing plug shown in FIG. 104.

[0118] FIG. 108 is a cross-sectional view of the fluid routing plug shown in FIG. 105, taken along line AE-AE.

[0119] FIG. 109 is a front perspective view of another embodiment of a fluid end section.

[0120] FIG. 110 is a rear perspective view of the fluid end section shown in FIG. 109.

[0121] FIG. 111 is a cross-sectional view of the fluid end section shown in FIGS. 109 and 110, taken along line AF-AF.

[0122] FIG. 112 is a cross-sectional view of the fluid end section shown in FIGS. 109 and 110, taken along line AG-AG.

[0123] FIG. 113 is an enlarged view of area AH shown in FIG. 111.

[0124] FIG. 114 is a front perspective view of the support element installed within the packing seal shown in FIG. 113.

[0125] FIG. 115 is a cross-sectional view of the support element and packing seal shown in FIG. 114, taken along line AI-AI.

[0126] FIG. 116 is an enlarged view of area AJ shown in FIG. 115.

[0127] FIG. 117 is a front perspective view of another embodiment of a fluid end section.

[0128] FIG. 118 is a rear perspective view of the fluid end section shown in FIG. 117.

[0129] FIG. 119 is a front perspective and partially exploded view of the fluid end section shown in FIG. 117.

[0130] FIG. 120 is a cross-sectional view of the fluid end assembly shown in FIG. 124, taken along line AK-AK.

[0131] FIG. 121 is an enlarged view of area AL shown in FIG. 120.

[0132] FIG. 122 is a cross-sectional view of the fluid end assembly shown in FIG. 124, taken along line AM-AM.

[0133] FIG. 123 is an enlarged view of area AN shown in FIG. 122.

[0134] FIG. 124 is a front perspective view of another embodiment of a high-pressure pump. The fluid end assembly uses the fluid end section shown in FIG. 117.

[0135] FIG. 125 is a front perspective view of another embodiment of a fluid end section.

[0136] FIG. 126 is a rear perspective view of the fluid end section shown in FIG. 125.

[0137] FIG. 127 is a cross-sectional view of the fluid end section shown in FIG. 126, taken along line AO-AO.

[0138] FIG. 128 is a front perspective view of the third section of the housing used with the fluid end section shown in FIG. 125.

[0139] FIG. 129 is a rear perspective view of the third section of the housing shown in FIG. 128.

[0140] FIG. 130 is a cross-sectional view of the third section of the housing shown in FIG. 128, taken along line AP-AP.

[0141] FIG. 131 is a cross-sectional view of the third section of the housing shown in FIG. 129, taken along line AQ-AQ.

[0142] FIG. 132 is a rear perspective view of a blind nut used with the fluid end section shown in FIG. 125.

[0143] FIG. 133 is a front perspective view of the blind nut shown in FIG. 132.

[0144] FIG. 134 is a rear elevational view of the blind nut shown in FIG. 132.

[0145] FIG. 135 is a cross-sectional view of the blind nut shown in FIG. 134, taken along line AR-AR.

[0146] FIG. 136 is a front perspective view of a reaction washer used with the fluid end section shown in FIG. 125.

[0147] FIG. 137 is a front elevational view of the reaction washer shown in FIG. 136.

[0148] FIG. 138 is a cross-sectional view of the fluid end section shown in FIG. 125, taken long line AS-AS.

[0149] FIG. 139 is an enlarged view of area AT, shown in FIG. 127.

[0150] FIG. 140 is an enlarged view of area AU, shown in FIG. 127.

[0151] FIG. 141 is an enlarged view of area AV, shown in FIG. 127.

[0152] FIG. 142 is a front perspective and exploded view of the fluid end section shown in FIG. 125.

[0153] FIG. 143 is a front perspective view of another embodiment of a fluid end section.

[0154] FIG. 144 is a rear perspective view of the fluid end section shown in FIG. 143.

[0155] FIG. 145 is a cross-sectional view of the fluid end section shown in FIG. 143, taken along line AW-AW, but the plunger is shown retracted.

[0156] FIG. 146 is a cross-sectional view of the fluid end section shown in FIG. 144, taken long line AX-AX.

[0157] FIG. 147 is a cross-sectional view of the fluid end section shown in FIG. 144, taken along line AY-AY.

[0158] FIG. 148 is a front perspective view of the housing used with the fluid end section shown in FIG. 143.

[0159] FIG. 149 is a rear perspective view of the housing shown in FIG. 148.

[0160] FIG. 150 is a cross-sectional view of the housing shown in FIG. 148, taken along line AZ-AZ.

[0161] FIG. 151 is a cross-sectional view of the housing shown in FIG. 149, taken along line BA-BA.

[0162] FIG. 152 is a front perspective view of the first section of the housing shown in FIG. 148.

[0163] FIG. 153 is a rear perspective view of the first section of the housing shown in FIG. 152.

[0164] FIG. 154 is a front perspective view of the first section of the housing shown in FIG. 152 with a plurality of stay rods attached thereto.

[0165] FIG. 155 is a rear perspective view of the first section of the housing and stay rods shown in FIG. 154.

[0166] FIG. 156 is a front perspective view of the second section of the housing shown in FIG. 148.

[0167] FIG. 157 is a rear perspective view of the second section of the housing shown in FIG. 156.

[0168] FIG. 158 is a front perspective view of the front spacer sleeve used with the housing shown in FIG. 148.

[0169] FIG. 159 is a rear perspective view of the front spacer sleeve shown in FIG. 158.

[0170] FIG. 160 is a front perspective and exploded view of the housing shown in FIG. 148.

[0171] FIG. 161 is a rear perspective and exploded view of the housing shown in FIG. 148.

[0172] FIG. 162 is a front perspective view of the rear spacer sleeve used with the fluid end section shown in FIGS. 145-147.

[0173] FIG. 163 is a rear perspective view of the rear spacer sleeve shown in FIG. 162.

[0174] FIG. 164 is an enlarged view of area BB, shown in FIG. 145.

[0175] FIG. 165 is an enlarged view of area BC, shown in FIG. 164.

[0176] FIG. 166 is an enlarged view of area BD, shown in FIG. 147.

[0177] FIG. 167 is a front perspective and exploded view of the fluid end section shown in FIG. 143.

[0178] FIG. 168 is a rear perspective and exploded view of the fluid end section shown in FIG. 167.

[0179] FIG. 169 is a front perspective view of another embodiment of a fluid end section.

[0180] FIG. 170 is a rear perspective view of the fluid end section shown in FIG. 169.

[0181] FIG. 171 is a cross-sectional view of the fluid end section shown in FIG. 170, taken along line BE.

[0182] FIG. 172 is the cross-sectional view of the fluid end section shown in FIG. 171 with the inner components shown installed therein.

[0183] FIG. 173 is a front perspective and exploded view of the fluid end section shown in FIG. 169.

[0184] FIG. 174 is a rear perspective and exploded view of the fluid end section shown in FIG. 173.

[0185] FIG. 175 is an enlarged view of area BF, shown in FIG. 165.

[0186] FIG. 176 is an enlarged view of area BG, shown in FIG. 48. All elements of the view behind the section plane have been removed for clarity.

[0187] FIG. 177 is an enlarged view of area BF, shown in FIG. 165. Another embodiment of the stress relief cutout is shown on the fluid routing plug.

[0188] FIG. 178 is an enlarged view of area BG, shown in FIG. 48. This is the same embodiment of stress relief cutout shown in FIG. 177. All elements of the view behind the section plane have been removed for clarity.

[0189] FIG. 179 is an enlarged view of area BF, shown in FIG. 165. Another embodiment of the stress relief cutout is shown on the fluid routing plug.

[0190] FIG. 180 is an enlarged view of area BG, shown in FIG. 48. This is the same embodiment of stress relief cutout shown in FIG. 179. All elements of the view behind the section plane have been removed for clarity.

[0191] FIG. 181 is an enlarged view of area BF, shown in FIG. 165. Another embodiment of the stress relief cutout is shown on the fluid routing plug.

[0192] FIG. 182 is an enlarged view of area BG, shown in FIG. 48. This is the same embodiment of stress relief cutout shown in FIG. 181. All elements of the view behind the section plane have been removed for clarity.

[0193] FIG. 183 is an enlarged view of area BF, shown in FIG. 165. Another embodiment of the stress relief cutout is shown on the fluid routing plug.

[0194] FIG. 184 is an enlarged view of area BG, shown in FIG. 48. This is the same embodiment of stress relief cutout shown in FIG. 183. All elements of the view behind the section plane have been removed for clarity.

[0195] FIG. 185 is an enlarged view of area BF, shown in FIG. 165. Another embodiment of stress relief cutout is shown on the hardened insert.

[0196] FIG. 186 is an enlarged view of area BH, shown in FIG. 63. This is the same embodiment of stress relief cutout shown in FIG. 185. All elements of the view behind the section plane have been removed for clarity.

DETAILED DESCRIPTION

Fluid End Assembly

[0197] Turning now to FIGS. 3-80, a high pressure pump 50 is shown in FIG. 3. The pump 50 comprises a fluid end assembly 52 joined to a power end assembly 54. The power end assembly 54 is described in more detail in U.S. patent application Ser. No. 17/884,691, authored by Keith et al., and filed on Aug. 10, 2022, the entire contents of which are incorporated herein by reference (hereinafter “the ‘691 application”). In alternative embodiments, the fluid end assembly 52 may be attached to other power end designs known in the art. The fluid end assembly 52 and its various embodiments are described herein.

Fluid End Section

[0198] Turning to FIGS. 4-8, the fluid end assembly 52 comprises a plurality of fluid end sections 56 positioned in a side-by-side relationship, as shown in FIGS. 6-8. Each fluid end section 56 is attached to the power end assembly 54 using a plurality of stay rods 58, as shown in FIGS. 4 and 5. Preferably, the fluid end assembly 52 comprises five fluid end sections 56 positioned adjacent one another. In alternative embodiments, the fluid end assembly 52 may comprise more or less than five fluid end sections 56. In operation, a single fluid end section 56 may be removed and replaced without removing the other fluid end sections 56 from the fluid end assembly 52.

Housing of Fluid End Section

[0199] Turning to FIGS. 10-16, each fluid end section 56 comprises a horizontally positioned housing 60 having a longitudinal axis 62 extending therethrough, as shown in FIGS. 10 and 11. The housing 60 has opposed front and rear surfaces 64 and 66 joined by an outer intermediate surface 68. A horizontal bore 70 is formed within the housing 60 and interconnects the front and rear surfaces 64 and 66, as shown in FIGS. 14 and 15. The horizontal bore 70 is sized to receive various components configured to route fluid throughout the housing 60, as shown in FIG. 9. The various components will be described in more detail later herein.

[0200] Continuing with FIGS. 10-16, the housing 60 is of multi-piece construction. The housing 60 comprises a first section 72 joined to a second section 74 and a third section 76 by a plurality of first fasteners 78, as shown in FIGS. 15 and 16. By making the housing 60 out of multiple pieces rather than a single, integral piece, any one of the sections 72, 74, and 76 may be removed and replaced with a new section 72, 74, and 76, without replacing the other sections. For example, if a portion of the second section 74 begins to erode or crack, the second section 74 can be replaced without having to replace the first or third sections 72 and 76. In contrast, if the housing 60 were one single piece, the entire housing would need to be replaced, resulting in much more costly repair to the fluid end assembly 52.

First Section of Housing

[0201] Turning to FIGS. 17-21, the first section 72 is positioned at the front end of the housing 60 and includes the front surface 64. During operation, fluid within the first section 72 remains at relatively the same high pressure. Thus, the first section 72 is considered the static or constant high pressure section of the housing 60. The first section 72 is configured to be attached to a plurality of the stay rods 58, as shown in FIGS. 19-21. Thus, each fluid end section 56 is attached to the power end assembly 54 via the first section 72 of the housing 60.

[0202] Continuing with FIGS. 17 and 18, each first section 72 comprises the front surface 64 joined to a rear surface 80. The surfaces 64 and 80 are interconnected by a portion of the outer intermediate surface 68 and a portion of the horizontal bore 70. The outer intermediate surface 68 of the first section 72 has the shape of a rectangular prism with a plurality of notches 82 formed within the front surface 64. A notch 82 is formed within each corner of the first section 72 such that the front surface 64 has a cross-sectional shape of a cross

sign having radiused corners. The notches **82** are configured to receive a first end **84** of each stay rod **58**, as shown in FIG. **21**.

[0203] With reference to FIGS. **17-21**, a plurality of passages **86** are formed in the first section **72**. Each passage **86** interconnects the rear surface **80** and a medial surface **88** of the first section **72**. The medial surface **88** is defined by the plurality of notches **82**. Each passage **86** comprises a counterbore **87** that opens on the rear surface **80**, as shown in FIG. **16**, and is configured to receive a corresponding one of the stay rods **58**. When installed within the first section **72**, the first end **84** of each stay rod **58** projects from the medial surface **88** and into the corresponding notch **82**, as shown in FIG. **21**.

[0204] Continuing with FIGS. **19-21**, a threaded nut **90** is installed on the first end **84** of each stay rod **58** within each notch **82**. The nut **90** is a three-piece nut, also known as a torque nut, which facilitates the application of high torque required to properly fasten the fluid end section **56** to the power end assembly **54**. The nut **90** is described in more detail in the '691 application. In alternative embodiments, a traditional 12-point flange nut similar to the flange nut **230**, shown in FIGS. **27** and **28**, may be installed on the first end **84** of each stay rod **58** instead of the nut **90**.

[0205] Continuing with FIGS. **19-22**, a sleeve **94** is disposed around a portion of each stay rod **58** and extends between the rear surface **80** of the first section **72** and the power end assembly **54**, as shown in FIG. **3**. A dowel sleeve **93** is inserted into each counterbore **87** formed in each passage **86**, as shown in FIG. **22**. When installed therein, a portion of the dowel sleeve **93** projects from the rear surface **80** of the first section **72**. A counterbore **95** is formed within the hollow interior of the sleeve **94** for receiving the projecting end of the dowel sleeve **93**, as shown in FIG. **22**. The dowel sleeve **93** aligns the sleeve **94** and the passage **86** concentrically. Such alignment maintains a planar engagement between the rear surface **80** of the first section **72** and the sleeve **94**. When the nut **90** is torqued against the medial surface **88** of the first section **72**, the sleeve **94** abuts the rear surface **80** of the first section **72**, rigidly securing the first section **72** to the stay rod **58**.

[0206] Turning back to FIGS. **14-16**, and **18**, a plurality of threaded openings **96** are formed in the rear surface **80** of the first section **72**. The openings **96** surround an opening of the horizontal bore **70**, as shown in FIG. **18**. Each opening **96** is configured to receive a corresponding one of the first fasteners **78** used to secure the sections **72**, **74**, and **76** together, as shown in FIGS. **15** and **16**. A plurality of dowel openings **98** are also formed in the rear surface **80** adjacent the openings **96**, as shown in FIGS. **14** and **18**. The dowel openings **98** are configured to receive first alignment dowels **100**, as shown in FIG. **14**. The first alignment dowels **100** assist in properly aligning the first section **72** and the second section **74** during assembly of the housing **60**.

[0207] Continuing with FIG. **14**, a pair of upper and lower discharge bores **102** and **104** are formed within the first section **72** and interconnect the intermediate surface **68** and the horizontal bore **70**. The upper and lower discharge bores **102** and **104** shown in FIG. **14** are collinear. In alternative embodiments, the bores **102** and **104** may be offset from one another and not collinear. Each bore **102** and **104** may include a counterbore **106** that opens on the intermediate surface **68**. Each counterbore **106** is sized to receive a portion of a discharge fitting adapter **504**, as shown in FIG.

9. The fitting adapter **504** spans between the discharge bore **102** or **104** and a discharge fitting **108** attached to the outer intermediate surface **68** of the first section **72**.

[0208] Continuing with FIG. **14**, a groove **110** may be formed in the side walls of the counterbore **106** for receiving a seal **112**. The seal **112** engages an outer surface of the fitting adapter **504** to prevent fluid from leaking between the first section **72** and the discharge fitting **108**, as shown in FIG. **9**.

[0209] With reference to FIGS. **17** and **18**, a plurality of threaded openings **114** are formed in the intermediate surface **68** and surrounding the opening of the upper and lower discharge bores **102** and **104**. The threaded openings **114** are configured to receive a plurality of threaded fasteners **116** configured to secure a discharge fitting **108** to the first section **72**, as shown in FIG. **9**.

[0210] Continuing with FIGS. **14** and **15**, the walls of the horizontal bore **70** within the first section **72** and positioned between the front surface **64** and the upper and lower discharge bores **102** and **104** are sized to receive a front retainer **118** and a discharge plug **120**, as shown in FIG. **9**. The discharge plug **120** seals fluid from leaking from the front surface **64** of the housing **60**, and the front retainer **118** secures the discharge plug **120** within the first section **72** of the housing **60**.

[0211] Continuing with FIGS. **14** and **15**, internal threads **122** are formed in the walls of the horizontal bore **70** of the first section **72** for mating with external threads **124**, shown in FIGS. **76** and **77**, formed on an outer surface of the front retainer **118**. In contrast, an outer surface of the discharge plug **120** faces the smooth walls of the horizontal bore **70** of the first section **72**. A small amount of clearance may exist between the plug **120** and the walls of the horizontal bore **70** of the first section **72**.

[0212] Continuing with FIGS. **14** and **15**, a groove **125** may be formed in such walls for receiving a seal **126** configured to engage an outer surface of the discharge plug **120**, as shown in FIG. **9**. The seal **126** prevents fluid from leaking around the discharge plug **120** during operation. A locating cutout **128** may further be formed in the walls that is configured to receive a locating dowel pin **130**. As will be described later herein, the locating dowel pin **130** is used to properly align the discharge plug **120** within the housing **60**.

[0213] Continuing with FIGS. **14** and **15**, the walls of the horizontal bore **70** positioned between the upper and lower discharge bores **102** and **104** and the rear surface **80** of the first section **72** are sized to receive a portion of a fluid routing plug **132**, as shown in FIG. **9**. This area of the walls of the horizontal bore **70** includes a counterbore **134** that opens on the rear surface **80**. The counterbore **134** is sized to receive a wear ring **136**, as shown in FIG. **9**. The wear ring **136** has an annular shape and is configured to engage a first seal **386** installed within an outer surface of the fluid routing plug **132**, as shown in FIG. **9**. In alternative embodiments, the first section **72** may not include the counterbore **134** or the wear ring **136** and instead may be sized to directly engage the first seal **386** installed within the fluid routing plug **132**.

[0214] Continuing with FIG. **9**, in addition to the above mentioned components, the first section **72** is also configured to house a discharge valve **138**. The components discussed above and installed within the first section **72** will be described in more detail later herein.

Second Section of Housing

[0215] Turning to FIGS. 23 and 24, the second section 74 of the housing 60 is configured to be positioned between the first and third sections 72 and 76 and has a cylindrical cross-sectional shape. During operation, fluid pressure within the second section 74 remains at relatively the same pressure. The pressure is lower than that within the first section 72. Thus, the second section 74 may be referred to as the static or constant low pressure section of the housing 60. The second section 74 comprises opposed front and rear surfaces 140 and 142 joined by a portion of the outer intermediate surface 68 and a portion of the horizontal bore 70.

[0216] Continuing with FIGS. 15, 16, 23, and 24, a plurality of passages 144 are formed in the second section 74. The passages 144 surround the horizontal bore 70 and interconnect the front and rear surfaces 140 and 142, as shown in FIGS. 15 and 16. Each passage 144 is configured to receive a corresponding one of the first fasteners 78 used to secure the sections 72, 74, and 76 of the housing 60 together.

[0217] Continuing with FIGS. 14, 23, and 24, a plurality of dowel openings 146 are formed in the front surface 140 of the second section 74, as shown in FIG. 23. The dowel openings 146 align with the dowel openings 98 formed in the rear surface 80 of the first section 72 and are configured to receive a portion of the first alignment dowels 100, as shown in FIG. 14. Likewise, a plurality of dowel openings 148 are formed in the rear surface 142 of the second section 74, as shown in FIG. 24. The dowel openings 148 are configured to receive a portion of second alignment dowels 150, as shown in FIG. 14. The second alignment dowels 150 are configured to align the second section 74 and the third section 76 during assembly.

[0218] Continuing with FIGS. 14, 15, 23 and 24, a first annular groove 152 is formed in the front surface 140 of the second section 74 such that it surrounds an opening of the horizontal bore 70, as shown in FIG. 23. The first groove 152 is positioned between the horizontal bore 70 and the plurality of passages 144 and is configured to receive a first seal 154, as shown in FIGS. 14 and 15. Likewise, a second annular groove 156 is formed in the rear surface 142 of the second section 74 and positioned between the horizontal bore 70 and the plurality of passages 144, as shown in FIG. 24. The second groove 156 is configured to receive a second seal 158, as shown in FIGS. 14 and 15. The seals 154 and 158 shown in FIGS. 14 and 15 are O-rings. The seals 154 and 158 prevent fluid from leaking between the first and second sections 72 and 74 and between the second and third sections 74 and 76 during operation.

[0219] Continuing with FIGS. 14, 23, and 24, a pair of upper and lower suction bores 160 and 162 are formed within the second section 74 and interconnect the intermediate surface 68 and the horizontal bore 70. The upper and lower suction bores 160 and 162 shown in FIG. 14 are collinear. In alternative embodiments, the bores 160 and 162 may be offset from one another and not collinear.

[0220] Continuing with FIGS. 9 and 14, the suction bores 160 and 162 are each configured to receive a suction conduit 166, as shown in FIG. 9. The suction conduit 166 comprises a first connection member 164 configured to mate with the housing 60. Each suction bore 160 and 162 opens into a counterbore 168 sized to receive a portion of the first connection member 164. Internal threads 170 are formed in

a portion of the walls of the counterbore 168 for mating with external threads 172, shown in FIG. 78, formed on the first connection member 164.

[0221] Continuing with FIGS. 9 and 14, a groove 174 is formed in the walls of the counterbore 168 and configured to receive a seal 176, as shown in FIG. 9. The seal 176 engages an outer surface of the first connection member 164 to prevent fluid from leaking from the housing 60 during operation. The suction conduits 166 will be described in more detail later herein.

[0222] Continuing with FIGS. 9, 14, and 15, the walls of the horizontal bore 70 within the second section 74 are configured to receive a majority of the fluid routing plug 132, as shown in FIG. 9. A small amount of clearance may exist between the walls of the second section 74 and an outer surface of the fluid routing plug 132.

Third Section of Housing

[0223] Turning to FIGS. 14-16, 25 and 26, the third section 76 of the housing 60 is positioned at the rear end of the housing 60 and includes the rear surface 66. The third section 76 has a generally cylindrical cross-sectional shape. Fluid pressure within the third section 76 varies during operation. Thus, the third section 76 may be referred to as the dynamic or variable pressure section of the housing 60.

[0224] Continuing with FIGS. 25 and 26, the third section 76 comprises a front surface 178 joined to the rear surface 66 of the housing 60 by a portion of the outer intermediate surface 68 and a portion of the horizontal bore 70. The outer intermediate surface 68 of the third section 76 varies in diameter such that the third section 76 comprises a front portion 180 joined to a rear portion 182.

[0225] Continuing with FIGS. 15, 16, 25, and 26, the front portion 180 has a constant outer diameter and has a plurality of passages 184 formed therein. The passages 184 interconnect the front surface 178 and a medial surface 186 of the third section 76. The passages 184 align with the plurality of passages 144 formed in the second section 74 and the threaded openings 96 formed in the first section 72 of the housing 60, as shown in FIGS. 14 and 15. The passages 184 are configured to receive the first fasteners 78 used to secure the sections 72, 74, and 76 together.

[0226] Continuing with FIGS. 14 and 25, a plurality of dowel openings 188 are formed in the front surface 178 of the third section 76, as shown in FIG. 25. The dowel openings 188 are configured to receive a portion of second alignment dowels 150, as shown in FIG. 14. The second alignment dowels 150 are configured to properly align the third section 76 within the second section 74 during assembly.

[0227] Continuing with FIGS. 25 and 26, the rear portion 182 of the third section 76 comprises a neck 190 joined to a shoulder 192. The neck 190 interconnects the front portion 180 and the shoulder 192. The shoulder 192 includes the rear surface 66 of the housing 60. The neck 190 has a smaller outer diameter than that of the front portion 180 and the shoulder 192 to provide clearance for the plurality of passages 184 formed in the front portion 180.

[0228] With reference to FIGS. 25 and 26, a plurality of first threaded openings 194 are formed in the rear surface 66 of the third section 76. The first threaded openings 194 are configured to receive a plurality of second fasteners 196, as shown in FIG. 29. The second fasteners 196 are configured to secure a stuffing box 198 and a rear retainer 200 to the

third section 76 of the housing 60, as shown in FIG. 29. The stuffing box 198 and the rear retainer 200 will be described in more detail later herein.

[0229] With reference to FIGS. 26 and 32, a plurality of second threaded openings 202 are also formed in the rear surface 66 of the third section 76, as shown in FIG. 26. The second threaded openings 202 are configured to receive a plurality of third fasteners 204. The third fasteners 204 are configured to secure a retention plate 206 to the rear surface of the housing 60, as shown in FIG. 32. The retention plate 206 will be described in more detail later herein. A plurality of dowel openings 207 are also formed in the rear surface 66 of the third section 76. The dowel openings 207 are configured to receive third alignment dowels 242, as shown in FIG. 39.

[0230] Turning back to FIGS. 9, 14, 15, and 25, a counterbore 208 is formed in the walls of the horizontal bore 70 within the third section 76 and opens on the front surface 178. The counterbore 208 is configured to receive a hardened insert 210, as shown in FIG. 9. The insert 210 will be described in more detail later herein. The insert 210 engages portions of the fluid routing plug 132 when the fluid routing plug 132 is installed within the housing 60, as shown in FIG. 9. The walls of the horizontal bore 70 between the counterbore 208 and the medial surface 186 of the third section 76 are further configured to receive a suction valve guide 212. A suction valve 214 is also installed within the third section 76 of the housing 60. The suction valve 214 and suction valve guide 212 will be described in more detail later herein. [0231] Continuing with FIGS. 9, 14, 15, 25, and 26, the walls of the horizontal bore 70 within the neck 190 of the rear portion 182 are sized to receive at least a portion of a reciprocating plunger 216, as shown in FIG. 9. The portion of the horizontal bore 70 extending through the neck 190 has a uniform diameter and opens into a first counterbore 218 formed in the shoulder 192, as shown in FIGS. 14 and 15. The first counterbore 218 is sized to receive a portion of the stuffing box 198, as shown in FIG. 9. The first counterbore 218 opens into a second counterbore 220, of which opens on the rear surface 66 of the housing 60, as shown in FIGS. 14, 15, and 26. The second counterbore 220 is sized to receive a wear ring 222 and a seal 224, as shown in FIG. 9. The wear ring 222 and the seal 224 each have an annular shape. When such components are installed within the housing 60, the wear ring 222 surrounds the seal 224, and the seal 224 engages an outer surface of the stuffing box 198.

Assembly of Housing

[0232] Turning to FIGS. 27-29, the housing 60 is assembled by threading a first end 226 of each of the first fasteners 78 into a corresponding one of the threaded openings 96 formed in the first section 72. Once installed therein, the first fasteners 78 project from the rear surface 80 of the first section 72. The second and third sections 74 and 76 may then be slid onto the fasteners 78 projecting from the first section 72 using the corresponding passages 144 and 184. The first and second alignment dowels 100 and 150 help to further align the sections 72, 74, and 76 together during assembly.

[0233] Continuing with FIGS. 27-29, when the second and third sections 74 and 76 are installed on the fasteners 78, a second end 228 projects from the medial surface 186 of the third section 76, as shown in FIG. 29. A flange nut 230 is installed on the second end 228 and torqued against the

medial surface 186, tightly securing the sections 72, 74, and 76 together. When the housing 60 is assembled, a footprint of the rear surface 142 of the second section 74 is entirely within a footprint of the rear surface 80 of the first section 72, as shown in FIG. 29.

[0234] Continuing with FIGS. 27-29, the first fastener 78 shown in the figures is a threaded stud. In alternative embodiments, other types of fasteners known in the art may be used instead of a threaded stud. For example, screws or bolts may be used to secure the sections together. In further alternative embodiments, the nut may comprise the three-piece nut 90, shown in FIGS. 19-21.

[0235] Continuing with FIGS. 27-29, to remove a section 72, 74, or 76, the nut 230 is unthreaded from the second end 228 of each first fastener 78. The sections 72, 74, and 76 may then be pulled apart, as needed. If the first section 72 is being replaced, the first fasteners 78 are also unthreaded from the threaded openings 96. The components installed within the housing 60 may also be removed, as needed, prior to disassembling the housing 60.

Components Attached to Rear Surface of Housing

[0236] Turning to FIGS. 29-32, in addition to the housing 60, the fluid end section 56 comprises a plurality of components attached to the rear surface 66 of the housing 60. Such components are configured to receive the plunger 216. The various components include the retention plate 206, the stuffing box 198, and the rear retainer 200, previously mentioned. The components further comprise a plunger packing 300, and a packing nut 290.

Retention Plate

[0237] Continuing with FIGS. 29-32, the retention plate 206 has a cylindrical cross-sectional shape and is sized to cover the rear surface 66 of the housing 60 and the wear ring 222 and the seal 224, as shown in FIG. 29. The retention plate 206 holds the wear ring 222 and the seal 224 within the housing 60 in the event the stuffing box 198 needs to be removed.

[0238] Continuing with FIGS. 29-32, the retention plate 206 comprises opposed front and rear surface 237 and 238 joined by a central opening 239 formed therein. A plurality of first passages 234 are formed in the retention plate 206 and surround the central opening 239 of the plate 206. The first passages 234 align with the first threaded openings 194 formed in the rear surface 66 of the housing 60 and are configured to receive the plurality of second fasteners 196.

[0239] Continuing with FIGS. 30-32, a plurality of second passages 236 are also formed in the retention plate 206. The second passages 236 align with the second threaded openings 202 formed in the rear surface 66 of the housing 60 and are configured to receive the third fasteners 204, as shown in FIG. 32. A third fastener 204 is threaded into one of the second threaded openings 202 and turned until it sits flush with the rear surface 238 of the retention plate 206, as shown in FIG. 32.

[0240] Continuing with FIGS. 30-32, a plurality of dowel openings 240 are formed in the retention plate 206 for receiving third alignment dowels 242, as shown in FIG. 39. The third alignment dowels 242 assist in properly aligning the retention plate 206 and the stuffing box 198 on the housing 60 during assembly.

[0241] Turning back to FIG. 29, since fluid does not contact the retention plate 206 during operation, the retention plate 206 may be made of a different and less costly material than that of the housing 60 or the stuffing box 198. For example, the retention plate 206 may be made of alloy steel, while the housing 60 and stuffing box 198 are made of stainless steel.

Stuffing Box

[0242] Turning to FIGS. 29, 33, 34, and 39, the stuffing box 198 comprises opposed front and rear surfaces 244 and 246 joined by an outer intermediate surface 248 and a central passage 250 formed therein. The stuffing box 198 further comprises a front portion 252 joined to a rear portion 254. The front portion 252 has a smaller outer diameter than the rear portion 254 such that a medial surface 256 is formed between the front and rear surfaces 244 and 246. The front portion 252 includes the front surface 244 of stuffing box 198, and the rear portion 254 includes the rear surface 246 of the stuffing box 198. An internal shoulder 272 is formed within the walls of the central passage 250 within the rear portion 254 of the stuffing box 198.

[0243] Continuing with FIGS. 33, 34, and 39, a plurality of passages 258 are formed within the rear portion 254 of the stuffing box 198 and interconnect the medial surface 256 and the rear surface 246. The passages 258 are configured to align with the plurality of first passages 234 formed in the retention plate 206 and the plurality of first threaded openings 194 formed in the rear surface 66 of the housing 60, as shown in FIG. 29.

[0244] Continuing with FIGS. 33, 34, and 39, a plurality of dowel openings 260 may be formed in the medial surface 256 of the stuffing box 198. The dowel openings 260 are configured to receive at least a portion of the third alignment dowels 242 to properly align the stuffing box 198 on the retention plate 206 and the housing 60 during assembly, as shown in FIG. 39. Likewise, a plurality of dowel openings 268 may be formed in the rear surface 246 of the stuffing box 198 for receiving fourth alignment dowels 270, as shown in FIG. 39. The fourth alignment dowels 270 assist in properly aligning the rear retainer 200 on the stuffing box 198 during assembly.

[0245] Continuing with FIG. 39, the stuffing box 198 is installed within the third section 76 of the housing 60 such that the front portion 252 is disposed within the horizontal bore 70 and the medial surface 256 abuts the rear surface 238 of the retention plate 206. The outer intermediate surface 248 of the front portion 252 of the stuffing box 198 engages the seal 224. The seal 224 prevents fluid from leaking between the housing 60 and the stuffing box 198.

[0246] Continuing with FIG. 39, during operation, the seal 224 wears against the outer intermediate surface 248 of the front portion 252. Should the front portion 252 begin to erode, the stuffing box 198 may be removed and replaced with a new stuffing box 198. Likewise, the seal 224 wears against the wear ring 222 during operation. The wear ring 222 is preferably made of a harder and more wear resistant material than the housing 60, such as tungsten carbide. Should the wear ring 222 begin to erode, the wear ring 222 can be removed and replaced with a new wear ring 222. Locating the seal 224 between replaceable parts protects the housing 60 over time.

Rear Retainer

[0247] Turning to FIGS. 29, 35, 36, and 39, the rear retainer 200 comprises opposed front and rear surfaces 276 and 278 joined by an outer intermediate surface 280 and a central passage 282 formed therein. A plurality of passages 284 are formed in the rear retainer 200 and surround the central passage 282. The passages 284 interconnect the front and rear surfaces 276 and 278 of the rear retainer 200 and are configured to align with the passages 258 formed in the rear portion 254 of the stuffing box 198, as shown in FIG. 29. A plurality of dowel openings 286 are formed in the front surface 276 of the rear retainer 200 for receiving a portion of the fourth alignment dowels 270, as shown in FIG. 39.

[0248] Continuing with FIGS. 35, 36, 39, and 40, an internal shoulder 279 is formed within the walls of the central passage 282 of the rear retainer 200. Internal threads 288 are formed in the walls of the central passage 282 and positioned between the internal shoulder 279 and the rear surface 278. The internal threads 288 are configured to receive a packing nut 290, as shown in FIG. 39. The walls positioned between the internal shoulder 279 and the front surface 276 are smooth and include one or more lubrication ports 292. The lubrication port 292 interconnects the central passage 282 and the outer intermediate surface 280 of the rear retainer 200, as shown in FIG. 40.

Plunger Packing and Packing Nut

[0249] Continuing with FIG. 39, fluid is prevented from leaking around the plunger 216 during operation by a plunger packing 300. The plunger packing 300 is installed within the stuffing box 198 and comprises a plurality of packing seals 302 sandwiched between first and second metal rings 304 and 306. The first metal ring 304 abuts the internal shoulder 272 formed within the stuffing box 198 and the second metal ring 306 extends into the central passage 282 formed in the rear retainer 200. The second metal ring 306 is known in the art as a “lantern ring”. One or more passages 303, shown in FIG. 29, may be formed in the second metal ring 306 and fluidly connect with the one or more lubrication ports 292 formed in the rear retainer 200. During operation, oil used to lubricate the plunger 216 and plunger packing 300 is supplied through the lubrication port 292 and second metal ring 306.

[0250] With reference to FIGS. 37-39, the plunger packing 300 is retained within the stuffing box 198 and the rear retainer 200 using the packing nut 290. The packing nut 290 comprises opposed front and rear surfaces 308 and 310 joined by an outer intermediate surface 312 and a central passage 314 formed therein. External threads 316 are formed in a portion of the outer intermediate surface 312 for engaging the internal threads 288 formed in the rear retainer 200, as shown in FIG. 39. When the packing nut 290 is installed within the rear retainer 200, the front surface 308 of the packing nut 290 engages and compresses the plunger packing 300, as shown in FIG. 39. When compressed, the packing seals 302 of the plunger packing 300 tightly seal against an outer surface of the plunger 216.

[0251] Continuing with FIG. 39, during operation, the packing nut 290 may be tightened, as needed, to ensure adequate compression of the packing seals 302 against the plunger 216. At least a portion of the packing nut 290 projects from the rear surface 278 of the rear retainer 200 to provide clearance to turn the packing nut 290, as needed.

The central passage 314 formed in the packing nut 290 is sized to closely receive the plunger 216. A groove 318 may be formed in the walls of the central passage 314 for receiving a seal 320. The seal 320 shown in FIG. 39 is an O-ring. The seal 320 prevents fluid and lubricant from leaking around the plunger 216 during operation.

Assembly of Components on Rear Surface of Housing

[0252] Turning back to FIG. 29, the front surface 276 of the rear retainer 200 abuts the rear surface 246 of the stuffing box 198 such that the plurality of passages 284 align with the plurality of passages 258 formed in the stuffing box 198. A second fastener 196 is installed within a corresponding one of the aligned first threaded openings 194 and passages 234, 258, and 284. A first end 294 of the second fastener 196 threads into the first threaded opening 194 and a second end 296 projects from the rear surface 278 of the rear retainer 200. A nut 298 is threaded onto the second end 296 and torqued against the rear surface 278, tightly securing the stuffing box 198 and the rear retainer 200 to the third section 76 of the housing 60.

[0253] Continuing with FIG. 29, the nut 298 shown in the figures is a 12-point flange nut. In alternative embodiments, the nut may comprise the three-piece nut 90, shown in FIGS. 19-21. The second fastener 196 shown in the figures is a threaded stud. In alternative embodiments, the second fastener 196 may comprise other fasteners known in the art, such as a bolt or screw.

[0254] Continuing with FIG. 29, the stuffing box 198 and rear retainer 200 are attached to the housing 60 after the retention plate 206 has first been attached to the rear surface 66 of the housing 60. The plunger packing 300 may be installed within stuffing box 198 either before or after the stuffing box 198 is attached to the housing 60. After all the components are assembled, the packing nut 290 is threaded into the rear retainer 200 until it engages the plunger packing 300.

[0255] With reference to FIGS. 29 and 39, when the retention plate 206, the stuffing box 198, and rear retainer 200 are attached to the housing 60, the central opening 239 of the retention plate 206 and the central passages 250 and 282 of the stuffing box 198 and the rear retainer 200 form an extension of the horizontal bore 70. Likewise, the interior of the plunger packing 300 and the central passage 314 of the packing nut 290 also form extensions of the horizontal bore 70. The plunger 216 is installed within the fluid end section 56 through the rear surface 310 of the packing nut 290. During operation, the plunger 216 reciprocates within the horizontal bore 70, creating the variance in fluid pressure within the fluid end section 56 during operation.

[0256] With reference to FIGS. 3, 39, and 40, during operation, reciprocal movement of the plunger 216 is driven by a pony rod 322 installed within the power end assembly 54. A clamp 324 secures the plunger 216 to the pony rod 322 such that the plunger 216 and pony rod 322 move in unison. Components Installed within the Housing

[0257] Turning to FIGS. 40-79, the various internal components of the housing 60 will now be described in more detail. Fluid is routed throughout the housing 60 by the fluid routing plug 132. The timing of movement throughout the fluid routing plug 132 is controlled by the suction valve 214 and the discharge valve 138. Movement of the valves 214 and 138 is guided by the suction valve guide 212 and the discharge plug 120.

Fluid Routing Plug

[0258] Turning to FIGS. 40-59, the fluid routing plug 132 comprises a body 330 having a suction surface 332 and an opposed discharge surface 334 joined by an outer intermediate surface 336. A central longitudinal axis 338 extends through the body 330 and the suction and discharge surfaces 332 and 334. When the fluid routing plug 132 is installed within the housing 60, at least a portion of the discharge surface 334 is positioned within the first section 72 of the housing 60, and at least a portion of the suction surface 332 is positioned within the third section 76 of the housing 60, as shown in FIGS. 40 and 41.

[0259] Continuing with FIGS. 43-59, the body 330 further comprises a plurality of suction fluid passages 340. The suction passages 340 interconnect the intermediate surface 336 and the suction surface 332 of the body 330, as shown in FIG. 48. The connection is formed within a blind bore 342 formed within the suction surface 332 of the body 330. The blind bore 342 may be referred to as an axially-blind bore 342 because it is blind along the longitudinal axis 338 of the body 330. During operation, fluid entering the housing 60 through the suction bores 160 and 162 flows into the suction passages 340 of the fluid routing plug 132 and into the axially-blind bore 342. From there, fluid flows towards the suction surface 332 of the body 330 and out of the fluid routing plug 132. Three suction fluid passages 340 are shown in FIGS. 43-59. In alternative embodiments, more or less than three suction fluid passages 340 may be formed within the body 330.

[0260] Continuing with FIGS. 49 and 50, each suction passage 340 has a generally oval or tear drop cross-sectional shape. An opening 344 of each suction passage 340 on the intermediate surface 336 comprises a first side wall 346 joined to a second side wall 348 by first and second ends 350 and 352. The first and second side walls 346 and 348 are straight lines of equal length S , and the first and second ends 350 and 352 are circular arcs, as shown in FIG. 50.

[0261] Continuing with FIG. 50, the first end 350 of the opening 344 has a radius of $R1$ with a center at $C1$, and the second end 352 has a radius of $R2$ with a center at $C2$. The first end 350 is larger than the second end 352 such that $R1 > R2$. The first and second side walls 346 and 348 are tangent to the first and second ends 350 and 352 and have an included angle, σ .

[0262] Continuing with FIG. 50, the opening 344 has a centerline 354 that connects the centers $C1$ and $C2$ of the first and second ends 350 and 352. The centerline 354 has a length E and is parallel with the central longitudinal axis 338. A cross-sectional shape of each suction passage 340 throughout the length of the body 330 corresponds with the shape of each opening 344, as shown in FIG. 55. Each suction passage 340 is sized and shaped to maximize fluid flow through the passage 340 and minimize fluid turbulence and stress to the body 330 of the fluid routing plug 132.

[0263] With reference to FIGS. 55 and 56, each suction fluid passage 340 extends between the axially-blind bore 342 and the suction surface 332 such that each suction passage 340 comprises a longitudinal axis 356. The longitudinal axis 356 extends through the center $C1$ of the first end 350 of the opening 344 and intersects the central longitudinal axis 338, as shown in FIG. 56.

[0264] Continuing with FIGS. 43-59, the body 330 further comprises a plurality of discharge fluid passages 360. The discharge passages 360 interconnect the suction surface 332

and the discharge surface 334 of the body 330 and do not intersect any of the suction passages 340. Rather, the discharge and suction passages 360 and 340 are in a spaced-relationship. In operation, fluid exiting the body 330 at the suction surface 332 is subsequently forced into the discharge passages 360, towards the discharge surface 334 of the body 330, and out of the fluid routing plug 132. Three discharge fluid passages 360 are shown in FIGS. 43-59. In alternative embodiments, more or less than three discharge fluid passages 360 may be formed within the body 330.

[0265] Continuing with FIGS. 43, 46, and 48, the suction surface 332 of the body 330 comprises an outer rim 362 joined to the axially-blind bore 342 by a tapered seating surface 366, as shown in FIGS. 46 and 48. Likewise, the discharge surface 334 comprises an outer rim 368 joined to a central base 370 by a tapered seating surface 372, as shown in FIGS. 43 and 48.

[0266] Continuing with FIGS. 43, 46, and 48, each discharge passage 360 opens at a first opening 374 on the outer rim 362 of the suction surface 332 and opens at a second opening 376 on the central base 370 of the discharge surface 334. The second openings 376 surround a blind bore 378 formed in the central base 370 of the discharge surface 334. The blind bore 378 is configured to engage a tool used to grip the fluid routing plug 132, as needed. For example, the walls of the blind bore 378 may be threaded. The central base 370 may also be slightly recessed from the tapered seating surface 372 such that a small counterbore 380 is created. The counterbore 380 helps further reduce any turbulence of fluid exiting the second openings 376.

[0267] Continuing with FIG. 54, a position of the first and second openings 374 and 376 of each discharge passage 360 may be determined relative to a plane containing a line 382 that is perpendicular to the central longitudinal axis 338. The first opening 374, when projected onto the plane, is positioned at a first distance F1 from the central longitudinal axis 338 and at a first angle $\varphi 1$ relative to the line 382. The second opening 376, when projected onto the plane, is positioned at a second distance F2 from the central longitudinal axis 338 and at a second angle $\varphi 2$ relative to the line 382.

[0268] The first and second distances F1 and F2 shown in FIG. 54 are different. Likewise, the first and second angles $\varphi 1$ and $\varphi 2$ shown in FIG. 54 are different. In alternative embodiments, the first and second angles $\varphi 1$ and $\varphi 2$ may be different, but the first and second distances F1 and F2 may be the same. In further alternative embodiments, the first and second angles $\varphi 1$ and $\varphi 2$ may be the same, but the first and second distances F1 and F2 may be different. In even further alternative embodiments, the first and second distances F1 and F2 may be the same, and the first and second angles $\varphi 1$ and $\varphi 2$ may be the same.

[0269] With reference to FIGS. 51-53 and 56-59, each discharge passage 360 has an arced cross-sectional shape. The length of the arc may gradually increase between the suction and discharge surfaces 332 and 334, as shown in FIGS. 51-53. In alternative embodiments, the discharge passages 360 may have different shapes and sizes.

[0270] Turning back to FIGS. 44 and 48, a first annular groove 384 is formed in the outer intermediate surface 336 of the body 330 for housing the first seal 386. The first groove 384 is positioned adjacent the discharge surface 334 and is characterized by two side walls 388 joined by a base 390. When the fluid routing plug 132 is installed within the

housing 60, the first seal 386 engages an inner surface of the wear ring 136 installed within the first section 72 of the housing 60, as shown in FIGS. 9, 40, and 41. During operation, the first seal 386 wears against the wear ring 136. If the wear ring 136 begins to erode, the wear ring 136 may be removed and replaced with a new wear ring 136. The wear ring 136 has an annular shape and may be made of a harder and more wear resistant material than the housing 60. For example, the housing 60 may be made of stainless steel and the wear ring 136 made of tungsten carbide.

[0271] With reference to FIGS. 42, 44, and 48, a second annular groove 392 is formed in the outer intermediate surface 336 of the body 330 for housing a second seal 394. The second groove 392 is positioned adjacent the suction surface 332 and is characterized by a plurality of side walls 396 joined by a base 398, as shown in FIG. 42. Four side walls 396 are shown in FIG. 42 such that the groove 392 has a rounded shape. When the fluid routing plug 132 is installed within the housing 60, the second seal 394 engages an inner intermediate surface 420 of the hardened insert 210, as shown in FIG. 42. During operation, the second seal 394 wears against the insert 210. If the insert 210 begins to erode, the insert 210 may be removed and replaced with a new insert 210.

[0272] Continuing with FIGS. 42, 44, and 48, the outer intermediate surface 336 of the body 330 further comprises an annular shoulder 400 formed in the body 330. The shoulder 400 is positioned between the opening 344 of the suction passages 340 and the second groove 392. When the fluid routing plug 132 is installed within the housing 60, the shoulder 400 abuts a front surface 416 of the insert 210, as shown in FIG. 42. Axial movement of the fluid routing plug 132 towards the rear surface 66 of the housing 60 is prevented by the engagement between the shoulder 400 and the insert 210. During operation, the shoulder 400 may wear against the insert 210. If either feature begins to wear, the fluid routing plug 132 and/or the insert 210 may be removed and replaced with a new fluid routing plug 132 and/or insert 210.

[0273] Continuing with FIGS. 40, 41, 44, and 48, the outer intermediate surface 336 of the body 330 adjacent the first groove 384 is characterized as a first cylindrical surface 404. Likewise, the outer intermediate surface 336 adjacent the annular shoulder 400 is characterized as a second cylindrical surface 406. The first cylindrical surface 404 has a maximum outer diameter that is equal or almost equal to a maximum outer diameter of the second cylindrical surface 406. The surfaces 404 and 406 are configured to closely face the walls of the horizontal bore 70 within the second section 74 of the housing 60, as shown in FIGS. 40 and 41.

[0274] Continuing with FIGS. 40, 41, 44, 48, and 49, the outer intermediate surface 336 of the body 330 further comprises a first bevel 408 joined to a transition surface 410 formed in the body 330. The first bevel 408 and the transition surface 410 are positioned between the first cylindrical surface 404 and the openings 344 of the suction passages 340. The outer intermediate surface 336 of the body 330 slowly tapers outward from the transition surface 410 to the second cylindrical surface 406.

[0275] Continuing with FIGS. 40 and 41, when the fluid routing plug 132 is installed within the housing 60, the first bevel 408 provides clearance between the outer intermediate surface 336 of the fluid routing plug 132 and an opening of the suction bores 160 and 162. Such clearance gives way to

an annular fluid channel 412 formed between the housing 60 and the fluid routing plug 132. The shape of the outer intermediate surface 336 of the fluid routing plug 132 between the first and second cylindrical surfaces 404 and 406 helps direct fluid flowing from the suction bores 160 and 162 into the openings 344 of the suction passages 340 while minimizing fluid turbulence.

[0276] Turning back to FIG. 42, the outer intermediate surface 336 of the body 330 further comprises a second bevel 414 formed in the body 330. The second bevel 414 is positioned between the suction surface 332 and the second groove 392. The second bevel 414 provides clearance to help install the fluid routing plug 132 within the housing 60 and the insert 210.

Hardened Insert

[0277] With reference to FIGS. 42 and 60-63, the insert 210 has an annular shape and comprises opposed front and rear surfaces 416 and 418 joined by inner and outer intermediate surfaces 420 and 422. The insert 210 further comprises a first bevel 426 formed in the inner intermediate surface 420 adjacent the front surface 416, as shown in FIG. 63. The first bevel 426 provides clearance to assist in installing the fluid routing plug 132 within the insert 210 within the housing 60, as shown in FIG. 42. The insert 210 also comprises a second bevel 424 formed in the outer intermediate surface 422 adjacent the rear surface 418. The second bevel 424 provides clearance to assist in installing the insert 210 within the counterbore 208 formed in the third section 76 of the housing 60, as shown in FIG. 42. The insert 210 is made of a harder and more wear resistant material than the housing 60. For example, if the housing 60 is made of stainless steel, the insert 210 may be made of tungsten carbide.

Suction and Discharge Valves

[0278] With reference to FIGS. 40, 41, and 64-67, the flow of fluid throughout the housing 60 and the fluid routing plug 132 is regulated by the suction and discharge valves 214 and 138. The suction valve 214 is configured to engage the suction surface 332, and the discharge valve 138 is configured to engage the discharge surface 334 of the fluid routing plug 132 such that the surfaces 332 and 334 function as valve seats. The valves 214 and 138 are similar in shape but may vary in size. As shown in FIGS. 40 and 41, the discharge valve 138 is slightly larger than suction valve 214.

[0279] Continuing with FIGS. 64-67, the discharge valve 138 is shown in more detail. The suction valve 214 has the same features as the discharge valve 138 so only the discharge valve 138 is shown in more detail in the figures. The discharge valve 138 comprises a stem 402 joined to a body 428. The body 428 comprises an outer rim 430 joined to a valve insert 432 by a tapered seating surface 434. An annular cutout 436 formed within the seating surface 434 is configured to house a seal 438, as shown in FIG. 67.

[0280] Continuing with FIGS. 40 and 41, during operation, the seating surface 434 and the seal 438 engage the seating surface 372 of the discharge surface 334 and block fluid from entering or exiting the discharge passages 360, as shown in FIG. 40. Likewise, the seating surface 434 and the seal 438 on the suction valve 214 engage the seating surface 366 of the suction surface 332 and block fluid from entering or exiting the suction passages 340, as shown in FIG. 41.

[0281] Continuing with FIGS. 40 and 41, when the seating surfaces 434 and 372 are engaged, the valve insert 432 extends partially into the counterbore 380 formed in the discharge surface 334. Fluid exiting the second openings 376 of the discharge passages 360 contacts the insert 432, pushing the discharge valve 138 away from the discharge surface 334 before flowing around the seating surface 434 of the discharge valve 138. Such motion enlarges the area for fluid to flow between the seating surfaces 372 and 434 before fluid reaches the surfaces 372 and 434, thereby reducing the velocity of fluid flow within such area. The lowered fluid velocity between the surfaces 372 and 434 causes any wear to the valve 138 or 214 to be concentrated at the insert 432 instead of the crucial sealing elements, thereby extending the life of the valve 138 or 214.

[0282] Likewise, the insert 432 on the suction valve 214 extends partially into the opening of the axially-blind bore 342. Fluid within the axially-blind bore 342 contacts the insert 432 before flowing around the seating surface 434 and seal 438 of the suction valve 214. Such motion enlarges the area for fluid to flow between the seating surfaces 366 and 434 before fluid reaches the surfaces 366 and 434, thereby reducing the velocity of fluid flow within such area.

[0283] Continuing with FIGS. 64-67, the stem 402 projects from a top surface 440 of the body 428 of the valve 138 or 214. The outer rim 430 surrounds the stem 402 and is spaced from the stem 402 by an annular void 442. A groove 444 is formed in the outer rim 430 for receiving a portion of a spring 446, as shown in FIGS. 40 and 41.

[0284] Continuing with FIGS. 40 and 41, during operation, the valves 138 and 214 move axially along the longitudinal axis 62 of the housing 60 between open and closed positions. In the closed position, the seating surface 434 and the seal 438 of each of the valves 138 and 214 tightly engage the corresponding seating surface 372 or 366 of the fluid routing plug 132 and the valve insert 432 is disposed within the corresponding bore 380 or 342. In the open position, the seating surface 434 and the seal 438 are spaced from the corresponding seating surface 372 or 366 of the fluid routing plug 132 and the valve insert 432 is spaced from the corresponding counterbore 380 or bore 342.

Suction Valve Guide

[0285] With reference to FIGS. 40, 41, and 68-71, axial movement of the suction valve 214 is guided by the suction valve guide 212. The suction valve guide 212 comprises a thin-walled skirt 448 joined to a body 450 by a plurality of support arms 452. The skirt 448 comprises a tapered upper section 454 joined to a cylindrical lower section 456. The plurality of arms 452 join the tapered upper section 454 to the body 450. A plurality of flow ports 458 are formed between adjacent arms 452 such that fluid may pass through the suction valve guide 212 during operation.

[0286] Continuing with FIGS. 40 and 41, the suction valve guide 212 is installed within the housing 60 such that the tapered upper section 454 engages a tapered surface 455 of the walls of the horizontal bore 70. Such engagement prevents further axial movement of the suction valve guide 212 within the housing 60. When the suction valve guide 212 is installed within the housing 60, the skirt 448 covers the walls of the housing 60 positioned between the flow ports 458 and the fluid routing plug 132. During operation, fluid wears against the skirt 448, thereby protecting the housing 60 from wear and erosion. If the skirt 448 begins to

erode, the suction valve guide 212 can be removed and replaced with a new guide 212.

[0287] Continuing with FIGS. 40, 41, and 68-71, the body 450 of the suction valve guide 212 is tubular and is centered within the skirt 448. A tubular insert 460 is installed within the body 450, as shown in FIG. 71. The insert 460 is configured to receive the stem 402 of the suction valve 214, as shown in FIGS. 40 and 41. During operation, the stem 402 moves axially within the insert 460 and wears against the insert 460. An annular cutout 462 formed in the stem 402, shown in FIGS. 66 and 67, provides space for any fluid or other material trapped between the stem 402 and the insert 460. The insert 460 is made of a harder and more wear resistant material than the body 450 thereby extending the life of the suction valve guide 212. For example, if the body 450 is made of stainless steel, the insert 460 may be made of tungsten carbide.

[0288] Continuing with FIGS. 40 and 41, a spring 446 is positioned between the outer rim 430 of the suction valve 214 and the plurality of arms 452 such that the spring 446 surrounds at least a portion of the body 450 of the suction valve guide 212. During operation, the spring 446 biases the suction valve 212 in a closed position, as shown in FIG. 41. Fluid pushing against the valve insert 432 moves the suction valve 214 axially to compress the spring 446 and move the suction valve 214 to an open position, as shown in FIG. 40.

Discharge Plug

[0289] With reference to FIGS. 40, 41, and 72-75, axial movement of the discharge valve 138 is guided by the discharge plug 120. The discharge plug 120 comprises a pair of legs 464 joined to a body 466. The body 466 comprises a front portion 468 joined to a rear portion 470 by a medial portion 472. The medial portion 472 has a larger outer diameter than both the front and rear portions 468 and 470. An outer surface of the medial portion 472 engages the seal 126 installed within the first section 72 of the housing 60, as shown in FIGS. 40 and 41. The pair of legs 464 are joined to the medial portion 472 and extend between the medial portion 472 and the discharge surface 334 of the fluid routing plug 132.

[0290] Continuing with FIGS. 40, 41, and 72-75, a dowel opening 474 is formed in the outer surface of the medial portion 472 for receiving the locating dowel pin 130. The discharge plug 120 is installed within the first section 72 of the housing 60 such that the locating dowel pin 130 is installed within the dowel opening 474 formed in the medial portion 472 and the locating cutout 128 formed in the first section 72 of the housing 60. Such installation aligns the discharge plug 120 within the housing 60 so that the pair of legs 464 do not block the openings of the upper and lower discharge bores 102 and 104.

[0291] Continuing with FIGS. 40 and 41, the locating cutout 128 may be large enough to provide sufficient clearance for installation of the locating dowel pin 130 within the locating cutout 128. The locating cutout 128 is sized to allow maximum clearance for assembly, but still maintain an acceptable rotational position of the discharge plug 120. For example, the cutout 128 may be a maximum of 15 degrees wide along the circumference of the horizontal bore 70.

[0292] Continuing with FIG. 75, an axially-blind bore 476 extends within the body 466 and opens on the rear portion 470 of the body 466. The bore 476 is sized to receive a tubular insert 478. The tubular insert 478 is similar to the

tubular insert 460 installed within the suction valve guide 212. The tubular insert 478 is configured to receive the stem 402 of the discharge valve 138, as shown in FIGS. 40 and 41.

[0293] Continuing with FIGS. 40, 41, and 75, during operation, the stem 402 moves axially within the tubular insert 478. A plurality of passages 480 are formed in the body 466 and interconnect the bore 476 and an outer surface of the medial portion 472. During operation, any fluid or other material trapped within the bore 476 exits the discharge plug 120 through the passages 480. A spring 446 is positioned between the medial portion 472 of the plug 120 and the outer rim 430 of the discharge valve 138, as shown in FIGS. 40 and 41. The spring 446 biases the discharge valve 138 in the closed position, as shown in FIG. 40. Fluid pushing against the valve insert 432 moves the discharge valve 138 axially to compress the spring 446 and move the discharge valve 138 to an open position, as shown in FIG. 41.

[0294] Continuing with FIGS. 40, 41, 72, and 75, the front portion 468 of the body 466 is sized to be disposed within a counterbore 482 formed within the front retainer 118. When disposed therein, a rear surface 484 of the front retainer 118 abuts an outer surface of the medial portion 472 of the discharge plug 120, as shown in FIGS. 40 and 42. Such engagement holds the discharge plug 120 in place between the front retainer 118 and the fluid routing plug 132. A blind bore 486 is formed in an outer surface of the front portion 468 of the plug 120. The blind bore 486 is configured to engage a tool used to help install or remove the plug 120 from the housing 60. For example, the bore 486 may have threaded walls.

Front Retainer

[0295] With reference to FIGS. 40, 41, 76, and 77, the front retainer 118 comprises opposed front and rear surfaces 488 and 484 joined by an outer surface having external threads 124 and a horizontal bore 490 formed therein. The horizontal bore 490 comprises a hex portion 492 that opens in the counterbore 482, as shown in FIGS. 40 and 41. The hex portion 492 is configured to mate with a tool used to thread the front retainer 118 into the housing 60 until it abuts the discharge plug 120, as shown in FIGS. 40 and 41. An annular void 494 is formed within the front surface 488 of the front retainer 118. The annular void 494 decreases the weight of the front retainer 118, making it easier to thread into the housing 60.

Discharge Conduits and Manifold

[0296] With reference to FIG. 41, each discharge fitting 108 comprises a support base 502 and a connection end 512. A discharge fitting adapter 504 is installed within the counterbore 106 formed in the upper and lower discharge bores 102 and 104. When installed, the seal 112 engages an outer surface of the fitting adapter 504. A groove 505 is formed within the discharge fitting 108 for receiving a second seal 507. The second seal 507 likewise engages an outer surface of the fitting adapter 504.

[0297] Continuing with FIG. 41, the support base 502 is sized to abut the outer intermediate surface 68 of the first section 72 of the housing 60. The support base 502 comprises a plurality of passages 506, shown in FIG. 29, configured to align with the threaded openings 114 formed

in the intermediate surface **68** and surrounding the discharge bores **102** and **104**. The threaded fasteners **116** are installed within the aligned passages **506** and openings **114** and tightened to secure the discharge fitting **108** to the first section **72**.

[0298] With reference to FIGS. **3** and **41**, the connection end **512** of the discharge fitting **108** is configured to mate within one or more discharge conduits **500** included in an upper or lower discharge manifold **514** or **516**, as shown in FIG. **3**. The upper and lower discharge manifolds **514** and **516** are supported on rack **518**, as shown in FIG. **3**. The fluid end assembly **52** is disposed within the interior open area of the rack **518**. The rack **518** supports the upper and lower discharge manifolds **514** and **516** in a spaced position from the discharge bores **102** and **104**. As a result, each discharge conduit **500** has an angled or bent shape. In operation, fluid discharges from the housing **60** through upper and lower discharge bores **102** and **104** and is carried to the corresponding upper or lower discharge manifolds **514** or **516** by the discharge fittings and conduits **108** and **500**.

Suction Conduits and Manifold

[0299] With reference to FIGS. **41** and **78**, each suction conduit **166** comprises the first connection member **164** joined to a second connection member **520** by threads, as shown in FIG. **78**. The first and second connection members **164** and **520** may be made of a metal or hardened material.

[0300] Continuing with FIG. **78**, the first connection member **164** comprises an upper portion **524** joined to a lower portion **526**. External threads **172** are formed on a portion of the lower portion **526** for mating with the internal threads **170** formed in the suction bores **160** or **162**. The seal **176** installed within the housing **60** engages a cylindrical outer surface of the lower portion **526** below the external threads **172**. The upper portion **524** has a larger outer diameter than the lower portion **526** and is positioned outside of the housing **60**. The lower portion **526** may abut the counterbore **168** of the suction bores **160** and **162** of the second section **74** of the housing **60**.

[0301] With reference to FIGS. **3** and **78**, the second connection member **520** is configured to mate with one or more connection members or hoses **528** formed on an upper or lower suction manifold **530** or **532**. The upper and lower suction manifolds **530** and **532** are supported on the rack **518** adjacent the discharge manifolds **514** and **516**. The connection members or hoses **528** may be flexible so that they may bend, as needed, to properly interconnect the suction conduits **166** and the suction manifolds **530** and **532**. In operation, fluid is drawn into the housing **60** from the suction manifolds **530** and **532** via the connection members **528**, the suction conduits **166**, and the upper and lower suction bores **160** and **162**.

Assembly of Fluid End Section and Assembly

[0302] Turning to FIGS. **9**, **29**, **79**, and **80**, prior to assembling the housing **60**, the wear ring **136** is preferably first pressed into the counterbore **134** formed in the first section **72** of the housing **60**. Likewise, the hardened insert **210** is pressed into the counterbore **208** formed in the third section **76** of the housing **60**. The seals **126**, **112**, and **176** may also be installed within the first and second sections **72** and **74** of the housing **60**. The wear ring **222** and seal **224**

may also be installed within the third section **76** of the housing **60** prior to assembling the housing **60**.

[0303] Following installation of the above described components, the housing **60** may be assembled as described above. Thereafter, the retention plates **206**, stuffing box **198**, rear retainer **200**, plunger packing **300**, and packing nut **290** may be attached to the rear surface **66** of the housing **60**. The inner components of the housing **60** are inserted within the housing **60** through the front surface **64** of the first section **72**. The inner components may be installed prior to attaching the components to the rear surface **66** of the housing **60**, if desired. Following assembly of each fluid end section **56**, each section **56** is attached to the power end assembly **54** using the stay rods **58**.

[0304] Each fluid end section **56** and its various components are heavy and cumbersome. Various tools or lifting mechanisms may be used to assemble the fluid end assembly **52** and attach it to the power end assembly **54**, creating the high pressure pump **50**.

Operation of Fluid End Assembly

[0305] Turning back to FIGS. **40** and **41**, in operation, retraction of the plunger **216** out of the housing **60** pulls fluid from the upper and lower suction bores **160** and **162** into the suction passages **340** within the fluid routing plug **132**. Fluid flowing through the suction passages **340** and into the axially-blind bore **342** pushes on the valve insert **432** of the suction valve **214**, causing the valve **214** to compress the spring **446** and move to an open position, as shown in FIG. **40**. When in the open position, fluid flows around the suction valve **214** and the suction valve guide **212** and into the open horizontal bore **70** within the third section **76** of the housing **60**.

[0306] Continuing with FIG. **41**, extension of the plunger **216** further into the housing **60** pushes against fluid within the open horizontal bore **70** and forces the fluid towards the suction surface **332** of the fluid routing plug **132**. Such motion also causes the suction valve **214** to move to a closed position, sealing the opening of the axially-blind bore **342**. Because the bore **342** is sealed, fluid is forced into the discharge passages **360**.

[0307] Fluid flowing through the discharge passages **360** contacts the valve insert **432** on the discharge valve **138**, causing the discharge valve **138** to compress the spring **446** and move into an open position, as shown in FIG. **41**. When in the open position, fluid flows around the discharge valve **138** and into the upper and lower discharge bores **102** and **104**. Because fluid exiting the discharge passages **360** has been compressed by extension of the plunger **216** into the housing **60**, such fluid has a higher fluid pressure than that entering the housing **60** through the suction bores **160** and **162**.

[0308] During operation, the plunger **216** continually reciprocates within the housing **60**, pressuring all fluid drawn into the housing **60** through the suction bores **160** and **162**. Pressurized fluid exiting the housing **60** through the upper and lower discharge bores **102** and **104** is delivered to the upper and lower discharge manifolds **514** and **516** in communication with each of the fluid end sections **56**. Pressurized fluid within the discharge manifolds **514** and **516** is eventually delivered to the wellhead **18**, as shown in FIG. **2**.

ALTERNATIVE EMBODIMENTS

[0309] Turning to FIGS. 81-124, alternative embodiments of fluid end sections that may be used to assemble a fluid end assembly, like the fluid end assembly 52 shown in FIG. 7, are shown. For ease of reference, those components that are the same or nearly the same as components making up the fluid end section 56 shown in FIG. 9 will be given the same reference numbers.

Fluid End Section 600

[0310] Turning to FIGS. 81-88, another embodiment of a fluid end section 600 is shown. The fluid end section 600 is generally identical to the fluid end section 56, with a few exceptions. The fluid end section 600 comprises a housing 602 having opposed front and rear surfaces 604 and 606 joined by an outer intermediate surface 608 and a horizontal bore 610 formed therein, as shown in FIG. 86. The housing 602 is identical to the housing 60, shown in FIGS. 10 and 11, but comprises another embodiment of a second section 612 of the housing 602.

[0311] Continuing with FIGS. 81-88, the second section 612 of the housing 602 comprises opposed front and rear surfaces 614 and 616 joined by a portion of the outer intermediate surface 608 and a portion of the horizontal bore 610, as shown in FIG. 86. The second section 612 has the shape of a rectangular prism and is the same height and width as the rear surface 80 of the first section 72.

[0312] Continuing with FIGS. 81-85, a plurality of first passages 618 are formed in the second section 612. Each passage 618 interconnects the front and rear surfaces 614 and 616 and is configured to align with the plurality of passages 86 formed in the first section 72 of the housing 602, as shown in FIG. 85. Each passage 618 is configured to each receive a corresponding one of the stay rods 58. The stay rods 58 pass through the aligned passages 86 and 618, such that both the first and second sections 72 and 612 of the housing 602 are attached to the power end assembly 54 using the stay rods 58.

[0313] Continuing with FIGS. 83-85, in contrast to the fluid end section 56, each sleeve 94 surrounding a corresponding stay rod 58 abuts the second section 612 of the housing 602, instead of the first section 72. Because the sleeves 94 abut the second section 612, the passages 86 formed in the first section 72 do not include the counterbore 87 for receiving the dowel sleeve 93, as shown in FIG. 22. Instead, each passage 618 includes a counterbore 620 that opens on the rear surface 616 of the second section 612 for receiving a corresponding one of the dowel sleeves 93, as shown in FIG. 85.

[0314] Continuing with FIG. 86, a plurality of second passages 622 are also formed in the second section 612. The second passages 622 surround the horizontal bore 610 and are configured to align with the threaded openings 96 formed in the first section 72 and the first passages 184 formed in the third section 76. The first fasteners 78 are installed within the aligned threaded openings 96 and passages 622 and 184 to secure the first, second, and third sections 72, 612, and 76 of the housing 602 together.

[0315] Continuing with FIG. 88, a plurality of dowel openings 624 may be formed in the rear surface 616 of the second section 612 for receiving the second alignment dowels 150. The first alignment dowels 100 may not be used between the first and second sections 72 and 612 since the

stay rods 58 extend through both sections 72 and 612. However, the sections 72 and 612 may be configured to receive the first alignment dowels 100, if desired.

[0316] Continuing with FIG. 88, the walls of the horizontal bore 610 within the second section 612 of the housing 602 are configured to receive another embodiment of a fluid routing plug 630. The fluid routing plug 630 is generally identical to the fluid routing plug 132, shown in FIGS. 43 and 44, but the fluid routing plug 630 comprises another embodiment of an outer intermediate surface 632.

[0317] With reference to FIGS. 88-94, a first groove 634 configured to receive the first seal 386 is formed in the intermediate surface 632 of the fluid routing plug 630 adjacent a discharge surface 636. The first seal 386 engages the wear ring 136 installed within the first section 72. Immediately adjacent the first groove 634, the intermediate surface 632 further comprises a cylindrical section 638 joined to landing bevel 640 and a transition section 641, as shown in FIG. 94. The cylindrical section 638 and the landing bevel 640 are configured to engage a landing bevel counterbore 642 formed in the second section 612 of the housing 602, as shown in FIG. 93.

[0318] Continuing with FIG. 93, the landing bevel 640 is configured so as to only contact the landing bevel counterbore 642 at a single point or at a very small engagement length. The cross-sectional profile of the landing bevel 640 is thus characterized as a splined curve. In operation, the varying fluid pressures in and around the fluid routing plug 630 cause the plug 630 to stretch or deform as the plunger 216 reciprocates. The landing bevel 640 and landing bevel counterbore 642 are sized so that the surfaces more fully engage as the fluid routing plug 630 stretches or deforms during operation. Such engagement supports the housing 602 and fluid routing plug 630 when in their highest state of loading, thereby reducing stress and increasing the life span of such components.

[0319] Continuing with FIGS. 88-92, in contrast to the fluid routing plug 132, the fluid routing plug 630 does not include a second groove 392 for receiving a second seal or an annular shoulder 400. Instead, the intermediate surface 632 adjacent a suction surface 644 comprises a sealing surface 646. The sealing surface 646 is configured to engage a second seal 648 installed within the counterbore 208 formed in the third section 76, as shown in FIG. 88.

[0320] Continuing with FIG. 88, in contrast to the fluid end section 56, the insert 210 is not installed within the counterbore 208. Rather, the second seal 648 and a second wear ring 650 are installed within the counterbore 208. Since the fluid routing plug 630 does not engage the insert 210, engagement of the landing bevel 640 with the landing bevel counterbore 642 prevents further axial movement of the fluid routing plug 630 within the housing 602 during installation. The fluid routing plug 630 is described in more detail in U.S. Pat. No. 11,686,296, authored by Son et al., and issued on Jun. 7, 2023, the entire contents of which are incorporated herein by reference (hereinafter “the ‘736 application”).

Fluid End Section 700

[0321] Turning to FIGS. 95-102, another embodiment of a fluid end section 700 is shown. The fluid end section 700 is similar to the fluid end section 600, with a few exceptions. The fluid end section 700 comprises a housing 702 having opposed front and rear surfaces 704 and 706 joined by an

outer intermediate surface 708 and a horizontal bore 710 formed therein, as shown in FIG. 102. In contrast to the housing 602, the housing 702 only comprises two sections—an integral first and second section 712 joined to the third section 76.

[0322] Continuing with FIGS. 95-102, the integral first and second section 712 is generally identical to the first section 72 and the second section 612 of the housing 602, but the sections are one-piece. The stay rods 58 extend through the entire integral section 712, as shown in FIG. 101. The integral section 712 has the shape of a rectangular prism and comprises opposed front and rear surfaces 714 and 716. A plurality of passages 718 are formed in the integral section 712 for receiving the stay rods 58. Each passage 718 interconnects the front and rear surfaces 714 and 716 of the integral section 712 and includes a counterbore 720 that opens on the rear surface 716. A dowel sleeve 93 is installed within each counterbore 720. The first end 84 of each stay rod 58 projects from the front surface 714 of the integral section 712. While not shown, the front surface 714 of the integral section 712 may include the notches 82 for receiving the first end 84 of each of the stay rods 58.

[0323] Continuing with FIGS. 98 and 101, the rear surface 716 of the integral section 712 comprise a plurality of threaded openings 722 and dowel pin openings 726. The threaded openings 722 align with the passages 184 formed in the third section 76, as shown in FIG. 101. The aligned openings 722 and passages 184 are configured to receive a plurality of first fasteners 724 configured to secure the sections 712 and 76 together. The first fasteners 724 are identical to the first fasteners 78, but the first fasteners 724 may be shorter in length. The plurality of dowel openings 726 may be formed in the rear surface 716 of the integral section 712 for receiving the second alignment dowels 150.

[0324] Continuing with FIG. 102, the components installed within the housing 702 are identical to those described with reference to the fluid end section 56, with the exception that another embodiment of a fluid routing plug 730 is shown installed therein. The fluid routing plug 730 is generally identical to the fluid routing plug 132, but the fluid routing plug 730 comprises another embodiment of an outer intermediate surface 732.

[0325] Continuing with FIGS. 103-108, the outer intermediate surface 732 comprises a first sealing surface 734 positioned adjacent a discharge surface 736 and a second sealing surface 738 positioned adjacent a suction surface 740. The first sealing surface 734 engages a first seal 742 installed within a first groove 744 formed in the integral section 712. In alternative embodiments, the first seal 742 may be surrounded by a wear ring also installed within the first groove 744. The second sealing surface 738 engages the second seal 648 installed within the counterbore 208 formed in the third section 76.

[0326] Continuing with FIGS. 103-108, the outer intermediate surface 732 further comprises a first bevel 746 positioned between the first sealing surface 734 and suction passages 748, and a second bevel 750 positioned between the suction passages 748 and the second sealing surface 738. The first bevel 746 engages a beveled surface 752 formed in the walls of the integral section 712, as shown in FIG. 103. The second bevel 750 faces a second beveled surface 754 formed in the walls of the integral section 712, as shown in FIG. 103. During operation, the second bevel 750 may engage the second beveled surface 754 as the fluid routing

plug 730 stretches or deforms. The fluid routing plug 730 is described in more detail in the '736 application.

[0327] Turning back to FIGS. 95 and 96, the fluid end section 700 further comprises alternative embodiments of a front and rear retainer 760 and 762. The front retainer 760 is identical to the front retainer 118, but the front retainer 760 does not include the annular void 494, shown in FIG. 76. In alternative embodiments, the front retainer 760 may include an annular void. The rear retainer 762 is identical to the rear retainer 200, shown in FIG. 36, but a plurality of passages 764 formed in the rear retainer 762 include a counterbore 766, as shown in FIG. 96. The first end of each second fastener 196 and the corresponding nut 298 installed thereon is positioned within a corresponding one of the counterbores 766.

Fluid End Section 800

[0328] Turning to FIGS. 109-116, another embodiment of a fluid end section 800 is shown. The fluid end section 800 comprises the housing 702, but it also comprises another embodiment of a packing seal system attached to the rear surface 706 of the housing 702. In contrast to the fluid end section 700, the fluid end section 800 does not comprise a stuffing box 198 attached to the rear surface 706 of the housing 702. Rather, the rear retainer 762 is attached directly to the rear surface 706 of the housing 702 using a plurality of second fasteners 802. The second fasteners 802 are identical to the second fasteners 196, but the second fasteners 802 may be shorter in length.

[0329] Continuing with FIGS. 111-113, instead of a stuffing box housing and plunger packing, the fluid end section 800 comprises one and only one packing seal 804 installed within the second counterbore 220 formed in the third section 76 adjacent the rear surface 706. The packing seal 804 is surrounded by a wear ring 806 also installed within the second counterbore 220. A rear surface 812 of the packing seal 804 engages the second metal ring 306 installed within the rear retainer 762.

[0330] Continuing with FIG. 113, the packing seal 804 is annular and has opposed front and rear surfaces 810 and 812 joined by inner and outer intermediate surfaces 814 and 816. The inner intermediate surface 814 of the packing seal 804 surrounds and engages an outer surface of the plunger 216, while the outer intermediate surface 814 seals against the wear ring 806.

[0331] Turning to FIGS. 114-116, the packing seal 804 comprises an energizing component 817 installed within an elastomeric body 818. The energizing component 817 is installed within the front surface 810 of the packing seal 804 and is configured to expand radially when compressed longitudinally. Such expansion causes the inner intermediate surface 814 to tightly seal against the outer surface of the plunger 216 and the outer intermediate surface 816 to tightly seal against the walls of the housing 702 or another component installed therein.

[0332] The energizing component 817 shown in FIG. 116 comprises a plurality of stacked metal pieces 820 having a V-shaped cross-section that function as a spring. Specifically, the packing seal 804 is known in the art as a multi-contact V-nested spring seal. In alternative embodiments, the energizing component may comprise other components known in the art that expand radially when compressed longitudinally. In further alternative embodiments, the ener-

gizing component may comprise one or more coiled springs configured to expand the seal regardless of any longitudinal compression.

[0333] The inner and outer surfaces **814** and **816** of the seal **804** further comprise a plurality of seal lips **822**. The seal lips **822** help effectuate sealing during operation. As the seal lips **822** wear over time, the energizing component **817** expands, pushing the lips **822** tighter against the plunger **216** and the wear ring **806**. Only one packing seal **804** is necessary because the packing seal **804** comprises the energizing component **817**, which allows the packing seal **804** to expand when compressed and further expand as it wears over time without applying any additional compression.

[0334] Continuing with FIGS. **113-116**, a support element **824** may also be installed within the second counterbore **220** and engage the front surface **810** and energizing component **817** of the packing seal **804**. The support element **824** comprises an annular base **826** joined to an annular protrusion **828**. The base **826** may be tapered and engage a tapered surface **830** of the second counterbore **220**, as shown in FIG. **113**. The protrusion **828** projects into the energizing component **817**, as shown in FIG. **116**. In operation, the protrusion **828** helps keep the energizing component **817** expanded so as to maintain a tight seal against the plunger **216**. The packing seal system used with the fluid end section **800** and other similar embodiments are described in U.S. Pat. No. **11,946,465**, authored by Barnett et al., and issued on Mar. **13, 2024**, the entire contents of which are incorporated herein by reference.

Fluid End Section **900**

[0335] Turning to FIGS. **117-124**, another embodiment of a fluid end section **900** is shown. The fluid end section **900** makes up the fluid end assembly **901**, shown in FIG. **124**. The fluid end section **900** is similar to the fluid end section **800**, with a few exceptions. The fluid end section **900** comprises a housing **902** having opposed front and rear surfaces **904** and **906** joined by an outer intermediate surface **908** and a horizontal bore **910** formed therein. The housing **902** is similar to the housing **702**, shown in FIGS. **95** and **96**, but the housing **902** comprises another embodiment of a third section **912**.

[0336] Continuing with FIGS. **117-119**, the third section **912** comprises a front portion **914** joined to a rear portion **916**. The front portion **914** has a rectangular cross-sectional shape and is the same height and width as the integral section **712**. A plurality of passages **918** are formed in the front portion **914** that align with a plurality of passages **718** formed in the integral section **712**. The aligned passages **918** and **718** are configured to receive the stay rods **58**. The integral section **712** and the third section **912** are held together by the stay rods **58**. Thus, in contrast to the housing **702**, the integral section **712** and the third section **912** do not comprise any passages for receiving the first fasteners **78**.

[0337] Turning to FIGS. **120-121**, the fluid end section **900** uses two dowel sleeves **93** in each pair of aligned passages **918** and **718**. One dowel sleeve **93** spans between the integral section **712** and the third section **912**. Another dowel sleeve **93** spans between the third section **912** and the sleeve **94** disposed around the stay rod **58**.

[0338] Turning to FIGS. **122** and **123**, the front portion **914** of the third section **912** includes the rear surface **906** of the housing **902**. Like the fluid end section **800**, the rear retainer **762** is attached directly to the rear surface **906** of the

housing **902** and holds the one and only one packing seal **804** within a counterbore **920** formed in the third section **912**. A support element **922** is shown installed within the packing seal **804**. The support element **922** is generally identical to the support element **824**, but it includes a cylindrical base **924**. The base **924** abuts an annular surface **926** of the counterbore **920**, as shown in FIG. **123**.

[0339] Continuing with FIGS. **122** and **124**, alternative embodiments of a discharge conduit **930** and a suction conduit **932** are shown attached to the housing **902**. Each discharge conduit **930** has a lower opening **934** joined to a horizontal passage **936**. A plurality of horizontally positioned conduits **938** interconnect each discharge conduit **930** to form a discharge manifold **942**, as shown in FIG. **124**. Upper and lower discharge bores **944** and **946** each comprise a counterbore **948** configured to receive the discharge fitting adapter **504**. The adapter **504** spans between a discharge bore **944** or **946** and a discharge conduit **930**.

[0340] Continuing with FIGS. **122** and **124**, upper and lower intake bores **962** and **964** do not include any counterbores or threaded surfaces for receiving the suction conduit **932**. Instead, the bores **962** and **964** have uniform diameters. The suction conduit **932** is attached directly to the intermediate surface **908** of the housing **902** aligning with a corresponding one of the bores **962** and **964** using a plurality of fasteners (not shown). Each suction conduit **932** extends vertically between the fluid end section **900** and a suction manifold **968**. Each suction conduit **932** has a curved shape that curves away from the discharge manifold **942** so as to provide clearance for the suction manifold **968**.

Fluid End Section **1000**

[0341] Turning to FIGS. **125-142**, another embodiment of a fluid end section **1000** is shown. The fluid end section **1000** is similar to the fluid end section **56** shown in FIGS. **8** and **9**, with a few exceptions. The fluid end section **1000** comprises a housing **1002** having opposed front and rear surfaces **1004** and **1006** joined by an outer intermediate surface **1008** and a horizontal bore **1010** formed therein, as shown in FIG. **127**. The housing **1002** is similar to the housing **60**, shown in FIGS. **9-11**, but the housing **1002** comprises another embodiment of a third section **1012**. The housing **1002** is configured to receive the same inner components as the housing **60**.

[0342] With reference to FIGS. **128-131**, the third section **1012** comprises a front surface **1014** joined to the rear surface **1006** of the housing **1002** by a portion of the intermediate surface **1008** and a portion of the horizontal bore **1010**. Like the third section **76**, the outer intermediate surface **1008** of the third section **1012** varies in diameter such that the third section **1012** comprises a front portion **1016** joined to a rear portion **1018**. The front and rear portions **1016** and **1018** may also be referred to as a front mounting flange **1016** and a rear mounting flange **1018**. A medial portion **1020** of the third section **1012** is positioned between the front and rear portion **1016** and **1018** and has lesser diameter than that of the front and rear portions **1016** and **1018**.

[0343] The front portion **1016** of the third section **1012** is identical to the front portion **180** of the third section **76** shown in FIGS. **25** and **26**. The front portion **1016** comprises the plurality of passages **184** for receiving the fasteners **78** and the fasteners **78** extend over the medial portion **1020** of

the third section **1016**, as shown in FIG. **127**. The nut **230** is then installed on the second end **228** of each first fastener **78**.

[0344] The rear portion **1018** of the third section **1012** is different from that of the third section **76** shown in FIGS. **25** and **26**. The rear portion **1018** comprises a plurality of passages **1022** that interconnect a front surface **1024** of the rear portion **1018** and the rear surface **1006** of the housing **1002**, as shown in FIG. **131**.

[0345] Instead of threading second fasteners into threaded openings formed in the third section, a plurality of second fasteners **1026** each extend all of the way through the corresponding passage **1022** and a first threaded end **1028** of each second fastener **1026** is positioned over the medial portion **1020** of the third section **1012**, as shown in FIG. **127**. A blind nut **1030** is installed on each first threaded end **1028** and is turned until it is tightly secured on the corresponding end **1028**.

[0346] Following installation of the second fasteners **1026** into the passages **1022** and assembling of the blind nuts **1030**, the retention plate **206**, stuffing box **198**, and rear retainer **200** are installed on the second fasteners **1026** in the same manner as the fluid end section **56**. The components are tightly secured together by installing a washer **1059** and a nut **1060** on a second threaded end **1062** of each second fastener **1026**, like it is done on the fluid end section **56**. The plunger packing **300**, packing nut **290**, and the plunger **216** are also installed within the fluid end section **1000** in the same manner as the fluid end section **56**.

[0347] As shown in FIGS. **125-127**, when the fluid end section **1000** is assembled, the second ends **228** of the first fasteners **78** and the corresponding nuts **230** and the first ends **1028** of the second fasteners **1026** and the corresponding nuts **1030** are in a spaced-relationship and face each other. Both ends **228** and **1026** and corresponding nuts **230** and **1030** are positioned between the front and rear portions **1016** and **1018** and both ends and corresponding nuts overlap the medial portion **1020** of the third section **1012**. When assembled, the second fasteners **1026** extend along an axis that is parallel to a longitudinal axis **1032** of the housing **1002**, as shown in FIG. **127**.

[0348] The rear portion **1018** of the third section **1012** also comprises the dowel openings **207** configured to receive third alignment dowels **242** used to align the retention plate **206** on the third section **1012**, as shown in FIGS. **129** and **142**. The rear portion **1018** further comprises the threaded openings **202** configured to receive the third fasteners **204** used to secure the retention plate **206** to the third section **1012** of the housing **1002**, as also shown in FIGS. **129** and **142**.

[0349] Turning back to FIG. **125**, a reaction washer **1034** is also installed on the blind nut **1030** during assembly of the fluid end section **1000**. The reaction washer **1034** provides a mechanism for resisting torque applied to the blind nuts **1030** when other parts of the fluid end section **1000** are assembled. The blind nut **1030** is a flanged nut and comprises a flanged section **1036** joined to a drive section **1038**, as shown in FIGS. **132-135**. The flanged section **1036** functions as a washer. The drive section **1038** comprises a standard 12-point drive, sometimes called a double hex drive. A series of grooves **1040** are formed in the outer surface of the drive section **1038** and extend radially around the periphery of the outer surface of the drive section **1038**. The grooves **1040** are configured to receive a seal **1042**, such as an O-ring seal. The seal **1042** is configured to hold the

reaction washer **1034** on the blind nut **1030**, as shown in FIGS. **138** and **139**. The blind nut **1030** further comprises a threaded blind bore **1044** having a base **1046**. An opening **1048** may be formed in the base **1046** for viewing the interior of the blind bore **1044**.

[0350] With reference to FIGS. **136** and **137**, the reaction washer **1034** comprises a pair of torque reaction arms **1054** positioned on opposite sides of a central opening **1052**. The reaction arms **1054** are joined by a concave surface **1056** sized to conform to the outer cylindrical surface of the medial portion **1020** of the third section **1012**, as shown in FIG. **138**. The central opening **1052** comprises a 12-point wall **1058** that is congruent to the 12-point drive section **1038** of the blind nut **1030**.

[0351] The reaction washer **1034** is shaped to slide over the drive section **1038** of the blind nut **1030** and is held on the drive section **1038** by the seal **1042**, as shown in FIG. **139**. When installed thereon, at least one of the reaction arms **1054** of the reaction washer **1034** engages the medial portion **1020** of the third section **1012**. When turning a nut **1060** on a second end **1062** of each second fastener **1026**, the reaction washers **1034** provide a mechanism for resisting the torque applied to the blind nuts **1030**, allowing the nuts **1060** to be tightened or loosened without the blind nuts **1030** spinning freely. Specifically, rotation of the blind nut **1030** is prevented by the engagement of one of the reaction arms **1054** with the medial portion **1020** of the housing **1002**, as shown in FIG. **138**. After assembly, the reaction washers **1034** may be removed from the blind nuts **1030**. Alternatively, the reaction washers **1034** may be left in place so as to prevent rotation of the blind nuts **1030** during operation, thereby reducing the likelihood of the second fasteners **1026** from coming loose.

[0352] Turning back to FIGS. **127** and **130**, the third section **1012** of the housing **1002** comprises the counterbore **208** for receiving the hardened insert **210**. The insert **210** may also be referred to as a wear ring **210**. In contrast to the third section **76** shown in FIGS. **9** and **14**, a seal groove **1064** is also formed within the third section **1012** at a corner of the counterbore **208**. The seal groove **1064** is configured to receive a seal **1066**, as shown in FIG. **140**. The seal **1066** shown in the figures is an O-ring. In alternative embodiments, the seal **1066** may comprise other types of seals known in the art. During operation, the seal prevents fluid from leaking around the hardened insert **210** when the insert **210** is installed within the counterbore **208**.

[0353] Similarly, a seal groove **1068** is formed in the corner of the second counterbore **220** formed in the third section **1012**. The seal groove **1068** is configured to receive a seal **1070**, such as an O-ring shown in FIG. **141**. During operation, the seal **1070** prevents fluid from leaking around the wear ring **222**.

Fluid End Section **1100**

[0354] Turning to FIGS. **143-168**, another embodiment of a fluid end section **1100** is shown. The fluid end section **1100** is similar to the fluid end section **56** shown in the FIGS. **7-9**, with a few exceptions. The fluid end section **1100** comprises a housing **1102** having opposed front and rear surfaces **1104** and **1106** joined by an outer intermediate surface **1108** and a horizontal bore **1110** formed therein, as shown in FIG. **145**. The housing **1102** has the same general shape as the housing **60**, but the housing **1102** comprises two sections instead of three. The housing **1102** comprises a first section **1112**

attached to a second section 1114 using a plurality of fasteners 1116, as shown in FIG. 146. The walls of the horizontal bore 1110 are shaped like those within the housing 60 and are configured to receive the same inner components as the housing 60, as shown in FIGS. 9 and 145.

[0355] The first section 1112 is identical to the first section 72 shown in FIGS. 17 and 18, but the first section 1112 is extended so as to also include the suction bores 160 and 162, as shown in FIG. 145. Thus, the middle or second section 74 making up the housing 60 has been removed. The first section 1112 comprises the plurality of passages 86 and corresponding notches 82 configured to receive a plurality of stay rods 58 in a one-to-one relationship, as shown in FIGS. 152-155. The first section 1112 further comprises the plurality of threaded openings 96 formed in its rear surface 1118 for receiving a first threaded end 1120 of each of the fasteners 1116, as shown in FIGS. 151 and 153.

[0356] The second section 1114 of the housing 1102 is similar in shape to the third section 76 of the housing 60 shown in FIGS. 25 and 26. The second section 1114 comprises a front surface 1122 joined to the rear surface 1106 by a portion of the outer intermediate surface 1108 and a portion of the horizontal bore 1110, as shown in FIG. 151. The outer intermediate surface 1108 comprises a front mounting flange 1124 positioned adjacent the front surface 1122 and a rear mounting flange 1126 positioned adjacent the rear surface 1106, as shown in FIGS. 156 and 157.

[0357] The front and rear mounting flanges 1124 and 1126 break up the intermediate surface 1108 of the second section 1114 such that a front projecting portion 1128 of the intermediate surface 1108 exists between the front surface 1122 and the front mounting flange 1124, and a rear projecting portion 1130 of the intermediate surface 1108 exists between the rear surface 1106 and the rear mounting flange 1126, as shown in FIGS. 156 and 157. Finally, a medial portion 1132 of the intermediate surface 1108 exists between the front and rear mounting flanges 1124 and 1126. The front and rear mounting flanges 1124 and 1126 each have a greater outer diameter than that of the front, rear, and medial portions 1128, 1130, and 1132 of the intermediate surface 1108, as shown in FIGS. 150 and 151. The front mounting flange 1124 further has a greater outer diameter than the rear mounting flange 1126.

[0358] Continuing with FIGS. 156 and 157, a plurality of first passages 1134 are formed within the front mounting flange 1124. The first passages 1134 surround the horizontal bore 1110. Each first passage 1134 is configured to receive one of the first fasteners 1116, as shown in FIG. 151. When the first and second sections 1112 and 1114 are joined together, a space exists around the front portion 1128 and between the rear surface 1118 of the first section 1112 and the front mounting flange 1124. As will be described herein, the space is sized to receive the front spacer sleeve 1136, as shown in FIG. 150. A plurality of clearance notches 1138 are also formed in the outer intermediate surface 1108 of the front mounting flange 1124, as shown in FIG. 156. Each clearance notch 1138 is sized to partially surround a corresponding sleeve 94 and stay rod 58 when the fluid end section 1100 is assembled.

[0359] Continuing with FIGS. 156 and 157, a plurality of second passages 1140 are formed within the rear mounting flange 1126. As will be described herein, the second passages 1140 are configured to receive a plurality of second fasteners 1026 in a one-to-one relationship, as shown in FIG.

147. When a stuffing box 198 is attached to the second section 1114, a space exists around the rear portion 1130 of the second section 1114 and between the rear surface 1106 and the rear mounting flange 1126. As will be described herein, the space is sized to receive a rear spacer sleeve 1144, as shown in FIGS. 145-147.

[0360] A plurality of dowel openings 1146 are also formed with the rear mounting flange 1126, each opening facing the rear surface 1106 of the housing 1102 and configured to receive a portion of a first dowel pin 1148, as shown in FIG. 151. Each first dowel pin 1148 is configured to extend between the rear mounting flange 1126 and the rear spacer sleeve 1144. A plurality of threaded openings 1150 are further formed in the rear mounting flange 1126, each opening facing the rear surface 1106 of the housing 1102. The threaded openings 1150 are preferably diametrically opposed to one another and are each configured to receive one of a plurality of third fasteners 1152, as shown in FIG. 146. The third fasteners 1152 are configured to secure the rear spacer sleeve 1144 to the second section 1114 of the housing 1102.

[0361] Continuing with FIGS. 150 and 151, like the third section 76 shown in FIG. 9, the second section 1114 comprises a counterbore 1115 for receiving a hardened insert 1123 and a counterbore 1117 for receiving a wear ring 1125, as shown in more detail in FIGS. 164-166. The counterbores 1115 and 1117 are like the counterbores 208 and 220 shown in FIGS. 140 and 141 but a side wall 1119 and 1121 of each corresponding counterbore 1115 and 1117 is tapered. The insert 1123 and wear ring 1125 are like the insert 210 and the wear ring 222 shown in FIGS. 164-166, but the insert 1123 and the wear ring 1125 each have a tapered outer surface to correspond with the taper of the corresponding counterbore 1115 or 1117. The corresponding tapers help secure or taper lock the components into the corresponding counterbore 1115 and 1117.

[0362] The second section 1114 further comprises the seal grooves 1064 and 1068, shown in FIGS. 164-166. However, the seal grooves 1064 and 1068 are formed in the center of a side wall 1119 and 1121 of each corresponding counterbore 1115 and 1117, instead of being formed in the corner of each counterbore 1115 and 1117. The seals 1066 and 1070 are installed within the corresponding seal grooves 1064 and 1068 and are configured to prevent fluid from leaking around the insert 1123 and the wear ring 1125.

[0363] Continuing with FIG. 165, a stress relief cutout 275 may be further formed in the outer surface 336 of the fluid routing plug 132 adjacent the annular shoulder 400. The annular shoulder 400 does not contact the insert 1123 in this area. The absence of contact reduces the stress on the annular shoulder 400 during operation, helping to extend the life of the fluid routing plug 132.

[0364] Turning back to FIGS. 148-151, the housing 1102 is also different from the housing 60 because it comprises the front spacer sleeve 1136. The front spacer sleeve 1136 comprises opposed front and rear surfaces 1172 and 1174 joined by a central opening 1176 and an outer intermediate surface 1178, as shown in FIGS. 158 and 159. A plurality of passages 1180 are formed in the front spacer sleeve 1136 surrounding the central opening 1176. The plurality of passages 1180 are each sized to receive one of the first fasteners 1116 and are positioned to correspond with the threaded openings 96 formed in the rear surface 1118 of the first section 1112, as shown in FIG. 151.

[0365] A plurality of clearance notches 1182 are formed in the outer intermediate surface 1178 of the front spacer sleeve 1136, as shown in FIGS. 158 and 159. Each clearance notch 1182 is sized to partially surround a corresponding sleeve 94 and stay rod 58 when the fluid end section 1100 is assembled. The clearance notches 1182 align with the clearance notches 1138 formed in the front mounting flange 1124 of the second section 1114 of the housing 1102 when the fluid end section 1100 is assembled, as shown in FIG. 144.

[0366] The central opening 1176 of the front spacer sleeve 1136 is sized to receive the front portion 1128 of the second section 1114 of the housing 1102 such that the front spacer sleeve 1136 surrounds the front portion 1128 of the second section 1114, as shown in FIGS. 150 and 151. When the front portion 1128 is installed within the central opening 1176, an outer diameter of the front spacer sleeve 1136 is equal or substantially equal to and aligned with an outer diameter of the front mounting flange 1124.

[0367] During operation, the central opening 1176 of the front spacer sleeve 1136 is sized so that it never applies any meaningful compressive force to the front portion 1128 of the second section 1114 of the housing 1102. Specifically, the front spacer sleeve 1136 does not constrain the expansion of the front portion 1128 of the second section 1114 when the annular insert 210 is pressed into the first counterbore 208. This results in more uniform deflection of the insert 210 during installation. The stress resulting from such deflection is therefore also more uniform. As a result, stress concentration areas within the insert 210 are eliminated or significantly reduced, thereby increasing the life of the insert and extending the time between maintenance intervals for the fluid end section 1100.

[0368] The front spacer sleeve 1136 is also sized so that the front surface 1172 of the spacer sleeve 1136 shares the compressive load with the front surface 1122 of the second section 1114 of the housing 1102. Sharing of the compressive load between the two components reduces the stress on the front surface 1122 of the second section 1114. The rear surface 1174 of the front spacer sleeve 1136 provides a reactionary surface for the front surface of the front mounting flange 1124. Providing such surface reduces the bending deflection of the front mounting flange 1124 and consequently the bending stress on the front mounting flange 1124. The benefits provided by the front spacer sleeve 1136 increase the life of the corresponding components, thereby extending the time between maintenance intervals for the fluid end section 1100.

[0369] With reference to FIGS. 145-147, prior to assembling the housing 1102, the seal 1066 is installed within the first seal groove 1064 formed in the second section 1114 and the insert 1123 is press-fit or interference fit into the first counterbore 208. The seal 1066 shown in the figures is an O-ring seal, but other types of seals known in the art may also be used. The insert 210 will be described in more detail later herein.

[0370] Following installation of the seal 1066 and the insert 210, the housing 1102 is assembled by threading a first threaded end 1120 of each of the first fasteners 1116 into a corresponding one of the threaded openings 96 formed in the first section 1112, as shown in FIG. 151. Once installed therein, the first fasteners 1116 project from the rear surface 1118 of the first section 1112. The front spacer sleeve 1136 is then slid onto the first fasteners 1116 using the corresponding passages 1180. The front spacer sleeve 1136 is

aligned such that the clearance notches 1182 correspond with the stay rod passages 86 formed within the first section 1112, as shown in FIG. 144.

[0371] After the front spacer sleeve 1136 is in place, the second section 1114 of the housing 1102 is slid onto the first fasteners 1116 using the corresponding passages 1134 formed in the front mounting flange 1124 as shown in FIG. 151. At the same time, the front portion 1128 of the second section 1122 is inserted into the central opening 1176 of the front spacer sleeve 1136 until the front surface 1122 of the second section 1114 abuts the rear surface 1118 of the first section 1112.

[0372] When the second section 1114 is brought together with the first section 1112, a second threaded end 1184 of each first fastener 1116 projects from the front mounting flange 1124 and extends over the medial portion 1132 of the intermediate surface 1108, as shown in FIGS. 144-147. At the same time, the front spacer sleeve 1136 is sandwiched between the rear surface 1118 of the first section 1112 and the front mounting flange 1124 such that the front and rear surfaces 1172 and 1174 of the front spacer sleeve 1136 abut the corresponding surfaces. The second section 1114 of the housing 1102 is aligned on the first fasteners 1116 such that the clearance notches 1138 formed in the front mounting flange 1124 align with the clearance notches 1182 formed in the front spacer sleeve 1136, as shown in FIG. 144.

[0373] The assembled sections 1112 and 1114 are secured together by installing a threaded nut 1186 onto the second end 1184 of each first fastener 1116, as shown in FIGS. 144 and 146. The nut 1186 shown in the figures is a flanged nut having a 12-point drive section. The nut 1186 is turned on the second end 1184 of the fastener 1116 to the desired specification or until it tightly engages the front mounting flange 1124. When installed thereon, each nut 1186 overlaps the medial portion 1132 and is positioned intermediate the front and rear mounting flanges 1124 and 1126 of the second section 1114 of the housing 1102.

[0374] Twelve first fasteners 1116 and corresponding threaded openings 96, passages 1180 and 1134, and nuts 1186 are shown in the figures. In alternative embodiments, more or less than twelve first fasteners 1116 and corresponding openings 96, passages 1180 and 1134, and nuts 1186 may be used to assemble the housing 1102. When the housing 1102 is assembled, the first fasteners 1116 are each positioned along an axis that is parallel to a longitudinal axis 1188 of the housing 1102.

[0375] The first fasteners 1116 are identical to the first fasteners 78 shown in FIG. 15 but are shorter in length because each fastener 1116 traverses a shorter distance. The first fastener 1116 shown in the figures is a threaded stud. In alternative embodiments, other types of fasteners known in the art may be used instead of a threaded stud. For example, screws or bolts may be used to secure the sections together. In further alternative embodiments, the nut 1186 may comprise the three-piece nut 90, shown in FIG. 19, or other types of threaded nuts known in the art.

[0376] To remove a section 1112 or 1114 or remove the front spacer sleeve 1136, the nut 1186 is unthreaded from the second end 1184 of each first fastener 1116. The sections 1112 and 1114 and the rear spacer sleeve 1144 may then be pulled apart, as needed. If the first section 1112 is being replaced, the first fasteners 1116 are also unthreaded from the threaded openings 96. The components installed within

the housing 1102 may also be removed, as needed, prior to disassembling the housing 1102.

[0377] With reference to FIG. 147, like the rear portion 1018 of the housing 1002 shown in FIG. 131, the rear mounting flange 1126 is configured to have the second fasteners 1026 extend entirely through the rear mounting flange 1126. Like the fluid end section 1000, the blind nut 1030 and reaction washer 1034 are installed on the first threaded end 1028 of the second fastener 1026. The stuffing box 198 and the rear retainer 200 are then installed on the second fasteners 1026 in the same manner as the fluid end section 1000.

[0378] Continuing with FIGS. 145-147, in contrast to the fluid end section 1000, the fluid end section 1100 uses the rear spacer sleeve 1144 instead of the retention plate 206. The washer 1059 and the nut 1060 are installed on the second threaded end 1062 of each second fastener 1026 to secure the components together. The plunger packing 300, the packing nut 290, and the plunger 216 are installed within the fluid end section 1100 in the same manner as the fluid end section 1000.

[0379] With reference to FIGS. 162 and 163, the rear spacer sleeve 1144 comprises opposed front and rear surfaces 1190 and 1192 joined by a central opening 1194 and an outer intermediate surface 1196. A plurality of passages 1198 are formed in the rear spacer sleeve 1144 surrounding the central opening 1194. The plurality of passages 1198 are each sized to receive one of the second fasteners 1026, as shown in FIG. 147. A plurality of counterbored passages 1200 are also formed within the rear spacer sleeve 1144. The counterbored passages 1200 are positioned to align with the threaded openings 1150 formed in the rear mounting flange 1126 and are configured to receive a corresponding third fastener 1152, as shown in FIG. 146. A counterbore 1202 of each counterbored passage 1200 opens on the rear surface 1192 of the rear spacer sleeve 1144. Finally, a plurality of dowel passages 1204 are formed within the rear spacer sleeve 1144. Each dowel passage 1204 is positioned to align with a corresponding dowel opening 1146 formed in the rear mounting flange 1126, as shown in FIG. 151.

[0380] The central opening 1194 of the rear spacer sleeve 1144 is sized to receive the rear portion 1130 of the second section 1114 of the housing 1102 such that the rear spacer sleeve 1144 surrounds the rear portion 1130 of the second section 1114, as shown in FIGS. 145-147. When the rear portion 1130 is installed within the central opening 1194, an outer diameter of the rear spacer sleeve 1144 is equal to or substantially equal to and aligned with an outer diameter of the rear mounting flange 1126.

[0381] Like the front spacer sleeve 1136, the central opening 1194 of the rear spacer sleeve 1144 is sized so that it never applies any meaningful compressive force to the rear portion 1130 of the second section 1114 of the housing 1102. Specifically, the rear spacer sleeve 1144 does not constrain the expansion of the rear portion 1130 of the second section 1114 when the wear ring 222 is pressed into the second counterbore 220. This results in more uniform deflection of the wear ring 222 during installation. The stress resulting from such deflection is therefore also more uniform. As a result, stress concentration areas within the wear ring 222 are eliminated or significantly reduced, thereby increasing the life of the wear ring 222 and extending the time between maintenance intervals for the fluid end section 1100.

[0382] Like the front spacer sleeve 1136, the rear spacer sleeve 1144 is also sized so that the rear surface 1192 of the spacer sleeve 1144 shares the compressive load with the rear surface 1106 of the housing 1102. Sharing of the compressive load between the two components reduces the stress on the rear surface 1106 of the housing 1102. The front surface 1190 of the rear spacer sleeve 1144 provides a reactionary surface for the rear surface of the rear mounting flange 1126. Providing such surface reduces the bending deflection of the rear mounting flange 1126 and consequently the bending stress on the rear mounting flange 1126. The benefits provided by the rear spacer sleeve 1144 increase the life of the corresponding components, thereby extending the time between maintenance intervals for the fluid end section 1100.

[0383] Continuing with FIGS. 145-147, to attach the rear spacer sleeve 1144 to the housing 1102, a first dowel pin 1148 is inserted into a corresponding one of the dowel openings 1146 formed on the rear mounting flange 1126 of the second section 1114, as shown in FIG. 151. The rear spacer sleeve 1144 is then brought together with the second section 1114 of the housing 1102 such that the first dowel pins 1148 are inserted into the dowel passages 1204, the second fasteners 1026 are disposed within the passages 1140, and the threaded openings 1150 are aligned with the counterbored passages 1200, as shown in FIGS. 146 and 147. When in such position, the front surface 1190 of the rear spacer sleeve 1144 abuts the rear surface 1106 of the housing 1102 and the rear portion 1130 of the second section 1114 is installed within the central opening 1194 of the rear spacer sleeve 1144.

[0384] The rear spacer sleeve 1144 is secured to the rear surface 1106 of the housing 1102 by installing a third fastener 1152 within a corresponding one of the aligned threaded openings 1150 and counterbored passages 1200, as shown in FIG. 146. The third fasteners 1152 shown in the figures are screws. The third fasteners 1152 are torqued within the threaded openings 1150 until a head 1206 of each fastener 1152 abuts a base 1208 of each counterbored passage 1200, thereby rigidly securing the rear spacer sleeve 1144 to the housing 1102. When the rear spacer sleeve 1144 is attached to the housing 1102, the rear surface 1106 of the rear spacer sleeve 1144 is positioned flush with the rear surface 1106 of the housing 1102.

[0385] As shown in FIGS. 143 and 144, when the fluid end section 1100 is assembled, the second ends 1184 of the first fasteners 1116 and the corresponding nuts 1186 and the first ends 1028 of the second fasteners 1026 and the corresponding nuts 1030 are in a spaced-relationship and face each other. Both ends 1184 and 1028 and corresponding nuts 1186 and 1030 are positioned between the first and rear mounting flanges 1124 and 1126 and both ends and corresponding nuts overlap the medial portion 1132 of the second section 1114. When assembled, the second fasteners 1026 extend along an axis that is parallel to the longitudinal axis 1188 of the housing 1102, as shown in FIG. 147.

Fluid End Section 1300

[0386] Turning to FIGS. 169-174, another embodiment of a fluid end section 1300 is shown. The fluid end section 1300 is similar to the fluid end section 1100 shown in FIGS. 143 and 144 with a few exceptions. The fluid end section 1300 comprises a housing 1302 having opposed front and rear surfaces 1304 and 1306 joined by an outer intermediate

surface **1308** and a horizontal bore **1310** formed therein, as shown in FIG. **171**. The housing **1302** has the same general shape as the housing **1102**, but the housing **1302** is of single-piece construction. The housing **1302** does not include multiple sections joined together by fasteners.

[0387] The walls of the horizontal bore **1310** are shaped identically to those within the housing **1102** and are configured to receive the same inner components as the housing **1102**, as shown in FIGS. **145** and **172**. However, the housing **1302** is shown with the counterbores **208** and **220** and corresponding inserts **210** and wear ring **222** installed therein. In alternative embodiments, the counterbores **1115** and **1117** may be formed within the housing **1302** and have the corresponding insert **1123** and wear ring **1125** installed therein.

[0388] The outer intermediate surface **1308** of the housing **1302** comprises a first section **1312** integrally joined to a second section **1314**. The first section **1312** is shaped identically to the first section **1112** and comprises the plurality of passages **86** and corresponding notches **82** configured to receive a plurality of stay rods **58** in a one-to-one relationship, as shown in FIGS. **169** and **170**. The second section **1314** of the housing **1302** comprises a cylindrical portion **1316** that transitions into the rear mounting flange **1126** and the rear projecting portion **1130** of the housing **1302**, as shown in FIGS. **171-174**. Because the housing **1302** is a single piece, the housing **1302** does not include the front spacer sleeve **1136**.

[0389] The cylindrical portion **1316** of the second section **1314** of the housing **1302** is configured to house the same components as the second section **1114** of the housing **1102**, shown in FIGS. **145** and **172**. The rear mounting flange **1126** and rear portion **1130** of the housing **1302** are configured to mate with the rear spacer sleeve **1144**, the stuffing box **198**, and the rear retainer **200** in the same manner as the fluid end section **1100**, shown in FIG. **145**.

[0390] Continuing with FIG. **171**, the second fasteners **1026** extend through the rear mounting flange **1126**, rear spacer sleeve **1144**, stuffing box **198**, and rear retainer **200** in the same manner as the fluid end section **1100**. The second fasteners **1026** are secured on opposite ends **1028** and **1062** using the blind nuts **1030** and the nuts **1060** and corresponding washers **1059**. While not shown, a reaction washer **1034** and seal **1042** may also be installed on each of the blind nuts **1030**. The plunger packing **300**, the packing nut **290**, and the plunger **216** are installed within the fluid end section **1300** in the same manner as the fluid end section **1100**, as shown in FIG. **172**. When the fluid end section **1300** is assembled, the ends **1028** of the second fasteners **1026** and the blind nuts **1030** are in a spaced-relationship with and face the cylindrical portion **1316** of the second section **1314** of the housing **1302**, as shown in FIGS. **169-172**.

Stress Relief Cutout

[0391] Referring now to FIGS. **175** and **176**, stress relief cutout **275** of the fluid routing plug **132** is shown in greater detail than previously depicted in FIGS. **145**, **164**, and **165**. Stress relief cutout **275** comprises a first radius **533**, a linear section **534**, and a second radius **535**. The first radius **533** is a convex radius tangent to the annular shoulder **400**, transitioning from the annular shoulder **400** to the linear section **534**. The second radius **535** is a concave radius that is tangent to both the outer intermediate surface **336** and the linear section **534**, smoothly transitioning between the two.

As mentioned above, the stress relief cutout **275** does not contact the hardened insert **1123**, or hardened insert **210** if the non-tapered embodiment is used.

[0392] While not discernible at the scale shown in the previous figures, FIG. **175** reveals that the first bevel **426** of the hardened insert **1123** comprises multiple sections, including a first radius **536**, a linear section **537**, and a second radius **538**. The first radius **536** is convex and tangent to both the inner intermediate surface **420** and the linear section **537**. The second radius **538** is also convex, larger than the first radius **536**, and tangent to both the linear section **537** and the front surface **416**. In other embodiments, the first bevel **426** may consist of only the linear section **537**, or the linear section **537** and a single radius, either **536** or **538**. Additionally, in other embodiments the first radius **536** may be the same size as, or larger than, the second radius **538**. It should be understood that “linear” in the context of portions of the ring-shaped insert, refers to the linear nature of the cross-section of the linear section **537** insert **1123**, rather than the surface itself.

[0393] The hardened insert **1123** contacts the annular shoulder **400** at the point where the first radius **533** joins the annular shoulder **400**, as shown in FIG. **175**. This point is defined as the minimum radial contact point **P1**. The minimum radial contact point **P1** is at a radial distance **D1** from the section of the outer intermediate surface **336** of the fluid routing plug **132** that contains the second annular groove **392**. The second annular groove **392** is shown in FIG. **165**. The minimum radial contact point **P1** represents a circular line of contact that extends around the front surface **416** of hardened insert **1123**.

[0394] In operation, the force generated by fluid pressure is distributed across the contacting surfaces of the annular shoulder **400** and the front surface **416** of the hardened insert **1123**. The radial distance **D1** locates the minimum radial contact point **P1** on the front surface **416** of the hardened insert **1123**, positioned radially outward from the inner intermediate surface **420**, toward the outer intermediate surface **422**, and away from any edge modifications, such as a first bevel **426**. Increasing the radial distance **D1** reduces or eliminates chipping or spalling of the front surface **416**. Additionally, the second radius **535** reduces the stress concentration at the transition between the linear section **534** and the section of outer intermediate surface **336** that contains the second annular groove **392**.

[0395] Another embodiment of a stress relief cutout **175** is shown in FIGS. **177** and **178**. Stress relief cutout **175** comprises a first radius **539** and a linear section **540**. The first radius **539** is a concave radius that is tangent to both the outer intermediate surface **336** and the linear section **540**, smoothly transitioning between the two. The linear section **540** is tangent to the first radius **539** and creates an angle $\beta 2$ with the annular surface **400** as shown in FIG. **178**. In this embodiment, the angle $\beta 2$ is 5 degrees but may be any angle between 0 and 90 degrees, exclusive of 0 degrees. The stress relief cutout **175** does not contact the hardened insert **1123**, or hardened insert **210** if the non-tapered embodiment is used.

[0396] The hardened insert **1123** contacts the annular shoulder **400** at the point where the linear section **540** joins the annular shoulder **400**, as shown in FIG. **177**. This point is defined as the minimum radial contact point **P2**. The minimum radial contact point **P2** is at a radial distance **D2** from the section of the outer intermediate surface **336** of the

fluid routing plug 132 that contains the second annular groove 392. The second annular groove 392 is shown in FIG. 165. The minimum radial contact point P2 represents a circular line of contact that extends around the front surface 416 of hardened insert 1123.

[0397] In operation, the radial distance D2 locates the minimum radial contact point P2 on the front surface 416 of the hardened insert 1123, positioned radially outward from the inner intermediate surface 420, toward the outer intermediate surface 422, and away from any edge modifications, such as a first bevel 426. This reduces or eliminates chipping or spalling of the hardened insert 1123. Additionally, the first radius 539 reduces the stress concentration at the transition between the linear section 540 and the section of outer intermediate surface 336 that contains the second annular groove 392. Further, the shallow transition angle $\beta 2$ reduces the stress concentration in the fluid routing plug 132 created by the transition from the annular surface 400 to the stress relief cutout 175.

[0398] Another embodiment of a stress relief cutout 375 is shown in FIGS. 179 and 180. The stress relief cutout 375 comprises a first radius 541 and a second radius 542. The first radius 541 is a convex radius that is tangent to both the annular shoulder 400 and the second radius 542, transitioning from the annular shoulder 400 to the second radius 542. The second radius 542 is a concave radius that is tangent to both the outer intermediate surface 336 and the first radius 541, smoothly transitioning between the two. The stress relief cutout 375 does not contact the hardened insert 1123, or hardened insert 210 if the non-tapered embodiment is used.

[0399] The hardened insert 1123 contacts the annular shoulder 400 at the point where the first radius 541 joins the annular shoulder 400, as shown in FIG. 179. This point is defined as the minimum radial contact point P3. The minimum radial contact point P3 is at a radial distance D3 from the section of the outer intermediate surface 336 of the fluid routing plug 132 that contains the second annular groove 392. The second annular groove 392 is shown in FIG. 165. The minimum radial contact point P3 represents a circular line of contact that extends around the front surface 416 of hardened insert 1123.

[0400] In operation, the radial distance D3 locates the minimum radial contact point P3 on the front surface 416 of the hardened insert 1123, positioned radially outward from the inner intermediate surface 420, toward the outer intermediate surface 422, and away from any edge modifications, such as a first bevel 426. This reduces or eliminates chipping or spalling of the hardened insert 1123. Additionally, the second radius 542 reduces the stress concentration at the transition between the first radius 541 and the section of outer intermediate surface 336 that contains the second annular groove 392. Further, the first radius 541 reduces the stress concentration in the fluid routing plug 132 created by the transition from the annular surface 400 to the stress relief cutout 375.

[0401] Another embodiment of a stress relief cutout 475 is shown in FIGS. 181 and 182. The stress relief cutout 475 comprises a first radius 543, a second radius 544, and a third radius 545. The first radius 543 is a convex radius that is tangent to both the annular shoulder 400 and the second radius 544, transitioning from the annular shoulder 400 to the second radius 544. The second radius 544 is a concave radius that is tangent to both the first radius 543 and the third

radius 545, smoothly transitioning between the two. The third radius 545 is a convex radius that is tangent to both the second radius 544 and the outer intermediate surface 336. The stress relief cutout 475 does not contact the hardened insert 1123, or hardened insert 210 if the non-tapered embodiment is used.

[0402] The hardened insert 1123 contacts the annular shoulder 400 at the point where the first radius 543 joins the annular shoulder 400, as shown in FIG. 181. This point is defined as the minimum radial contact point P4. The minimum radial contact point P4 is at a radial distance D4 from the section of the outer intermediate surface 336 of the fluid routing plug 132 that contains the second annular groove 392. The second annular groove 392 is shown in FIG. 165. The minimum radial contact point P4 represents a circular line of contact that extends around the front surface 416 of hardened insert 1123.

[0403] In operation, the radial distance D4 locates the minimum radial contact point P4 on the front surface 416 of the hardened insert 1123, positioned radially outward from the inner intermediate surface 420, toward the outer intermediate surface 422, and away from any edge modifications, such as a first bevel 426. This reduces or eliminates chipping or spalling of the hardened insert 1123. Additionally, the second radius 544 reduces the stress concentration at the transition between the first radius 543 and the third radius 545. Further, the first radius 543 reduces the stress concentration in the fluid routing plug 132 created by the transition from the annular surface 400 to the stress relief cutout 475 and the third radius 545 reduces the stress concentration in the fluid routing plug 132 created by the transition from the section of outer intermediate surface 336 that contains the second annular groove 392 to the stress relief cutout 475.

[0404] Another embodiment of a stress relief cutout 575 is shown in FIGS. 183 and 184. The stress relief cutout 575 comprises a first linear section 546, a second linear section 547, a first radius 548, a third linear section 549, and a second radius 550. The first linear section 546 is perpendicular to the annular surface 400 and connects the annular surface 400 to the second linear section 547. The second linear section 547 is parallel to the annular surface 400, thus perpendicular to the first linear section 546, and tangent to the first radius 548 connecting the first linear section 546 to the first radius 548. The first radius 548 is a concave radius that is tangent to both the second linear section 547 and the third linear section 549. The third linear section 549 forms an angle $\beta 5$ with the second linear section, is tangent to both the first and second radii 548 and 550, and connects the first radius 548 to the second radius 550. The second radius 550 is a concave radius that is tangent to both the third linear section 549 and the outer intermediate surface 336. The stress relief cutout 575 does not contact the hardened insert 1123, or hardened insert 210 if the non-tapered embodiment is used. In this embodiment, the angle $\beta 5$ is 45 degrees but may vary from 0 to 90 degrees, exclusive of both.

[0405] The hardened insert 1123 contacts the annular shoulder 400 at the point where the first linear section 546 joins the annular shoulder 400, as shown in FIG. 183. This point is defined as the minimum radial contact point P5. The minimum radial contact point P5 is at a radial distance D5 from the section of the outer intermediate surface 336 of the fluid routing plug 132 that contains the second annular groove 392. The second annular groove 392 is shown in FIG. 165. The minimum radial contact point P5 represents

a circular line of contact that extends around the front surface **416** of hardened insert **1123**.

[0406] In operation, the radial distance **D5** locates the minimum radial contact point **P5** on the front surface **416** of the hardened insert **1123**, positioned radially outward from the inner intermediate surface **420**, toward the outer intermediate surface **422**, and away from any edge modifications, such as a first bevel **426**. This reduces or eliminates chipping or spalling of the hardened insert **1123**. Additionally, the first radius **548**, third linear section **549**, and second radius **550** combine to reduce the stress concentration at the transition between the second linear section **547** and the outer intermediate surface **336**.

[0407] The key feature in all the disclosed embodiments of the stress relief cutouts **175**, **275**, **375**, **475**, and **575** is the location of the minimum radial contact points **P1**, **P2**, **P3**, **P4**, and **P5** radially outward from the inner intermediate surface **420**, toward the outer intermediate surface **422**, and away from any edge modifications, such as a first bevel **426**. Though the specific geometries of each embodiment vary, all the embodiments accomplish the placement through their specific designs. As such, any combination of geometric entities used to create a stress relief cutout that accomplishes this placement is contemplated herein. Specifically, but not limiting, it is contemplated that the linear sections may be modified into spline curves, radii, multiple radii of differing magnitudes, or any combination thereof. Likewise, it is contemplated that and of the radii may be modified into spline curves, multiple radii of differing magnitudes, combinations of radii and straight or splined segments, or any combination thereof.

[0408] Another embodiment of a stress relief cutout **75** is shown in FIGS. **185** and **186**. Stress relief cutout **75** is formed on the hardened insert **1123**. In this embodiment the stress relief cutout **75** comprises a linear section **551**. The linear section **551** is tangent to the second radius **538** of the first bevel **426** and creates an angle $\beta 6$ with the front surface **416** as shown in FIG. **186**. In this embodiment, the angle $\beta 6$ is 5 degrees but may be other non-zero values. Depending on the other values chosen, other modifications to the first bevel **426** may be required. The stress relief cutout **75** does not contact the annular shoulder **400**. In this embodiment the fluid routing plug **132** has only a transition radius **152** between the outer intermediate surface **336** and the annular shoulder **400**. The fluid routing plug **132** does not have a stress relief cutout.

[0409] The hardened insert **1123** contacts the annular shoulder **400** at the point where the linear section **551** joins the second radius **538**, as shown in FIG. **185**. This point is defined as the minimum radial contact point **P6**. The minimum radial contact point **P6** is at a radial distance **D6** from the section of the outer intermediate surface **336** of the fluid routing plug **132** that contains the second annular groove **392**. The second annular groove **392** is shown in FIG. **165**. The minimum radial contact point **P6** represents a circular line of contact that extends around the front surface **416** of hardened insert **1123**.

[0410] In operation, the radial distance **D6** locates the minimum radial contact point **P6** on the front surface **416** of the hardened insert **1123**, positioned radially outward from the inner intermediate surface **420**, toward the outer intermediate surface **422**. This reduces or eliminates chipping or spalling of the hardened insert **1123**.

[0411] Stress relief cutout **75** may be applied to hardened insert **210** if a non-tapered embodiment is used. While this embodiment of stress relief cutout **75** comprises only one component, the linear section **551**, other embodiments comprising multiple sections having geometric constructions similar to the embodiments **175**, **275**, **375**, **475**, and **575** may be used. As with the embodiments employed on the fluid routing plug **132** the key feature of this improvement is to locate the minimum radial contact point **P6** as far as practical from the inner intermediate surface **420** of the hardened insert **1123** or **210**.

[0412] The fluid end sections described herein have various embodiments of housings, inner components, and components attached to the various housings. While not specifically shown in a figure herein, various features from one fluid end section embodiment may be included in another fluid end section embodiment.

[0413] In alternative embodiments, features of one of the various housings, inner components, or components attached to the various housings and described in the '552 application, may be included in one or more of the fluid end sections disclosed herein. One of skill in the art will appreciate that the various housing and components described herein may have different shapes and sizes, depending on the shape and size of the various components chosen to assemble each fluid end section.

[0414] One or more kits may be useful in assembling a fluid end assembly out of the various fluid end sections described herein. A single kit may comprise a plurality of one of the various embodiments of housings and fasteners described herein. The kit may further comprise a plurality of one or more of the various inner components described herein. The kit may even further comprise a plurality of one or more of the various components attached to the various housings described herein.

[0415] The various features and alternative details of construction of the apparatuses described herein for the practice of the present technology will readily occur to the skilled artisan in view of the foregoing discussion. It is to be understood that even though numerous characteristics and advantages of various embodiments of the present technology have been set forth in the foregoing description, together with details of the structure and function of various embodiments of the technology, this detailed description is illustrative only. Changes may be made in detail, especially in matters of structure and arrangements of parts within the principles of the present technology to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

1. An apparatus, comprising:
 - a fluid routing plug, comprising:
 - an outer surface;
 - an annular shoulder; and
 - a stress relief cutout defined on a surface of the fluid routing plug between the outer surface and the annular shoulder;
 - in which the annular shoulder is configured to abut an insert, the insert comprising a beveled surface.
2. A fluid end, comprising:
 - the apparatus of claim 1;
 - an insert which is the insert of claim 1; and
 - a housing containing both the apparatus of claim 1 and the insert.

3. The fluid end of claim 2, in which the stress relief cutout is in a spaced relationship with the insert.

4. The fluid end of claim 2, in which the stress relief cutout does not engage the insert.

5. The fluid end of claim 2, in which the insert is made of a harder material than a material used to make the housing.

6. The fluid end of claim 2, in which the insert is made of tungsten carbide.

7. The fluid end of claim 6, in which the housing is made of stainless steel.

8. A fluid routing plug configured to be situated within a fluid end, the fluid routing plug comprising:

- a body having a first surface and an opposed second surface joined by an outer intermediate surface, the outer intermediate surface comprising:
 - an annular shoulder formed therein;
 - an annular groove formed therein; and
 - a stress relief cutout formed therein and situated intermediate the annular shoulder and the annular groove;
- a longitudinal axis extending through the body and both the first and second surfaces;
- a plurality of first fluid passages interconnecting the first surface and the outer intermediate surface; and
- a plurality of second fluid passages interconnecting the first surface and the second surface.

9. The fluid routing plug of claim 8, in which the annular groove is configured to receive an annular seal.

10. The fluid routing plug of claim 8, in which the annular shoulder is configured to abut a wear ring; in which the stress relief cutout is in a spaced relationship with the wear ring.

11. The fluid routing plug of claim 8, in which the annular groove is defined as a first annular groove; in which the body further comprises a second annular groove formed on the intermediate surface, the second annular groove configured to receive an annular seal.

12. The fluid routing plug of claim 8, in which the stress relief cutout comprises:

- a planar section; and
 - a curved section;
- in which the curved section is situated intermediate the outer intermediate surface and the planar section.

13. The fluid routing plug of claim 12, in which the planar section is situated intermediate the annular shoulder and the curved section.

14. A fluid end, comprising:

- a fluid end housing;
- the fluid routing plug of claim 8; and
- a wear ring abutting the annular shoulder of the body.

15. The fluid end of claim 14, in which the wear ring does not contact the stress relief cutout.

16. The fluid end of claim 14, in which the wear ring comprises:

- a first surface engaging the annular shoulder;
- a second surface opposed to the first surface and engaging a portion of the fluid end housing;

an inner intermediate surface bound between the first and second surfaces;

an outer intermediate surface bound between the first and second surfaces; and

a first bevel formed intermediate the first surface and the outer intermediate surface.

17. The fluid end of claim 16, in which the wear ring further comprises:

- a second bevel situated intermediate the inner intermediate surface and the second surface.

18. The fluid end of claim 17, in which the first bevel is larger than the second bevel.

19. The fluid end of claim 14, in which the fluid end housing comprises:

- a longitudinal bore formed therein; and
- a valve pair situated within the longitudinal bore, the valve pair configured to engage the fluid routing plug.

20. A fluid routing plug configured for use within a fluid end assembly comprising a housing and a wear ring, the fluid routing plug comprising:

- an annular seal engaging the wear ring; and
- a body defining an annular shoulder, the annular shoulder comprising:
 - a cutout section;
 - in which the cutout section does not contact the annular seal or the wear ring.

21. An apparatus, comprising:

- a fluid end comprising a housing;
- a fluid routing plug situated within the housing; and
- a wear ring situated within the housing and configured to abut the fluid routing plug, the wear ring comprising:
 - a first surface configured to abut the fluid routing plug;
 - a second surface opposed to the first surface;
 - an intermediate surface situated between the first and second surfaces; and
 - a stress relief cutout situated between the intermediate surface and the first surface, in which the stress relief cutout does not contact the fluid routing plug.

22. The apparatus of claim 21, in which the wear ring further comprises:

- a bevel situated between the intermediate surface and the stress relief cutout, in which the bevel does not contact the fluid routing plug.

23. The apparatus of claim 22, in which the bevel comprises:

- a first radius joining the stress relief cutout;
- a second radius joining the intermediate surface; and
- a linear section situated between the first and second radii.

24. The apparatus of claim 21, in which the stress relief cutout comprises:

- a planar section;
- in which the planar section is angled with respect to the first surface at a non-zero angle.

25. The apparatus of claim 24, in which the non-zero angle is 5 degrees.

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